FINAL
TRIM AND STABILITY BOOKLET
OF
ARCADIA SUPARSHVA
(OFFICIAL NO. JMR-0024)

SEE LETTER E-210766-279820

APPROVED
FOR COMPLIANCE WITH AS MENTIONED (SEE LETTER0 REQUIREMENTS
ON BEHALF OF GOVT. OF INDIA
INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 02-Jan-2024

AS AMENDED
ON PAGE 4,9,15,20,52,53,
54,58,59,79,85,103,109

Rev.	Description	Date	Prepared	Review

Issued for : Approval

Client: **SMT. PARUL A. SHAH** Builder: **CHISTI MARINE ENGINEERING WORKS**	Project: **PONTOON** **(YARD NO. CHISTI/036)**

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Refer IRS Letter E-210766-279820 dated, January 02, 2024

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<u>FOR 400T CRAWLER CRANE</u>

<u>FOR 350T CRAWLER CRANE</u>

Refer IRS Letter F-210766-279820 dated, January 02, 2024

FOR 250T CRAWLER CRANE

FOR 150T CRAWLER CRANE

Appendix

Refer IRS Letter E-210766-279820 dated, January 02, 2024

GENERAL PARTICULARS

Name of the Vessel	ARCADIA SUPARSHVA
Client	SMT. PARUL A. SHAH
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/036
Type of Vessel	PONTOON
Official No.	JMR-0024
Class Notation	卐SU,PONTOON,**INDIAN COASTAL SERVICE** “DECK STRENGTHENED FOR 10T/M^2” “FOR CRANE OPERATION”

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1893.06 T
Lightship	447.10 T
Deadweight	1445.96 T

Refer IRS Letter E-210766-279820 dated January 02, 2024

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.

2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

3) The minimum range of stability should be:
   For L <= 100m: 20^0
   For L >= 150m: 15^0
   For intermediate length by interpolation

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta C \leq 15°$
- $GZC \leq 0,6\ GZMAX$
- $A1 \geq 0.4\ ATOT$

where:

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC, GZMAX: Defined in Fig 1

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle equal to the lesser of:

- heeling angle θR of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θF, corresponding to flooding of unprotected openings

ATOT : Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b=(Pd-Zz)/\Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z: Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM.The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

Refer IRS Letter E-210766-279820 dated January 02, 2024

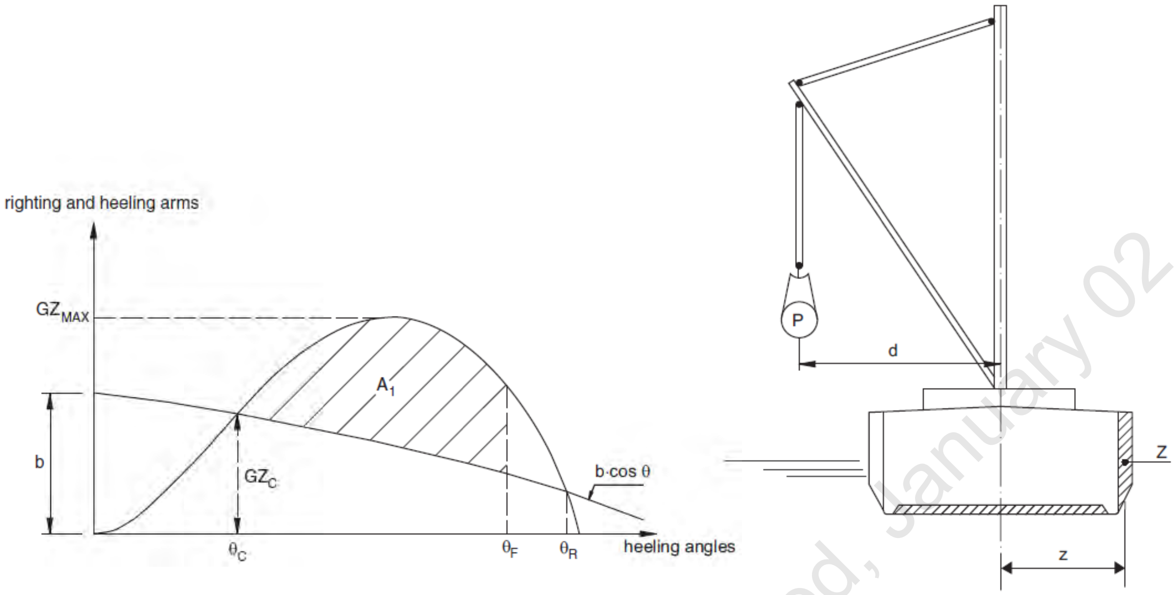


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A2 / A1 \geq 1$
- $\theta 2 - \theta 3 \geq 20°$

where:

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ1 to the heeling angle θC (see Fig 2)

A2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ2 (see Fig 2)

Refer IRS Letter E-210/66-27980 dated, January 02, 2024

A3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle $\theta 3$ (see Fig 2)

$\theta 1$: Heeling angle of equilibrium during lifting (see Fig 2)

$\theta 2$: Heeling angle corresponding to the lesser of θR and θF

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

$\theta 3$: Maximum heeling angle due to roll, at which A3 = A1, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θR : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θF : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$b = Zz \cos\theta/\Delta$

HEELING AND RIGHTING ARMS

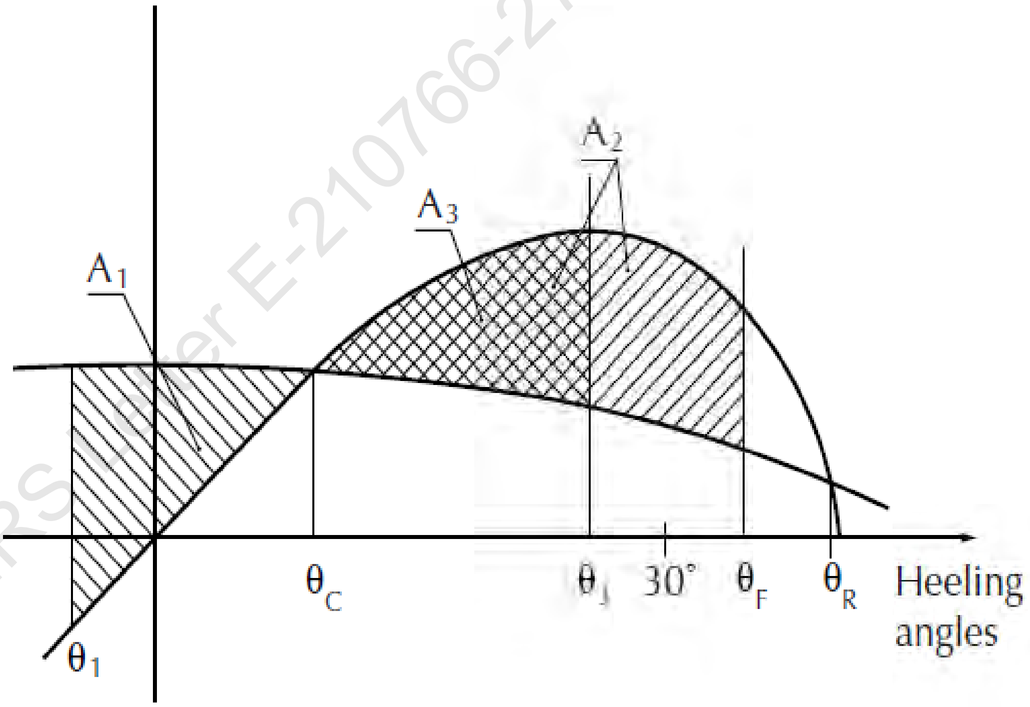


FIGURE 2: CARGO LOSS

Refer IRS letter E-210766-279820 dated, January 02, 2024

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.

2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship’s stability in various loading conditions.

3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.

4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.

5. In this booklet hydrostatic particulars and cross curves are given at following trims.

   -1.00m,-0.50m, 0.00m,0.50m, 1.00m

   These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

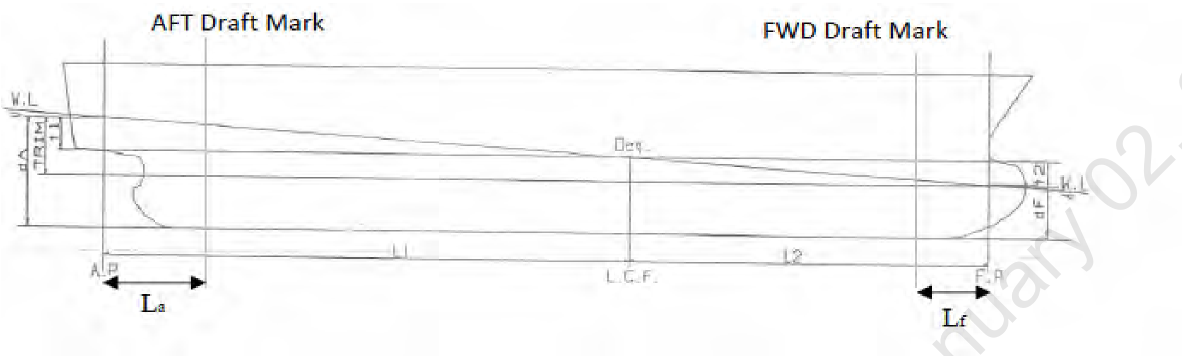
The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Note:
1. During unmanned tow minimum forward draft of 1.94m to be maintained at all times for compliance with slamming requirements or else it is recommended that unmanned tow of vessel should be carried out in fair weather and against favorable weather forecast.

2.Deck strength is examined considering Deck loading of 25 T/m² in way of crane operating area & 10 T/m² in way of other deck area as marked in Structural Plan. The selected crane's ground pressure as well as Deck cargo shall not exceed above considered values for any orientation and its stated capacity.

Refer RS Letter E-210766-79820 dated, January 02, 2024

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

b) Longitudinal centre of buoyancy, LCB

c) Displacement, Δ

d) Moment to change trim, MCT1cm

e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (LCB-LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * Trim}{LBP.} + T\ (extreme)$$

$$T_{FP} = T_{AP} - Trim$$

Refer IRS Letter E-210766-270820 dated, January 02, 2024

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{Trim * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{Trim * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

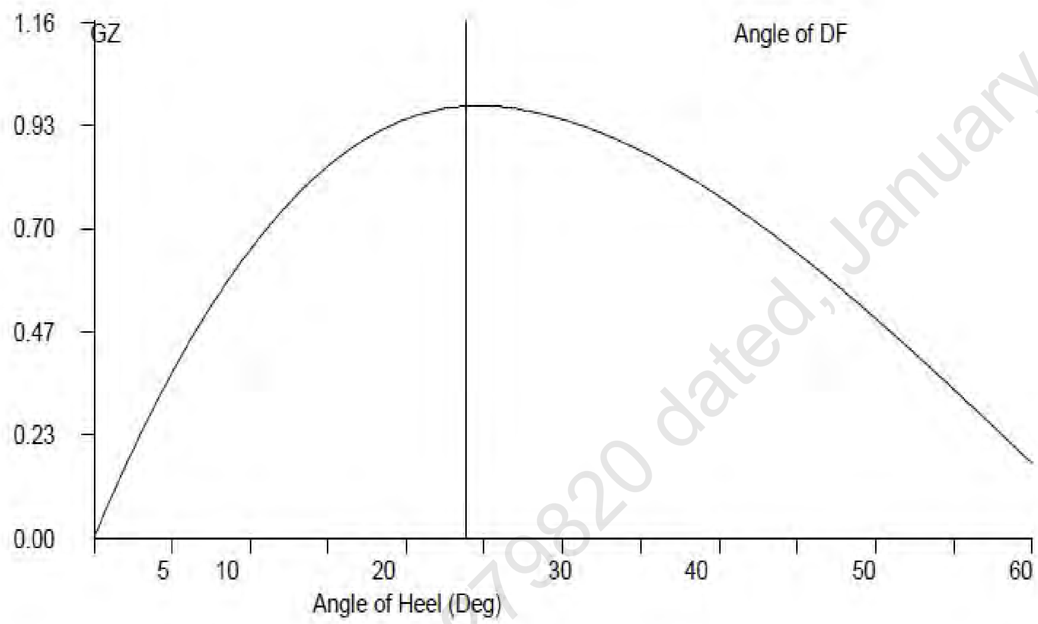
GZ Calculation:

1) GZ = KN - KGo Sin (theta)
2) Area under 0-10 = 10*0.01745*[GZ0 + GZ10] / 2
3) Area under 10-20 = 10*0.01745*[GZ10 + GZ20] / 2
4) Area under 20-30 = 10*0.01745*[GZ20 + GZ30] / 2
5) Area under 30-40 = 10*0.01745*[GZ30 + GZ40] / 2
6) Area under 40-50 = 10*0.01745*[GZ40 + GZ50] / 2
7) Area under 50-60 = 10*0.01745*[GZ50 + GZ60] / 2
8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



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Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	1081.97	Tonnes
LCG	24.250	M Fwd of AP
VCG (KG)	3.938	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

Free Surface moment for a Tank = Moment of Inertia (M^4) for a Tank x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

Trim = (LCB-LCG) X displacement / (MCT x 100) + Trim*

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Refer IRS Letter E-210766-27920 dated, January 02, 2024

Condition 1 : Lightship + Crane + Ballast

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
06_WBTK.C	132.36	1.025	135.67	42.251	5732.19	0.000	0.00	1.499	203.37	135.61
07_WBTK.P	48.39	1.025	49.60	47.660	2363.94	-5.971	-296.16	1.880	93.25	72.32
07_WBTK.S	48.39	1.025	49.60	47.660	2363.94	5.971	296.16	1.880	93.25	72.32
TOTAL TANK			234.87	44.536	10460.07	0.000	0.00	1.660	389.87	280.25

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			447.10	25.000	11,177.50	0.000	0.00	3.000	1,341.30
CRANE			400.00	11.500	4,600.00	0.000	0.00	6.325	2,530.00
TOTAL FIXED WEIGHT			847.10	18.625	15,777.50	0.000	0.00	4.570	3,871.30

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			234.870	44.536	10460.065	0.000	0.000	1.660	389.865
TOTAL FIXED WEIGHT			847.100	18.625	15777.500	0.000	0.000	4.570	3871.300
TOTAL WEIGHT			1081.970	24.250	26237.565	0.000	0.000	3.938	4261.165

Refer IRS Letter E-210766-27080 dated January 02, 2024

Condition 1 : Lightship + Crane + Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.259	m
VCGc	4.197	m

From the MaxVCG table,

MAX VCG DOES NOT COVER CRANE OPERATIONS/ LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION

Displacement	1081.97	T
Allowable Max VCG	14.620	m

$\boxed{\text{VCGc < Maxvcg}}$ PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1081.970	T
Draft ,Tm @ 25.0f =	1.316	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	33.105	T-m/cm
KMT =	22.563	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.245 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid 1.316 m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2
1.193 m

Draft@0(AP),Taft =Tm + Trim/2
1.439 m

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Condition 1 : Lightship + Crane + Ballast

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	33.105	T-m	Draft (Mean)	1.316 m
LCB	25.000	m	Draft AP (USK)	1.439 m
			Draft FP (USK)	1.193 m
Metacentric Height				
KMt	22.563	m		
KG	3.938	m		
KG0	4.197	m		
GMt	18.366	m	**Trim**	0.245 m

The condition shown above is first worked
using Even Keel hydrostatic perticulars
Program then interpolates the hydrostatic perticulars
from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	32.280	T-m	Draft (Mean)	1.315 m
LCB	24.233	m	Draft AP (USK)	1.441 m
			Draft FP (USK)	1.190 m
Metacentric Height				
KMt	22.574	m		
KG	3.938	m		
KG0	4.197	m		
GMt	18.377	m	**Trim**	0.252 m

Trim is calculaed as follows:
Trim = (LCB - LCG) X displacement / (MCT X 100) + Trim*
Where Trim* is trim at which hydrostatics were calculated.

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Condition 1 : Lightship + Crane + Ballast

Intact Stability (Primary Criterion)

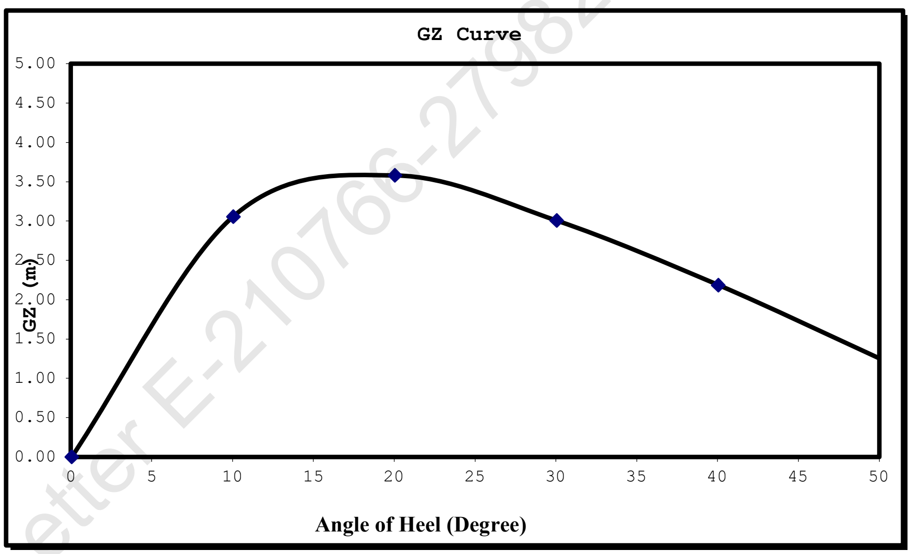
			Minimum Required
Displacement	1081.97	T	
KG0	4.197	M	
GMt	18.377	M	0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	0.000	3.788	5.018	5.107	4.885	4.468
GZ (M)	0.000	3.056	3.579	3.005	2.184	1.251

	Computed values		Minimum Required	
Residual Area from abs 0 deg to MAxRA	0.6908	m-rad	0.080	m-rad
Angle from Equilibrium to RAZero or flood	LARGE	m-rad	20.00	Deg.
Angle from Equilibrium to 50% DK Imm Angle	5.05	m-rad	0.00	Deg.

Plot GZ values against the angles of heel as above.
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates.Join the point to the origin.
This is the Tangent to the GZ Curve at the Origin.Find the Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding
If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Refer IRS Letter E-210766-279820 dated, January 02, 2024

Worked Out Example Contd.

Mean Draft	1.315	M
LCF from AP	24.766	M
LCB from AP	24.233	M
MCT	32.280	T-M/Cm
Trim	0.252	M
Draft (AP)	1.441	M
Draft (FP)	1.190	M
KG	3.938	M
KGo	4.197	M
GoMt	18.377	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Refer IRS Letter E-210766-279820 dated, January 02, 2024

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			447.100	25.000	11,177.500	0.000	0.000	3.000	1,341.300
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK									
TOTAL FIXED WEIGHT									
TOTAL WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

m

VCGc m

From the MaxVCG table,

#MAX VCG DOES NOT COVER CRANE OPERATIONS/ LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED

Displacement T

Allowable Max VCG m

$\boxed{\text{VCGc < Maxvcg}}$

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact

Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.0f = m

LCB = m

LCF = m

MCT = T-m/cm

KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)

m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2

m

Draft@0(AP),Taft =Tm + Trim/2

m

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Refer IRS Letter E-210766-279820 dated January 02, 2024

Metric Conversion Table

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	TO OBTAIN	**TO CONVERT FROM**	**MULTIPLY**

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Refer IRS Letter E-21076-279020 dated, January 02, 2024

Angle of Downflooding

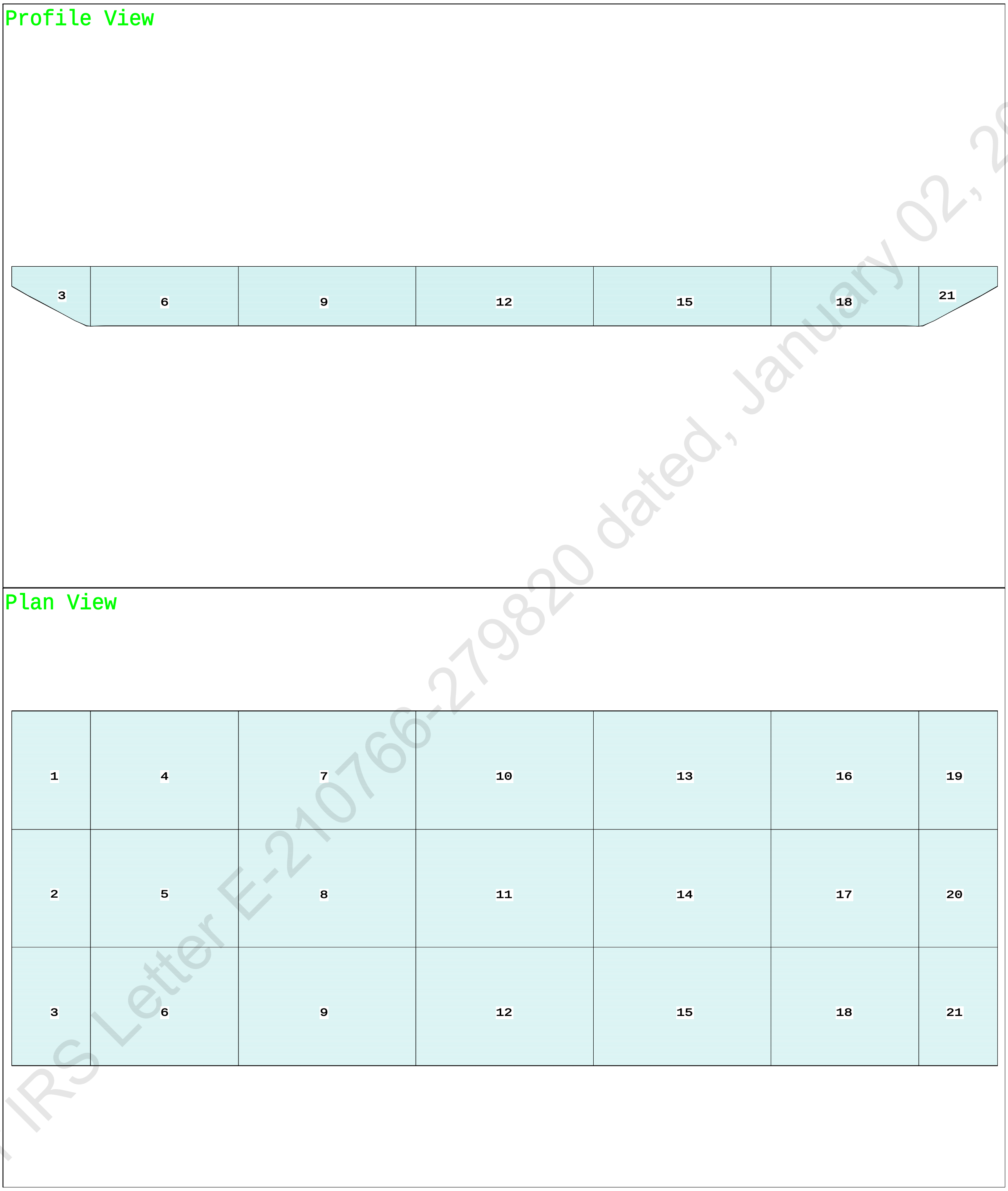
1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill off this opening is a `potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
   If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

   As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

Refer IRS Letter E-21076-27820 dated, January 02, 2024

01-08-20 21:16:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

CAPACITY PLAN AND TABLE



Tanks

1 01_WBTK.P	4 02_WBTK.P	8 03_WBTK.C	12 04_WBTK.S	16 06_WBTK.P	20 07_WBTK.C
2 01_WBTK.C	5 02_WBTK.C	9 03_WBTK.S	13 05_WBTK.P	17 06_WBTK.C	21 07_WBTK.S
3 01_WBTK.S	6 02_WBTK.S	10 04_WBTK.P	14 05_WBTK.C	18 06_WBTK.S	
	7 03_WBTK.P	11 04_WBTK.C	15 05_WBTK.S	19 07_WBTK.P	

Refer IRS Letter E-210766-279820 dated, January 02, 2024

01-08-20 21:16:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

TANK STATUS
Trim: zero, Heel: zero

Part	Cu.M.	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
01_WBTK.P	48.4	1.025	49.60	2.340f	5.971p	1.880	72.32*
01_WBTK.C	48.9	1.025	50.11	2.336f	0.000	1.872	72.33*
01_WBTK.S	48.4	1.025	49.60	2.340f	5.971s	1.880	72.32*
02_WBTK.P	131.4	1.025	134.72	7.749f	5.980p	1.509	135.61*
02_WBTK.C	132.4	1.025	135.67	7.749f	0.000	1.499	135.61*
02_WBTK.S	131.4	1.025	134.72	7.749f	5.980s	1.509	135.61*
03_WBTK.P	157.7	1.025	161.60	16.000f	5.980p	1.509	162.73*
03_WBTK.C	158.8	1.025	162.73	16.000f	0.000	1.500	162.73*
03_WBTK.S	157.7	1.025	161.60	16.000f	5.980s	1.509	162.73*
04_WBTK.P	157.7	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	158.8	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	157.7	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.P	157.7	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.C	158.8	1.025	162.73	34.000f	0.000	1.500	162.73*
05_WBTK.S	157.7	1.025	161.60	34.000f	5.980s	1.509	162.73*
06_WBTK.P	131.4	1.025	134.72	42.251f	5.980p	1.509	135.61*
06_WBTK.C	132.4	1.025	135.67	42.251f	0.000	1.499	135.61*
06_WBTK.S	131.4	1.025	134.72	42.251f	5.980s	1.509	135.61*
07_WBTK.P	48.4	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	48.9	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	48.4	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			2,566.63	25.000f	0.000	1.549	2712.14

Distances in METERS.--------------------------------------Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Refer IRS Letter E-210766-279820 dated January 02, 2024

Refer IRS Letter E-210766-279820 dated, January 02, 2024

HYDROSTATIC PARTICULARS

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

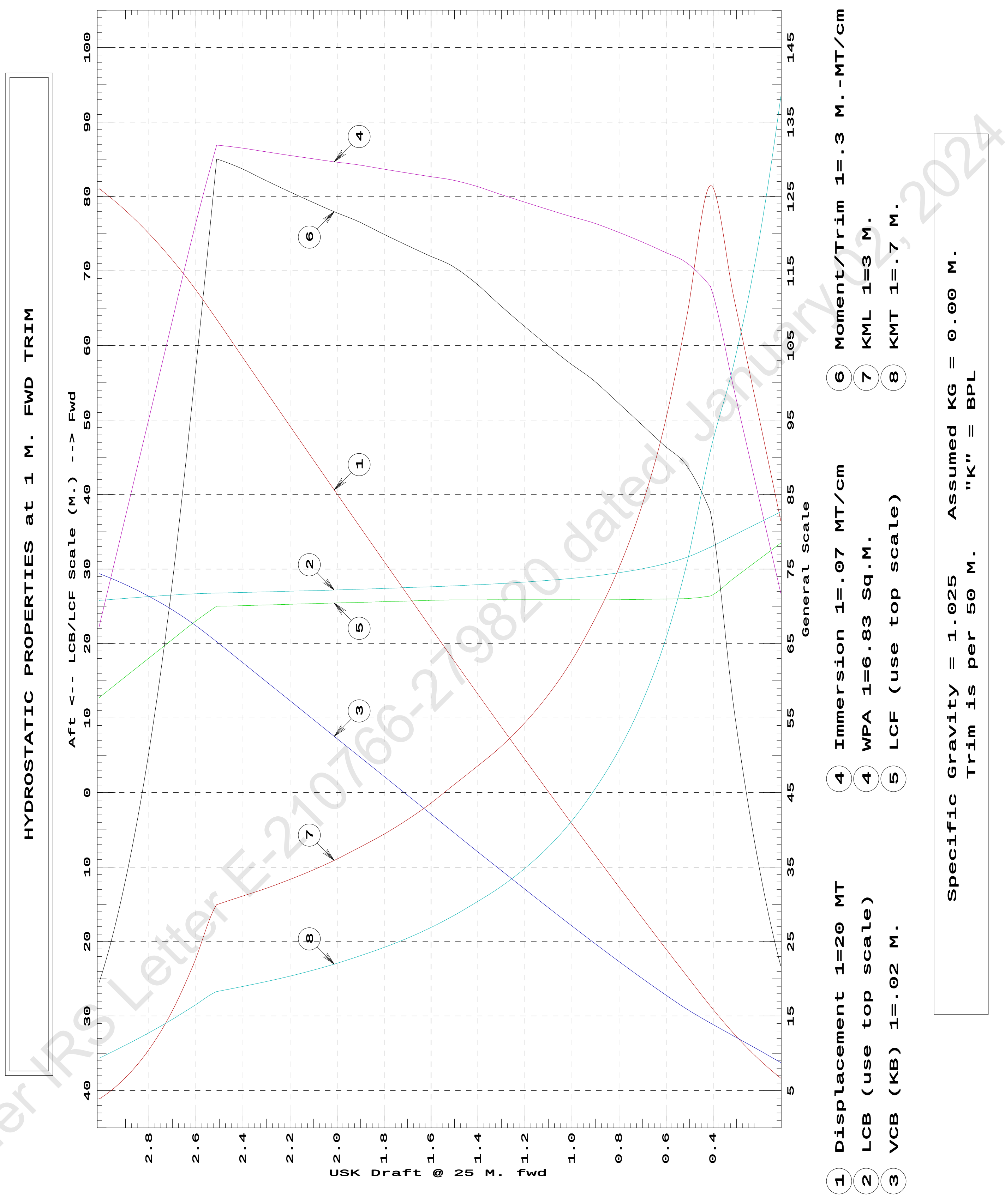
HYDROSTATIC PROPERTIES
Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	133.29	37.689f	0.175	5.02	33.490f	6.52	244.36	96.881
0.212	188.30	36.107f	0.211	5.98	31.124f	10.90	289.46	82.456
0.312	252.92	34.523f	0.246	6.94	28.751f	16.92	334.36	71.772
0.412	327.16	32.938f	0.281	7.91	26.375f	24.82	379.30	63.564
0.512	407.76	31.583f	0.319	8.13	26.034f	26.65	326.70	53.012
0.612	489.54	30.651f	0.361	8.23	25.975f	27.51	280.91	45.226
0.712	572.35	29.971f	0.406	8.33	25.923f	28.38	247.92	39.491
0.812	656.10	29.452f	0.452	8.42	25.886f	29.26	222.97	35.068
0.912	740.70	29.044f	0.499	8.50	25.870f	30.12	203.29	31.535
1.012	826.02	28.717f	0.548	8.57	25.880f	30.81	186.46	28.540
1.112	912.01	28.450f	0.597	8.63	25.878f	31.57	173.02	26.147
1.212	998.69	28.226f	0.646	8.70	25.875f	32.35	161.94	24.171
1.312	1,086.07	28.037f	0.696	8.78	25.872f	33.17	152.69	22.519
1.412	1,174.15	27.875f	0.746	8.85	25.869f	34.03	144.90	21.124
1.512	1,262.91	27.735f	0.797	8.91	25.870f	34.71	137.41	19.834
1.612	1,352.12	27.609f	0.848	8.94	25.783f	35.14	129.91	18.700
1.712	1,441.67	27.493f	0.899	8.97	25.694f	35.58	123.36	17.712
1.812	1,531.58	27.384f	0.950	9.01	25.603f	36.03	117.61	16.848
1.912	1,621.83	27.283f	1.000	9.05	25.508f	36.51	112.52	16.088
2.012	1,712.42	27.187f	1.051	9.08	25.436f	36.88	107.68	15.374
2.112	1,803.32	27.097f	1.102	9.11	25.351f	37.30	103.41	14.750
2.212	1,894.54	27.011f	1.153	9.14	25.263f	37.74	99.58	14.190
2.312	1,986.08	26.928f	1.203	9.17	25.174f	38.19	96.11	13.689
2.412	2,077.95	26.849f	1.254	9.21	25.081f	38.65	92.98	13.238
2.512	2,170.15	26.772f	1.305	9.23	25.013f	39.01	89.86	12.803
2.612	2,258.72	26.664f	1.353	8.40	22.760f	29.59	65.48	11.408
2.712	2,338.10	26.490f	1.395	7.48	20.257f	21.09	45.09	10.043
2.812	2,408.27	26.272f	1.431	6.56	17.755f	14.46	30.01	8.792
2.912	2,469.20	26.032f	1.462	5.63	15.252f	9.46	19.15	7.632
3.012	2,520.92	25.786f	1.488	4.71	12.750f	5.86	11.62	6.543

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON



Refer IRS Letter E-210766-279820 dated January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

HYDROSTATIC PROPERTIES
Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

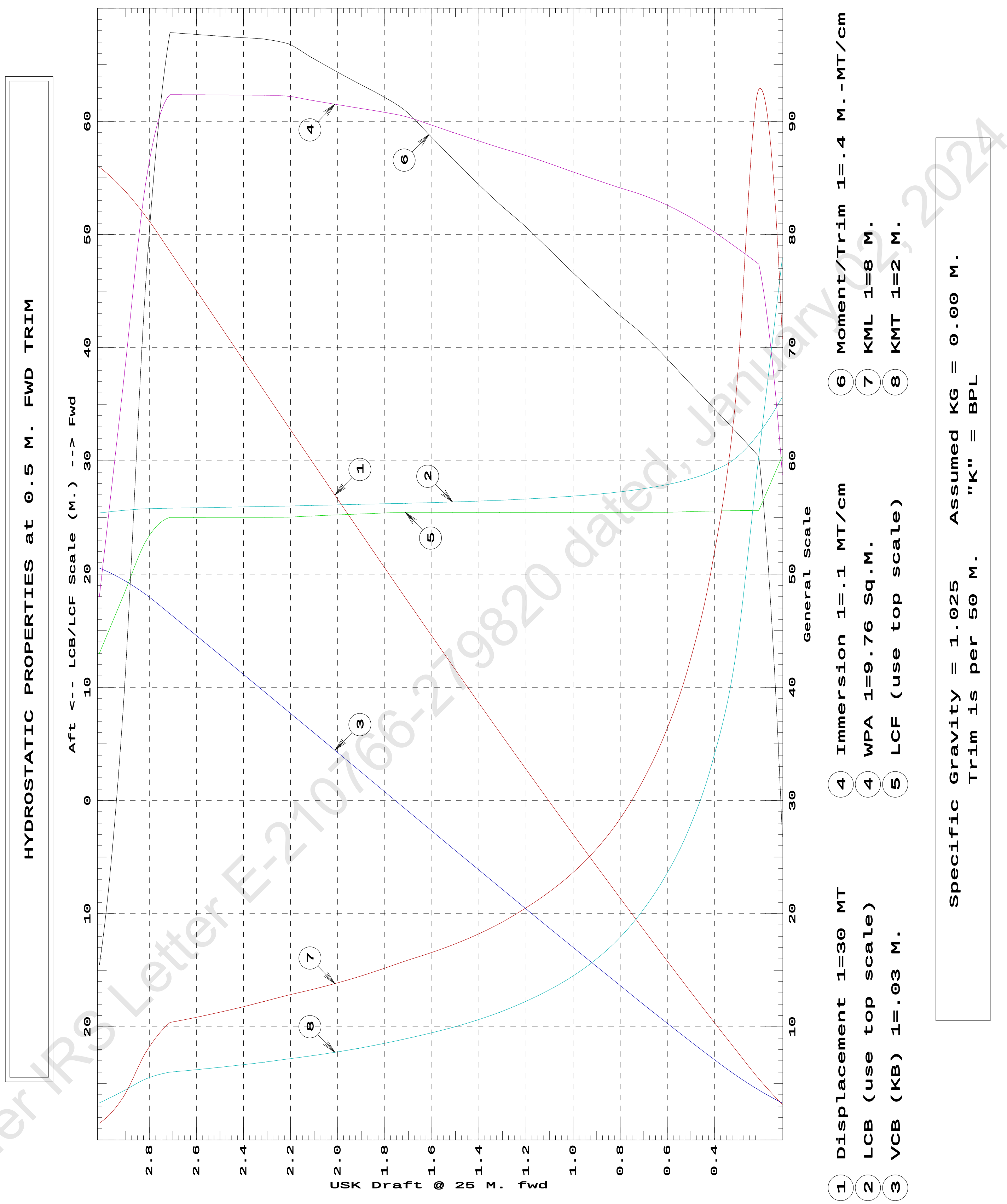
Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	94.61	35.657f	0.098	5.88	30.439f	10.72	566.32	155.843
0.212	162.65	32.417f	0.132	7.74	25.617f	24.15	742.47	120.816
0.312	240.79	30.208f	0.173	7.89	25.602f	25.06	520.34	85.097
0.412	320.44	29.059f	0.218	8.04	25.566f	25.94	404.70	66.389
0.512	401.45	28.349f	0.266	8.16	25.509f	26.81	333.87	54.610
0.612	483.61	27.862f	0.315	8.27	25.457f	27.70	286.38	46.395
0.712	566.72	27.508f	0.365	8.35	25.449f	28.52	251.60	40.187
0.812	650.57	27.242f	0.415	8.42	25.440f	29.21	224.49	35.343
0.912	735.11	27.034f	0.466	8.49	25.439f	29.96	203.76	31.632
1.012	820.35	26.868f	0.517	8.56	25.438f	30.74	187.34	28.689
1.112	906.31	26.733f	0.567	8.63	25.436f	31.56	174.08	26.309
1.212	992.99	26.620f	0.619	8.71	25.451f	32.35	162.89	24.309
1.312	1,080.33	26.524f	0.670	8.77	25.439f	33.07	153.05	22.576
1.412	1,168.34	26.443f	0.722	8.83	25.438f	33.86	144.89	21.132
1.512	1,257.02	26.372f	0.773	8.90	25.437f	34.68	137.92	19.898
1.612	1,346.39	26.310f	0.825	8.97	25.435f	35.53	131.95	18.836
1.712	1,436.46	26.255f	0.877	9.04	25.448f	36.37	126.58	17.889
1.812	1,527.11	26.206f	0.929	9.08	25.389f	36.90	120.81	16.991
1.912	1,618.11	26.157f	0.981	9.12	25.304f	37.34	115.37	16.198
2.012	1,709.44	26.109f	1.033	9.15	25.217f	37.79	110.53	15.496
2.112	1,801.11	26.062f	1.085	9.19	25.127f	38.26	106.21	14.872
2.212	1,893.13	26.014f	1.137	9.22	25.034f	38.75	102.33	14.317
2.312	1,985.42	25.968f	1.189	9.23	25.012f	38.91	97.97	13.759
2.412	2,077.73	25.925f	1.240	9.23	25.009f	38.96	93.75	13.255
2.512	2,170.06	25.886f	1.292	9.23	25.006f	39.02	89.90	12.799
2.612	2,262.39	25.850f	1.343	9.23	25.004f	39.08	86.36	12.384
2.712	2,354.73	25.817f	1.394	9.24	25.001f	39.14	83.10	12.006
2.812	2,445.61	25.772f	1.445	8.50	23.000f	30.67	62.71	10.846
2.912	2,521.33	25.617f	1.485	6.65	18.000f	15.11	29.97	8.622
3.012	2,578.59	25.395f	1.516	4.80	13.000f	6.19	12.01	6.555

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Refer IRS Letter E-21076 279820 dated January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

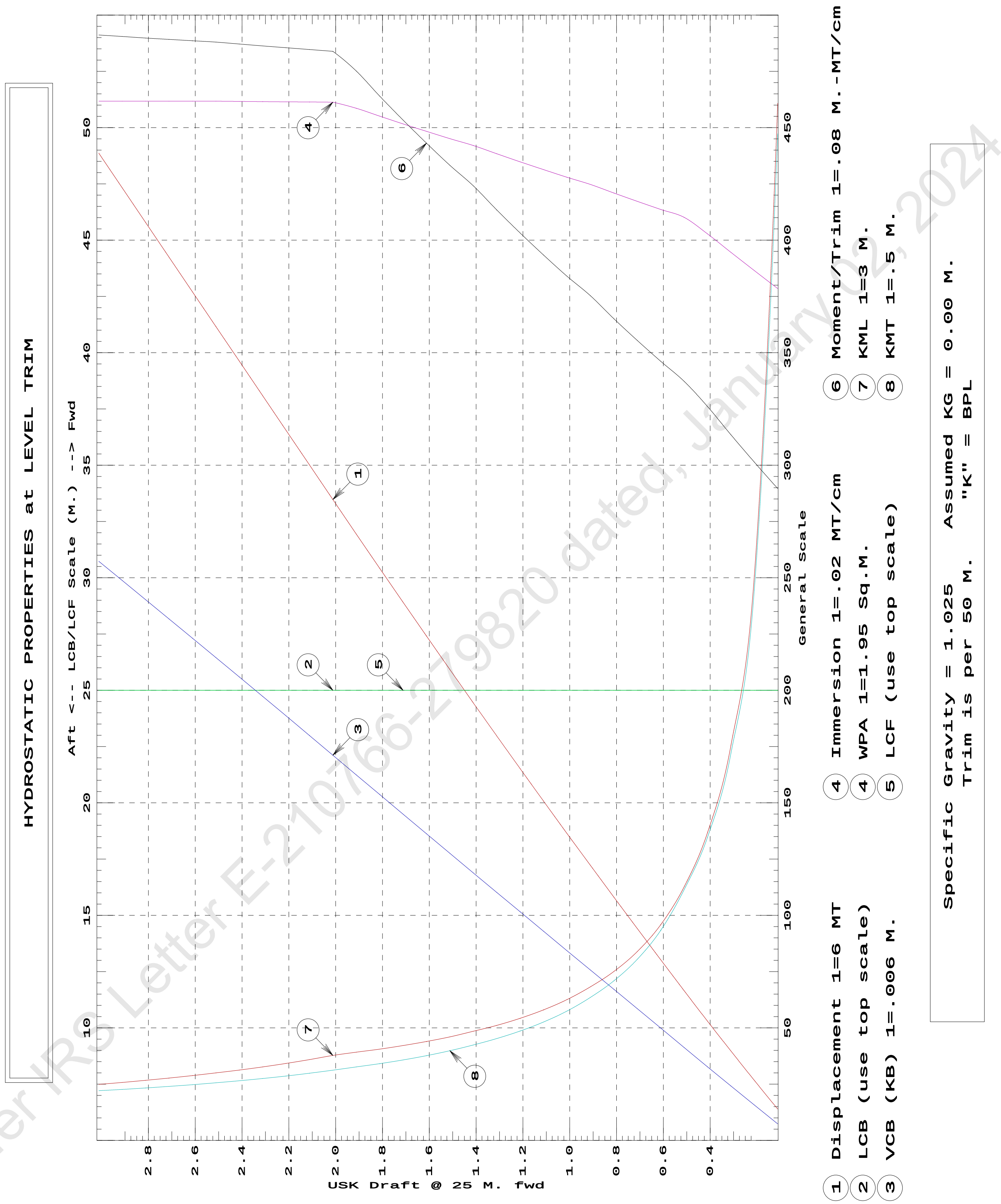
Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	83.79	25.000f	0.044	7.57	25.000f	23.17	1382.58	223.643
0.212	160.29	25.000f	0.095	7.73	25.000f	24.11	752.15	122.246
0.312	238.38	25.000f	0.145	7.89	25.000f	25.08	526.13	85.934
0.412	318.09	25.000f	0.196	8.06	25.000f	26.09	410.04	67.317
0.512	399.40	25.000f	0.248	8.20	25.000f	26.98	337.80	55.685
0.612	481.78	25.000f	0.300	8.27	25.000f	27.70	287.43	46.641
0.712	564.87	25.000f	0.351	8.35	25.000f	28.44	251.70	40.223
0.812	648.69	25.000f	0.403	8.42	25.000f	29.21	225.14	35.449
0.912	733.24	25.000f	0.454	8.50	25.000f	30.02	204.72	31.774
1.012	818.51	25.000f	0.506	8.56	25.000f	30.72	187.65	28.736
1.112	904.44	25.000f	0.558	8.63	25.000f	31.47	173.99	26.304
1.212	991.06	25.000f	0.610	8.70	25.000f	32.26	162.74	24.297
1.312	1,078.37	25.000f	0.661	8.77	25.000f	33.07	153.35	22.621
1.412	1,166.40	25.000f	0.713	8.84	25.000f	33.93	145.46	21.208
1.512	1,255.11	25.000f	0.765	8.90	25.000f	34.66	138.07	19.915
1.612	1,344.46	25.000f	0.818	8.97	25.000f	35.45	131.83	18.822
1.712	1,434.46	25.000f	0.870	9.03	25.000f	36.27	126.41	17.872
1.812	1,525.13	25.000f	0.922	9.10	25.000f	37.12	121.69	17.042
1.912	1,616.48	25.000f	0.974	9.17	25.000f	38.01	117.57	16.315
2.012	1,708.51	25.000f	1.027	9.23	25.000f	38.71	113.30	15.610
2.112	1,800.77	25.000f	1.079	9.23	25.000f	38.78	107.66	14.924
2.212	1,893.06	25.000f	1.132	9.23	25.000f	38.84	102.58	14.309
2.312	1,985.36	25.000f	1.184	9.23	25.000f	38.90	97.98	13.756
2.412	2,077.68	25.000f	1.236	9.23	25.000f	38.97	93.78	13.257
2.512	2,170.03	25.000f	1.287	9.24	25.000f	39.04	89.95	12.804
2.612	2,262.38	25.000f	1.339	9.24	25.000f	39.09	86.38	12.385
2.712	2,354.73	25.000f	1.390	9.24	25.000f	39.14	83.10	12.003
2.812	2,447.09	25.000f	1.442	9.24	25.000f	39.19	80.07	11.654
2.912	2,539.44	25.000f	1.493	9.24	25.000f	39.24	77.26	11.334
3.012	2,631.79	25.000f	1.544	9.24	25.000f	39.29	74.65	11.040

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Refer IRS Letter E-2107... 279820 dated January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

HYDROSTATIC PROPERTIES
Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

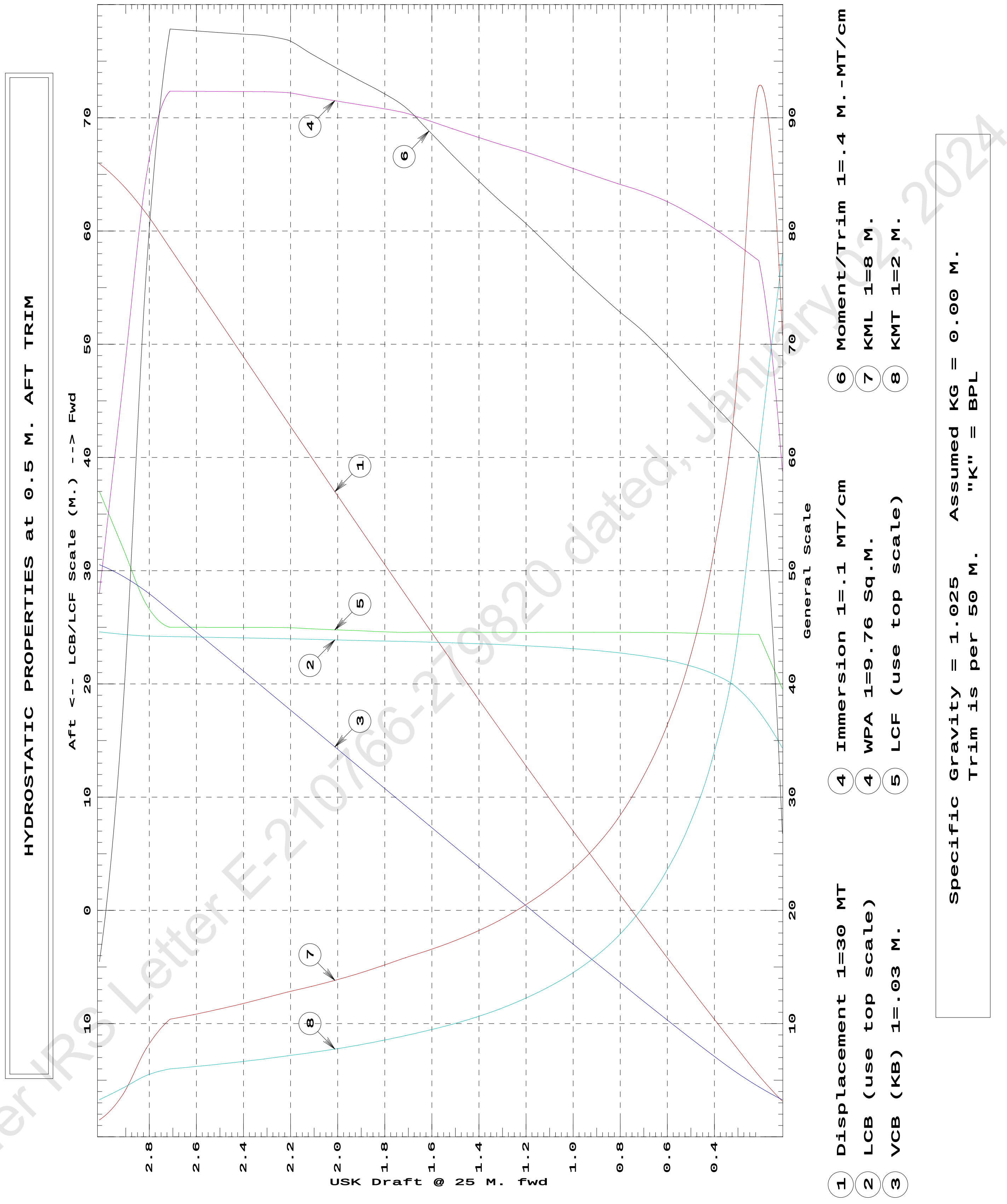
Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	94.61	14.343f	0.098	5.88	19.561f	10.72	566.32	155.836
0.212	162.65	17.583f	0.132	7.74	24.383f	24.15	742.47	120.811
0.312	240.79	19.792f	0.173	7.89	24.398f	25.06	520.34	85.098
0.412	320.44	20.941f	0.218	8.04	24.434f	25.94	404.70	66.390
0.512	401.45	21.651f	0.266	8.16	24.491f	26.81	333.87	54.611
0.612	483.61	22.138f	0.315	8.27	24.543f	27.70	286.38	46.395
0.712	566.72	22.492f	0.365	8.35	24.551f	28.52	251.60	40.186
0.812	650.57	22.758f	0.415	8.42	24.560f	29.21	224.49	35.343
0.912	735.11	22.966f	0.466	8.49	24.561f	29.96	203.76	31.632
1.012	820.35	23.132f	0.517	8.56	24.562f	30.74	187.34	28.689
1.112	906.31	23.267f	0.567	8.63	24.564f	31.56	174.08	26.309
1.212	992.99	23.380f	0.619	8.71	24.549f	32.35	162.89	24.308
1.312	1,080.33	23.476f	0.670	8.77	24.561f	33.07	153.05	22.576
1.412	1,168.34	23.557f	0.722	8.83	24.562f	33.86	144.89	21.132
1.512	1,257.02	23.628f	0.773	8.90	24.563f	34.68	137.92	19.898
1.612	1,346.39	23.690f	0.825	8.97	24.565f	35.53	131.95	18.836
1.712	1,436.46	23.745f	0.877	9.04	24.552f	36.37	126.58	17.889
1.812	1,527.11	23.794f	0.929	9.08	24.611f	36.90	120.81	16.991
1.912	1,618.10	23.843f	0.981	9.12	24.696f	37.34	115.37	16.198
2.012	1,709.44	23.891f	1.033	9.15	24.783f	37.79	110.53	15.496
2.112	1,801.11	23.938f	1.085	9.19	24.873f	38.26	106.21	14.872
2.212	1,893.13	23.986f	1.137	9.22	24.966f	38.75	102.33	14.317
2.312	1,985.42	24.032f	1.189	9.23	24.988f	38.91	97.97	13.758
2.412	2,077.73	24.075f	1.240	9.23	24.991f	38.96	93.75	13.255
2.512	2,170.06	24.114f	1.292	9.23	24.994f	39.02	89.90	12.799
2.612	2,262.39	24.150f	1.343	9.23	24.996f	39.08	86.36	12.384
2.712	2,354.73	24.183f	1.394	9.24	24.999f	39.14	83.10	12.006
2.812	2,445.61	24.228f	1.445	8.50	27.000f	30.67	62.71	10.846
2.912	2,521.33	24.383f	1.485	6.65	32.000f	15.11	29.97	8.622
3.012	2,578.59	24.605f	1.516	4.80	37.000f	6.19	12.01	6.555

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Refer IRS Letter E-210768-279820 dated January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

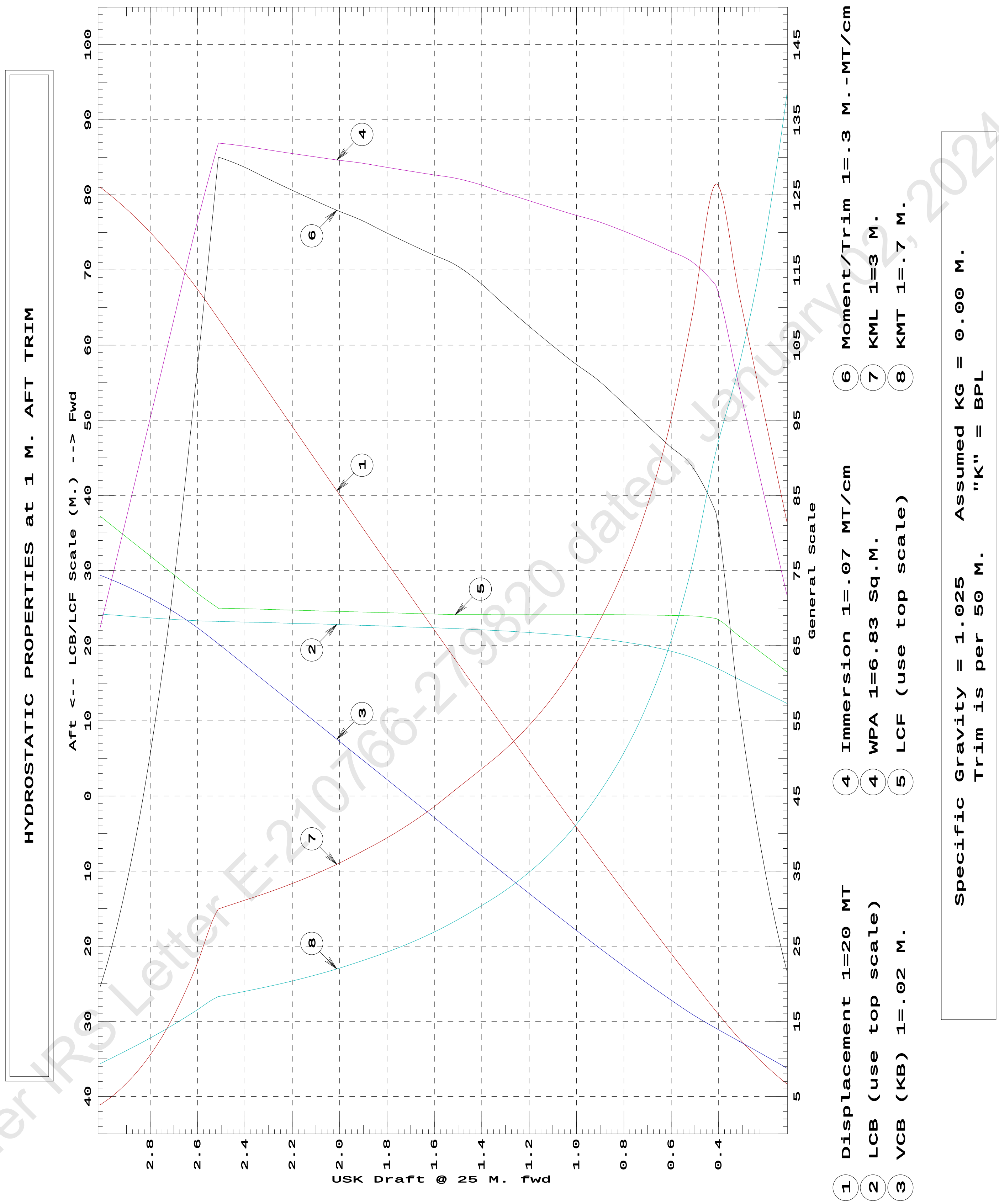
HYDROSTATIC PROPERTIES
Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	133.29	12.311f	0.175	5.02	16.510f	6.52	244.36	96.880
0.212	188.30	13.893f	0.211	5.98	18.876f	10.90	289.46	82.454
0.312	252.92	15.477f	0.246	6.94	21.249f	16.92	334.36	71.771
0.412	327.16	17.062f	0.281	7.91	23.625f	24.82	379.31	63.562
0.512	407.76	18.417f	0.319	8.13	23.966f	26.65	326.70	53.012
0.612	489.54	19.349f	0.361	8.23	24.026f	27.51	280.91	45.226
0.712	572.35	20.029f	0.406	8.33	24.077f	28.38	247.92	39.491
0.812	656.10	20.548f	0.452	8.42	24.114f	29.26	222.97	35.069
0.912	740.70	20.956f	0.499	8.50	24.130f	30.12	203.29	31.535
1.012	826.02	21.283f	0.548	8.57	24.120f	30.81	186.46	28.540
1.112	912.01	21.550f	0.597	8.63	24.122f	31.57	173.02	26.147
1.212	998.69	21.774f	0.646	8.70	24.125f	32.35	161.94	24.171
1.312	1,086.07	21.963f	0.696	8.78	24.128f	33.17	152.69	22.519
1.412	1,174.15	22.125f	0.746	8.85	24.131f	34.03	144.90	21.125
1.512	1,262.91	22.265f	0.797	8.91	24.130f	34.71	137.41	19.834
1.612	1,352.12	22.391f	0.848	8.94	24.217f	35.14	129.91	18.700
1.712	1,441.67	22.507f	0.899	8.97	24.306f	35.58	123.36	17.712
1.812	1,531.58	22.616f	0.950	9.01	24.397f	36.03	117.61	16.848
1.912	1,621.83	22.717f	1.000	9.05	24.492f	36.51	112.52	16.088
2.012	1,712.42	22.813f	1.051	9.08	24.564f	36.88	107.68	15.374
2.112	1,803.32	22.903f	1.102	9.11	24.649f	37.30	103.41	14.749
2.212	1,894.54	22.989f	1.153	9.14	24.737f	37.74	99.58	14.190
2.312	1,986.08	23.072f	1.203	9.17	24.826f	38.19	96.11	13.689
2.412	2,077.95	23.151f	1.254	9.21	24.919f	38.65	92.98	13.238
2.512	2,170.15	23.228f	1.305	9.23	24.987f	39.01	89.86	12.803
2.612	2,258.72	23.336f	1.353	8.40	27.240f	29.59	65.48	11.408
2.712	2,338.10	23.510f	1.395	7.48	29.743f	21.09	45.09	10.042
2.812	2,408.27	23.728f	1.431	6.56	32.245f	14.46	30.01	8.792
2.912	2,469.21	23.968f	1.462	5.63	34.748f	9.46	19.15	7.632
3.012	2,520.92	24.214f	1.488	4.71	37.250f	5.86	11.62	6.543

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

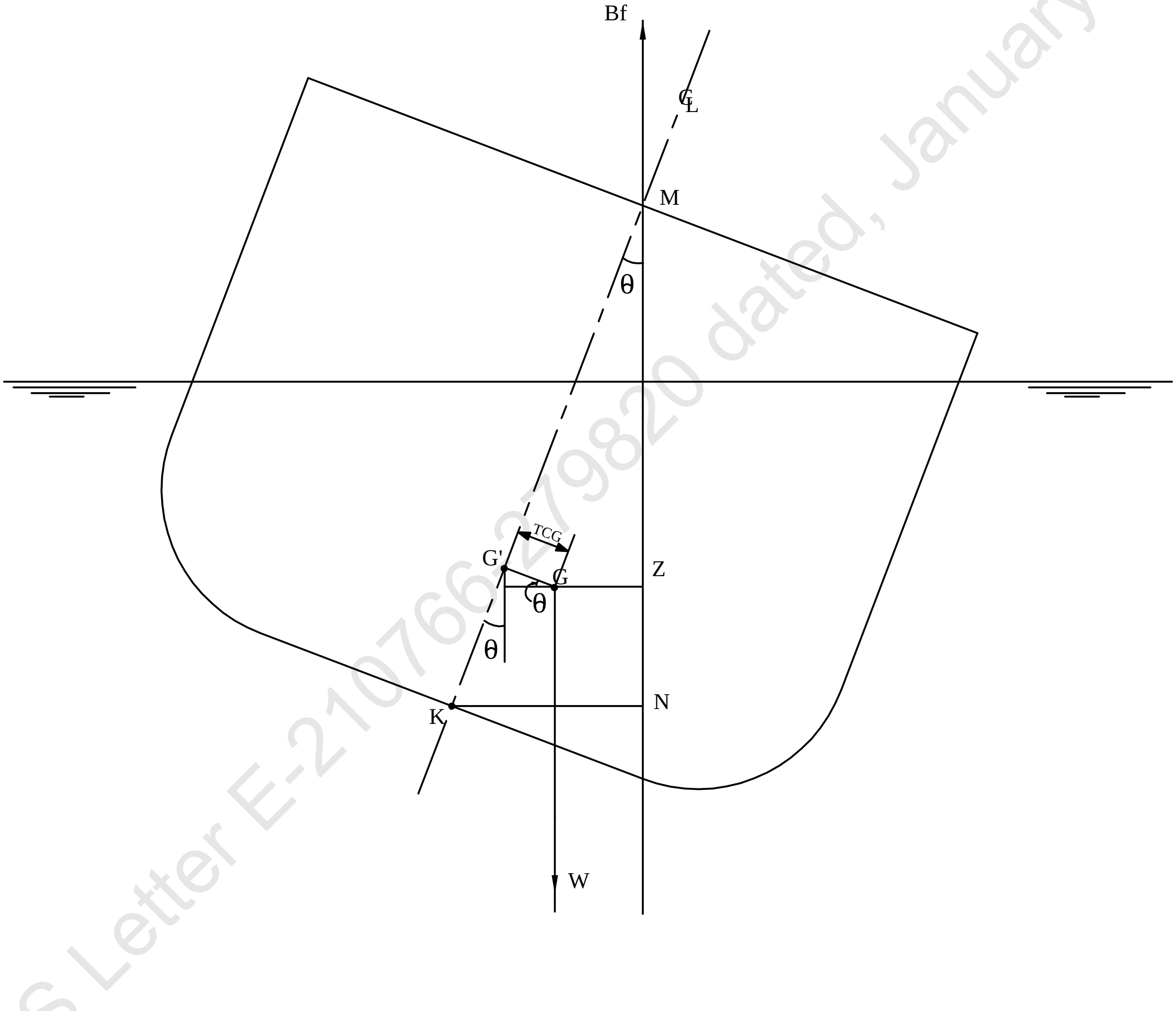
Draft is from USK.

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated January 02, 2024

KN CONCEPT



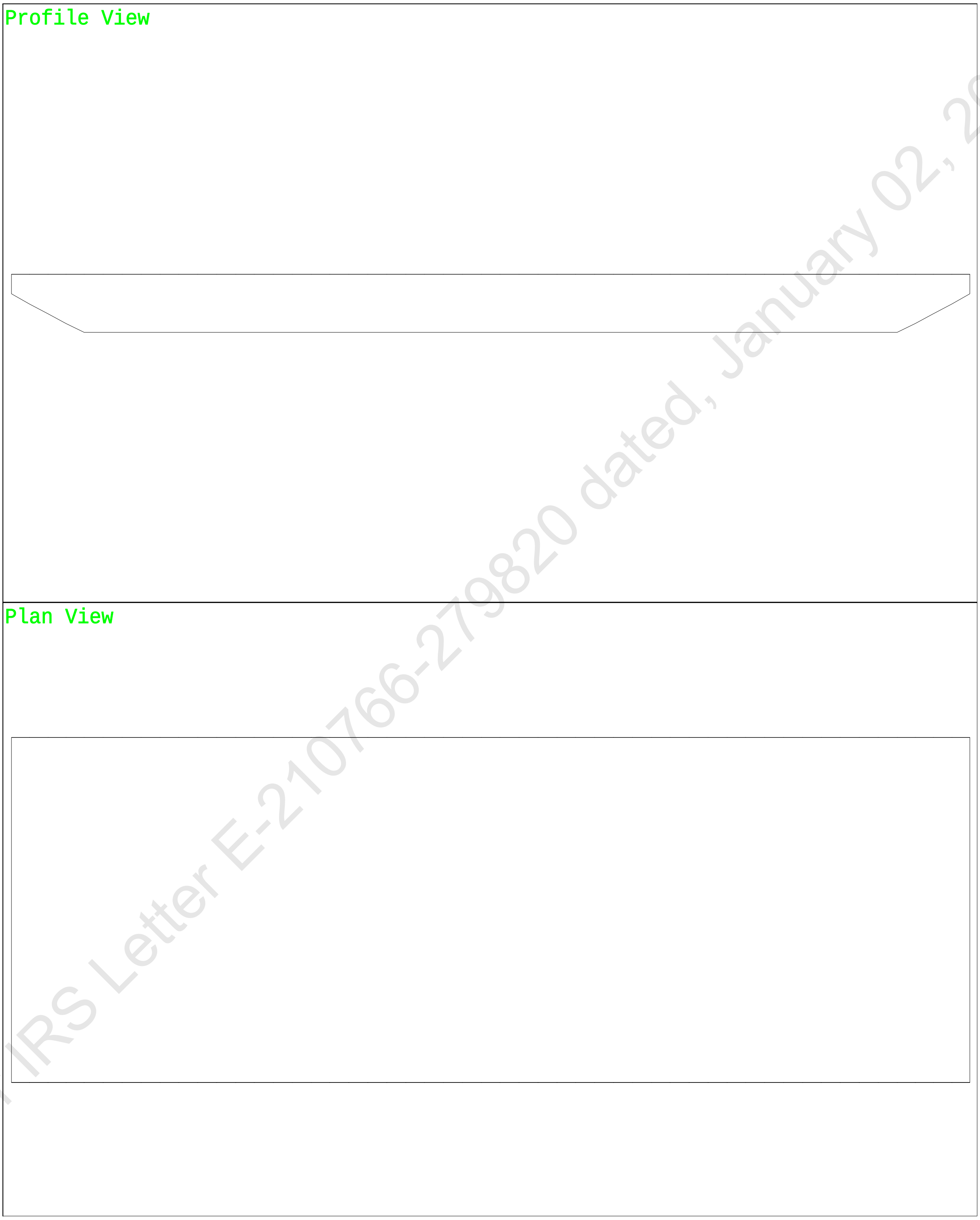
$$GZ = KN - KG' \sin\theta - TCG \cos\theta$$

Refer IRS Letter E-2107662/79820 dated, January 02, 2024

01-08-20 21:16:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

VOLUME USED FOR KN COMPUTATION

Condition Graphic



Refer IRS Letter E-210766-279820 dated, January 02, 2024

Refer IRS Letter E-210766-279820 dated, January 02, 2024

CROSS CURVE TABLES

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

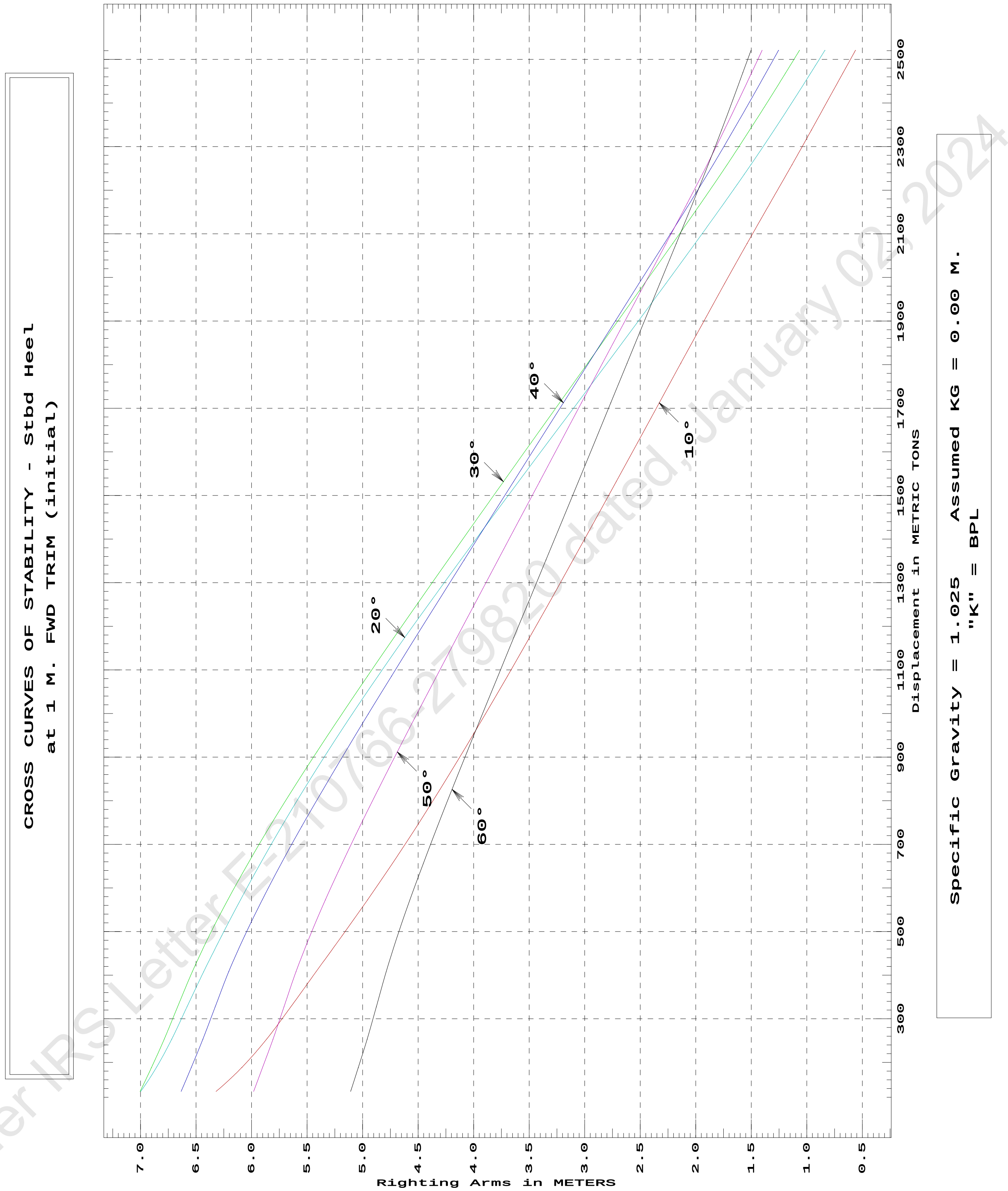
CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.320s	6.999s	7.003s	6.634s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.865s	6.719s	6.786s	6.438s	5.806s	4.959s
327.16	5.646s	6.581s	6.666s	6.327s	5.707s	4.876s
407.76	5.419s	6.431s	6.534s	6.205s	5.600s	4.788s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.859s	4.194s
912.01	4.097s	5.316s	5.412s	5.155s	4.687s	4.059s
998.69	3.895s	5.094s	5.186s	4.949s	4.512s	3.921s
1,086.07	3.695s	4.861s	4.955s	4.739s	4.333s	3.781s
1,174.15	3.497s	4.620s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.371s	4.475s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.086s	3.781s	3.349s
1,441.67	2.910s	3.858s	3.982s	3.864s	3.594s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.641s	3.406s	3.057s
1,621.83	2.523s	3.331s	3.480s	3.417s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.765s
1,803.32	2.134s	2.798s	2.972s	2.965s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.649s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.459s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.897s	1.888s
2,338.10	0.958s	1.294s	1.514s	1.660s	1.743s	1.767s
2,408.27	0.806s	1.115s	1.338s	1.502s	1.610s	1.664s
2,469.20	0.674s	0.963s	1.190s	1.368s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.066s	1.255s	1.402s	1.502s

Distances in METERS.---Specific Gravity = 1.025.-------------

Refer IRS Letter E-21768-2020 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210/66-279820 dated, January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

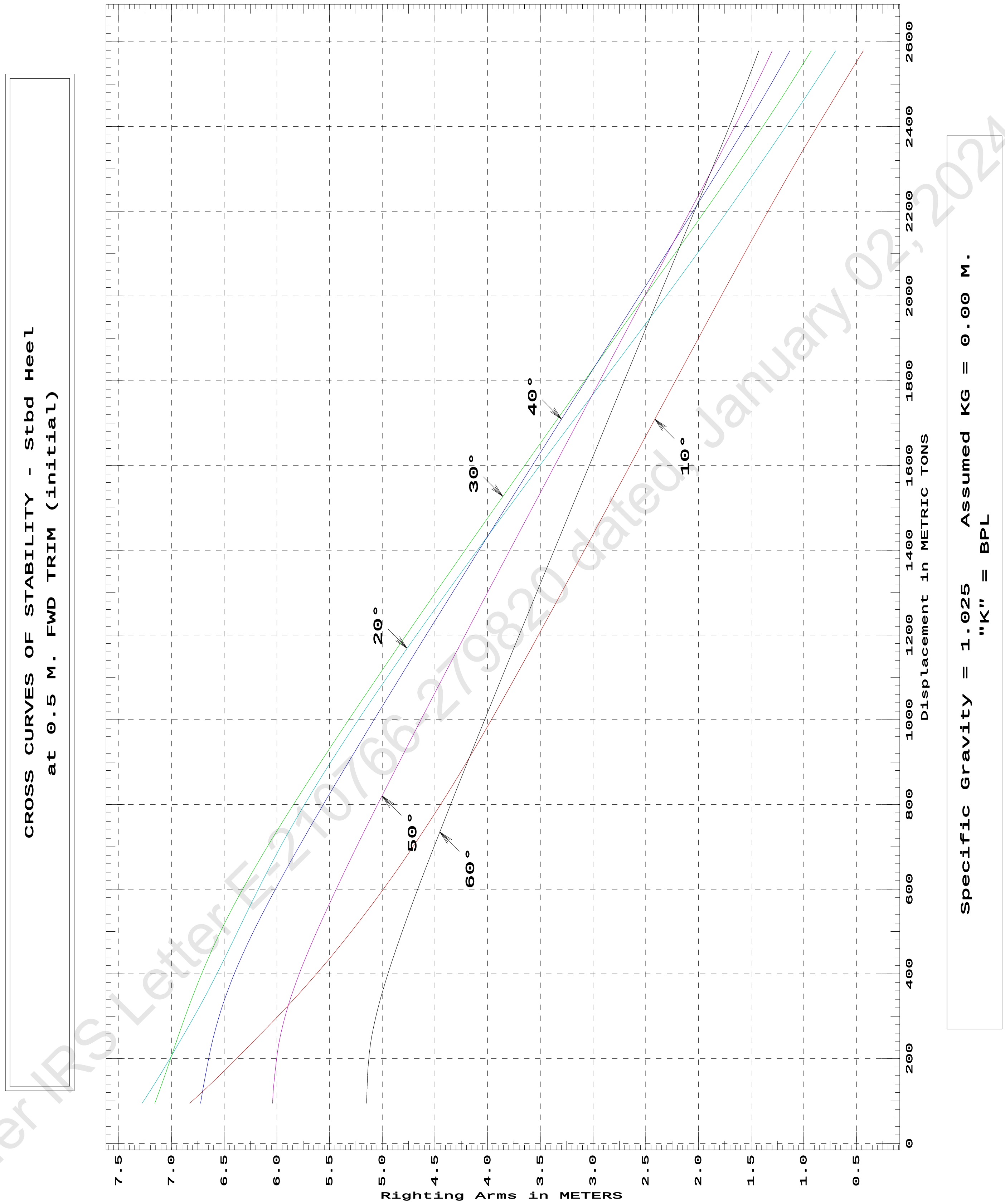
CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.827s	7.278s	7.158s	6.723s	6.041s	5.149s
162.65	6.533s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.221s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.039s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.404s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.512s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.217s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.856s	3.761s	3.519s	3.157s
1,618.11	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.199s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.699s	0.931s	1.133s	1.301s	1.428s

Distances in METERS.---Specific Gravity = 1.025.-------------

Refer IRS Letter E-21766 279830 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

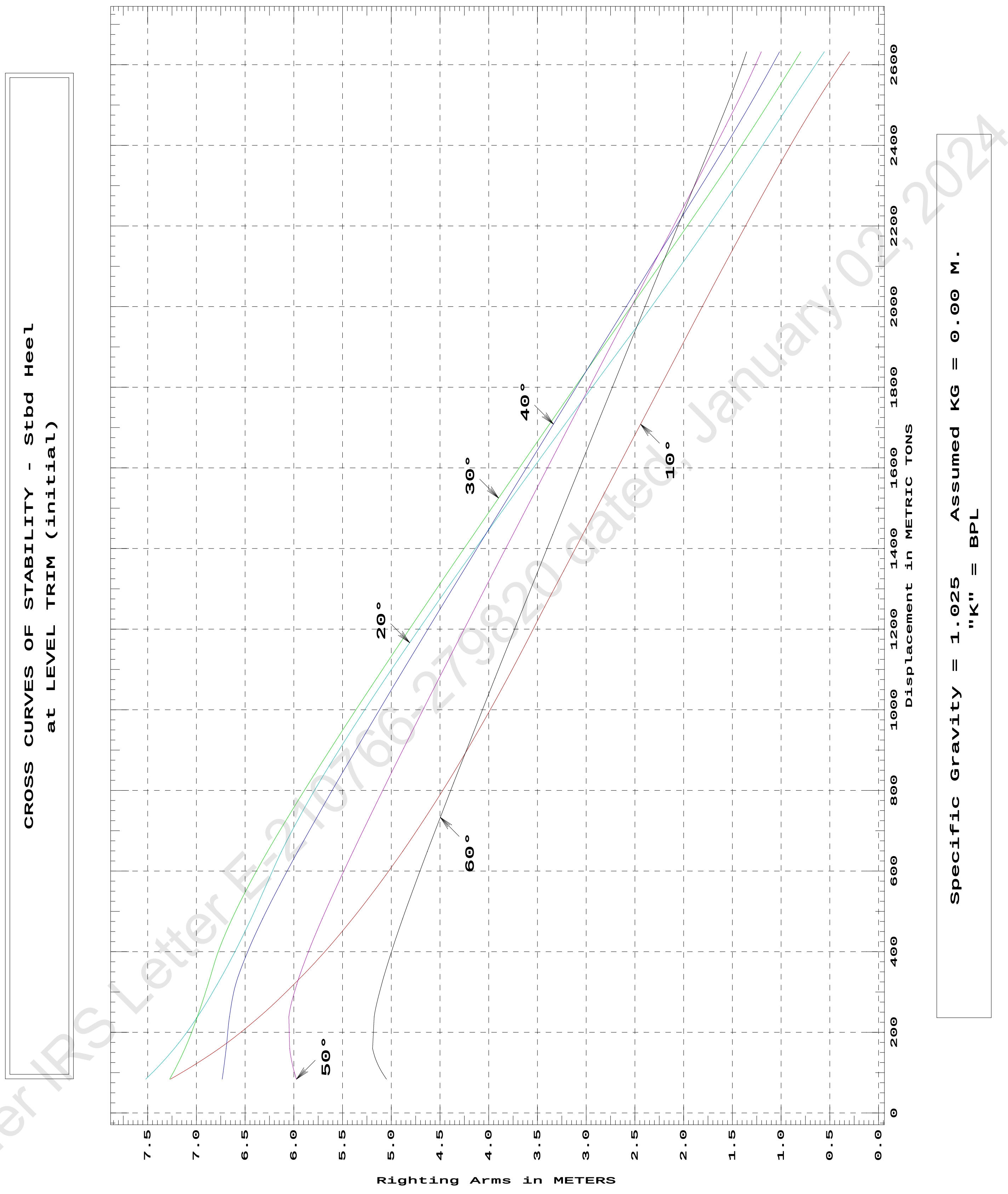
CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: zero at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
83.79	7.268s	7.519s	7.274s	6.735s	5.973s	5.049s
160.29	6.763s	7.226s	7.116s	6.694s	6.043s	5.192s
238.38	6.354s	6.990s	6.990s	6.660s	6.051s	5.176s
318.09	6.003s	6.788s	6.881s	6.597s	5.971s	5.100s
399.40	5.685s	6.606s	6.776s	6.474s	5.851s	4.999s
481.78	5.397s	6.438s	6.634s	6.319s	5.709s	4.883s
564.87	5.132s	6.283s	6.462s	6.146s	5.555s	4.760s
648.69	4.884s	6.127s	6.269s	5.960s	5.392s	4.630s
733.24	4.648s	5.945s	6.061s	5.764s	5.224s	4.497s
818.51	4.425s	5.743s	5.844s	5.562s	5.050s	4.360s
904.44	4.211s	5.525s	5.619s	5.355s	4.873s	4.221s
991.06	4.006s	5.297s	5.387s	5.143s	4.693s	4.080s
1,078.37	3.807s	5.058s	5.149s	4.927s	4.509s	3.936s
1,166.40	3.615s	4.812s	4.907s	4.708s	4.324s	3.791s
1,255.11	3.427s	4.559s	4.660s	4.486s	4.135s	3.643s
1,344.46	3.231s	4.300s	4.410s	4.261s	3.945s	3.495s
1,434.46	3.036s	4.035s	4.156s	4.033s	3.753s	3.345s
1,525.13	2.840s	3.767s	3.899s	3.803s	3.559s	3.193s
1,616.48	2.642s	3.493s	3.640s	3.571s	3.363s	3.040s
1,708.51	2.442s	3.216s	3.377s	3.336s	3.165s	2.886s
1,800.77	2.241s	2.937s	3.113s	3.100s	2.966s	2.731s
1,893.06	2.041s	2.656s	2.848s	2.864s	2.767s	2.576s
1,985.36	1.840s	2.374s	2.582s	2.627s	2.568s	2.421s
2,077.68	1.638s	2.100s	2.316s	2.390s	2.369s	2.265s
2,170.03	1.433s	1.832s	2.049s	2.153s	2.169s	2.110s
2,262.38	1.225s	1.570s	1.786s	1.916s	1.969s	1.954s
2,354.73	1.010s	1.314s	1.529s	1.679s	1.769s	1.798s
2,447.09	0.789s	1.063s	1.280s	1.449s	1.571s	1.642s
2,539.44	0.554s	0.813s	1.038s	1.228s	1.380s	1.491s
2,631.79	0.298s	0.558s	0.799s	1.016s	1.204s	1.354s

Distances in METERS.---Specific Gravity = 1.025.-------------

Refer IRS Letter E-21766-2/380 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated: January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

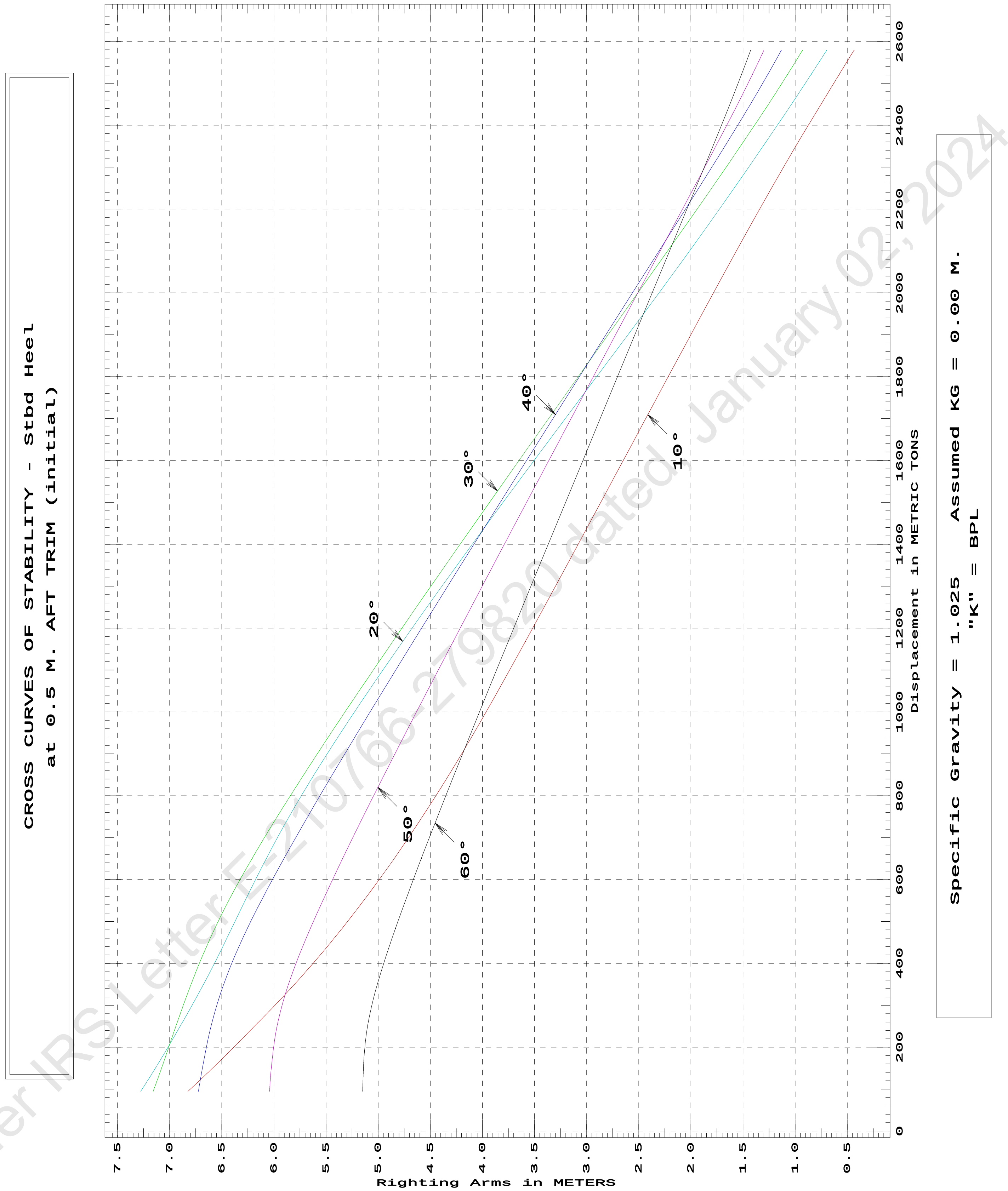
CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.824s	7.277s	7.158s	6.723s	6.041s	5.149s
162.65	6.535s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.222s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.040s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.405s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.511s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.216s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.855s	3.761s	3.519s	3.157s
1,618.10	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.198s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.698s	0.931s	1.133s	1.301s	1.428s

Distances in METERS.---Specific Gravity = 1.025.-------------

Refer IRS Letter E-21766 2180 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210766-279820 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

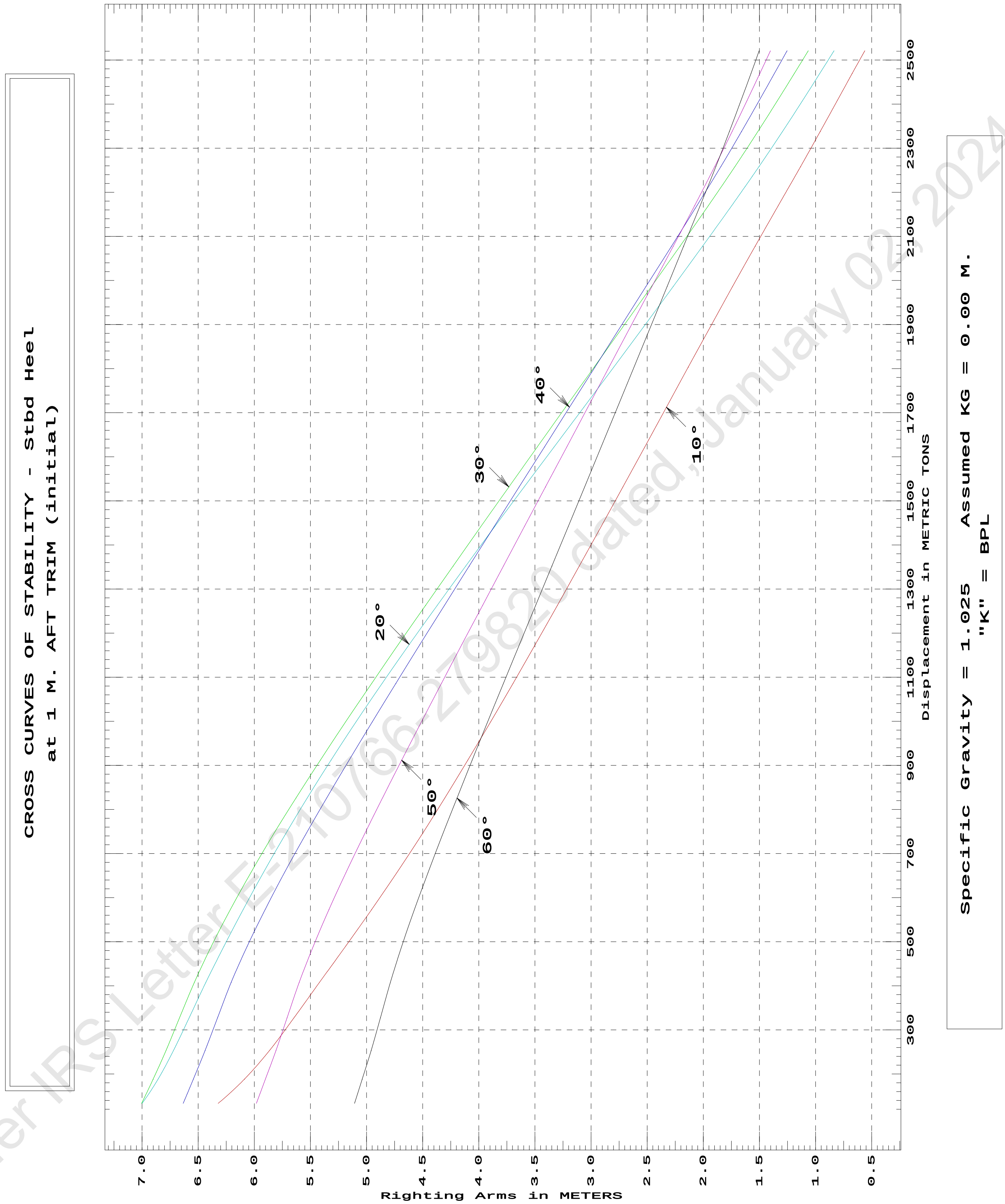
CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.322s	6.999s	7.004s	6.633s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.864s	6.719s	6.786s	6.438s	5.807s	4.959s
327.16	5.646s	6.580s	6.666s	6.327s	5.708s	4.876s
407.76	5.419s	6.431s	6.533s	6.205s	5.600s	4.787s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.858s	4.194s
912.01	4.097s	5.316s	5.411s	5.155s	4.687s	4.059s
998.69	3.895s	5.093s	5.186s	4.949s	4.511s	3.921s
1,086.07	3.695s	4.861s	4.954s	4.738s	4.333s	3.780s
1,174.15	3.497s	4.619s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.370s	4.474s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.085s	3.780s	3.348s
1,441.67	2.910s	3.858s	3.981s	3.863s	3.593s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.640s	3.406s	3.057s
1,621.83	2.523s	3.330s	3.480s	3.416s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.764s
1,803.32	2.134s	2.798s	2.973s	2.966s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.650s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.458s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.898s	1.888s
2,338.10	0.958s	1.294s	1.513s	1.660s	1.743s	1.768s
2,408.27	0.806s	1.115s	1.339s	1.502s	1.610s	1.664s
2,469.21	0.674s	0.963s	1.189s	1.367s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.065s	1.254s	1.402s	1.503s

Distances in METERS.---Specific Gravity = 1.025.-------------

Refer IRS Letter E-210766-279320 dated January 02, 2024

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**



Refer IRS Letter E-210/66-279820 dated, January 02, 2024

DEADWEIGHT TABLE

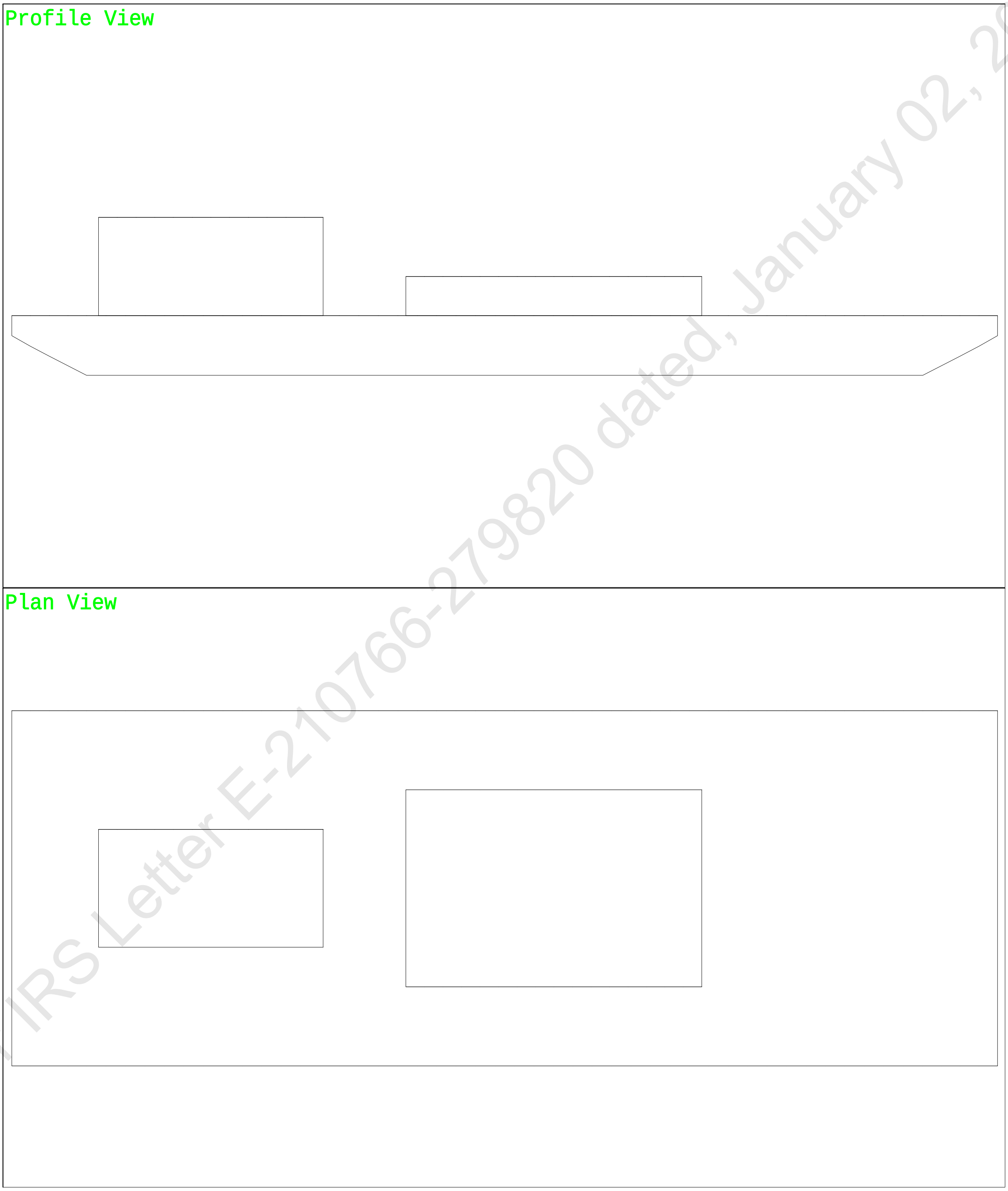
Draft M	Disp T	Dwt T	
0.612	481.78	34.68	
0.712	564.87	117.77	
0.812	648.69	201.59	
0.912	733.24	286.14	
1.012	818.51	371.41	
1.112	904.44	457.34	
1.212	991.06	543.96	
1.312	1078.37	631.27	
1.412	1166.40	719.30	
1.512	1255.11	808.01	
1.612	1344.46	897.36	
1.712	1434.46	987.36	
1.812	1525.13	1078.03	
1.912	1616.48	1169.38	
2.012	1708.51	1261.41	
2.112	1800.77	1353.67	
2.212	**1893.06**	**1445.96**	**Full Load Displacement**

Refer IRS Letter E-210766-279520 dated, January 02, 2024

01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic



Refer IRS Letter E-210766-279820 dated, January 02, 2024

01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind
Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	~~18.489~~ 18.466	122%	0d	7d
600.00	17.349	115%	0d	6d
800.00	16.087	112%	0d	5d
1,000.00	14.631	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	~~11.429~~ 11.407	30%	0d	3d
1,600.00	~~9.770~~ 9.697	12%	0d	2d
1,800.00	~~7.778~~ 7.619	0%	1d	1d
~~2,000.00~~	~~5.433~~	~~0%~~	~~5d~~	~~1d~~

Heeling moment is present from: wind
Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	18.792	138%	0d	10d
600.00	17.834	103%	0d	7d
800.00	16.573	83%	0d	6d
1,000.00	15.128	70%	0d	5d
1,200.00	13.510	57%	0d	4d
1,400.00	11.824	35%	0d	3d
1,600.00	10.137	7%	0d	3d
1,800.00	7.959	0%	2d	2d
~~2,000.00~~	~~5.618~~	~~3%~~	~~6d~~	~~1d~~

Heeling moment is present from: wind
Trim = zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	18.899	137%	0d	14d
600.00	17.990	117%	0d	10d
800.00	16.745	80%	0d	7d
1,000.00	15.270	69%	0d	6d
1,200.00	13.683	59%	0d	5d
1,400.00	11.943	37%	0d	4d
1,600.00	10.176	8%	0d	3d
1,800.00	8.007	0%	2d	3d
~~2,000.00~~	~~5.806~~	~~0%~~	~~5d~~	~~2d~~

Heeling moment is present from: wind
Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	18.793	138%	0d	10d
600.00	17.835	103%	0d	7d
800.00	16.573	83%	0d	6d
1,000.00	15.129	70%	0d	5d
1,200.00	13.510	57%	0d	4d

Refer IRS Letter E-21766-279820 dated, January 02, 2024

01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

1,400.00	11.825	35%	0d	3d
1,600.00	10.136	7%	0d	3d
1,800.00	7.958	0%	2d	2d
~~2,000.00~~	~~5.599~~	~~4%~~	~~6d~~	~~1d~~

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
400.00	~~18.491~~ 18.452	119%	0d	7d
600.00	17.392	114%	0d	6d
800.00	16.095	112%	0d	5d
1,000.00	14.632	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	~~11.430~~ 11.408	31%	0d	3d
1,600.00	~~9.769~~ 9.699	12%	0d	2d
1,800.00	~~7.781~~ 7.661	0%	1d	1d
~~2,000.00~~	~~5.434~~	~~0%~~	~~5d~~	~~1d~~

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM	IS CODE 2008 CRITERION		Min/Max
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00 deg
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg

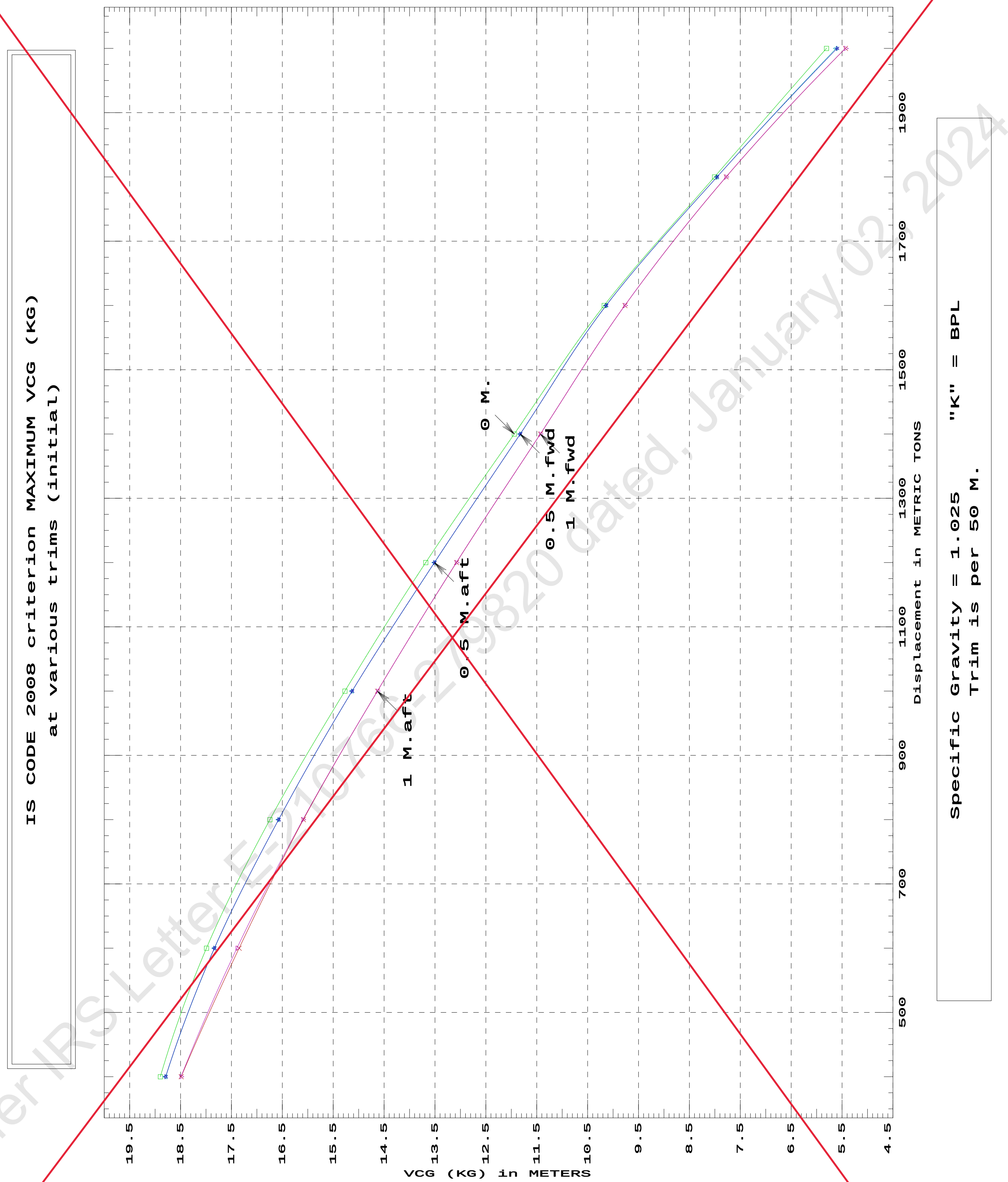
MAX VCG DOES NOT COVER CRANE OPERATIONS/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW,WITH CRANE IN STOWED POSITION.

MAX DECK CARGO HEIGHT = 2.00M ABOVE MAIN DECK

MAX VCG VALUES ARE GENERATED BASED ON WINDAGE AREA AS SHOWN IN PAGE 51 OF THE DOCUMENT.

Refer IRS Letter E-210766-279820 dated, January 02, 2024

01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON



Refer IRS Letter E-210760-279820 dated, January 02, 2024

LOADING CONDITIONS

Refer IRS Letter E-210766-279820 dated, January 02, 2024

SUMMARY OF CONDITIONS				
CONDITION NO.	1	2	3	4
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Crane + Deck Cargo + Ballast / Maximum Draft Condition	Condition 3 : Lightship + Deck Cargo / Maximum Deck Cargo Condition
TOTAL TANK LOAD (T)	0.000	234.870	234.870	0.000
DWT CONST.(T)	0.000	400.000	1211.090	1445.960
DEADWEIGHT (T)	0.000	634.870	1445.960	1445.960
LIGHT WEIGHT (T)	447.100	447.100	447.100	447.100
DISPLACEMENT (T)	447.100	1081.970	1893.060	1893.060
FWD DRAFT (M)	0.570	1.190	2.205	2.212
AFT DRAFT (M)	0.570	1.441	2.219	2.212
MEAN DRAFT (M)	0.570	1.316	2.212	2.212
TRIM (M)	0.000	0.252a	0.013a	0.000
L.C.G. (FROM AP) (M)	25.000f	24.250f	24.974f	25.000f
L.C.B. (FROM AP) (M)	25.000f	24.233f	24.973f	25.000f
M.C.T. (T-CM)	27.130	32.280	37.340	37.410
KG (M)	3.000	3.938	3.965	3.764
GM_T (M)	47.038	18.377	10.196	10.545
MAX. GZ (M)	5.459	3.639	1.362	1.451

Refer IRS Letter E-210766-279820 dated: January 02, 2024

SUMMARY OF CONDITIONS				
CONDITION NO.	5	6	7	8
CONDITION	Condition 4 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius aft	Condition 5 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius stbd	Condition 6 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius fwd	Condition 7 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius aft
TOTAL TANK LOAD (T)	284.980	284.980	284.980	422.400
DWT CONST.(T)	440.000	440.000	440.000	390.000
DEADWEIGHT (T)	724.980	724.980	724.980	812.400
LIGHT WEIGHT (T)	447.100	447.100	447.100	447.100
DISPLACEMENT (T)	1172.080	1172.080	1172.080	1259.500
FWD DRAFT (M)	1.194	1.385	1.574	1.291
AFT DRAFT (M)	1.639	1.451	1.261	1.739
MEAN DRAFT (M)	1.417	1.418	1.418	1.515
TRIM (M)	0.444a	0.066a	0.312f	0.448a
L.C.G. (FROM AP) (M)	23.758f	24.816f	25.874f	23.802f
L.C.B. (FROM AP) (M)	23.721f	24.810f	25.900f	23.773f
M.C.T. (T-CM)	32.740	32.840	32.750	33.680
KG (M)	4.897	4.897	4.897	4.040
GM_T (M)	15.881	15.929	15.882	15.455
MAX. GZ (M)	3.100	3.134	3.117	3.074

Refer IRS Letter E-210766-270820 dated, January 02, 2024

SUMMARY OF CONDITIONS				
CONDITION NO.	9	10	11	12
CONDITION	Condition 8 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius stbd	Condition 9 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius fwd	Condition 10 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius aft	Condition 11 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius stbd
TOTAL TANK LOAD (T)	422.400	422.400	422.400	422.400
DWT CONST.(T)	390.000	390.000	286.600	286.600
DEADWEIGHT (T)	812.400	812.400	709.000	709.000
LIGHT WEIGHT (T)	447.100	447.100	447.100	447.100
DISPLACEMENT (T)	1259.500	1259.500	1156.100	1156.100
FWD DRAFT (M)	1.501	1.707	1.321	1.422
AFT DRAFT (M)	1.533	1.324	1.480	1.378
MEAN DRAFT (M)	1.517	1.516	1.401	1.400
TRIM (M)	0.032a	0.383f	0.159a	0.044f
L.C.G. (FROM AP) (M)	~~33.680~~ **24.914f**	26.026f	24.552f	25.122f
L.C.B. (FROM AP) (M)	~~24.914f~~ **24.912f**	26.051f	24.538f	25.127f
M.C.T. (T-CM)	~~24.912f~~ **33.68**	33.690	32.550	32.600
KG (M)	4.040	4.040	5.305	5.305
GM_T (M)	15.444	15.455	15.621	15.651
MAX. GZ (M)	3.108	3.083	3.038	3.042

Refer IRS Letter E-210766-279820 dated, January 02, 2024

SUMMARY OF CONDITIONS					
CONDITION NO.	13	14	15	16	17
CONDITION	Condition 12 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius fwd	Condition 13 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius aft	Condition 14 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius stbd	Condition 15 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius fwd	Condition 16 : Lightship + Crane + Ballast TOWING CONDITION
TOTAL TANK LOAD (T)	422.400	323.200	323.200	323.200	924.210
DWT CONST.(T)	286.600	201.300	201.300	201.300	400.000
DEADWEIGHT (T)	709.000	524.500	524.500	524.500	1324.210
LIGHT WEIGHT (T)	447.100	447.100	447.100	447.100	447.100
DISPLACEMENT (T)	1156.100	971.600	971.600	971.600	1771.310
FWD DRAFT (M)	1.523	0.991	1.091	1.190	2.040
AFT DRAFT (M)	1.277	1.385	1.288	1.190	2.120
MEAN DRAFT (M)	1.400	1.188	1.190	1.190	2.080
TRIM (M)	0.246f	0.394a	0.197a	0.000	0.080a
L.C.G. (FROM AP) (M)	25.692f	23.734f	24.368f	25.002f	24.831f
L.C.B. (FROM AP) (M)	25.715f	23.704f	24.352f	25.001f	24.828f
M.C.T. (T-CM)	32.540	31.230	31.230	31.220	37.570
KG (M)	5.305	4.439	4.439	4.439	2.935
GM_T (M)	15.614	~~19.949~~ **19.972**	19.949	19.941	11.582
MAX. GZ (M)	3.031	3.761	3.761	3.768	1.853

Refer IRS Letter E-210766-279820 dated: January 02, 2024

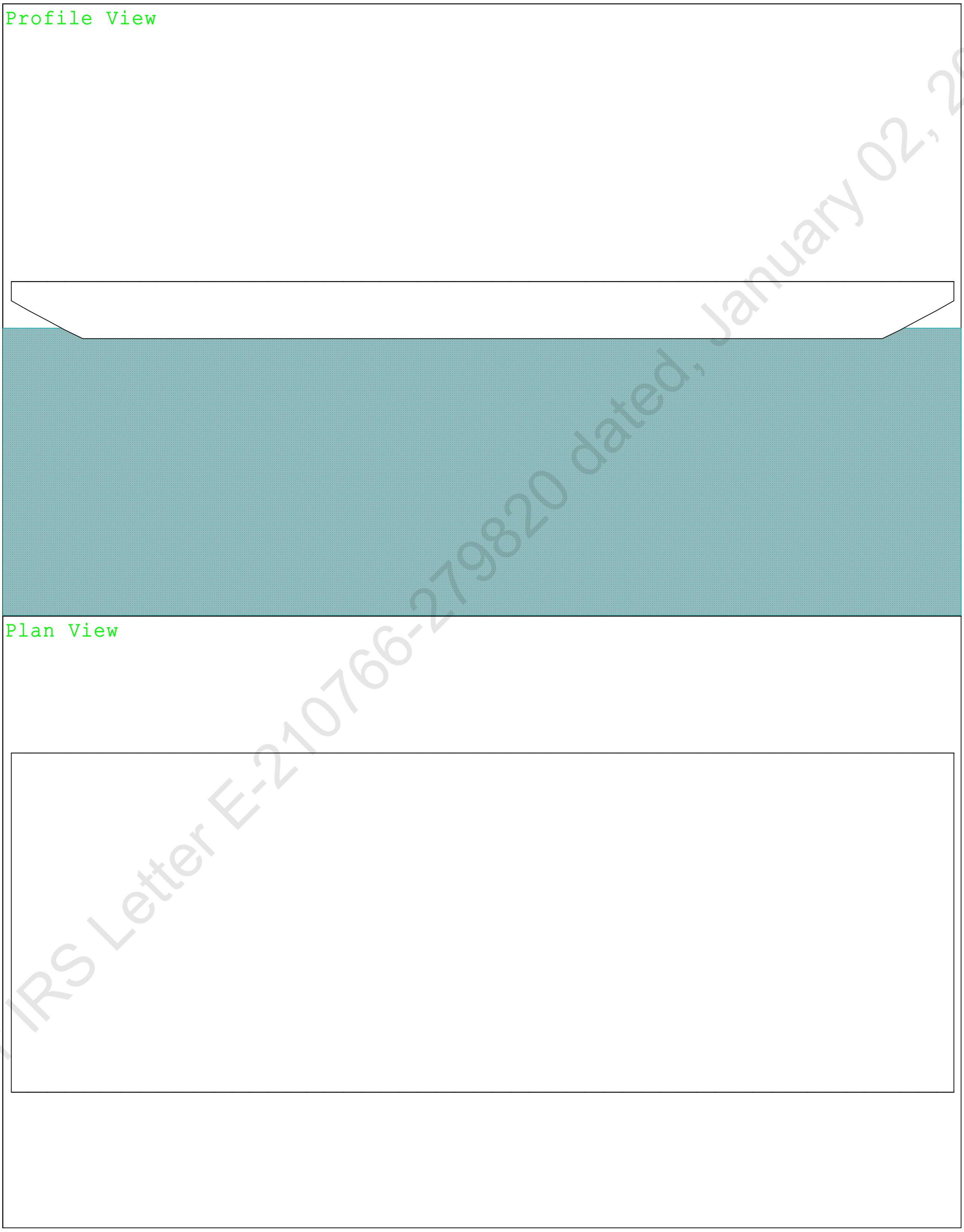
12/21/23 09:38:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 00 : Lightship

Lightship condition

Condition Graphic - Draft: 0.570 Trim: zero Heel: zero



Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:38:53    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
                            Condition 00 : Lightship
                               Lightship condition

                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 0.570 @ 50.00f, 0.570 @ 25.00f, 0.570 @ 0.00
                            Trim: zero,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
WEIGHT                               447.10  25.000f  0.000   3.000
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                0.00                                0.00
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        447.10  25.000f  0.000   0.278
-------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         804.2  25.000f  0.000    306.07  49.760
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.24           27.13    303.34  47.038
Distances in METERS.-----------------------------------------Moments in m.-MT.

                              SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER

DRAFT AT FP (M) =    0.570
DRAFT AT MS (M) =    0.570
DRAFT AT AP (M) =    0.570
DRAFT AT FWD MARK (M) =    0.570
DRAFT AT AFT MARK (M) =    0.570
                            HYDROSTATIC PROPERTIES
                         No Trim, No Heel, VCG = 3.000

    LCF   Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  0.570       447.10  25.000f  0.278     8.24  25.000f   27.13  303.34  47.038
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                              Trim is per 50.00m.
 Draft is from USK.

                         HEELING MOMENT specification
                             Lateral Plane Method
                          Wind pressure toward starboard
          USK draft: 0.570 @ 50.00f, 0.570 @ 25.00f, 0.570 @ 0.00
                            Trim: zero,  Heel: zero

    Part---------------LPA*SF------HCP------Arm----Pressure----------Moment
    HULL                118.8    1.251    1.532     0.05500            10.01
    Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT
```

Refer IRS Letter E-210766279820 dated, January 02, 2024

12/21/23 09:38:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 00 : Lightship

Lightship condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
0.558	0.00	0.00	447.18	0.000	-0.022	0.0000	12.741
0.558	0.00	0.03s	447.10	0.000	0.000	-0.0000	12.714
0.519	0.00	5.03s	447.23	0.000	3.710	0.1619	7.714
0.257	0.00	10.03s	447.10	0.000	4.975	0.5522	2.714
0.053	0.00	12.74s	447.10	0.000	5.244	0.7954	0.000
-0.140	0.00	15.03s	446.96	0.000	5.371	1.0072	-2.286
-0.573	0.00	19.69s	446.85	0.000	5.459	1.4511	-6.952
-0.604	0.00	20.03s	447.10	0.000	5.459	1.4829	-7.286
-1.112	0.00	25.03s	447.10	0.000	5.384	1.9572	-12.286
-1.632	0.00	30.03s	447.30	0.000	5.175	2.4189	-17.286
-2.141	0.00	35.03s	447.15	0.000	4.845	2.8569	-22.286
-2.634	0.00	40.03s	446.95	0.000	4.435	3.2624	-27.286
-3.105	0.00	45.03s	447.14	0.000	3.964	3.6293	-32.286
-3.555	0.00	50.03s	446.97	0.000	3.447	3.9530	-37.286
-3.976	0.00	55.03s	446.98	0.000	2.893	4.2299	-42.286
-4.366	0.00	60.03s	447.13	0.000	2.309	4.4572	-47.286

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 10.01 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	1.4511 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	12.71 P

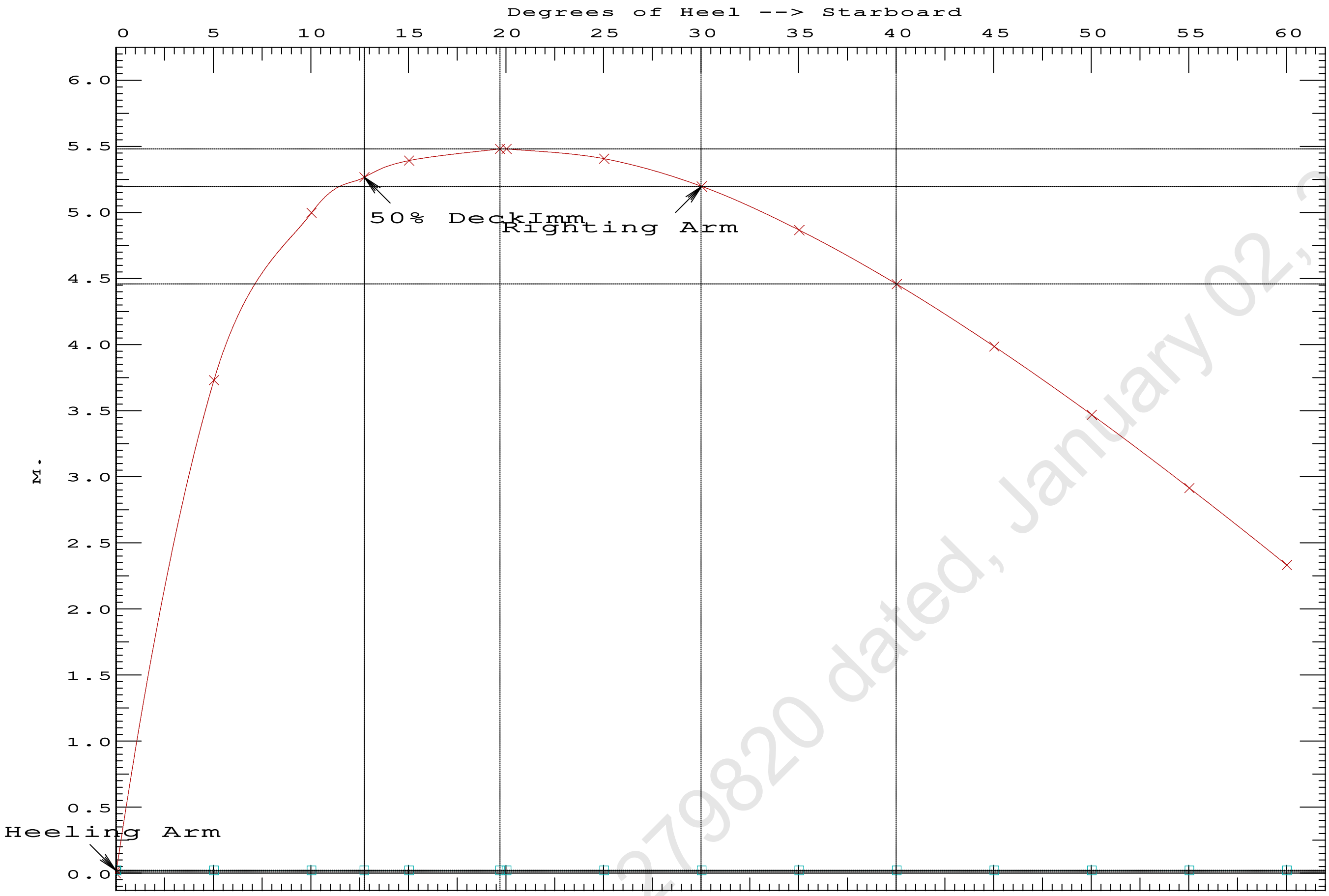
Refer IRS Letter E-21076b 279320 dated, January 02, 2024

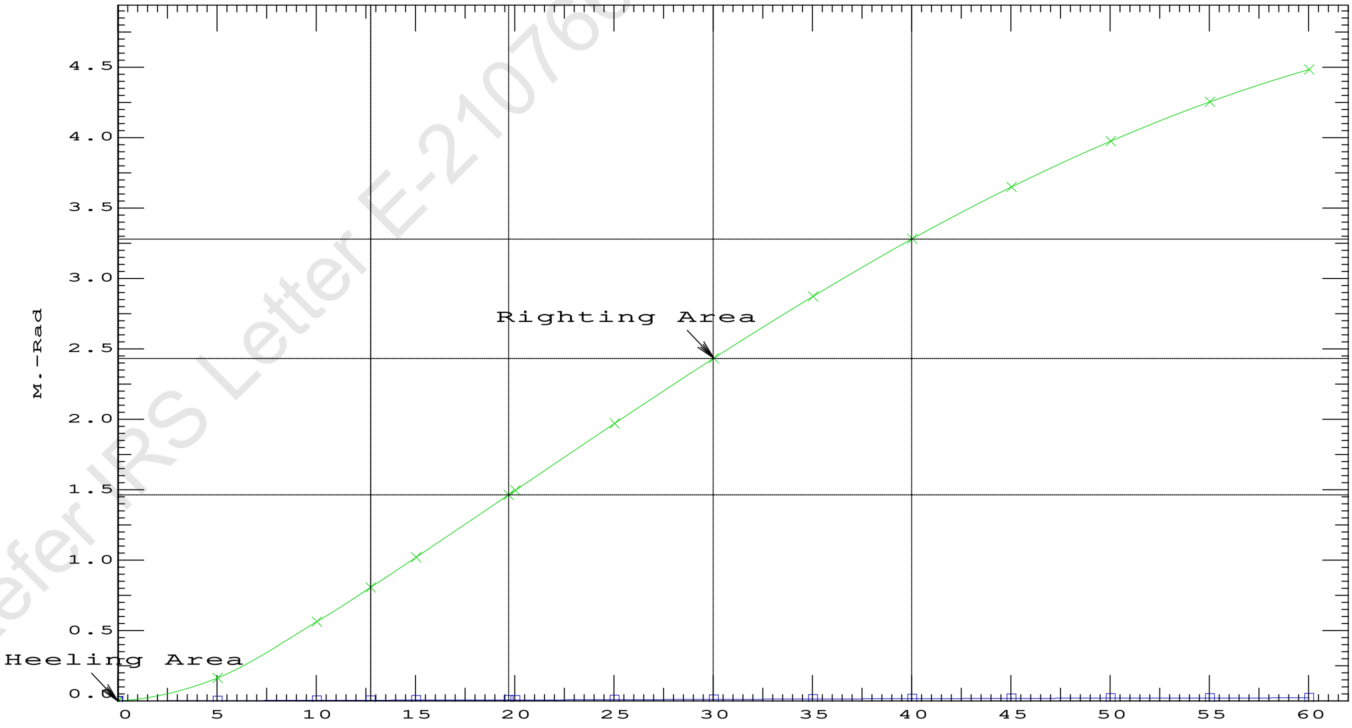
12/21/23 09:38:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 00 : Lightship

Lightship condition



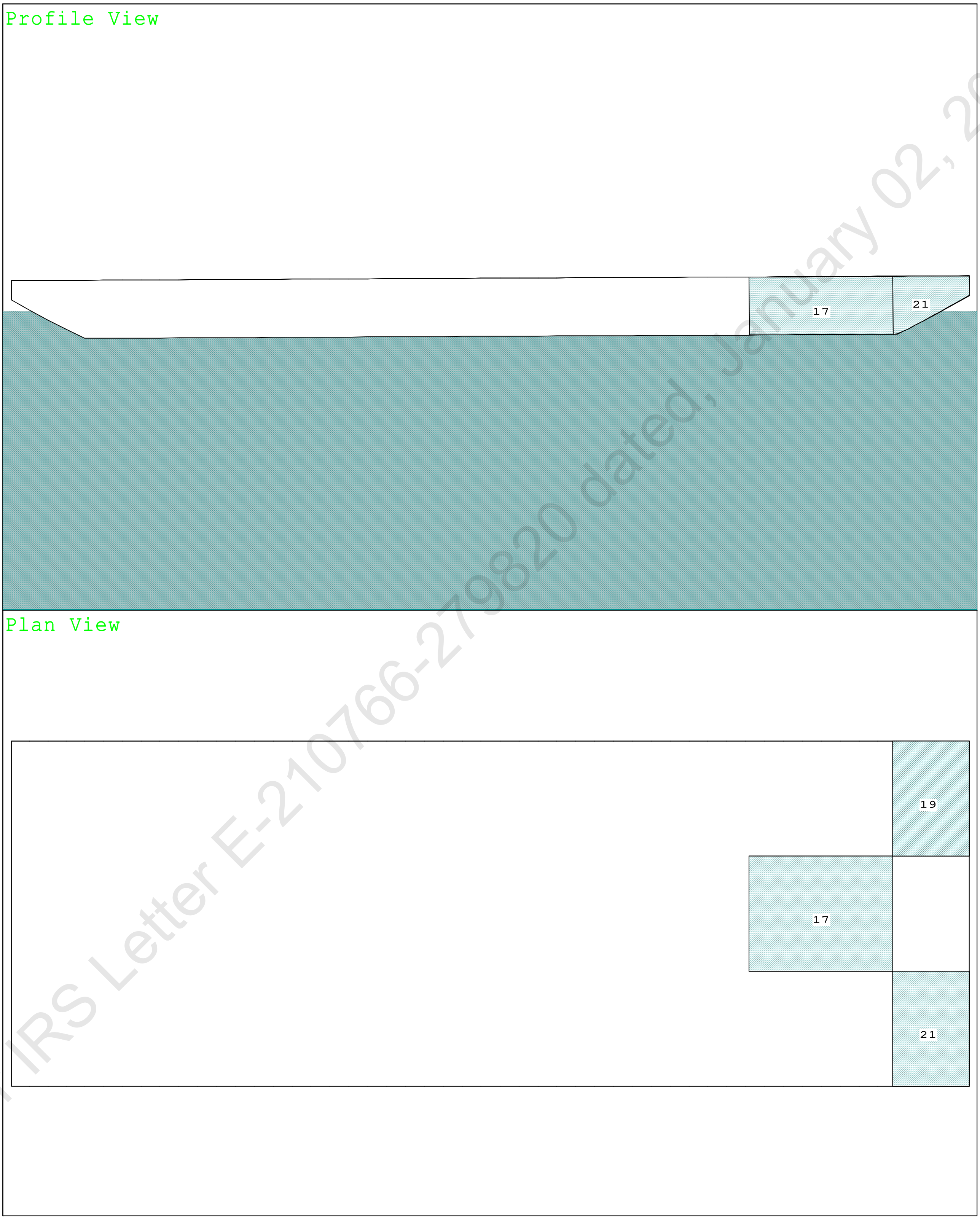


Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:38:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 01 : Lightship + Crane + Ballast

Condition Graphic - Draft: 1.190 @ 50.000f, 1.441 @ 0.000 Heel: zero



Tanks
19 07_WBTK.P.100% SALT WATER Intact
17 06_WBTK.C.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:38:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 01 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.190 @ 50.00f, 1.315 @ 25.00f, 1.441 @ 0.00
Trim: Aft 0.252/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Total Fixed--------->			847.10	18.625f	0.000	4.570	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			234.87	44.536f	0.000	1.660	280.25
Total Weight-------->			1,081.97	24.250f	0.000	3.938	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,081.92	24.233f	0.000	0.665

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	855.9	24.766f	0.000	152.45	21.650*
		MT/cm		m.-MT/cm	GML	GMT
		8.77		32.28	149.17	18.377

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

DRAFT AT FP (M) = 1.190
DRAFT AT MS (M) = 1.315
DRAFT AT AP (M) = 1.441
DRAFT AT FWD MARK (M) = 1.210
DRAFT AT AFT MARK (M) = 1.421

HYDROSTATIC PROPERTIES
Trim: Aft 0.252/50.000, No Heel, VCG = 3.938

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.317	1,081.92	24.233f	0.665	8.77	24.766f	32.28	149.17	18.377

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:38:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 01 : Lightship + Crane + Ballast

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.190 @ 50.00f, 1.315 @ 25.00f, 1.441 @ 0.00
Trim: Aft 0.252/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	84.4	0.862	1.508	0.05500	7.00
CRANE	57.0	4.121	4.767	0.05500	14.95
Total wind heeling moment to starboard------------->					21.94

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.250f TCG = 0.000 VCG = 3.938
Free Surface Adjustment: 0.259
Adjusted CG: LCG = 24.251f TCG = 0.000 VCG = 4.197

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.429	0.29a	0.00	1,082.0	0.000	-0.020	0.0000	5.105
1.429	0.29a	0.05s	1,082.0	0.000	0.000	-0.0000	5.055
1.414	0.29a	5.05s	1,082.0	0.000	1.609	0.0702	0.055
1.414	0.29a	5.11s	1,081.9	0.000	1.627	0.0717	0.000
1.368	0.32a	10.05s	1,081.9	0.000	3.056	0.2750	-4.945
1.265	0.45a	15.05s	1,082.0	0.000	3.615	0.5726	-9.945
1.225	0.51a	16.91s	1,082.0	0.000	3.639	0.6908	-11.808
1.157	0.60a	20.05s	1,082.0	0.000	3.579	0.8893	-14.945
1.043	0.76a	25.05s	1,082.0	0.000	3.340	1.1937	-19.945
0.925	0.92a	30.05s	1,082.0	0.000	3.005	1.4712	-24.945
0.803	1.08a	35.05s	1,082.0	0.000	2.614	1.7168	-29.945
0.677	1.24a	40.05s	1,082.0	0.000	2.184	1.9264	-34.945
0.549	1.39a	45.05s	1,082.0	0.000	1.727	2.0973	-39.945
0.418	1.54a	50.05s	1,082.0	0.000	1.251	2.2274	-44.945
0.285	1.67a	55.05s	1,082.0	0.000	0.760	2.3153	-49.945
0.149	1.79a	60.05s	1,082.0	0.000	0.261	2.3599	-54.945

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.94 (constant)

Refer IRS Letter E-210766279820 dated, January 02, 2024

12/21/23 09:38:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66
PONTOON

Condition 01 : Lightship + Crane + Ballast

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.6908 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg      LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     5.05 P
-----------------------------------------------------------------------------
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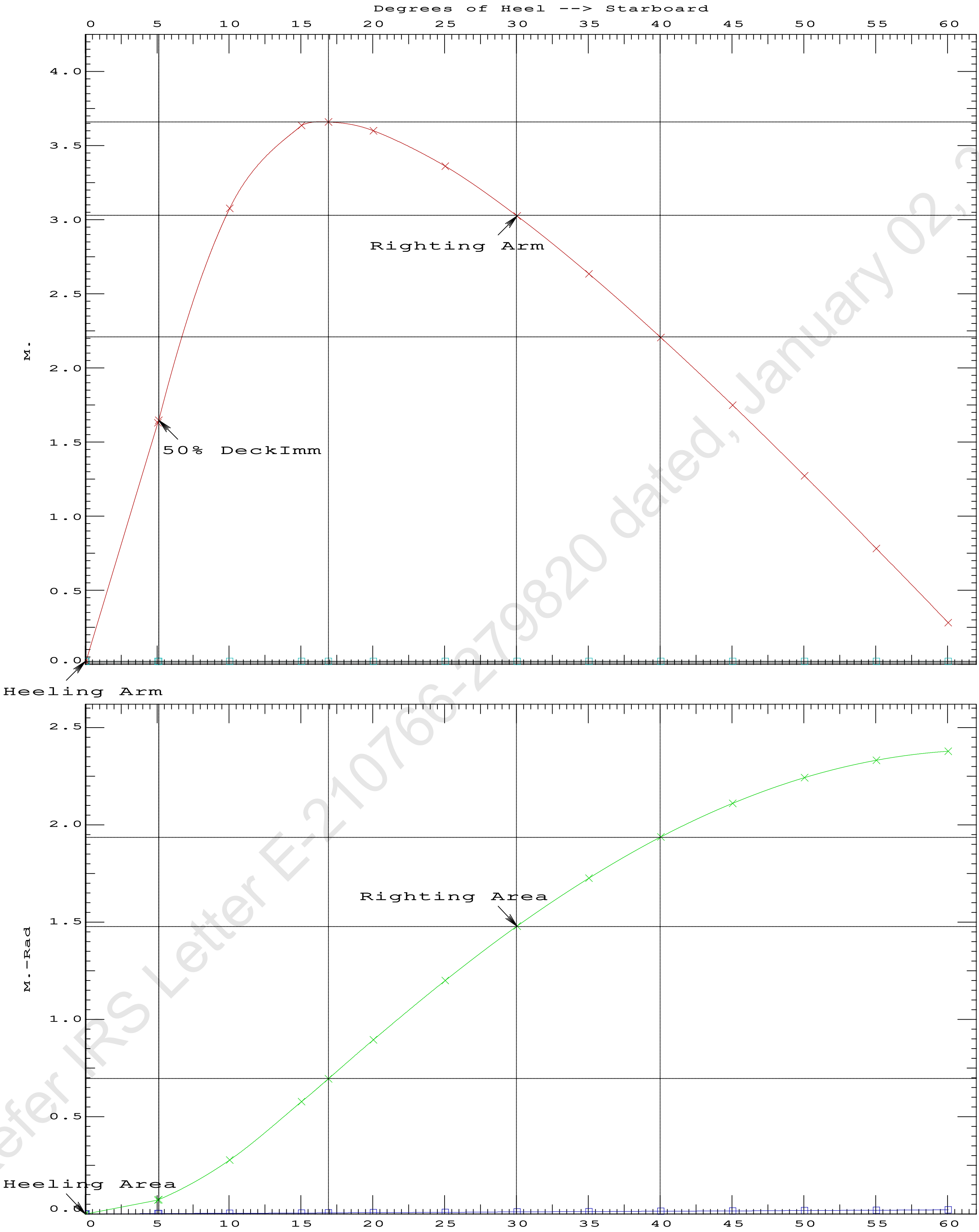
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:38:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 01 : Lightship + Crane + Ballast



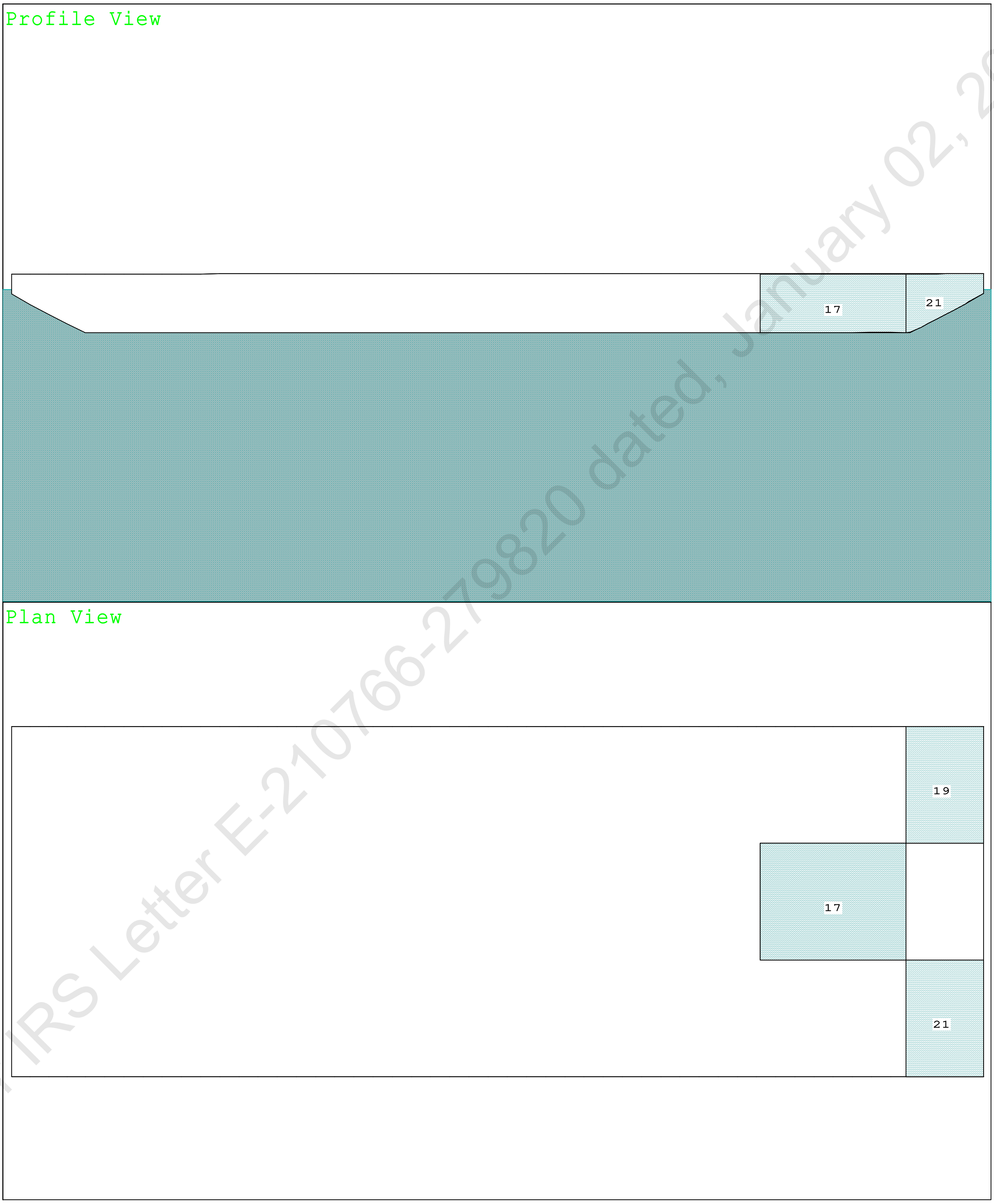
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:42:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

Condition Graphic - Draft: 2.205 @ 50.000f, 2.219 @ 0.000 Heel: zero



Refer IRS Letter E-210766-279820 dated, January 02, 2024

Tanks
19 07_WBTK.P.100% SALT WATER Intact
17 06_WBTK.C.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact

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12/21/23 09:42:46    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
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Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.205 @ 50.00f, 2.212 @ 25.00f, 2.219 @ 0.00
                     Trim: Aft 0.013/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            447.10  25.000f  0.000   3.000
Crane                                 400.00  11.500f  0.000   6.325
Deck Cargo                            811.09  25.940f  0.000   4.000
  Total Fixed--------->             1,658.19  22.203f  0.000   4.291
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025         135.67  42.251f  0.000   1.499   135.61*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880    72.32*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880    72.32*
  Total Tanks--------->               234.87  44.536f  0.000   1.660   280.25
  Total Weight-------->             1,893.06  24.974f  0.000   3.965
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,893.06  24.973f  0.000   1.132
------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          900.4  25.000f  0.000    101.45  13.029*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        9.23           37.34     98.62  10.196
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           SUMMARY OF LOADING
       229.1 Cu.M.   (9%) SALT WATER

      400.00 MT  Crane                       811.09 MT  Deck Cargo

DRAFT AT FP (M) =    2.205
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.219
DRAFT AT FWD MARK (M) =    2.206
DRAFT AT AFT MARK (M) =    2.218
                          HYDROSTATIC PROPERTIES
                   Trim: Aft 0.013/50.000, No Heel, VCG = 3.965

   LCF   Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  2.212     1,893.06  24.973f  1.132      9.23  25.000f   37.34   98.62  10.196
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                           Trim is per 50.00m.
 Draft is from USK.                             Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 280.2 m.-MT
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Refer IRS Letter E-20766279820 dated January 02, 2024

12/21/23 09:42:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.205 @ 50.00f, 2.212 @ 25.00f, 2.219 @ 0.00
Trim: Aft 0.013/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.296	4.369	0.05500	13.70
DECKCARGO.C	29.4	1.801	2.874	0.05500	4.65
Total wind heeling moment to starboard------------->					21.64

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.974f TCG = 0.000 VCG = 3.965
Free Surface Adjustment: 0.148
Adjusted CG: LCG = 24.974f TCG = 0.000 VCG = 4.113

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.207	0.02a	0.00	1,893.1	0.000	-0.011	0.0000	2.554
2.207	0.02a	0.06s	1,893.1	0.000	0.000	-0.0000	2.492
2.204	0.02a	2.55s	1,893.1	0.000	0.448	0.0097	0.000
2.196	0.02a	5.06s	1,893.1	0.000	0.892	0.0391	-2.508
2.295	0.02a	10.06s	1,893.1	0.000	1.318	0.1406	-7.508
2.439	0.03a	13.38s	1,893.1	0.000	1.362	0.2190	-10.824
2.524	0.03a	15.06s	1,893.1	0.000	1.350	0.2589	-12.508
2.798	0.04a	20.06s	1,893.1	0.000	1.235	0.3734	-17.508
3.064	0.05a	25.06s	1,893.1	0.000	1.030	0.4729	-22.508
3.307	0.06a	30.06s	1,893.1	0.000	0.776	0.5520	-27.508
3.525	0.07a	35.06s	1,893.1	0.000	0.498	0.6078	-32.508
3.716	0.08a	40.06s	1,893.1	0.000	0.205	0.6386	-37.508
3.830	0.08a	43.49s	1,893.1	0.000	0.000	0.6448	-40.936
3.878	0.09a	45.06s	1,893.1	0.000	-0.095	0.6435	-42.508
4.012	0.09a	50.06s	1,893.1	0.000	-0.398	0.6220	-47.508
4.115	0.10a	55.06s	1,893.1	0.000	-0.702	0.5740	-52.508
4.186	0.11a	60.06s	1,893.1	0.000	-1.001	0.4997	-57.508

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.64 (constant)

Refer IRS Letter E-2107662798 20 dated, January 02, 2024

12/21/23 09:42:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.2190 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    43.43 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     2.49 P
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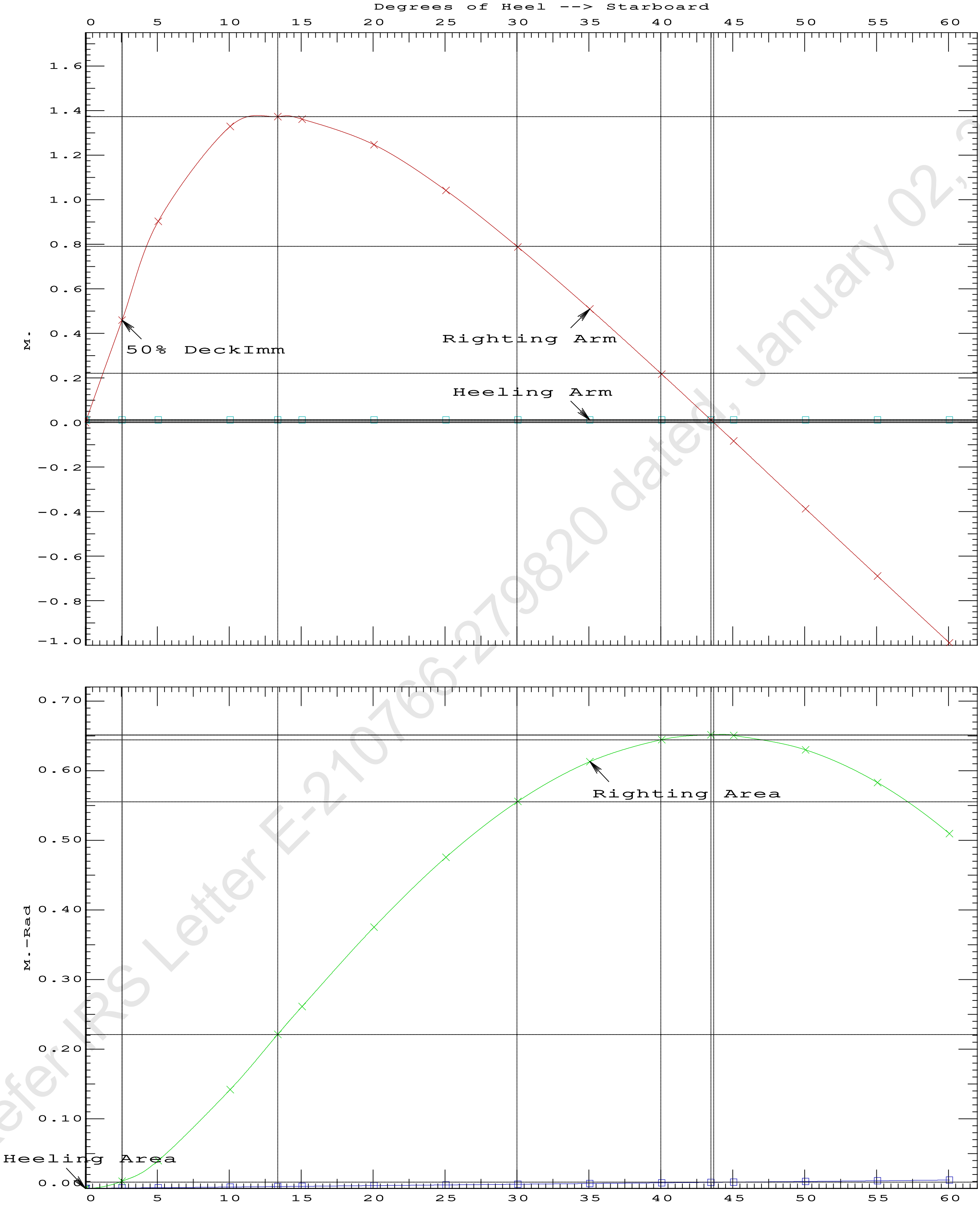
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:42:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 02 : Lightship + Crane + Deck Cargo + Ballast

Maximum Draft Condition



Refer IRS Letter E-210766-279820 dated, January 02, 2024

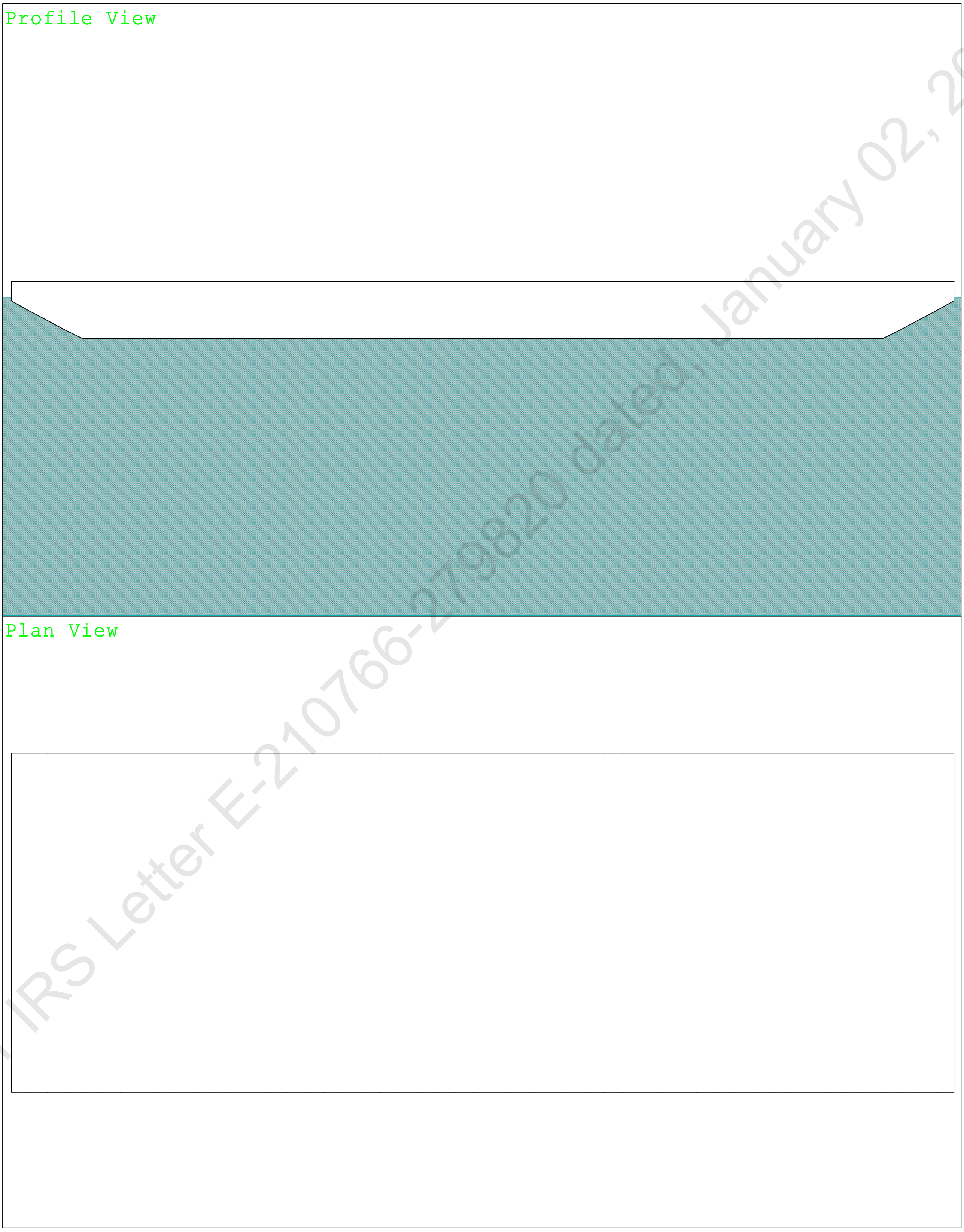
12/21/23 09:44:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:44:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
                          Trim: zero,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
LIGHT SHIP                            447.10  25.000f  0.000   3.000
Deck Cargo                          1,445.96  25.000f  0.000   4.000
  Total Weight-------->             1,893.06  25.000f  0.000   3.764
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                 0.00                              0.00
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,893.06  25.000f  0.000   1.132
-------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          900.4  25.000f  0.000    101.45  13.177
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        9.23            37.41     98.82  10.545
Distances in METERS.----------------------------------------Moments in m.-MT.

                              SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER

    1,445.96 MT  Deck Cargo

DRAFT AT FP (M) =    2.212
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.212
DRAFT AT FWD MARK (M) =    2.212
DRAFT AT AFT MARK (M) =    2.212
                            HYDROSTATIC PROPERTIES
                        No Trim, No Heel, VCG = 3.764

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  2.212     1,893.06  25.000f  1.132     9.23  25.000f   37.41   98.82  10.545
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                              Trim is per 50.00m.
 Draft is from USK.

                        HEELING MOMENT specification
                            Lateral Plane Method
                        Wind pressure toward starboard
           USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
                          Trim: zero,  Heel: zero

    Part---------------LPA*SF------HCP------Arm----Pressure----------Moment
    HULL                  40.5    0.405    1.478     0.05500            3.29
    DECKCARGO.C           29.4    1.800    2.873     0.05500            4.65
      Total wind heeling moment to starboard------------->              7.94
    Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT
```

Refer IRS Letter E-21766 279820 dated, January 02, 2024

12/21/23 09:44:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.764

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.200	0.00	0.00	1,893.1	0.000	-0.004	0.0000	2.576
2.200	0.00	0.02s	1,893.1	0.000	0.000	-0.0000	2.554
2.197	0.00	2.58s	1,893.3	0.000	0.474	0.0106	0.000
2.189	0.00	5.02s	1,893.1	0.000	0.923	0.0404	-2.446
2.283	0.00	10.02s	1,893.1	0.000	1.384	0.1462	-7.446
2.466	0.00	14.18s	1,893.1	0.000	1.451	0.2508	-11.603
2.509	0.00	15.02s	1,893.1	0.000	1.448	0.2721	-12.446
2.780	0.00	20.02s	1,893.1	0.000	1.363	0.3965	-17.446
3.041	0.00	25.02s	1,893.1	0.000	1.186	0.5084	-22.446
3.280	0.00	30.02s	1,893.1	0.000	0.960	0.6025	-27.446
3.494	0.00	35.02s	1,893.1	0.000	0.708	0.6754	-32.446
3.681	0.00	40.02s	1,893.1	0.000	0.439	0.7256	-37.446
3.840	0.00	45.02s	1,893.1	0.000	0.161	0.7519	-42.446
3.918	0.00	47.88s	1,893.1	0.000	0.000	0.7559	-45.307
3.969	0.00	50.02s	1,893.1	0.000	-0.121	0.7537	-47.446
4.069	0.00	55.02s	1,893.1	0.000	-0.406	0.7307	-52.446
4.138	0.00	60.02s	1,893.1	0.000	-0.689	0.6829	-57.446

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.94 (constant)

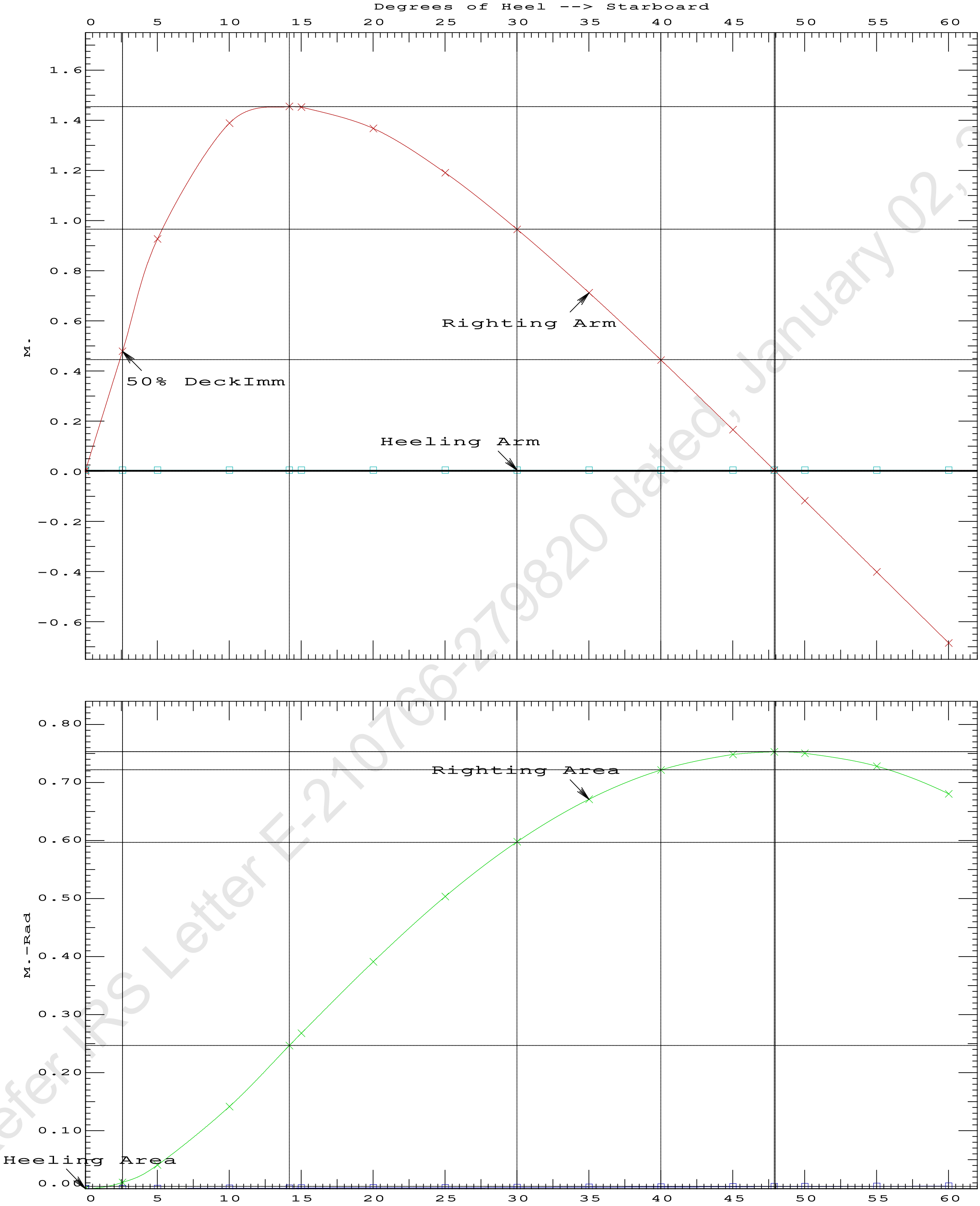
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.2508	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	47.86	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.55	P

Refer IRS Letter E-21076-270820 dated, January 02, 2024

12/21/23 09:44:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 03 : Lightship + Deck Cargo

Maximum Deck Cargo Condition



Refer IRS Letter E-210766-279820 dated, January 02, 2024

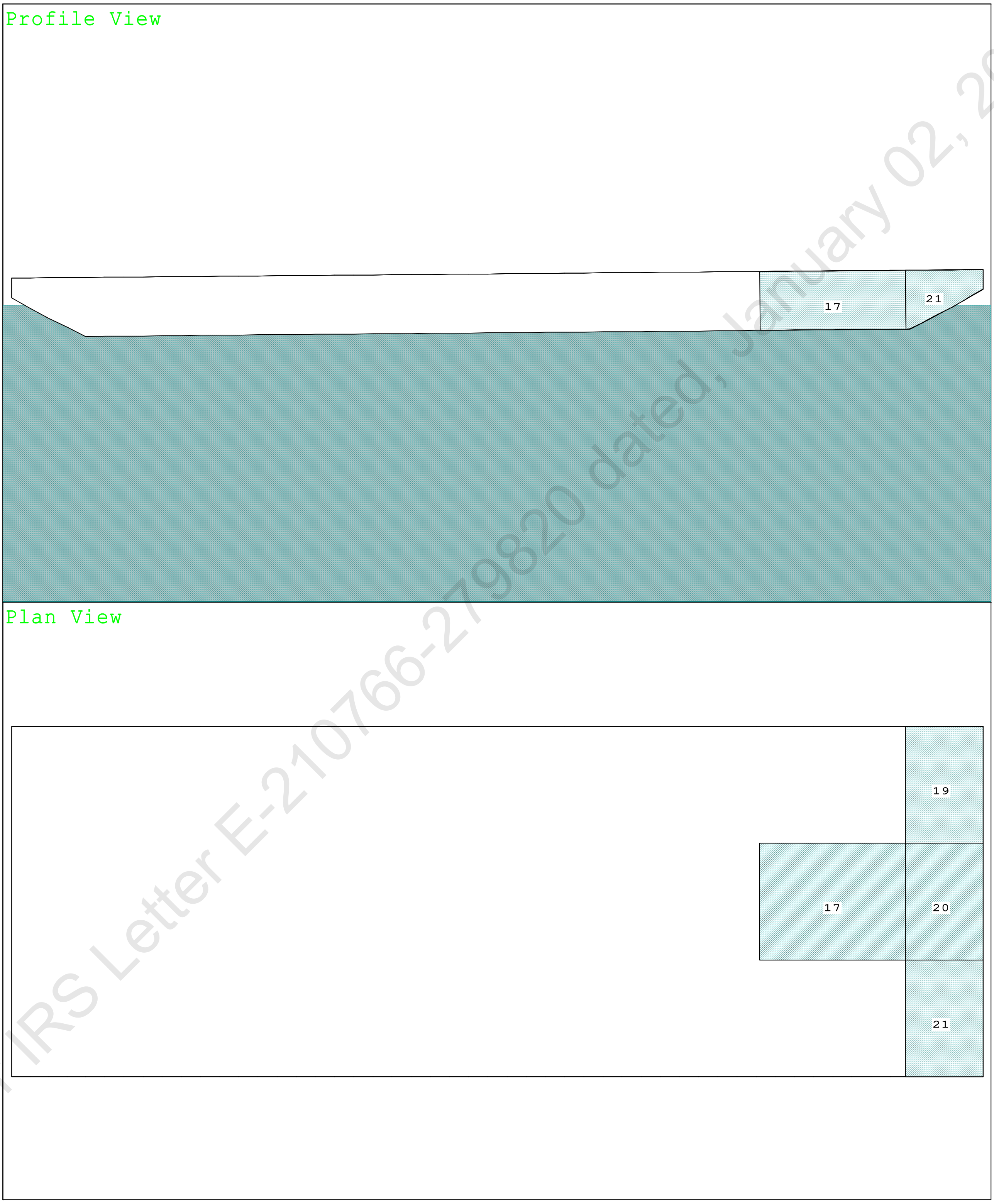
12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 04 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Aft

Condition Graphic - Draft: 1.194 @ 50.000f, 1.639 @ 0.000 Heel: zero



Tanks
19 07_WBTK.P.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
17 06_WBTK.C.100% SALT WATER Intact 20 07_WBTK.C.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.194 @ 50.00f, 1.417 @ 25.00f, 1.639 @ 0.00
Trim: Aft 0.444/50.000, Heel: zero

Part	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~19.500a~~ 18.500a	0.000	34.602	
Total Fixed--------->			887.10	16.906f	0.000	5.924	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			284.98	45.086f	0.000	1.697	352.57
Total Weight-------->			1,172.08	23.758f	0.000	4.897	

Part	SpGr	Displ (MT)	LCB	TCB	VCB
HULL	1.025	1,171.97	23.721f	0.000	0.722

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	862.2	24.611f	0.000	143.87	20.055*

MT/cm	m.-MT/cm	GML	GMT
8.84	32.74	139.69	15.881

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING
278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.194
DRAFT AT MS (M) = 1.417
DRAFT AT AP (M) = 1.639
DRAFT AT FWD MARK (M) = 1.230
DRAFT AT AFT MARK (M) = 1.603

HYDROSTATIC PROPERTIES
Trim: Aft 0.444/50.000, No Heel, VCG = 4.897

LCF Draft	Displacement Weight (MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.420	1,171.97	23.721f	0.722	8.84	24.611f	32.74	139.69	15.881

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

Refer IRS Letter E-240766 279820 dated January 02, 2024

12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.194 @ 50.00f, 1.417 @ 25.00f, 1.639 @ 0.00
Trim: Aft 0.444/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.5	0.814	1.511	0.05500	6.60
CRANE	57.0	3.963	4.660	0.05500	14.61
Total wind heeling moment to starboard------------->					21.21

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 23.758f TCG = 0.000 VCG = 4.897
Free Surface Adjustment: 0.301
Adjusted CG: LCG = 23.761f TCG = 0.000 VCG = 5.197

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% Area	DeckImm Angle
1.627	0.51a	0.00	1,172.1	0.000	-0.018	0.0000	4.421
1.627	0.51a	0.06s	1,172.1	0.000	0.000	-0.0000	4.360
1.615	0.51a	4.42s	1,172.1	0.000	1.217	0.0463	0.000
1.613	0.51a	5.06s	1,172.1	0.000	1.396	0.0609	-0.640
1.581	0.56a	10.06s	1,172.0	0.000	2.676	0.2394	-5.640
1.584	0.82a	15.06s	1,172.1	0.000	3.099	0.4984	-10.640
1.584	0.83a	15.34s	1,172.1	0.000	3.100	0.5133	-10.917
1.584	1.10a	20.06s	1,172.1	0.000	2.961	0.7665	-15.640
1.579	1.40a	25.06s	1,172.1	0.000	2.639	1.0122	-20.640
1.568	1.69a	30.06s	1,172.1	0.000	2.232	1.2254	-25.640
1.551	1.99a	35.06s	1,172.1	0.000	1.778	1.4007	-30.640
1.528	2.29a	40.06s	1,172.1	0.000	1.293	1.5349	-35.640
1.497	2.57a	45.06s	1,172.1	0.000	0.789	1.6259	-40.640
1.457	2.84a	50.06s	1,172.1	0.000	0.273	1.6723	-45.640
1.431	2.98a	52.69s	1,172.1	0.000	0.000	1.6786	-48.266
1.405	3.09a	55.06s	1,172.1	0.000	-0.248	1.6735	-50.640
1.340	3.31a	60.06s	1,172.1	0.000	-0.770	1.6291	-55.640

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.21 (constant)

Refer IRS Letter E-210766 279820 dated January 12, 2024

12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5133 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    52.63 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.36 P
-----------------------------------------------------------------------------
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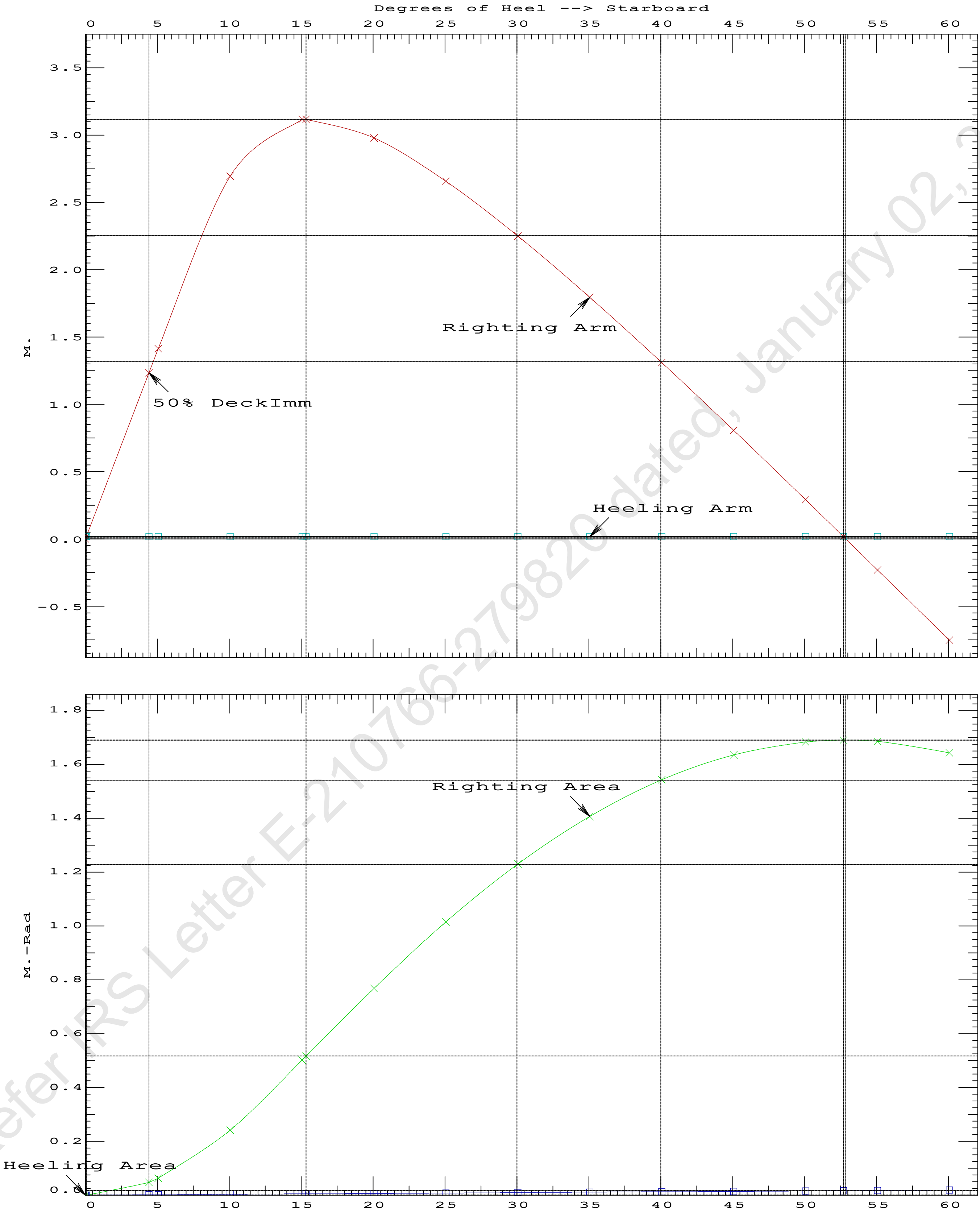
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft



Refer IRS Letter E-210766-279826 dated, January 02, 2024

12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.758f TCG = 0.000 VCG = 4.897
Free Surface Adjustment: 0.301
Adjusted CG: LCG = 23.761f TCG = 0.000 VCG = 5.197

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.627	0.51a	0.01s	1,172.1	0.000	0.000	0.0000
1.613	0.51a	5.01s	1,172.1	0.000	1.396	0.0609
1.581	0.55a	10.01s	1,172.0	0.000	2.680	0.2395
1.584	0.82a	15.01s	1,172.1	0.000	3.112	0.4993
1.584	0.83a	15.35s	1,172.1	0.000	3.113	0.5176
1.584	1.10a	20.01s	1,172.1	0.000	2.977	0.7687
1.579	1.39a	25.01s	1,172.1	0.000	2.656	1.0159
1.568	1.69a	30.01s	1,172.1	0.000	2.250	1.2306
1.552	1.99a	35.01s	1,172.1	0.000	1.796	1.4075
1.528	2.28a	40.01s	1,172.1	0.000	1.312	1.5433
1.497	2.57a	45.01s	1,172.1	0.000	0.808	1.6359
1.457	2.84a	50.01s	1,172.1	0.000	0.292	1.6840
1.430	2.98a	52.82s	1,172.1	0.000	0.000	1.6911
1.406	3.09a	55.01s	1,172.1	0.000	-0.229	1.6867
1.341	3.31a	60.01s	1,172.1	0.000	-0.751	1.6440

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

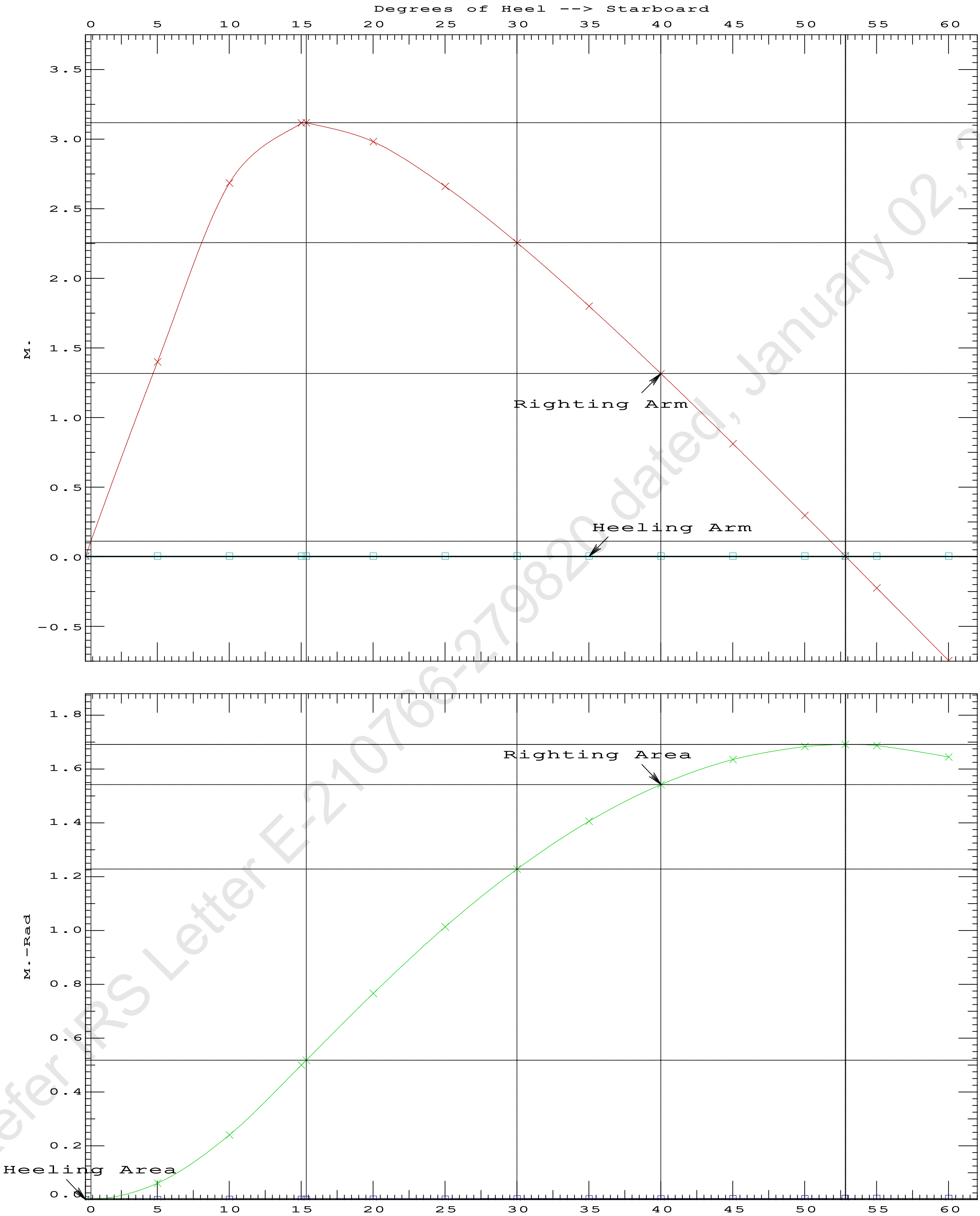
LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	72981.3	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		431.139	P

Refer IRS Letter E-210766-23020 dated January 02, 2024

12/21/23 09:39:04 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 04 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Aft



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:06 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.195 @ 50.00f, 1.417 @ 25.00f, 1.639 @ 0.00
Trim: Aft 0.444/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~19.500a~~ 18.500a	0.000	34.602	
Total Fixed--------->			887.10	16.906f	0.000	5.924	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			284.98	45.086f	0.000	1.697	352.57
Total Weight-------->			1,172.08	23.758f	0.000	4.897	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,172.08	23.721f	0.001s	0.722

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	859.5	24.579f	0.021s	142.53	19.951*
		MT/cm		m.-MT/cm	GML	GMT
		8.81		32.43	138.36	15.777

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:06    Cybermarine Technologies PTE. Ltd.
GHS 18.66                           PONTOON

                          After hook load drop
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
            USK draft: 1.423 @ 50.00f, 1.373 @ 25.00f, 1.323 @ 0.00
                   Trim: Fwd 0.099/50.000,  Heel: 0.00 deg.
Part----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          447.10  25.000f  0.000   3.000
Crane                               400.00  11.500f  0.000   6.325
  Total Fixed--------->             847.10  18.625f  0.000   4.570
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025       135.67  42.251f  0.000   1.499    135.61*
07_WBTK.P      1.000    1.025        49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025        50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      1.000    1.025        49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->             284.98  45.086f  0.000   1.697    352.57
  Total Weight-------->           1,132.08  25.286f  0.000   3.847
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025     1,132.09  25.293f  0.005s  0.693
------------------------------------------------------------------------------
               Righting Arms:                0.000   0.005s
               External Arms:                0.000   0.004s
      Residual Righting Arms:                0.000   0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          849.0  25.027f  0.068s   142.25  20.465*
                                       MT/cm
                                        8.70
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

              Theta3 is 0.07 degrees, as demonstrated below.
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Refer IRS Letter F-2/0768-279820 dated January 02, 2024

12/21/23 09:39:06 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.625f TCG = 0.000 VCG = 4.570

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.311	0.11f	0.00	1,132.1	0.000	0.001	0.0000
1.311	0.11f	0.07s	1,132.2	0.000	0.022	0.0000
1.296	0.11f	5.00s	1,132.1	0.000	1.566	0.0683
1.240	0.12f	10.00s	1,132.1	0.000	3.016	0.2691
1.080	0.18f	15.00s	1,132.1	0.000	3.600	0.5641
1.003	0.21f	17.25s	1,132.1	0.000	3.631	0.7066
0.906	0.24f	20.00s	1,132.1	0.000	3.587	0.8804
0.724	0.30f	25.00s	1,132.1	0.000	3.376	1.1866
0.536	0.37f	30.00s	1,132.1	0.000	3.072	1.4686

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 847.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:06 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.625f TCG = 0.000 VCG = 4.570

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.311	0.11f	0.00	1,132.1	0.000	0.001	0.0000
1.311	0.11f	0.07s	1,132.2	0.000	0.022	0.0000
1.296	0.11f	5.00s	1,132.1	0.000	1.566	0.0683
1.240	0.12f	10.00s	1,132.1	0.000	3.016	0.2691
1.080	0.18f	15.00s	1,132.1	0.000	3.600	0.5641
1.003	0.21f	17.25s	1,132.1	0.000	3.631	0.7066
0.906	0.24f	20.00s	1,132.1	0.000	3.587	0.8804
0.724	0.30f	25.00s	1,132.1	0.000	3.376	1.1866
0.536	0.37f	30.00s	1,132.1	0.000	3.072	1.4686
0.342	0.43f	35.00s	1,132.1	0.000	2.712	1.7214
0.144	0.49f	40.00s	1,132.1	0.000	2.315	1.9410
-0.056	0.55f	45.00s	1,132.1	0.000	1.889	2.1247
-0.256	0.61f	50.00s	1,132.1	0.000	1.443	2.2702
-0.454	0.67f	55.00s	1,132.1	0.000	0.982	2.3761
-0.649	0.72f	60.00s	1,132.1	0.000	0.510	2.4413
-0.839	0.76f	65.00s	1,132.1	0.000	0.032	2.4650
-0.851	0.76f	65.34s	1,132.1	0.000	0.000	2.4651
-1.021	0.79f	70.00s	1,132.1	0.000	-0.447	2.4469

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

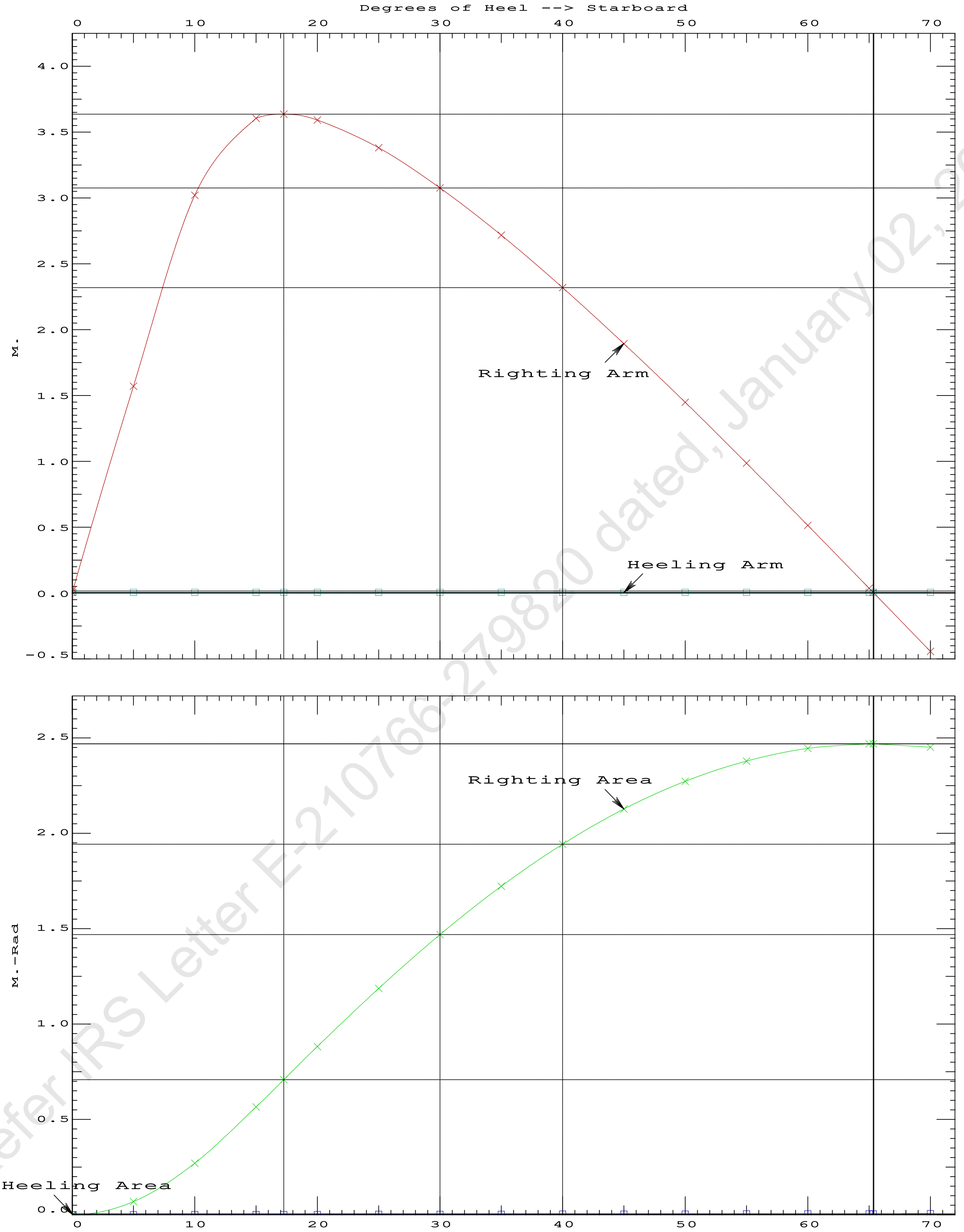
Note: The Center of Gravity shown above is for the Fixed Weight of 847.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM	RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1)	Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2)	Angle from abs 0.1 deg to RAzero	>	20.00	deg	65.27 P

Refer IRS Letter E-218766-279820 dated January 02, 2024

12/21/23 09:39:06 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

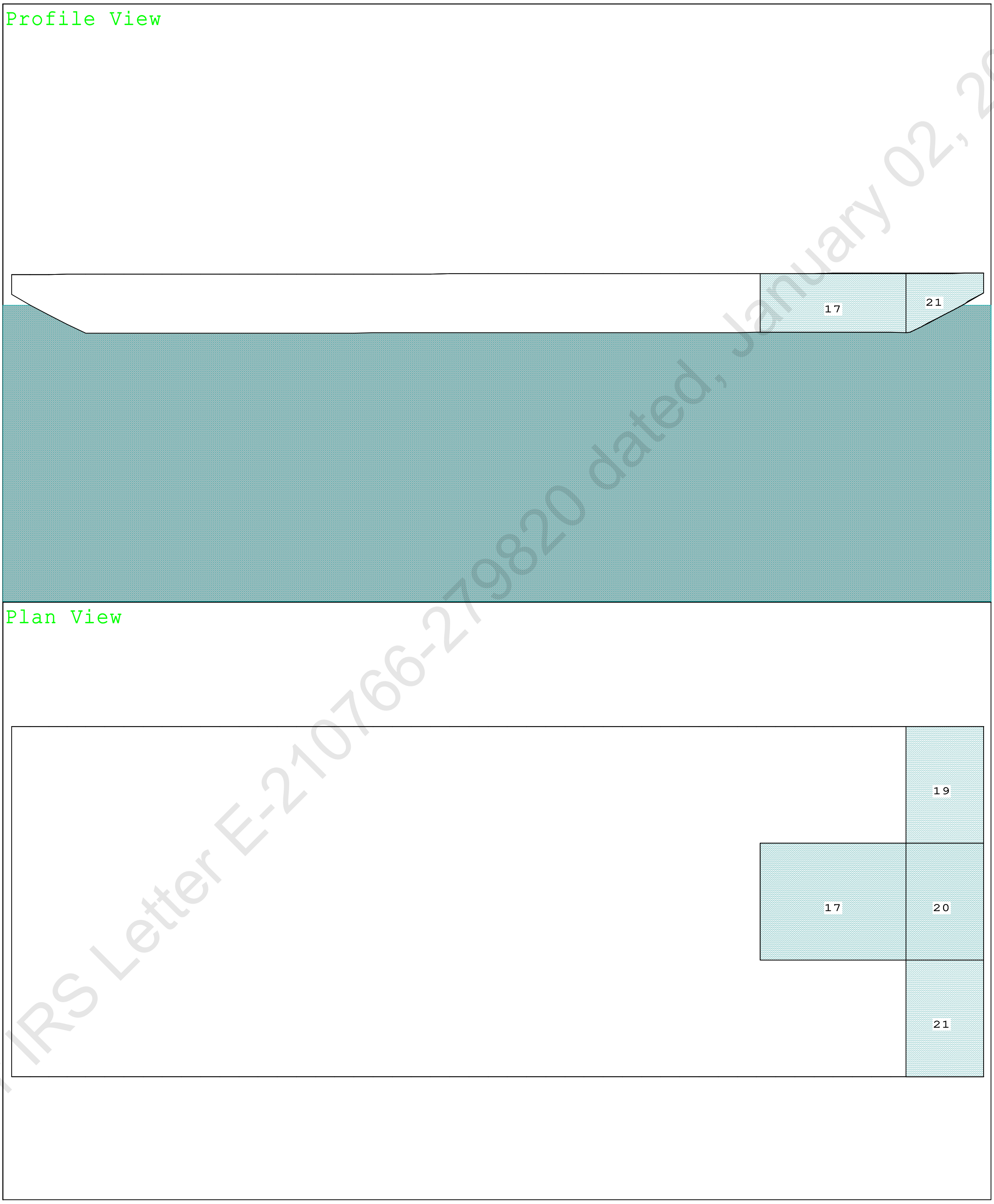
12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 05 : Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Stbd

Condition Graphic - Draft: 1.385 @ 50.000f, 1.451 @ 0.000 Heel: zero



Refer IRS Letter E-210766-279820 dated, January 02, 2024

Tanks
19 07_WBTK.P.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
17 06_WBTK.C.100% SALT WATER Intact 20 07_WBTK.C.100% SALT WATER Intact

12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.385 @ 50.00f, 1.418 @ 25.00f, 1.451 @ 0.00
Trim: Aft 0.066/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed--------->			887.10	18.304f	0.000	5.924	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			284.98	45.086f	0.000	1.697	352.57
Total Weight-------->			1,172.08	24.816f	0.000	4.897	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	1,172.08	24.810f	0.000	0.717	

Righting Arms:			0.000	0.000			
Part	**SpGr**	**WPA**	**LCF**	**TCF**	**BML**	**BMT**	
Total Waterplane---->	1.025	863.1	24.936f	0.000	144.29	20.109*	
		MT/cm		**m.-MT/cm**	**GML**	**GMT**	
		8.85		32.84	140.11	15.929	

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.385
DRAFT AT MS (M) = 1.418
DRAFT AT AP (M) = 1.451
DRAFT AT FWD MARK (M) = 1.391
DRAFT AT AFT MARK (M) = 1.446

HYDROSTATIC PROPERTIES
Trim: Aft 0.066/50.000, No Heel, VCG = 4.897

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.418	1,172.08	24.810f	0.717	8.85	24.936f	32.84	140.11	15.929

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.385 @ 50.00f, 1.418 @ 25.00f, 1.451 @ 0.00
Trim: Aft 0.066/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.5	0.807	1.502	0.05500	6.57
CRANE	57.0	4.074	4.768	0.05500	14.95
Total wind heeling moment to starboard------------->					21.52

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.816f TCG = 0.000 VCG = 4.897
Free Surface Adjustment: 0.301
Adjusted CG: LCG = 24.816f TCG = 0.000 VCG = 5.197

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.439	0.08a	0.00	1,172.1	0.000	-0.018	0.0000	5.074
1.439	0.08a	0.04s	1,172.1	0.000	0.000	-0.0000	5.030
1.425	0.08a	5.04s	1,172.1	0.000	1.391	0.0607	0.030
1.424	0.08a	5.07s	1,172.1	0.000	1.399	0.0614	0.000
1.377	0.08a	10.04s	1,172.1	0.000	2.692	0.2395	-4.970
1.280	0.12a	15.04s	1,172.1	0.000	3.133	0.5011	-9.970
1.274	0.12a	15.35s	1,172.1	0.000	3.134	0.5179	-10.278
1.174	0.16a	20.04s	1,172.1	0.000	2.997	0.7723	-14.970
1.061	0.21a	25.04s	1,172.1	0.000	2.676	1.0211	-19.970
0.940	0.25a	30.04s	1,172.1	0.000	2.270	1.2376	-24.970
0.813	0.30a	35.04s	1,172.1	0.000	1.815	1.4161	-29.970
0.680	0.34a	40.04s	1,172.1	0.000	1.330	1.5536	-34.970
0.543	0.38a	45.04s	1,172.1	0.000	0.824	1.6477	-39.970
0.402	0.42a	50.04s	1,172.1	0.000	0.307	1.6972	-44.970
0.318	0.44a	52.97s	1,172.1	0.000	0.000	1.7050	-47.898
0.258	0.46a	55.04s	1,172.1	0.000	-0.218	1.7011	-49.970
0.112	0.49a	60.04s	1,172.1	0.000	-0.743	1.6592	-54.970

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.52 (constant)

Refer IRS Letter E-2107662798?? dated., January 02, 2024

12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

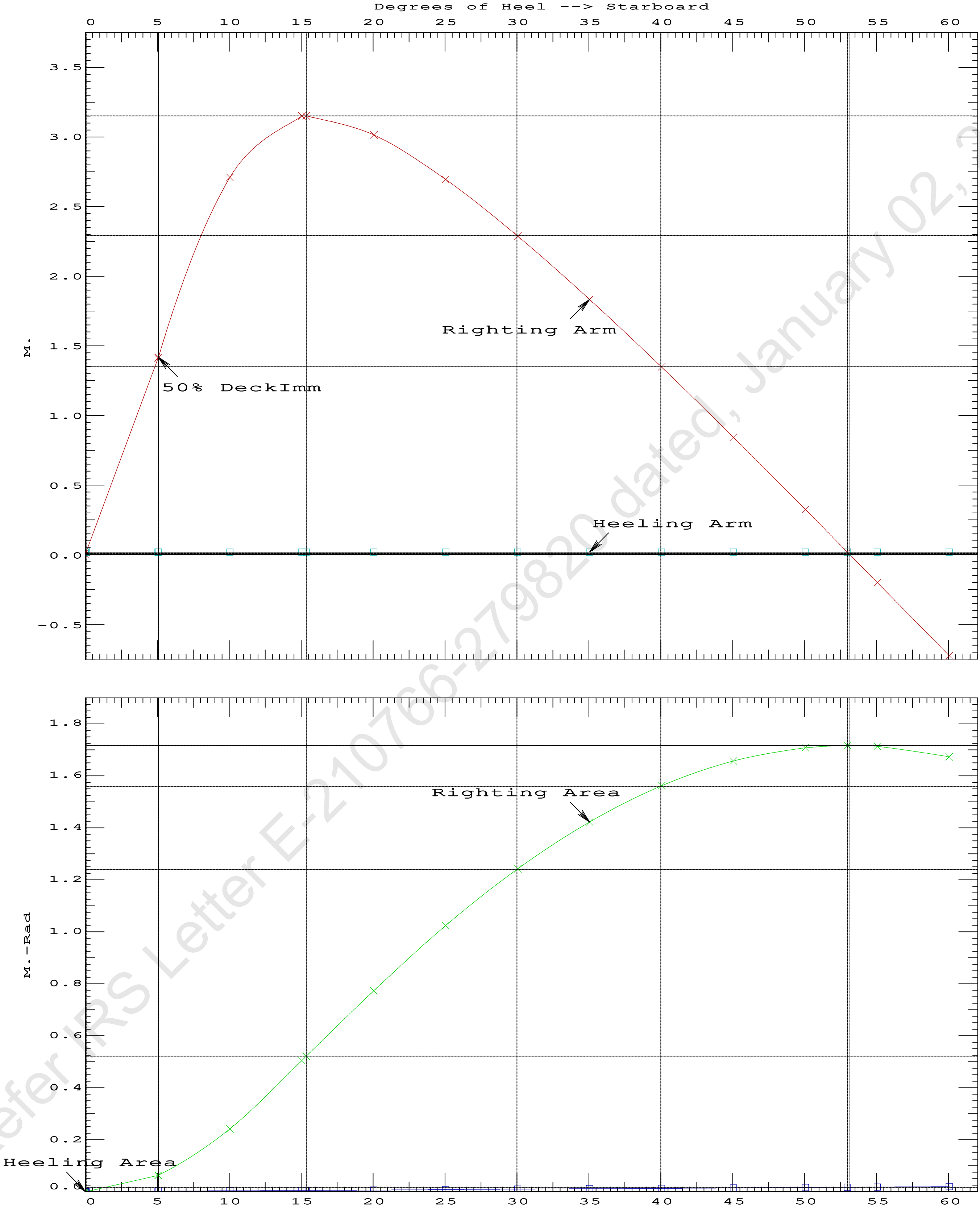
Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5179 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    52.93 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     5.03 P
-----------------------------------------------------------------------------
```

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

```
   Total CG:  LCG = 24.816f  TCG = 0.000   VCG =  4.897
                      Free Surface Adjustment:   0.301
Adjusted CG:  LCG = 24.816f  TCG = 0.019p  VCG =  5.197
```

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.431	0.07a	3.59s	1,172.1	0.000	0.000	0.0000
1.397	0.08a	8.59s	1,172.1	0.000	1.355	0.0591
1.309	0.11a	13.59s	1,172.1	0.000	2.112	0.2197
1.269	0.13a	15.58s	1,172.1	0.000	2.146	0.2940
1.206	0.15a	18.59s	1,172.1	0.000	2.076	0.4061
1.095	0.19a	23.59s	1,172.1	0.000	1.793	0.5780
0.976	0.24a	28.59s	1,172.1	0.000	1.405	0.7183
0.850	0.28a	33.59s	1,172.1	0.000	0.962	0.8220
0.719	0.33a	38.59s	1,172.1	0.000	0.483	0.8853
0.588	0.37a	43.41s	1,172.1	0.000	0.000	0.9058
0.583	0.37a	43.59s	1,172.1	0.000	-0.018	0.9057
0.443	0.41a	48.59s	1,172.1	0.000	-0.534	0.8817
0.300	0.45a	53.59s	1,172.1	0.000	-1.059	0.8123
0.155	0.48a	58.59s	1,172.1	0.000	-1.586	0.6969
0.007	0.51a	63.59s	1,172.1	0.000	-2.111	0.5356

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1200.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

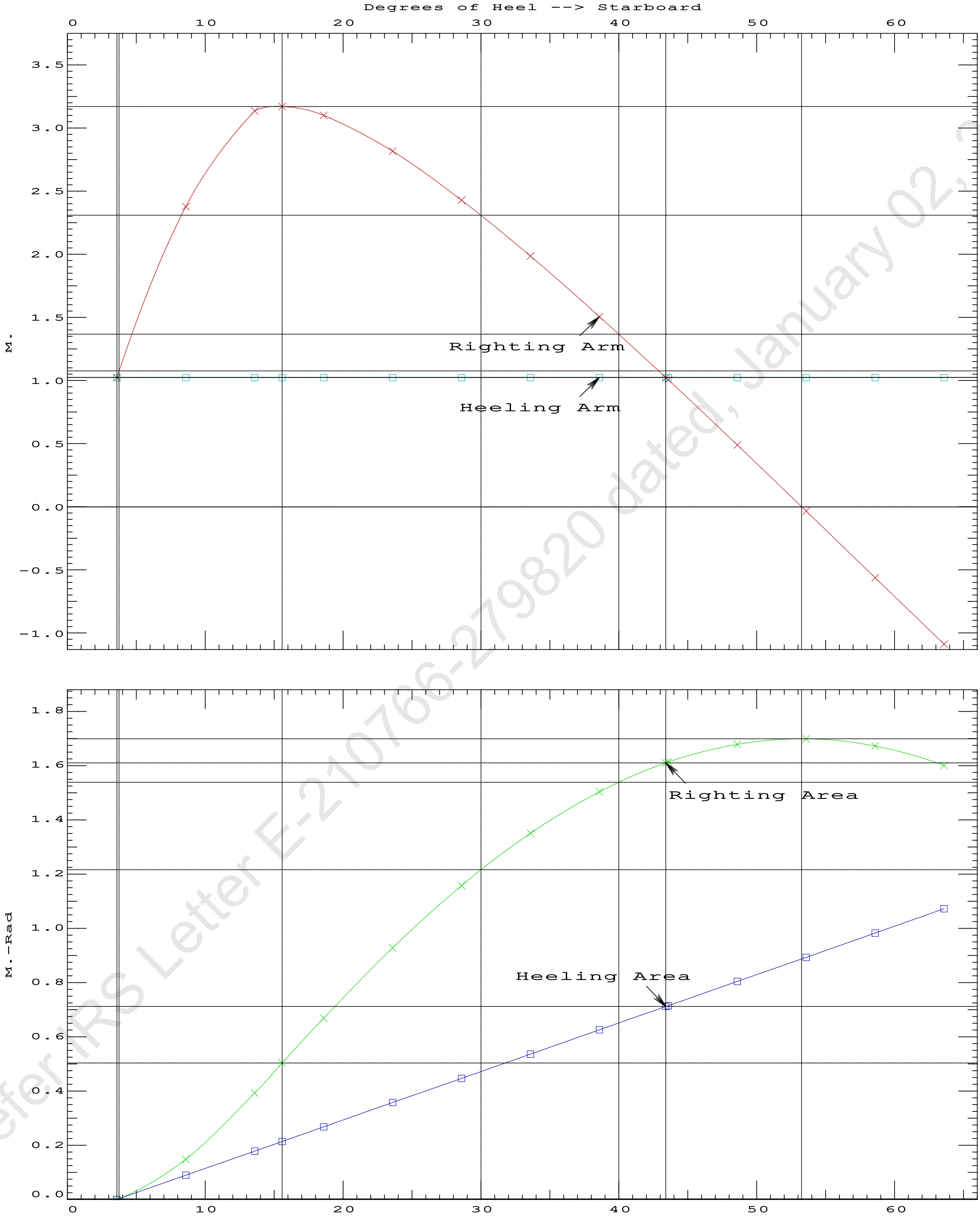
LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	3.59 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	209.6 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2.273 P

Refer IRS Letter E-21076-23620 dated January 02, 2024

12/21/23 09:39:08 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 05 : Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Stbd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 05 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.380 @ 50.00f, 1.413 @ 25.00f, 1.446 @ 0.00
Trim: Aft 0.065/50.000, Heel: Stbd 3.59 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed--------->			887.10	18.304f	0.000	5.924	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			284.98	45.086f	0.000	1.697	352.57
Total Weight-------->			1,172.08	24.816f	0.000	4.897	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	1,172.09	24.810f	1.285s	0.757	

Righting Arms:	0.000	1.024s
External Arms:	0.000	1.024s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	862.5	24.930f	0.151s	143.78	20.118*
		MT/cm				
		8.84				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.59

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:11    Cybermarine Technologies PTE. Ltd.
GHS 18.66                         PONTOON

                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.423 @ 50.00f, 1.373 @ 25.00f, 1.323 @ 0.00
                    Trim: Fwd 0.099/50.000,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            447.10  25.000f  0.000   3.000
Crane                                 400.00  11.500f  0.000   6.325
  Total Fixed--------->               847.10  18.625f  0.000   4.570
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025         135.67  42.251f  0.000   1.499    135.61*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025          50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->               284.98  45.086f  0.000   1.697    352.57
  Total Weight-------->             1,132.08  25.286f  0.000   3.847
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,132.06  25.294f  0.000   0.693
------------------------------------------------------------------------------
             Righting Arms:                  0.001f  0.000
             External Arms:                  0.000   0.001
      Residual Righting Arms:                0.001f -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          859.8  25.095f  0.000    147.70  20.728*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.81            32.73    144.54  17.575
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 4070.21; t3: 30.0; t: 15.0
limmarg: 1502.44; t3: 15.0; t: 6.0
limmarg: 524.44; t3: 9.0; t: 2.4
limmarg: 238.30; t3: 6.6; t: 1.0
limmarg: 143.83; t3: 5.6; t: 0.4
limmarg: 110.28; t3: 5.2; t: 0.2
limmarg: 94.43; t3: 5.0; t: 0.1
limmarg: 86.76; t3: 4.9; t: 0.0
limmarg: 86.76; t3: 4.9; t: 0.0

               Theta3 is 4.90 degrees, as demonstrated below.

                    RESIDUAL RIGHTING ARMS vs HEEL ANGLE
          Fixed CG:  LCG = 18.625f  TCG = 0.000   VCG =  4.570

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.304   0.11f   3.59s     1,132.1     0.000   -1.127    0.0000
       1.311   0.11f   0.00      1,132.1     0.000    0.001   -0.0353
       1.297   0.11f   4.90p     1,132.1     0.000    1.539    0.0306
       1.296   0.11f   5.00p     1,132.1     0.000    1.570    0.0333
       1.240   0.12f  10.00p     1,132.1     0.000    3.021    0.2345
       1.080   0.18f  15.00p     1,132.1     0.000    3.605    0.5300
       1.003   0.21f  17.25p     1,132.1     0.000    3.637    0.6728
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Refer IRS Letter E-2170760 279820 dated January 02, 2024

12/21/23 09:39:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

0.906	0.24f	20.00p	1,132.1	0.000	3.592	0.8467
0.724	0.30f	25.00p	1,132.1	0.000	3.381	1.1534
0.536	0.37f	30.00p	1,132.1	0.000	3.077	1.4359

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 847.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Res. Area Ratio from abs -3.6 deg to abs 4.9   >    1.000         1.868 P
-----------------------------------------------------------------------------
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Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:11 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.625f TCG = 0.000 VCG = 4.570

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.304	0.11f	3.59s	1,132.1	0.000	-1.127	0.0000
1.311	0.11f	0.00	1,132.1	0.000	0.001	-0.0353
1.297	0.11f	4.90p	1,132.1	0.000	1.539	0.0306
1.296	0.11f	5.00p	1,132.1	0.000	1.570	0.0333
1.240	0.12f	10.00p	1,132.1	0.000	3.021	0.2345
1.080	0.18f	15.00p	1,132.1	0.000	3.605	0.5300
1.003	0.21f	17.25p	1,132.1	0.000	3.637	0.6728
0.906	0.24f	20.00p	1,132.1	0.000	3.592	0.8467
0.724	0.30f	25.00p	1,132.1	0.000	3.381	1.1534
0.536	0.37f	30.00p	1,132.1	0.000	3.077	1.4359
0.342	0.43f	35.00p	1,132.1	0.000	2.718	1.6892
0.144	0.49f	40.00p	1,132.1	0.000	2.320	1.9093
-0.056	0.55f	45.00p	1,132.1	0.000	1.895	2.0934
-0.256	0.61f	50.00p	1,132.1	0.000	1.449	2.2394
-0.454	0.67f	55.00p	1,132.1	0.000	0.987	2.3458
-0.649	0.72f	60.00p	1,132.1	0.000	0.515	2.4114
-0.839	0.76f	65.00p	1,132.1	0.000	0.038	2.4356
-0.853	0.76f	65.39p	1,132.1	0.000	0.000	2.4357
-1.021	0.79f	70.00p	1,132.1	0.000	-0.442	2.4180

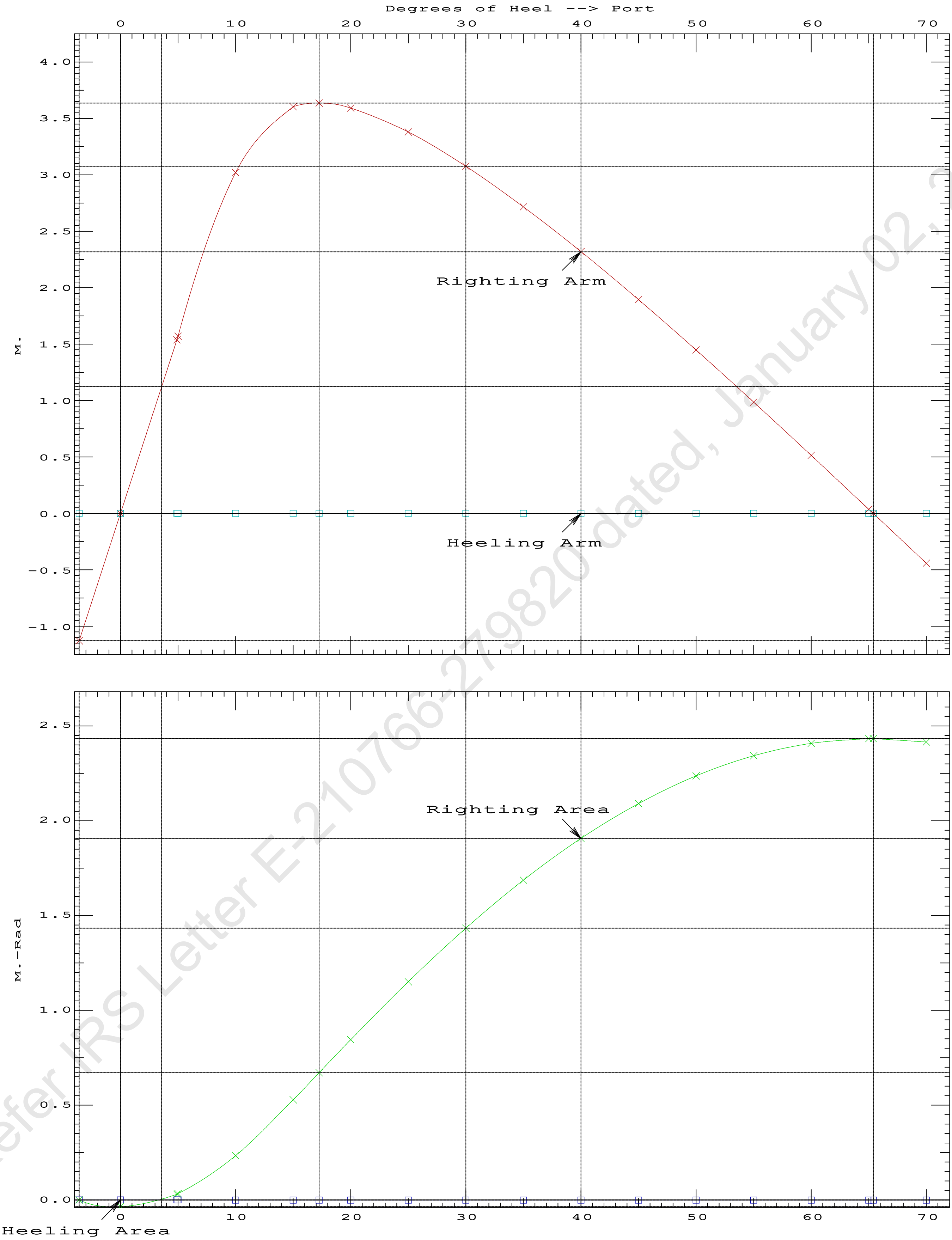
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 847.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -3.6 deg to RAzero	>	1.000		70.049 P
(2) Angle from abs 4.9 deg to RAzero	>	20.00	deg	60.49 P

Refer IRS Letter E-2107/66 27/0020 dated January 02, 2024



Refer IRS Letter E-210766-279820 dated, January 02, 2024

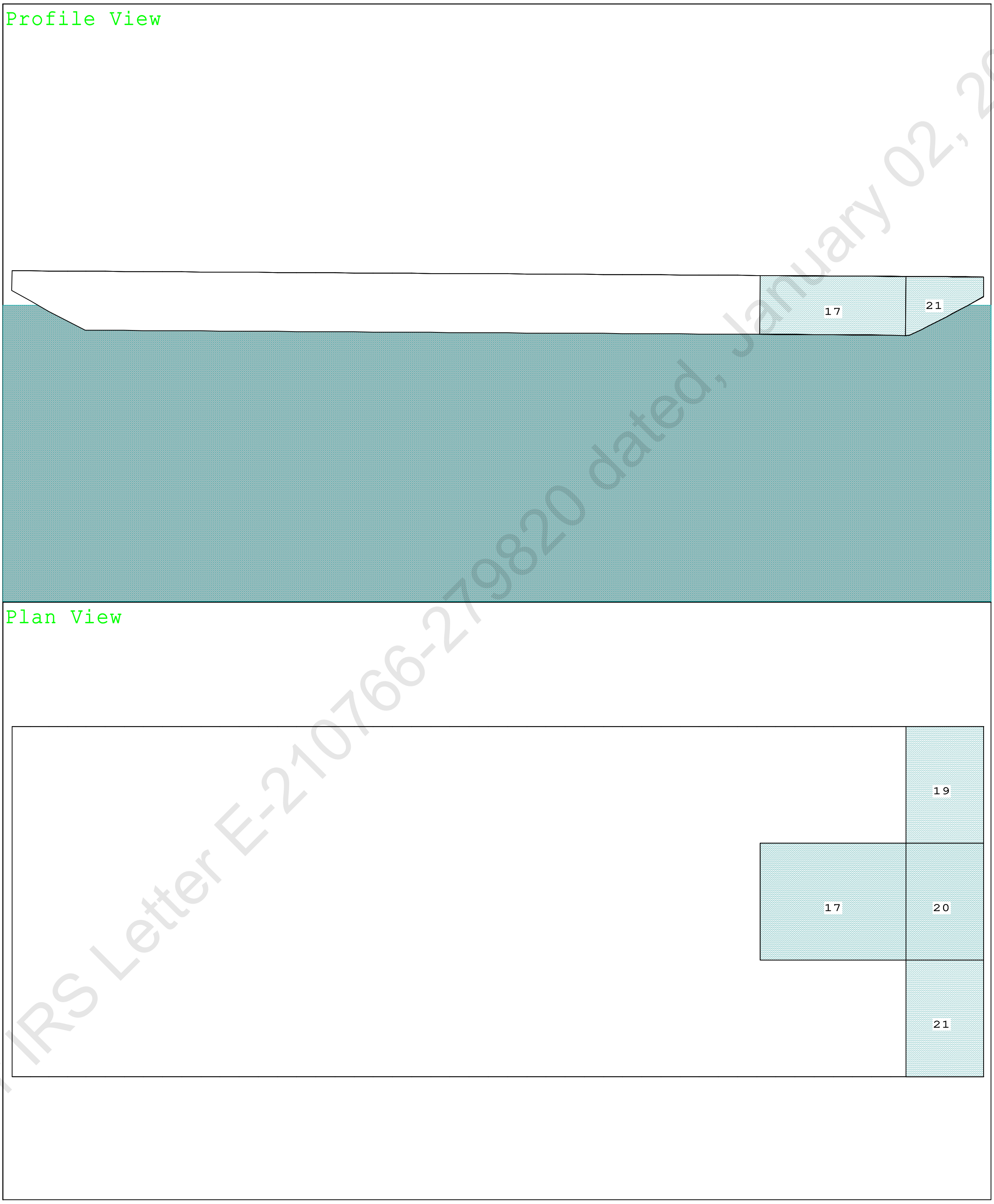
12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd

Condition Graphic - Draft: 1.574 @ 50.000f, 1.261 @ 0.000 Heel: zero



Tanks
19 07_WBTK.P.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
17 06_WBTK.C.100% SALT WATER Intact 20 07_WBTK.C.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.574 @ 50.00f, 1.418 @ 25.00f, 1.261 @ 0.00
Trim: Fwd 0.312/50.000, Heel: zero

Part	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~42.500f~~ 41.500f	0.000	34.602	
Total Fixed--------->			887.10	19.702f	0.000	5.924	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			284.98	45.086f	0.000	1.697	352.57
Total Weight-------->			1,172.08	25.874f	0.000	4.897	
			Displ (MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	1,172.02	25.900f	0.000	0.719	

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	862.3	25.273f	0.000	143.91	20.059*
		MT/cm		**m.-MT/cm**	**GML**	**GMT**
		8.84		32.75	139.73	15.882

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING
278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.574
DRAFT AT MS (M) = 1.418
DRAFT AT AP (M) = 1.261
DRAFT AT FWD MARK (M) = 1.549
DRAFT AT AFT MARK (M) = 1.286

HYDROSTATIC PROPERTIES
Trim: Fwd 0.312/50.000, No Heel, VCG = 4.897

LCF Draft	Displacement Weight (MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.419	1,172.02	25.900f	0.719	8.84	25.273f	32.75	139.73	15.882

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

Refer IRS Letter E-240766 279820 dated January 02, 2024

12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.574 @ 50.00f, 1.418 @ 25.00f, 1.261 @ 0.00
Trim: Fwd 0.312/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	79.5	0.811	1.506	0.05500	6.58
CRANE	57.0	4.188	4.883	0.05500	15.31
Total wind heeling moment to starboard------------->					21.89

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.874f TCG = 0.000 VCG = 4.897
Free Surface Adjustment: 0.301
Adjusted CG: LCG = 25.872f TCG = 0.000 VCG = 5.197

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.249	0.36f	0.00	1,172.0	0.000	-0.018	0.0000	4.642
1.249	0.36f	0.06s	1,172.1	0.000	0.000	-0.0000	4.580
1.236	0.36f	4.64s	1,172.1	0.000	1.279	0.0511	0.000
1.233	0.36f	5.06s	1,172.1	0.000	1.396	0.0609	-0.420
1.171	0.39f	10.06s	1,172.1	0.000	2.686	0.2398	-5.420
0.976	0.58f	15.06s	1,172.1	0.000	3.116	0.5001	-10.420
0.964	0.59f	15.35s	1,172.1	0.000	3.117	0.5155	-10.704
0.764	0.78f	20.06s	1,172.1	0.000	2.979	0.7698	-15.420
0.541	0.98f	25.06s	1,172.1	0.000	2.657	1.0171	-20.420
0.309	1.19f	30.06s	1,172.1	0.000	2.250	1.2318	-25.420
0.071	1.40f	35.06s	1,172.1	0.000	1.796	1.4087	-30.420
-0.171	1.61f	40.06s	1,172.1	0.000	1.311	1.5445	-35.420
-0.414	1.81f	45.06s	1,172.1	0.000	0.806	1.6370	-40.420
-0.656	2.00f	50.06s	1,172.1	0.000	0.289	1.6848	-45.420
-0.788	2.10f	52.83s	1,172.1	0.000	0.000	1.6918	-48.189
-0.893	2.18f	55.06s	1,172.1	0.000	-0.234	1.6873	-50.420
-1.121	2.33f	60.06s	1,172.1	0.000	-0.757	1.6440	-55.420

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.89 (constant)

Refer IRS Letter E-2107662798..., dated January 02, 2024

12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 06 :Operating Condition of 400T Crane

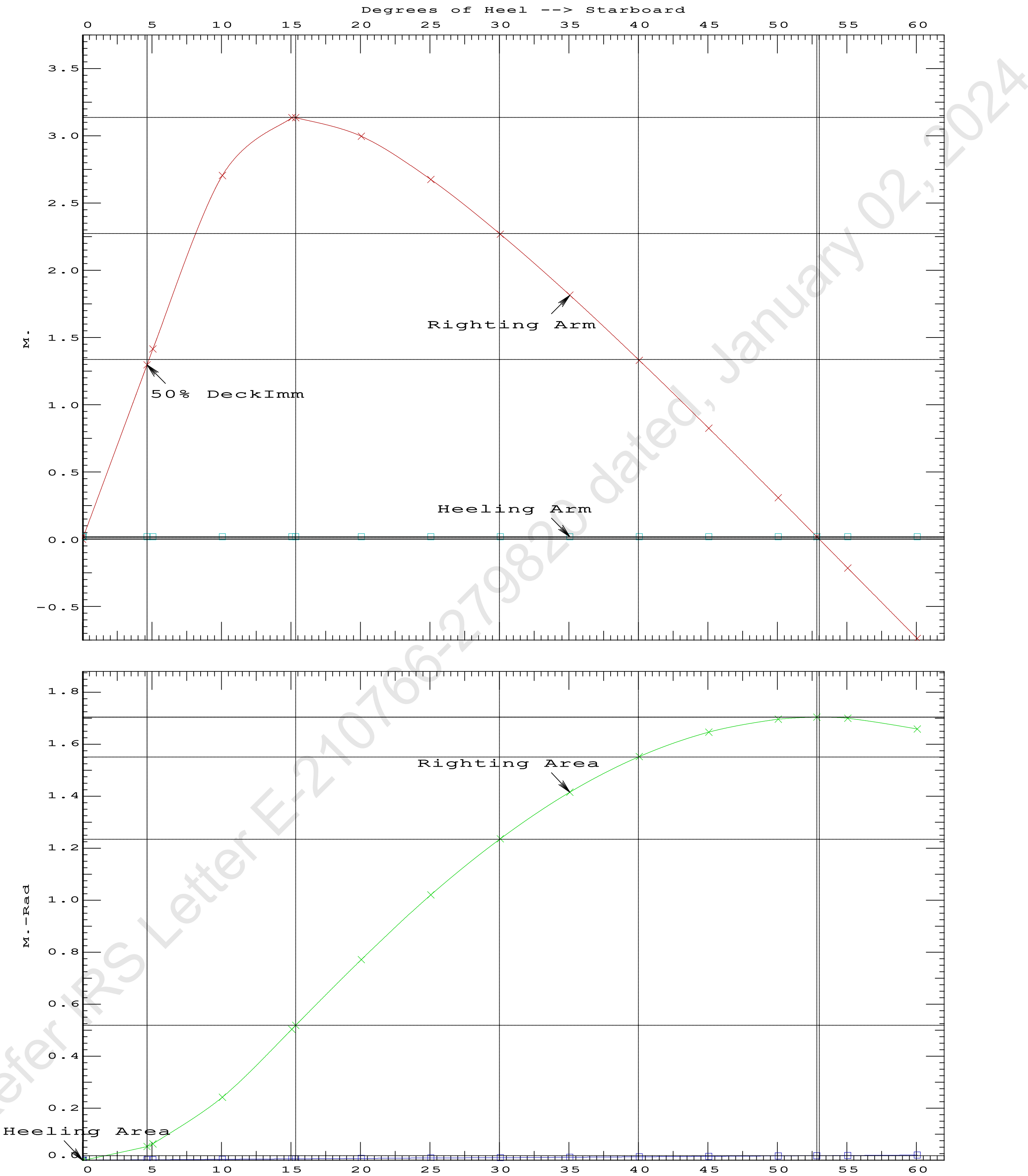
Crane Lifting 40.0T @ 30m Radius Fwd

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5155 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    52.77 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.58 P
-----------------------------------------------------------------------------
```

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.874f TCG = 0.000 VCG = 4.897
Free Surface Adjustment: 0.301
Adjusted CG: LCG = 25.872f TCG = 0.000 VCG = 5.197

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.249	0.36f	0.00	1,172.1	0.000	0.000	0.0000
1.234	0.36f	5.00s	1,172.0	0.000	1.396	0.0609
1.172	0.39f	10.00s	1,172.1	0.000	2.691	0.2400
0.978	0.57f	15.00s	1,172.1	0.000	3.133	0.5014
0.964	0.59f	15.36s	1,172.1	0.000	3.134	0.5210
0.766	0.77f	20.00s	1,172.1	0.000	3.000	0.7727
0.544	0.98f	25.00s	1,172.1	0.000	2.679	1.0218
0.312	1.19f	30.00s	1,172.1	0.000	2.273	1.2385
0.074	1.40f	35.00s	1,172.1	0.000	1.819	1.4175
-0.168	1.61f	40.00s	1,172.1	0.000	1.334	1.5553
-0.411	1.81f	45.00s	1,172.1	0.000	0.830	1.6499
-0.653	2.00f	50.00s	1,172.1	0.000	0.313	1.6998
-0.796	2.11f	53.00s	1,172.1	0.000	0.000	1.7080
-0.890	2.18f	55.00s	1,172.1	0.000	-0.210	1.7044
-1.119	2.33f	60.00s	1,172.1	0.000	-0.733	1.6632

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	367381	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2165.25	P

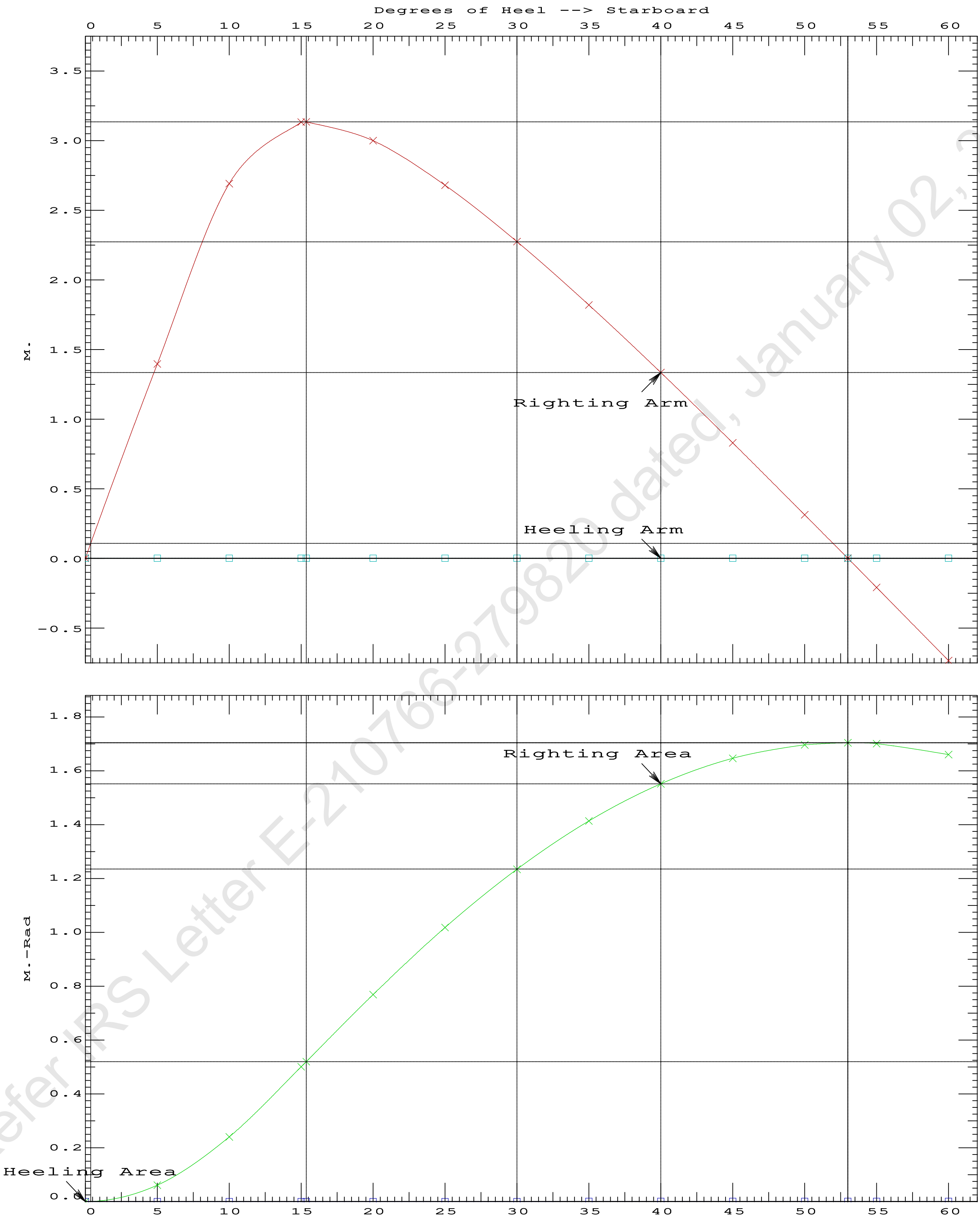
Refer IRS Letter E-21076-2 0620 dated January 02, 2024

12/21/23 09:39:13 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:15 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 06 :Operating Condition of 400T Crane

Crane Lifting 40.0T @ 30m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.574 @ 50.00f, 1.418 @ 25.00f, 1.261 @ 0.00
Trim: Fwd 0.312/50.000, Heel: Stbd 0.01 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	~~42.500f~~ 41.500f	0.000	34.602	
Total Fixed--------->			887.10	19.702f	0.000	5.924	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			284.98	45.086f	0.000	1.697	352.57
Total Weight-------->			1,172.08	25.874f	0.000	4.897	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,172.08	25.900f	0.005s	0.719

Righting Arms:	0.000	0.004s
External Arms:	0.000	0.004s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	859.1	25.348f	0.024s	142.31	19.956*
		MT/cm 8.81				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:15    Cybermarine Technologies PTE. Ltd.
GHS 18.66                         PONTOON

                           After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.423 @ 50.00f, 1.373 @ 25.00f, 1.323 @ 0.00
                  Trim: Fwd 0.099/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          447.10  25.000f  0.000   3.000
Crane                               400.00  11.500f  0.000   6.325
  Total Fixed--------->             847.10  18.625f  0.000   4.570
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
06_WBTK.C      1.000    1.025       135.67  42.251f  0.000   1.499    135.61*
07_WBTK.P      1.000    1.025        49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.C      1.000    1.025        50.11  47.663f  0.000   1.872     72.33*
07_WBTK.S      1.000    1.025        49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->             284.98  45.086f  0.000   1.697    352.57
  Total Weight-------->           1,132.08  25.286f  0.000   3.847
                                Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025     1,132.09  25.292f  0.005s  0.693
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.005s
               External Arms:                  0.000   0.004s
      Residual Righting Arms:                  0.000   0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          849.0  25.027f  0.068s   142.25  20.465*
                                       MT/cm
                                        8.70
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 365220544.00; t3: 30; t: 15.00
limmarg: 140335440.00; t3: 15; t: 13.50
limmarg: 1545946.25; t3: 2; t: 12.15
limmarg: ; t3: -10; t: 10.94
limmarg: 613356.25; t3: 1; t: 9.84
limmarg: ; t3: -9; t: 8.86
limmarg: ; t3: 0; t: 7.97
limmarg: 43077152.00; t3: 8; t: 7.17
limmarg: 479708.06; t3: 1; t: 6.46
limmarg: ; t3: -5; t: 5.81
limmarg: 457179.78; t3: 1; t: 5.23
limmarg: ; t3: -4; t: 4.71
limmarg: 352710.34; t3: 1; t: 4.24
limmarg: ; t3: -3; t: 3.81
limmarg: 457179.78; t3: 1; t: 3.43
limmarg: ; t3: -2; t: 3.09
limmarg: 821454.56; t3: 1; t: 2.78
limmarg: ; t3: -2; t: 2.50
limmarg: 177591.89; t3: 1; t: 2.25
limmarg: ; t3: -1; t: 2.03
limmarg: 734670.31; t3: 1; t: 1.82
limmarg: ; t3: -1; t: 1.64
limmarg: 287677.50; t3: 1; t: 1.48
limmarg: ; t3: 0; t: 1.33
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Refer IRS Letter F-210766-279820 dated January 02, 2024

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limmarg: ; t3: -1; t: 0.79
limmarg: ; t3: 0; t: 0.71
limmarg: 352710.34; t3: 1; t: 0.64
limmarg: 94128.28; t3: 0; t: 0.57
limmarg: ; t3: -1; t: 0.52
limmarg: ; t3: 0; t: 0.46
limmarg: 151032.06; t3: 0; t: 0.42
limmarg: ; t3: 0; t: 0.38
limmarg: 104422.68; t3: 0; t: 0.34
limmarg: ; t3: 0; t: 0.30
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limmarg: ; t3: 0; t: 0.25
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limmarg: ; t3: 0; t: 0.20
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limmarg: ; t3: 0; t: 0.16
limmarg: 20232.62; t3: 0; t: 0.15
limmarg: ; t3: 0; t: 0.13
limmarg: 13796.02; t3: 0; t: 0.12
limmarg: ; t3: 0; t: 0.11
limmarg: 10179.55; t3: 0; t: 0.10
limmarg: ; t3: 0; t: 0.09
limmarg: 7102.84; t3: 0; t: 0.08
limmarg: ; t3: 0; t: 0.07
limmarg: 4565.33; t3: 0; t: 0.06
limmarg: ; t3: 0; t: 0.06
limmarg: 3498.73; t3: 0; t: 0.05
limmarg: ; t3: 0; t: 0.05
limmarg: 2566.87; t3: 0; t: 0.04
limmarg: ; t3: 0; t: 0.04
limmarg: 1769.74; t3: 0; t: 0.03
limmarg: ; t3: 0; t: 0.03
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limmarg: 484.84; t3: 0; t: 0.02
limmarg: 0.00; t3: 0; t: 0.02
```

Theta3 is -0.02 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.625f TCG = 0.000 VCG = 4.570

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.312	0.11f	0.01s	1,132.1	0.000	-0.004	0.0000
1.311	0.11f	0.00	1,132.1	0.000	-0.001	-0.0000
1.297	0.11f	5.00p	1,132.1	0.000	1.573	0.0686
1.240	0.12f	10.00p	1,132.1	0.000	3.025	0.2702
1.080	0.18f	15.00p	1,132.1	0.000	3.609	0.5659
1.003	0.21f	17.25p	1,132.1	0.000	3.640	0.7090
0.906	0.24f	20.00p	1,132.1	0.000	3.595	0.8830

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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GHS 18.66 **PONTOON**

0.724	0.30f	25.00p	1,132.1	0.000	3.385	1.1899
0.536	0.37f	30.00p	1,132.1	0.000	3.081	1.4727

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 847.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0	>	1.000	1.000 P

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:15 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.625f TCG = 0.000 VCG = 4.570

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.312	0.11f	0.01s	1,132.1	0.000	-0.004	0.0000
1.311	0.11f	0.00	1,132.1	0.000	-0.001	-0.0000
1.297	0.11f	5.00p	1,132.1	0.000	1.573	0.0686
1.240	0.12f	10.00p	1,132.1	0.000	3.025	0.2702
1.080	0.18f	15.00p	1,132.1	0.000	3.609	0.5659
1.003	0.21f	17.25p	1,132.1	0.000	3.640	0.7090
0.906	0.24f	20.00p	1,132.1	0.000	3.595	0.8830
0.724	0.30f	25.00p	1,132.1	0.000	3.385	1.1899
0.536	0.37f	30.00p	1,132.1	0.000	3.081	1.4727
0.342	0.43f	35.00p	1,132.1	0.000	2.721	1.7263
0.144	0.49f	40.00p	1,132.1	0.000	2.324	1.9467
-0.056	0.55f	45.00p	1,132.1	0.000	1.898	2.1312
-0.256	0.61f	50.00p	1,132.1	0.000	1.452	2.2775
-0.454	0.67f	55.00p	1,132.1	0.000	0.991	2.3842
-0.649	0.72f	60.00p	1,132.1	0.000	0.519	2.4502
-0.839	0.76f	65.00p	1,132.1	0.000	0.041	2.4747
-0.855	0.76f	65.43p	1,132.1	0.000	0.000	2.4748
-1.021	0.79f	70.00p	1,132.1	0.000	-0.438	2.4574

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

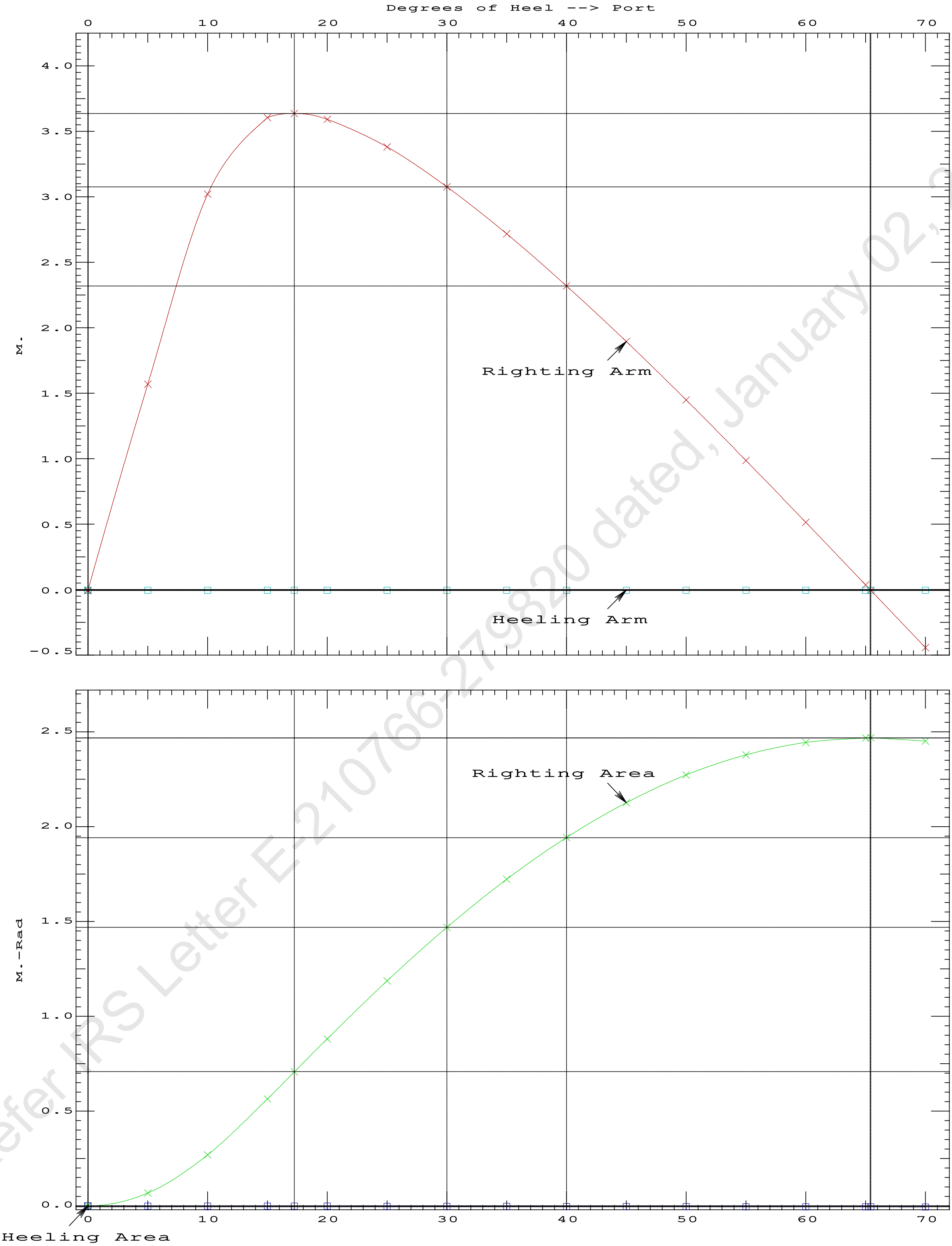
Note: The Center of Gravity shown above is for the Fixed Weight of 847.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		6137253 P
(2) Angle from abs 0 deg to RAzero	>	20.00	deg	65.44 P

Refer IRS Letter E-21766-279820 dated January 02, 2024

12/21/23 09:39:15 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

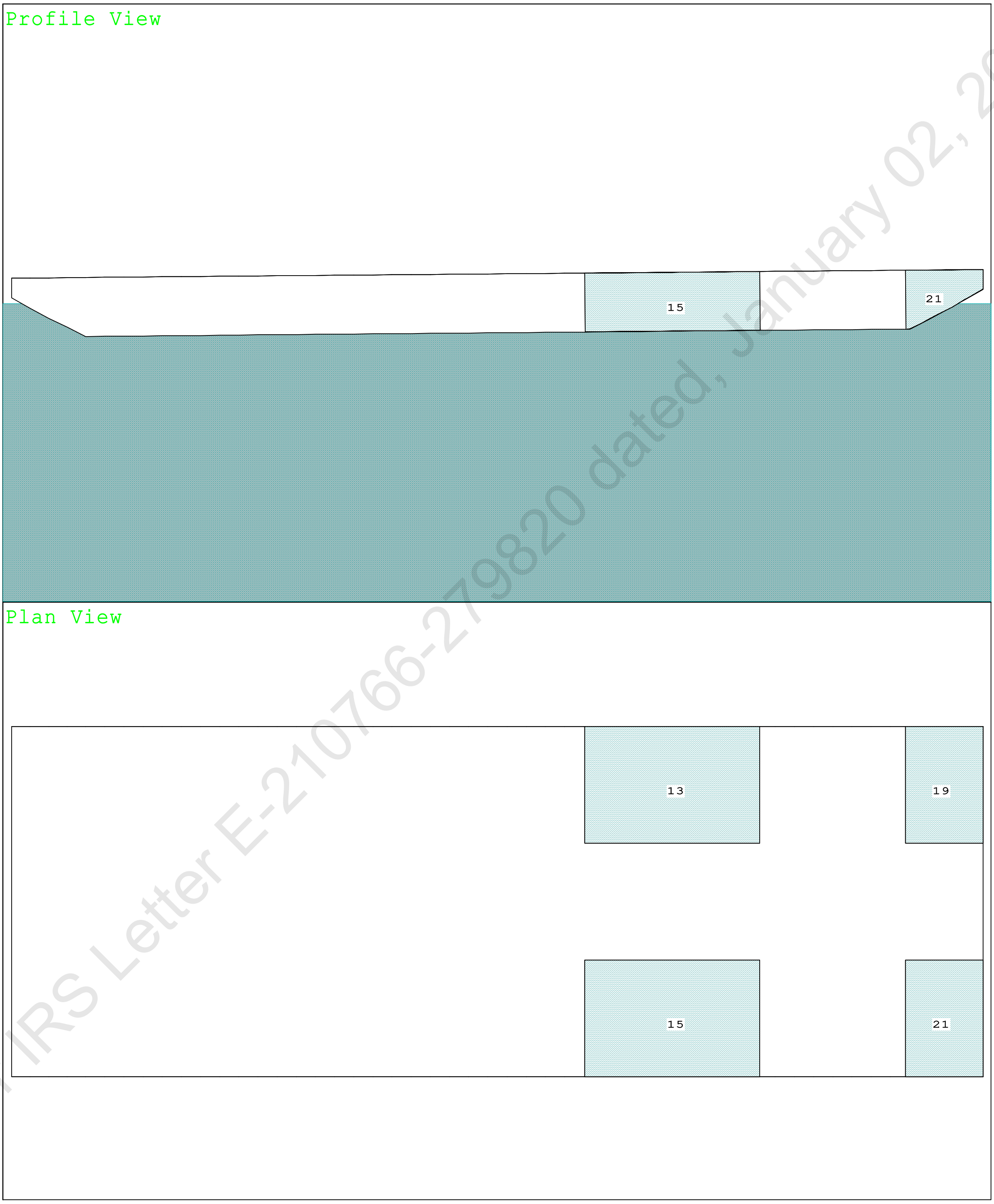


Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:17 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Condition Graphic - Draft: 1.291 @ 50.000f, 1.739 @ 0.000 Heel: zero



Tanks
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:39:17    Cybermarine Technologies PTE. Ltd.
GHS 18.66                         PONTOON
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Condition 07 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Aft

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                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.291 @ 50.00f, 1.515 @ 25.00f, 1.739 @ 0.00
                     Trim: Aft 0.448/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                350.00  11.500f  0.000   6.325
Crane Load                            40.00  23.500a  0.000  21.478
  Total Fixed--------->              837.10  17.038f  0.000   5.273
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596    470.09
  Total Weight-------->            1,259.50  23.802f  0.000   4.040
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,259.38  23.773f  0.000   0.773
-------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         868.8  24.611f  0.000    136.99  18.722*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.91            33.68    133.73  15.455
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                             SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      350.00 MT  Crane                        40.00 MT  Crane Load

DRAFT AT FP (M) =    1.291
DRAFT AT MS (M) =    1.515
DRAFT AT AP (M) =    1.739
DRAFT AT FWD MARK (M) =    1.327
DRAFT AT AFT MARK (M) =    1.703
                            HYDROSTATIC PROPERTIES
                  Trim: Aft 0.448/50.000, No Heel, VCG = 4.040

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.519     1,259.38  23.773f  0.773      8.91  24.611f   33.68  133.73  15.455
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
 Draft is from USK.                                  Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
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Refer IRS Letter E-210766 279820 dated January 02, 2024

12/21/23 09:39:17 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.291 @ 50.00f, 1.515 @ 25.00f, 1.739 @ 0.00
Trim: Aft 0.448/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	74.7	0.764	1.508	0.05500	6.20
CRANE	57.0	3.863	4.608	0.05500	14.44
Total wind heeling moment to starboard------------->					20.64

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 23.802f TCG = 0.000 VCG = 4.040
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 23.806f TCG = 0.000 VCG = 4.413

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.727	0.51a	0.00	1,259.5	0.000	-0.016	0.0000	4.091
1.727	0.51a	0.05s	1,259.5	0.000	0.000	-0.0000	4.037
1.717	0.51a	4.09s	1,259.5	0.000	1.095	0.0386	0.000
1.714	0.52a	5.05s	1,259.5	0.000	1.355	0.0592	-0.963
1.691	0.56a	10.05s	1,259.4	0.000	2.621	0.2333	-5.963
1.748	0.84a	15.05s	1,259.5	0.000	3.068	0.4882	-10.963
1.759	0.90a	16.02s	1,259.5	0.000	3.074	0.5399	-11.927
1.801	1.13a	20.05s	1,259.5	0.000	2.983	0.7551	-15.963
1.846	1.43a	25.05s	1,259.5	0.000	2.725	1.0063	-20.963
1.883	1.73a	30.05s	1,259.5	0.000	2.385	1.2299	-25.963
1.912	2.04a	35.05s	1,259.5	0.000	1.999	1.4215	-30.963
1.932	2.34a	40.05s	1,259.5	0.000	1.582	1.5779	-35.963
1.941	2.63a	45.05s	1,259.5	0.000	1.144	1.6970	-40.963
1.938	2.90a	50.05s	1,259.5	0.000	0.692	1.7772	-45.963
1.921	3.16a	55.05s	1,259.5	0.000	0.232	1.8176	-50.963
1.907	3.28a	57.55s	1,259.5	0.000	0.000	1.8227	-53.460
1.888	3.39a	60.05s	1,259.5	0.000	-0.233	1.8176	-55.963

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.64 (constant)

Refer IRS Letter E-21076 279820 dated January 02, 2024

12/21/23 09:39:17 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

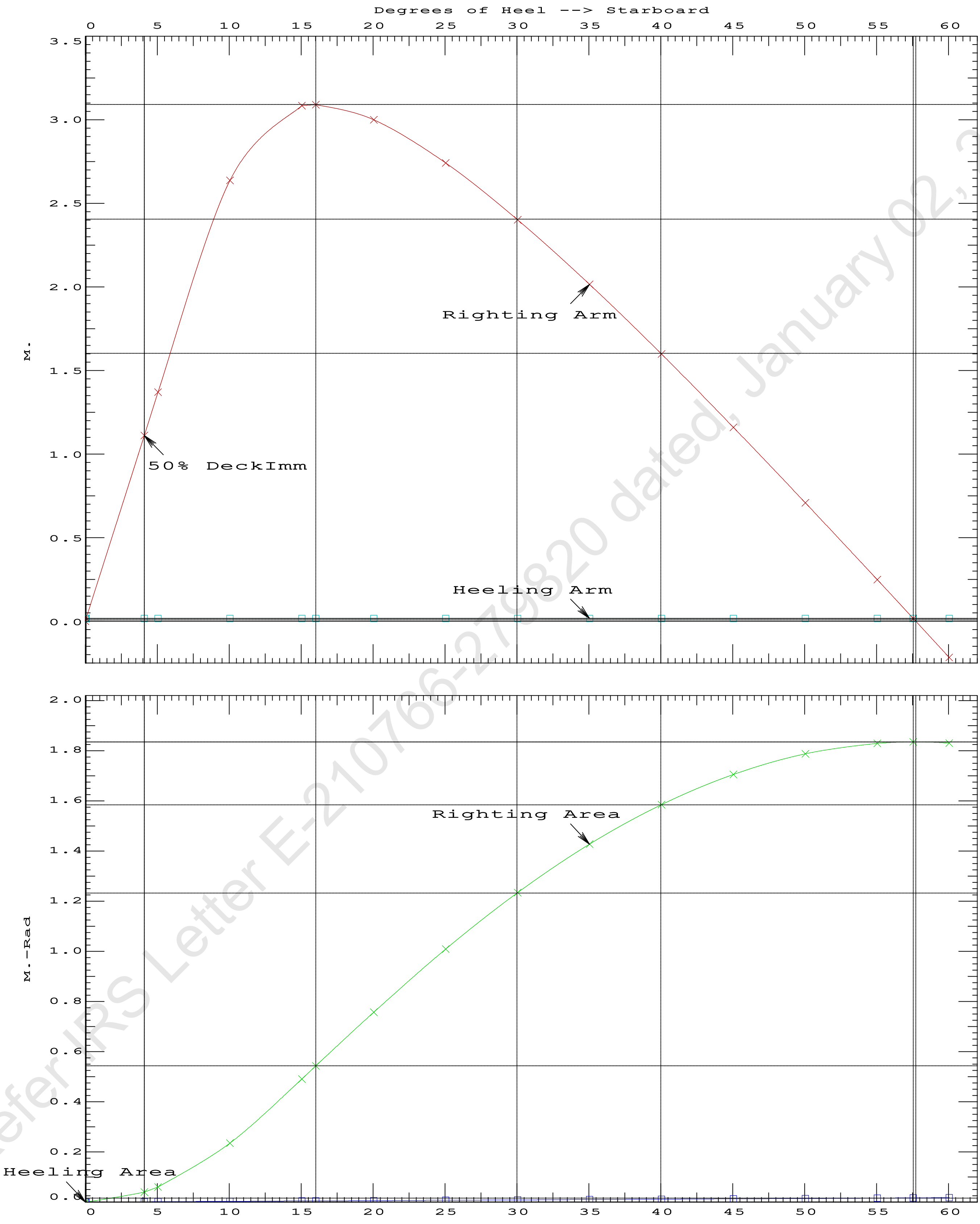
```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5399 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    57.50 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.04 P
-----------------------------------------------------------------------------
```

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:17 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 07 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Aft



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:17 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

```
     Total CG:  LCG = 23.802f  TCG = 0.000   VCG =  4.040
                       Free Surface Adjustment:   0.373
  Adjusted CG:  LCG = 23.806f  TCG = 0.000   VCG =  4.413
```

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.727	0.51a	0.00	1,259.5	0.000	-0.001	0.0000
1.714	0.52a	5.00s	1,259.5	0.000	1.356	0.0591
1.691	0.56a	10.00s	1,259.5	0.000	2.626	0.2335
1.748	0.84a	15.00s	1,259.5	0.000	3.082	0.4893
1.759	0.90a	16.03s	1,259.5	0.000	3.089	0.5446
1.800	1.13a	20.00s	1,259.5	0.000	3.000	0.7576
1.845	1.43a	25.00s	1,259.5	0.000	2.743	1.0104
1.882	1.73a	30.00s	1,259.5	0.000	2.404	1.2356
1.912	2.03a	35.00s	1,259.5	0.000	2.018	1.4289
1.931	2.33a	40.00s	1,259.5	0.000	1.602	1.5871
1.941	2.62a	45.00s	1,259.5	0.000	1.164	1.7079
1.938	2.90a	50.00s	1,259.5	0.000	0.712	1.7899
1.922	3.16a	55.00s	1,259.5	0.000	0.252	1.8320
1.906	3.29a	57.72s	1,259.5	0.000	0.000	1.8380
1.889	3.39a	60.00s	1,259.5	0.000	-0.212	1.8338

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

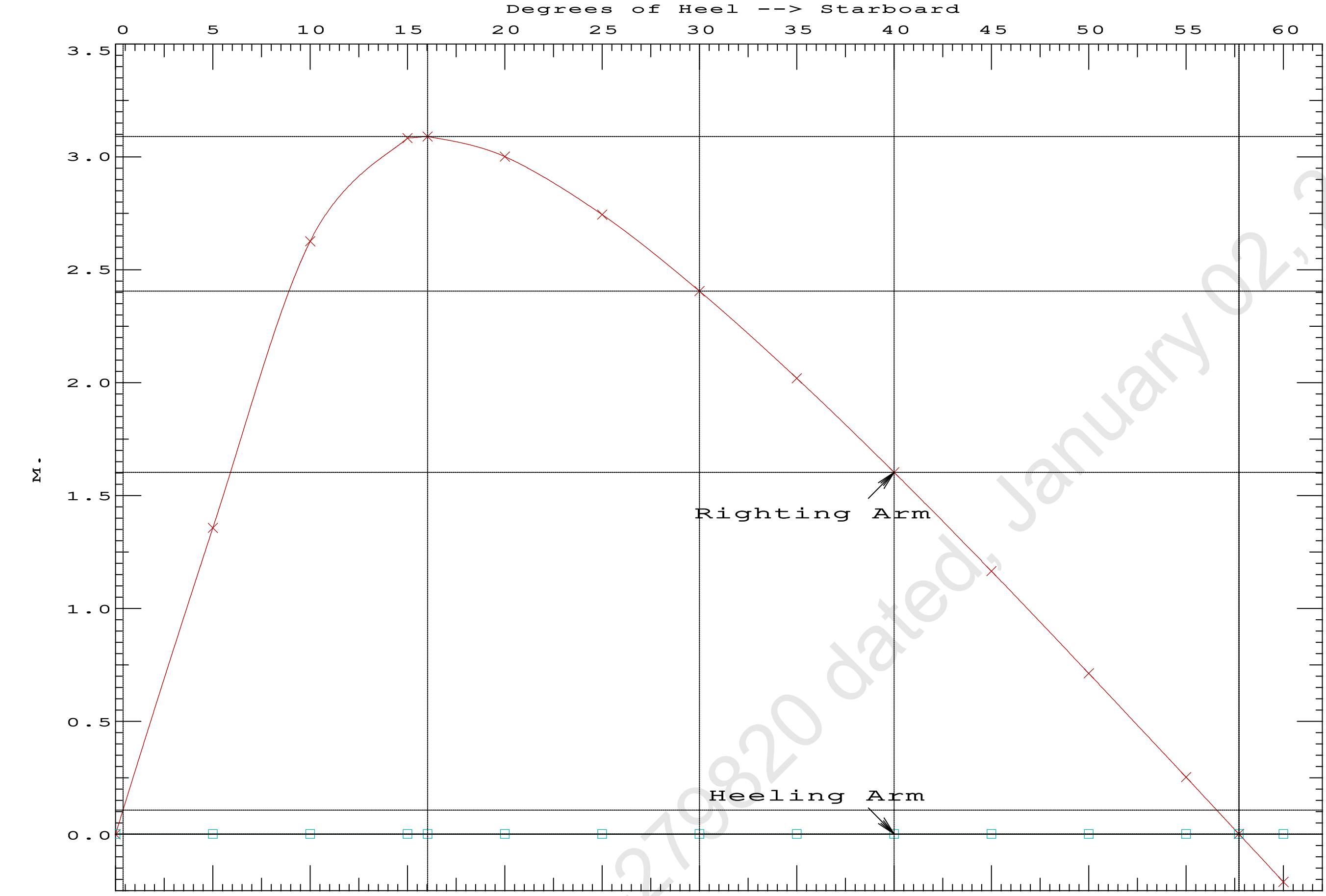
LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	389112	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2299.11	P

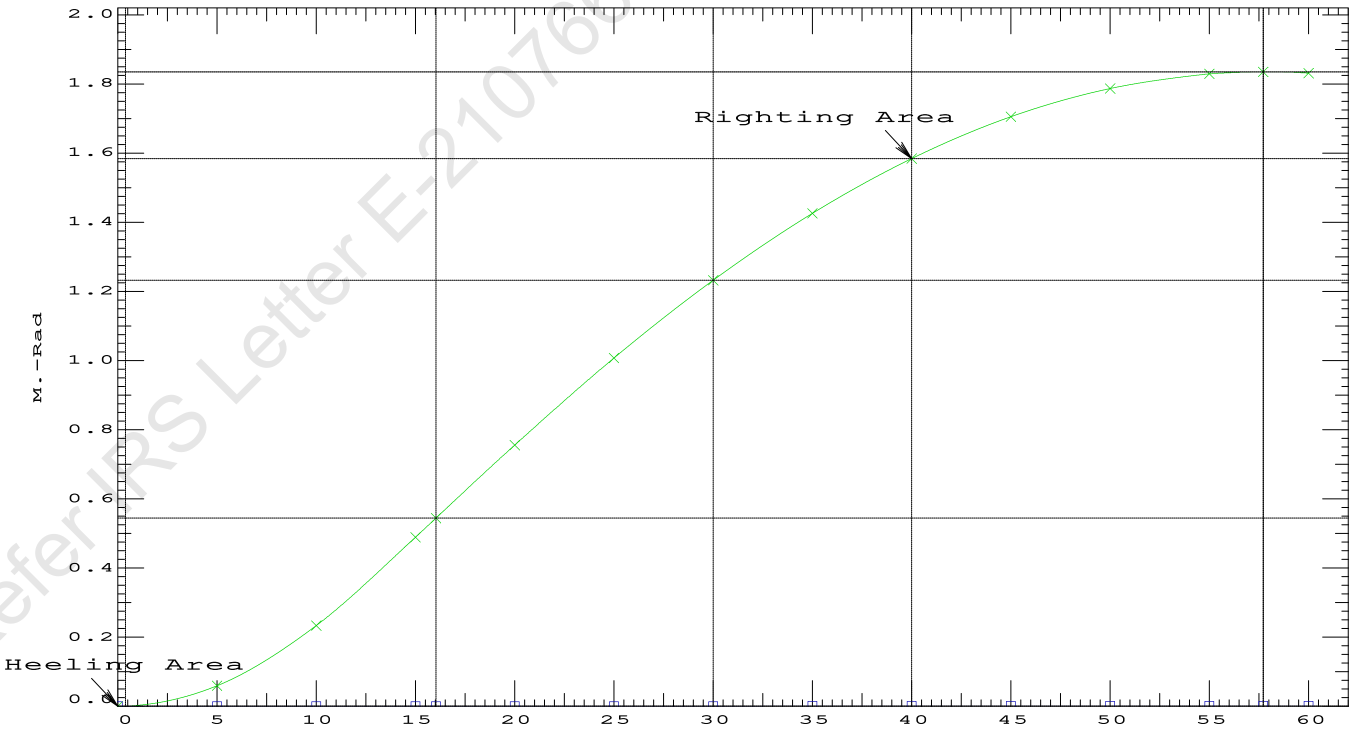
Refer IRS Letter E-210766-23020 dated January 02, 2024

12/21/23 09:39:17 Cybermarine Technologies PTE. Ltd.
GHS 18.66 PONTOON

Condition 07 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Aft





Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:20 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.291 @ 50.00f, 1.515 @ 25.00f, 1.739 @ 0.00
Trim: Aft 0.448/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed--------->			837.10	17.038f	0.000	5.273	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,259.50	23.802f	0.000	4.040	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,259.49	23.773f	0.000	0.773	

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	-0.001s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	868.8	24.611f	0.000	136.98	18.720*
		MT/cm		m.-MT/cm	GML	GMT
		8.91		33.68	133.72	15.453

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:20    Cybermarine Technologies PTE. Ltd.
GHS 18.66                        PONTOON

                       After hook load drop
            WEIGHT and DISPLACEMENT and WATERPLANE STATUS
       USK draft: 1.536 @ 50.00f, 1.472 @ 25.00f, 1.407 @ 0.00
                  Trim: Fwd 0.129/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                            447.10  25.000f  0.000   3.000
Crane                                 350.00  11.500f  0.000   6.325
  Total Fixed--------->               797.10  19.072f  0.000   4.460
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025         161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025         161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025          49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025          49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->               422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->             1,219.50  25.354f  0.000   3.468
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       1,219.49  25.361f  0.000   0.745
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          866.2  25.103f  0.000    140.18  19.284*
                                       MT/cm---------m.-MT/cm-------GML-----GMT
                                        8.88            33.53    137.46  16.561
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 0.9; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.2; t: 0.12
limmarg: INFINITY; t3: 0.1; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.0; t: 0.01

                  Theta3 is 0.03 degrees, as demonstrated below.
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Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:20 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.072f TCG = 0.000 VCG = 4.460

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.395	0.15f	0.00	1,219.5	0.000	-0.001	0.0000
1.395	0.15f	0.03s	1,219.5	0.000	0.010	0.0000
1.380	0.15f	5.00s	1,219.5	0.000	1.487	0.0649
1.330	0.16f	10.00s	1,219.5	0.000	2.898	0.2568
1.208	0.24f	15.00s	1,219.5	0.000	3.468	0.5406
1.141	0.28f	17.56s	1,219.5	0.000	3.504	0.6970
1.075	0.32f	20.00s	1,219.5	0.000	3.471	0.8455
0.932	0.40f	25.00s	1,219.5	0.000	3.290	1.1428
0.780	0.49f	30.00s	1,219.5	0.000	3.022	1.4188

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 797.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0	>	1.000	LARGE

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:20 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.072f TCG = 0.000 VCG = 4.460

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.395	0.15f	0.00	1,219.5	0.000	-0.001	0.0000
1.395	0.15f	0.03s	1,219.5	0.000	0.010	0.0000
1.380	0.15f	5.00s	1,219.5	0.000	1.487	0.0649
1.330	0.16f	10.00s	1,219.5	0.000	2.898	0.2568
1.208	0.24f	15.00s	1,219.5	0.000	3.468	0.5406
1.141	0.28f	17.56s	1,219.5	0.000	3.504	0.6970
1.075	0.32f	20.00s	1,219.5	0.000	3.471	0.8455
0.932	0.40f	25.00s	1,219.5	0.000	3.290	1.1428
0.780	0.49f	30.00s	1,219.5	0.000	3.022	1.4188
0.621	0.57f	35.00s	1,219.5	0.000	2.700	1.6689
0.455	0.65f	40.00s	1,219.5	0.000	2.342	1.8892
0.285	0.73f	45.00s	1,219.5	0.000	1.957	2.0770
0.111	0.81f	50.00s	1,219.5	0.000	1.551	2.2302
-0.063	0.88f	55.00s	1,219.5	0.000	1.129	2.3472
-0.238	0.95f	60.00s	1,219.5	0.000	0.696	2.4269
-0.410	1.01f	65.00s	1,219.5	0.000	0.256	2.4685
-0.508	1.04f	67.89s	1,219.5	0.000	0.000	2.4750
-0.578	1.06f	70.00s	1,219.5	0.000	-0.187	2.4715

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

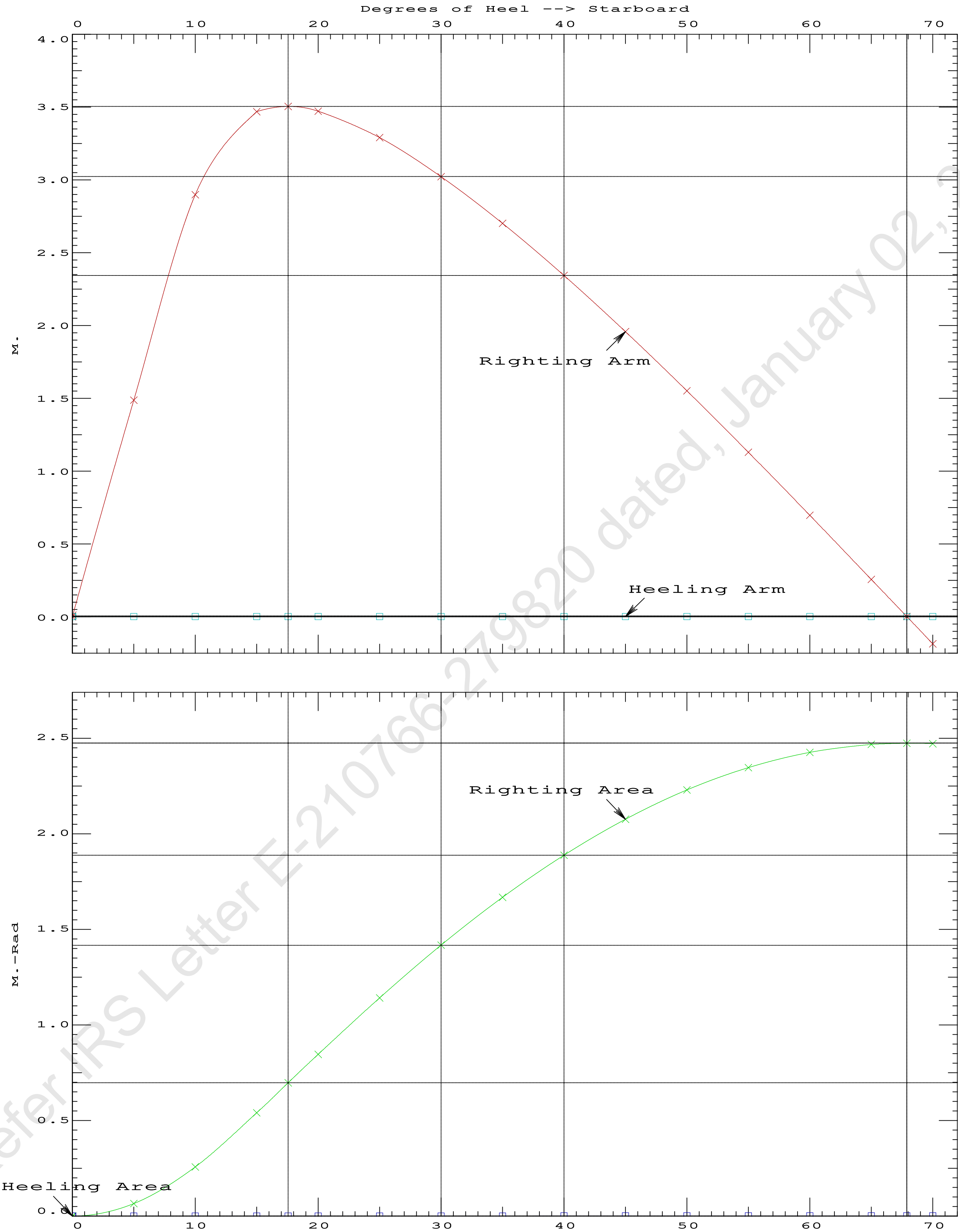
Note: The Center of Gravity shown above is for the Fixed Weight of 797.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0 deg to RAzero	>	20.00	deg	67.86 P

Refer IRS Letter E-218766-279820 dated January 02, 2024

12/21/23 09:39:20 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

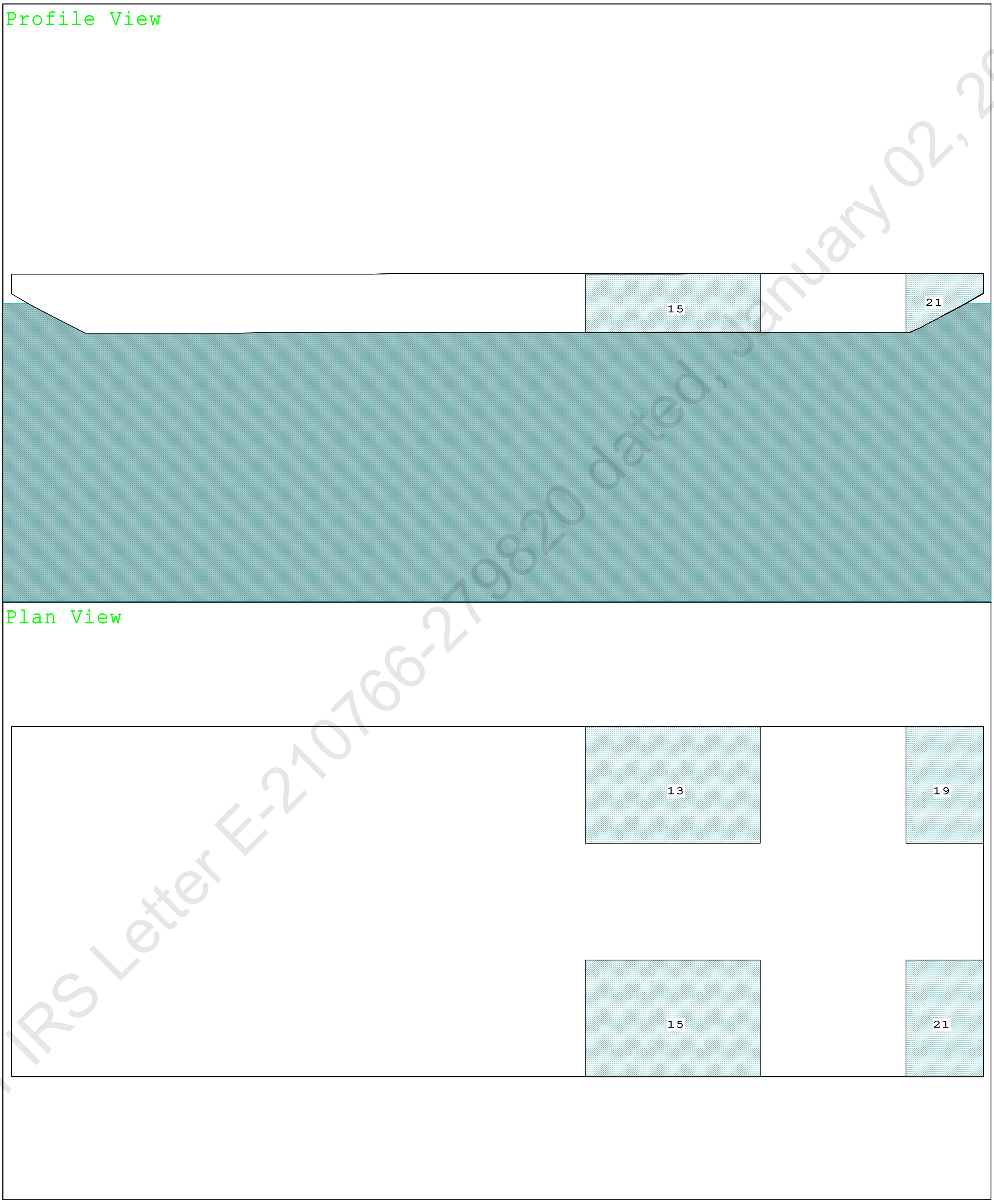
12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 08 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Stbd

Condition Graphic - Draft: 1.501 @ 50.000f, 1.533 @ 0.000 Heel: zero



Tanks
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.501 @ 50.00f, 1.517 @ 25.00f, 1.533 @ 0.00
Trim: Aft 0.032/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	21.478	
Total Fixed--------->			837.10	18.710f	0.000	5.273	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,259.50	24.914f	0.000	4.040	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,259.50	24.912f	0.000	0.768

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	868.8	24.973f	0.000	136.98	18.717*

MT/cm	m.-MT/cm	GML	GMT
8.91	33.68	133.70	15.444

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.501
DRAFT AT MS (M) = 1.517
DRAFT AT AP (M) = 1.533
DRAFT AT FWD MARK (M) = 1.503
DRAFT AT AFT MARK (M) = 1.530

HYDROSTATIC PROPERTIES
Trim: Aft 0.032/50.000, No Heel, VCG = 4.040

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.517	1,259.50	24.912f	0.768	8.91	24.973f	33.68	133.70	15.444

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

Refer IRS Letter E-2410766 279820 dated January 02, 2024

12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.501 @ 50.00f, 1.517 @ 25.00f, 1.533 @ 0.00
Trim: Aft 0.032/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	74.8	0.756	1.497	0.05500	6.16
CRANE	57.0	3.985	4.726	0.05500	14.82
Total wind heeling moment to starboard------------->					20.98

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.914f TCG = 0.000 VCG = 4.040
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 24.914f TCG = 0.000 VCG = 4.413

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% Area	DeckImm Angle
1.521	0.04a	0.00	1,259.5	0.000	-0.016	0.0000	4.791
1.521	0.04a	0.06s	1,259.5	0.000	0.000	-0.0000	4.733
1.507	0.04a	4.79s	1,259.5	0.000	1.285	0.0531	0.000
1.506	0.04a	5.06s	1,259.5	0.000	1.357	0.0592	-0.267
1.463	0.04a	10.06s	1,259.5	0.000	2.647	0.2344	-5.267
1.408	0.06a	15.06s	1,259.5	0.000	3.102	0.4923	-10.267
1.396	0.06a	16.03s	1,259.5	0.000	3.108	0.5449	-11.238
1.342	0.08a	20.06s	1,259.5	0.000	3.018	0.7622	-15.267
1.266	0.10a	25.06s	1,259.5	0.000	2.760	1.0164	-20.267
1.181	0.12a	30.06s	1,259.5	0.000	2.421	1.2431	-25.267
1.088	0.15a	35.06s	1,259.5	0.000	2.034	1.4378	-30.267
0.987	0.17a	40.06s	1,259.5	0.000	1.617	1.5973	-35.267
0.878	0.19a	45.06s	1,259.5	0.000	1.177	1.7193	-40.267
0.764	0.21a	50.06s	1,259.5	0.000	0.723	1.8024	-45.267
0.643	0.23a	55.06s	1,259.5	0.000	0.260	1.8454	-50.267
0.574	0.24a	57.84s	1,259.5	0.000	0.000	1.8517	-53.049
0.517	0.24a	60.06s	1,259.5	0.000	-0.208	1.8477	-55.267

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.98 (constant)

Refer IRS Letter E-2107__279820 dated January 12, 2024

12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5449 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    57.78 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.73 P
-----------------------------------------------------------------------------
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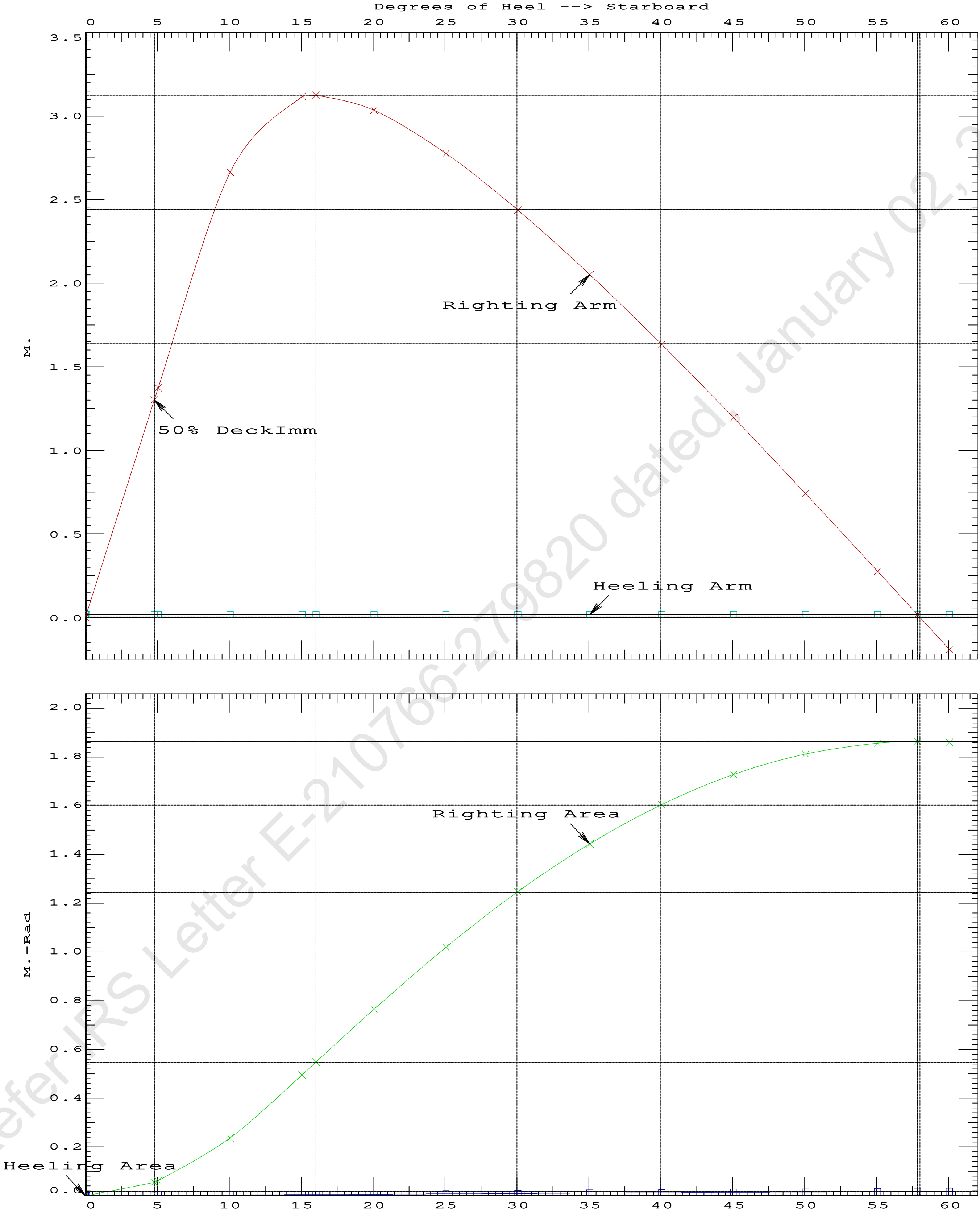
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.914f TCG = 0.000 VCG = 4.040
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 24.914f TCG = 0.026p VCG = 4.412

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.511	0.04a	3.99s	1,259.5	0.000	0.000	0.0000
1.476	0.04a	8.99s	1,259.5	0.000	1.319	0.0576
1.420	0.06a	13.99s	1,259.5	0.000	2.005	0.2117
1.394	0.07a	16.16s	1,259.5	0.000	2.038	0.2889
1.357	0.08a	18.99s	1,259.5	0.000	1.985	0.3889
1.283	0.10a	23.99s	1,259.5	0.000	1.752	0.5547
1.200	0.12a	28.99s	1,259.5	0.000	1.426	0.6941
1.109	0.14a	33.99s	1,259.5	0.000	1.047	0.8023
1.009	0.16a	38.99s	1,259.5	0.000	0.633	0.8759
0.902	0.18a	43.99s	1,259.5	0.000	0.197	0.9123
0.853	0.19a	46.18s	1,259.5	0.000	0.000	0.9160
0.789	0.21a	48.99s	1,259.5	0.000	-0.256	0.9098
0.669	0.22a	53.99s	1,259.5	0.000	-0.720	0.8673
0.545	0.24a	58.99s	1,259.5	0.000	-1.189	0.7841
0.416	0.26a	63.99s	1,259.5	0.000	-1.659	0.6598

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1400.00

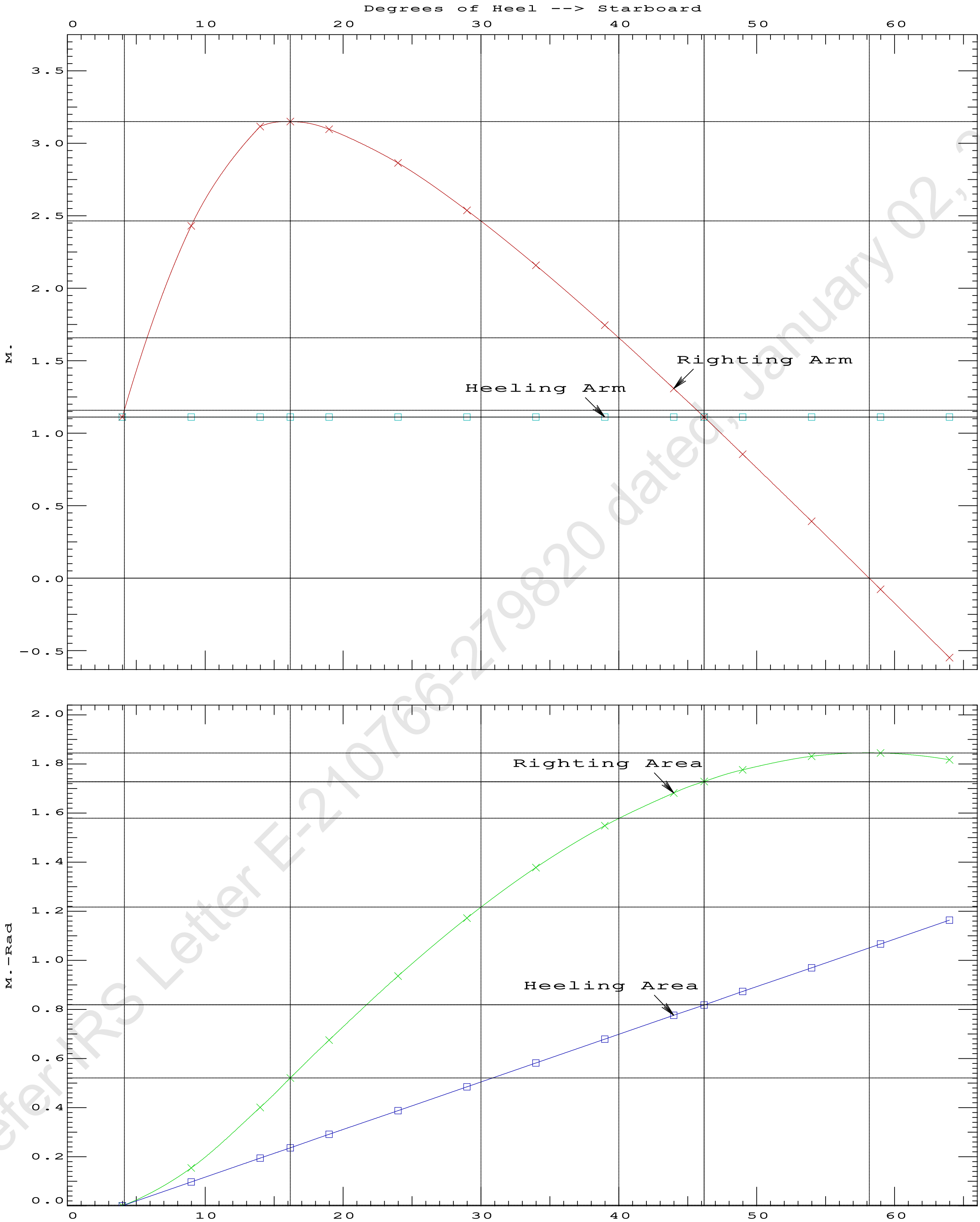
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	3.99	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	183.4	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2.119	P

Refer IRS Letter E-210766-23020 dated January 02, 2024

12/21/23 09:39:22 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:24 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.494 @ 50.00f, 1.511 @ 25.00f, 1.527 @ 0.00
Trim: Aft 0.032/50.000, Heel: Stbd 3.99 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	21.478	
Total Fixed--------->			837.10	18.710f	0.000	5.273	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,259.50	24.914f	0.000	4.040	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,259.50	24.912f	1.339s	0.815

Righting Arms:	0.000	1.112s
External Arms:	0.000	1.112s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	869.2	24.970f	0.155s	136.84	18.796*
		MT/cm 8.91				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.99

Refer IRS Letter E-21076627983O dated, January 02, 2024

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12/21/23 09:39:24    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

                         After hook load drop
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.536 @ 50.00f, 1.472 @ 25.00f, 1.407 @ 0.00
                    Trim: Fwd 0.129/50.000,  Heel: zero
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                350.00  11.500f  0.000   6.325
  Total Fixed--------->              797.10  19.072f  0.000   4.460
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->            1,219.50  25.354f  0.000   3.468
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,219.49  25.361f  0.000   0.745
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          866.2  25.103f  0.000    140.18  19.284*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.88           33.53    137.46  16.561
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 3332.69; t3: 30.0; t: 15.0
limmarg: 1208.19; t3: 15.0; t: 7.5
limmarg: 252.77; t3: 7.5; t: 3.8
limmarg: -13.74; t3: 3.7; t: 1.9
limmarg: 97.09; t3: 5.6; t: 0.9
limmarg: 38.94; t3: 4.7; t: 0.5
limmarg: 11.06; t3: 4.2; t: 0.2
limmarg: 0.78; t3: 4.0; t: 0.1
limmarg: -4.19; t3: 3.9; t: 0.1
limmarg: 0.78; t3: 4.0; t: 0.0
limmarg: 0.78; t3: 4.0; t: 0.0

              Theta3 is 4.00 degrees, as demonstrated below.
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Refer IRS Letter E-210766 279820 dated January 02, 2024

12/21/23 09:39:24 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.072f TCG = 0.000 VCG = 4.460

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.385	0.15f	3.99s	1,219.5	0.000	-1.188	0.0000
1.396	0.15f	0.00	1,219.5	0.000	0.001	-0.0414
1.385	0.15f	4.00p	1,219.5	0.000	1.193	0.0003
1.380	0.15f	5.00p	1,219.5	0.000	1.488	0.0237
1.330	0.16f	10.00p	1,219.5	0.000	2.899	0.2156
1.208	0.24f	15.00p	1,219.5	0.000	3.469	0.4996
1.141	0.28f	17.56p	1,219.5	0.000	3.505	0.6560
1.075	0.32f	20.00p	1,219.5	0.000	3.473	0.8046
0.932	0.40f	25.00p	1,219.5	0.000	3.292	1.1020
0.780	0.49f	30.00p	1,219.5	0.000	3.023	1.3782

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 797.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs -4 deg to abs 4	>	1.000	1.008 P

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:24 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.072f TCG = 0.000 VCG = 4.460

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.385	0.15f	3.99s	1,219.5	0.000	-1.188	0.0000
1.396	0.15f	0.00	1,219.5	0.000	0.001	-0.0414
1.385	0.15f	4.00p	1,219.5	0.000	1.193	0.0003
1.380	0.15f	5.00p	1,219.5	0.000	1.488	0.0237
1.330	0.16f	10.00p	1,219.5	0.000	2.899	0.2156
1.208	0.24f	15.00p	1,219.5	0.000	3.469	0.4996
1.141	0.28f	17.56p	1,219.5	0.000	3.505	0.6560
1.075	0.32f	20.00p	1,219.5	0.000	3.473	0.8046
0.932	0.40f	25.00p	1,219.5	0.000	3.292	1.1020
0.780	0.49f	30.00p	1,219.5	0.000	3.023	1.3782
0.621	0.57f	35.00p	1,219.5	0.000	2.702	1.6284
0.455	0.65f	40.00p	1,219.5	0.000	2.344	1.8488
0.285	0.73f	45.00p	1,219.5	0.000	1.959	2.0368
0.111	0.81f	50.00p	1,219.5	0.000	1.552	2.1901
-0.063	0.88f	55.00p	1,219.5	0.000	1.131	2.3073
-0.238	0.95f	60.00p	1,219.5	0.000	0.698	2.3872
-0.410	1.01f	65.00p	1,219.5	0.000	0.258	2.4289
-0.508	1.04f	67.91p	1,219.5	0.000	0.000	2.4354
-0.578	1.06f	70.00p	1,219.5	0.000	-0.186	2.4320

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

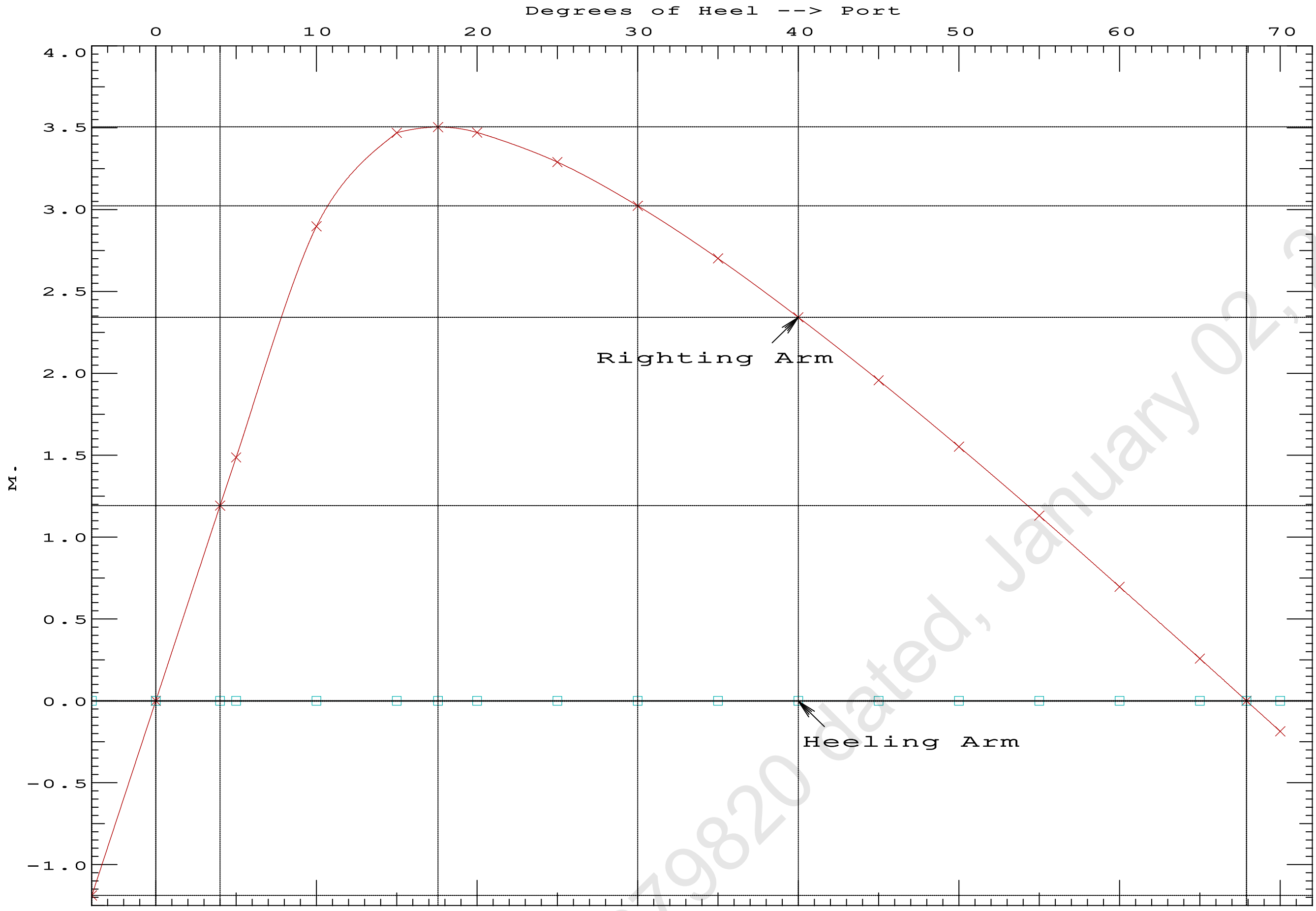
Note: The Center of Gravity shown above is for the Fixed Weight of 797.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

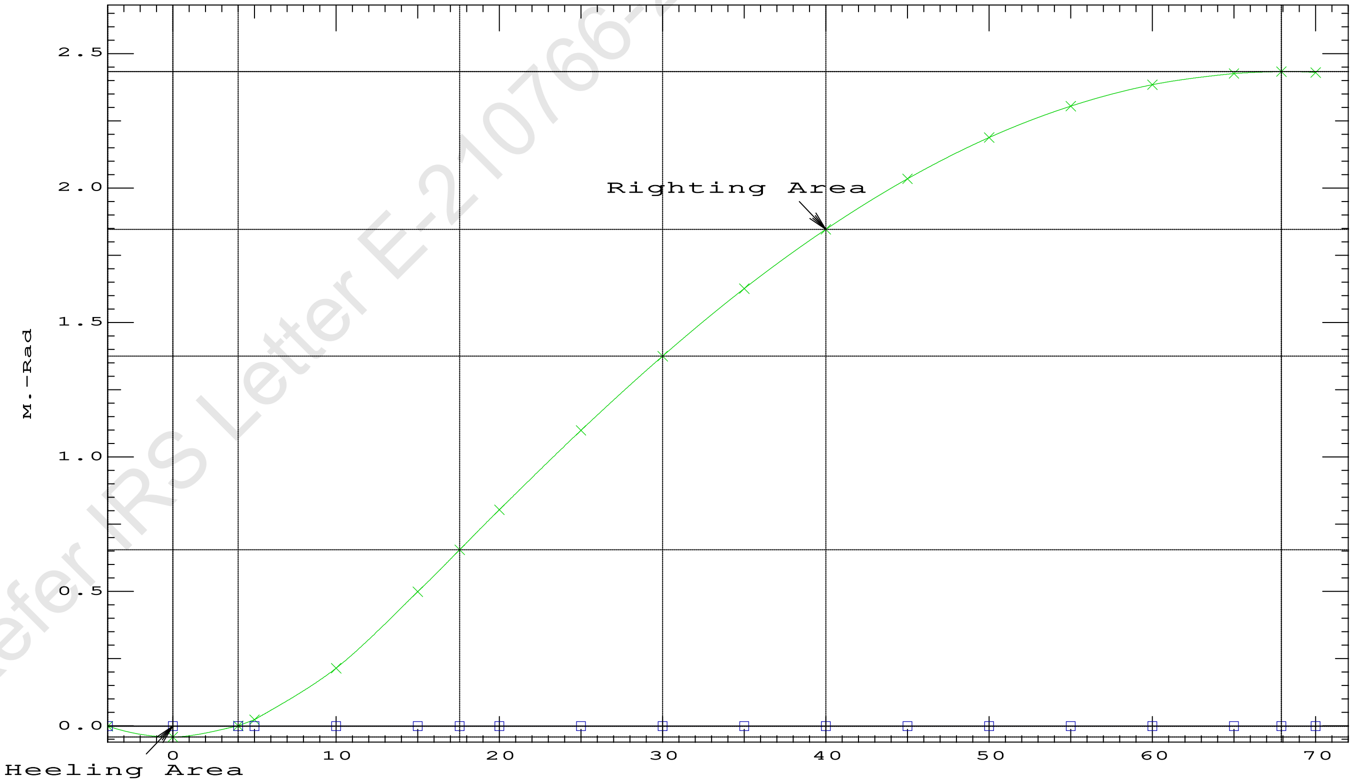
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs -4 deg to RAzero	>	1.000		59.893 P
(2) Angle from abs 4 deg to RAzero	>	20.00	deg	63.91 P

Refer IRS Letter E-27076 270820 dated January 02, 2024

12/21/23 09:39:24 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



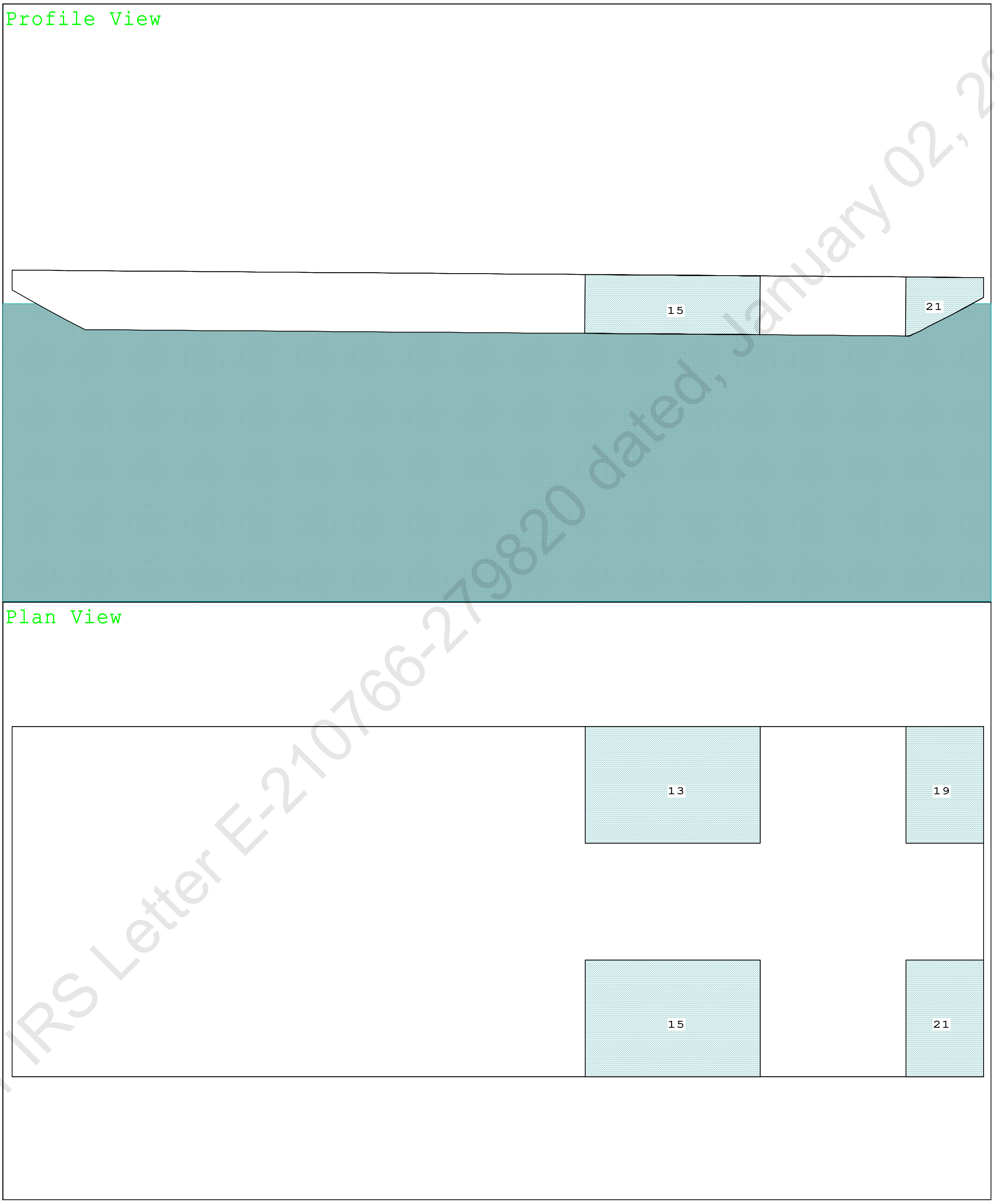


Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

Condition Graphic - Draft: 1.707 @ 50.000f, 1.324 @ 0.000 Heel: zero



Refer IRS Letter E-210766-279820 dated, January 02, 2024

Tanks
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.707 @ 50.00f, 1.516 @ 25.00f, 1.324 @ 0.00
Trim: Fwd 0.383/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed--------->			837.10	20.383f	0.000	5.273	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,259.50	26.026f	0.000	4.040	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,259.41	26.051f	0.000	0.772	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	868.8	25.330f	0.000	137.01	18.724*
		MT/cm		m.-MT/cm	GML	GMT
		8.91		33.69	133.74	15.455

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane 40.00 MT Crane Load

DRAFT AT FP (M) = 1.707
DRAFT AT MS (M) = 1.516
DRAFT AT AP (M) = 1.324
DRAFT AT FWD MARK (M) = 1.677
DRAFT AT AFT MARK (M) = 1.355

HYDROSTATIC PROPERTIES
Trim: Fwd 0.383/50.000, No Heel, VCG = 4.040

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.518	1,259.41	26.051f	0.772	8.91	25.330f	33.69	133.74	15.455

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

Refer IRS Letter E-2107606 279820 dated January 02, 2024

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.707 @ 50.00f, 1.516 @ 25.00f, 1.324 @ 0.00
Trim: Fwd 0.383/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	74.7	0.762	1.505	0.05500	6.19
CRANE	57.0	4.111	4.854	0.05500	15.22
Total wind heeling moment to starboard------------->					21.40

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 26.026f TCG = 0.000 VCG = 4.040
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 26.023f TCG = 0.000 VCG = 4.413

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.312	0.44f	0.00	1,259.4	0.000	-0.017	0.0000	4.195
1.312	0.44f	0.05s	1,259.5	0.000	0.000	-0.0000	4.140
1.301	0.44f	4.20s	1,259.5	0.000	1.123	0.0406	0.000
1.296	0.44f	5.05s	1,259.5	0.000	1.355	0.0592	-0.860
1.236	0.48f	10.05s	1,259.5	0.000	2.627	0.2335	-5.860
1.067	0.72f	15.05s	1,259.5	0.000	3.076	0.4892	-10.860
1.032	0.77f	16.02s	1,259.5	0.000	3.083	0.5411	-11.825
0.882	0.97f	20.05s	1,259.5	0.000	2.992	0.7568	-15.860
0.686	1.23f	25.05s	1,259.5	0.000	2.733	1.0088	-20.860
0.479	1.48f	30.05s	1,259.5	0.000	2.394	1.2331	-25.860
0.263	1.74f	35.05s	1,259.5	0.000	2.008	1.4255	-30.860
0.040	2.00f	40.05s	1,259.5	0.000	1.591	1.5828	-35.860
-0.187	2.25f	45.05s	1,259.5	0.000	1.152	1.7026	-40.860
-0.414	2.49f	50.05s	1,259.5	0.000	0.700	1.7835	-45.860
-0.639	2.71f	55.05s	1,259.5	0.000	0.239	1.8245	-50.860
-0.752	2.81f	57.62s	1,259.5	0.000	0.000	1.8299	-53.427
-0.857	2.91f	60.05s	1,259.5	0.000	-0.227	1.8251	-55.860

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.40 (constant)

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.5411 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    57.57 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     4.14 P
-----------------------------------------------------------------------------
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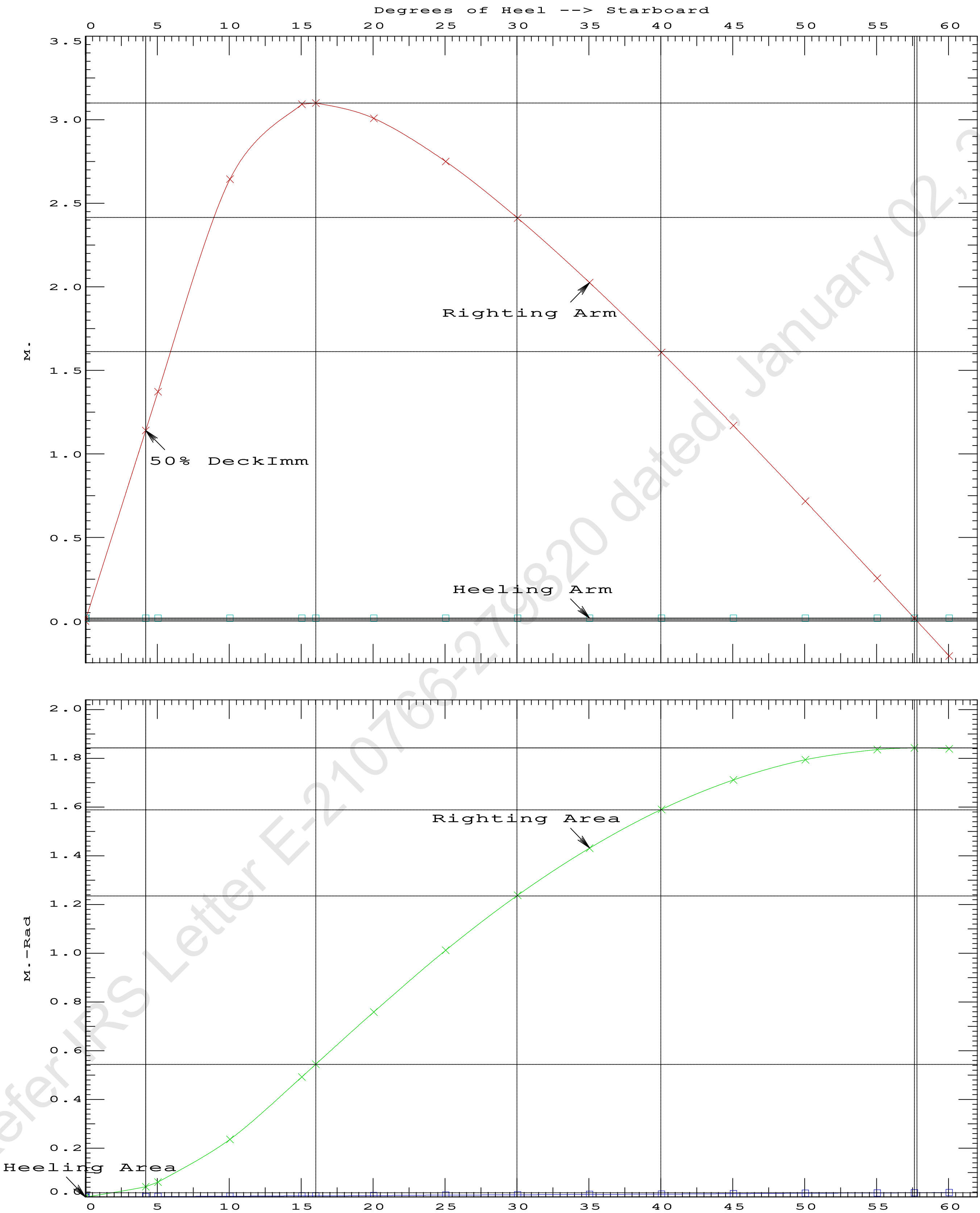
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 09 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.026f TCG = 0.000 VCG = 4.040
Free Surface Adjustment: 0.373
Adjusted CG: LCG = 26.023f TCG = 0.000 VCG = 4.413

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.312	0.44f	0.00	1,259.5	0.000	0.001	0.0000
1.296	0.44f	5.00s	1,259.5	0.000	1.356	0.0592
1.237	0.48f	10.00s	1,259.5	0.000	2.631	0.2337
1.069	0.72f	15.00s	1,259.5	0.000	3.091	0.4903
1.032	0.77f	16.03s	1,259.5	0.000	3.098	0.5458
0.885	0.97f	20.00s	1,259.5	0.000	3.009	0.7593
0.688	1.22f	25.00s	1,259.5	0.000	2.752	1.0128
0.481	1.48f	30.00s	1,259.5	0.000	2.413	1.2388
0.265	1.74f	35.00s	1,259.5	0.000	2.027	1.4329
0.043	2.00f	40.00s	1,259.5	0.000	1.610	1.5918
-0.184	2.25f	45.00s	1,259.5	0.000	1.172	1.7134
-0.411	2.49f	50.00s	1,259.5	0.000	0.720	1.7960
-0.636	2.71f	55.00s	1,259.5	0.000	0.259	1.8388
-0.759	2.82f	57.79s	1,259.5	0.000	0.000	1.8451
-0.855	2.90f	60.00s	1,259.5	0.000	-0.206	1.8411

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 2.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

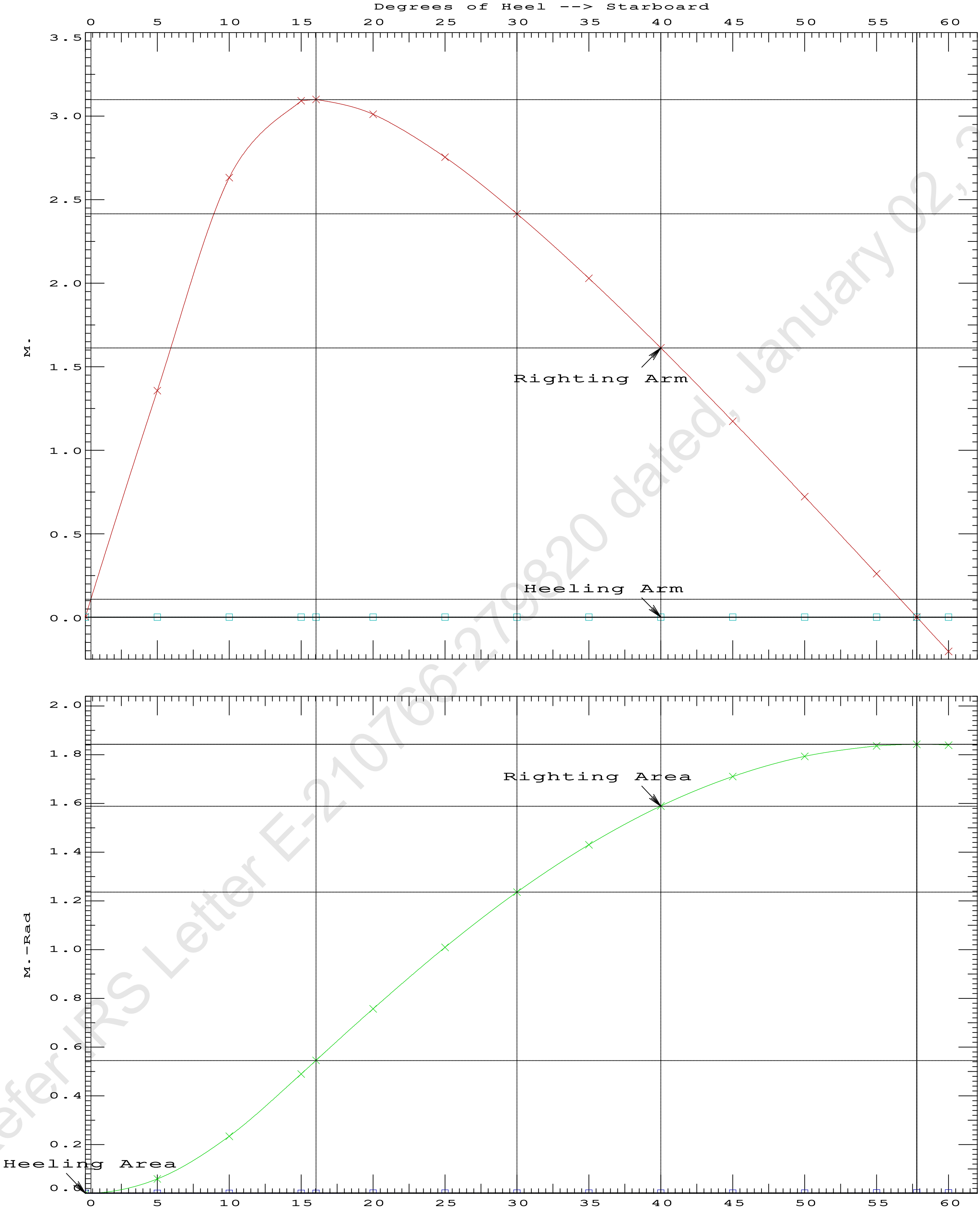
LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	195091	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1153.13	P

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:26 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 09 : Operating Condition of 350T Crane

Crane Lifting 40.0T @ 35m Radius Fwd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:28 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.707 @ 50.00f, 1.516 @ 25.00f, 1.324 @ 0.00
Trim: Fwd 0.383/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed--------->			837.10	20.383f	0.000	5.273	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,259.50	26.026f	0.000	4.040	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,259.49	26.051f	0.000	0.772

Righting Arms:	0.000	0.000
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	-0.001s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	868.9	25.330f	0.000	137.00	18.722*

MT/cm	m.-MT/cm	GML	GMT
8.91	33.69	133.73	15.454

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:28    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

                          After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.536 @ 50.00f, 1.472 @ 25.00f, 1.407 @ 0.00
                     Trim: Fwd 0.129/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                350.00  11.500f  0.000   6.325
  Total Fixed--------->              797.10  19.072f  0.000   4.460
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->            1,219.50  25.354f  0.000   3.468
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,219.49  25.361f  0.000   0.745
------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.000
               External Arms:                   0.000   0.001
      Residual Righting Arms:                   0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         866.2  25.103f  0.000    140.18  19.284*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.88           33.53    137.46  16.561
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

               Theta3 is 0.97 degrees, as demonstrated below.
```

Refer IRS Letter F-21076 279820 dated January 02, 2024

12/21/23 09:39:28 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.072f TCG = 0.000 VCG = 4.460

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.395	0.15f	0.00	1,219.5	0.000	-0.001	0.0000
1.393	0.15f	0.97s	1,219.5	0.000	0.291	0.0025
1.380	0.15f	5.00s	1,219.5	0.000	1.487	0.0650
1.330	0.16f	10.00s	1,219.5	0.000	2.898	0.2568
1.208	0.24f	15.00s	1,219.5	0.000	3.468	0.5407
1.141	0.28f	17.56s	1,219.5	0.000	3.504	0.6971
1.075	0.32f	20.00s	1,219.5	0.000	3.471	0.8456
0.932	0.40f	25.00s	1,219.5	0.000	3.290	1.1428
0.780	0.49f	30.00s	1,219.5	0.000	3.022	1.4189

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 797.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 1	>	1.000	LARGE

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:28 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 19.072f TCG = 0.000 VCG = 4.460

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.395	0.15f	0.00	1,219.5	0.000	-0.001	0.0000
1.393	0.15f	0.97s	1,219.5	0.000	0.291	0.0025
1.380	0.15f	5.00s	1,219.5	0.000	1.487	0.0650
1.330	0.16f	10.00s	1,219.5	0.000	2.898	0.2568
1.208	0.24f	15.00s	1,219.5	0.000	3.468	0.5407
1.141	0.28f	17.56s	1,219.5	0.000	3.504	0.6971
1.075	0.32f	20.00s	1,219.5	0.000	3.471	0.8456
0.932	0.40f	25.00s	1,219.5	0.000	3.290	1.1428
0.780	0.49f	30.00s	1,219.5	0.000	3.022	1.4189
0.621	0.57f	35.00s	1,219.5	0.000	2.700	1.6689
0.455	0.65f	40.00s	1,219.5	0.000	2.342	1.8892
0.285	0.73f	45.00s	1,219.5	0.000	1.957	2.0770
0.111	0.81f	50.00s	1,219.5	0.000	1.551	2.2302
-0.063	0.88f	55.00s	1,219.5	0.000	1.129	2.3473
-0.238	0.95f	60.00s	1,219.5	0.000	0.696	2.4270
-0.410	1.01f	65.00s	1,219.5	0.000	0.256	2.4686
-0.508	1.04f	67.89s	1,219.5	0.000	0.000	2.4750
-0.578	1.06f	70.00s	1,219.5	0.000	-0.187	2.4716

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

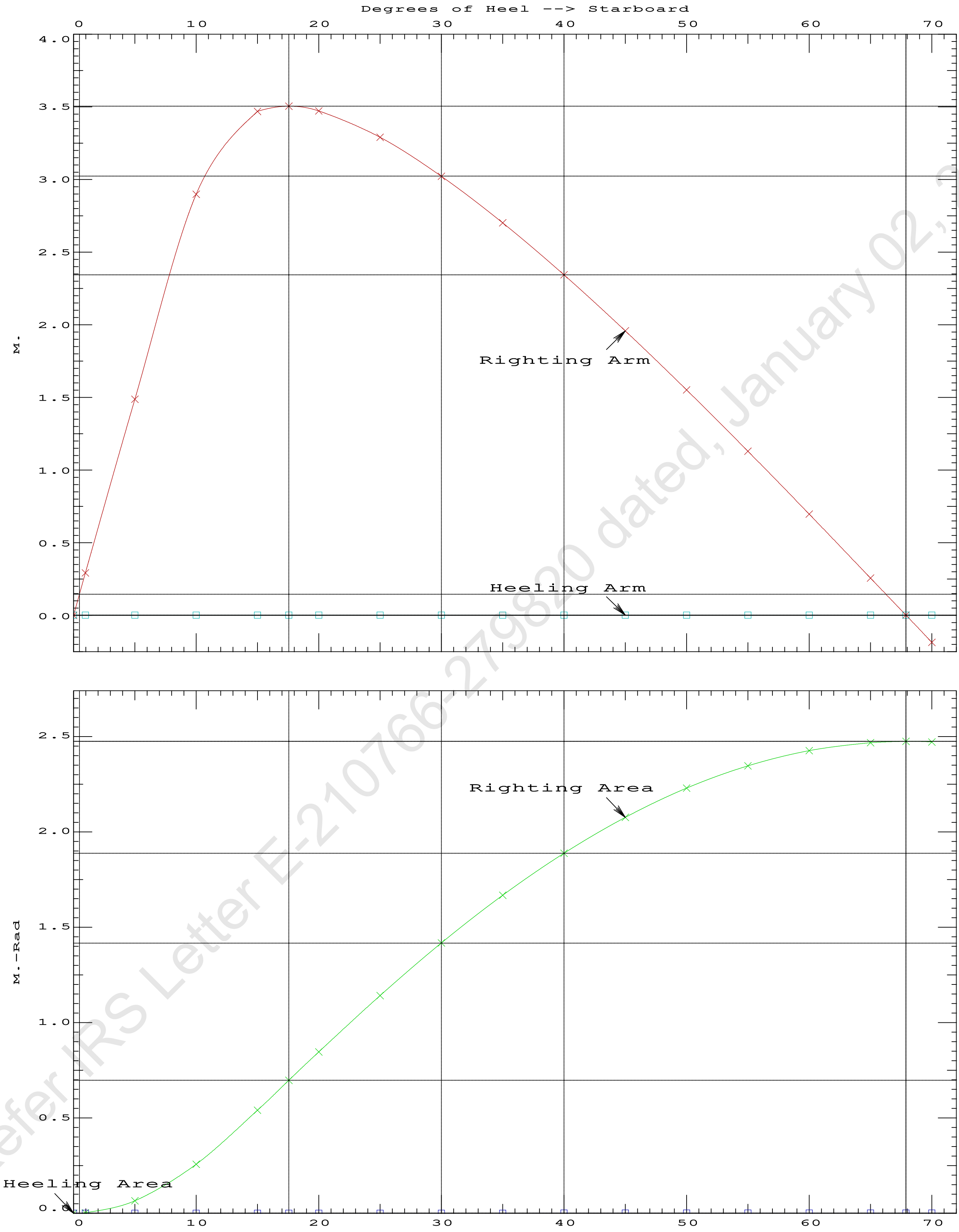
Note: The Center of Gravity shown above is for the Fixed Weight of 797.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 1 deg to RAzero	>	20.00	deg	66.92 P

Refer IRS Letter E-21876-279820 dated January 02, 2024

12/21/23 09:39:28 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



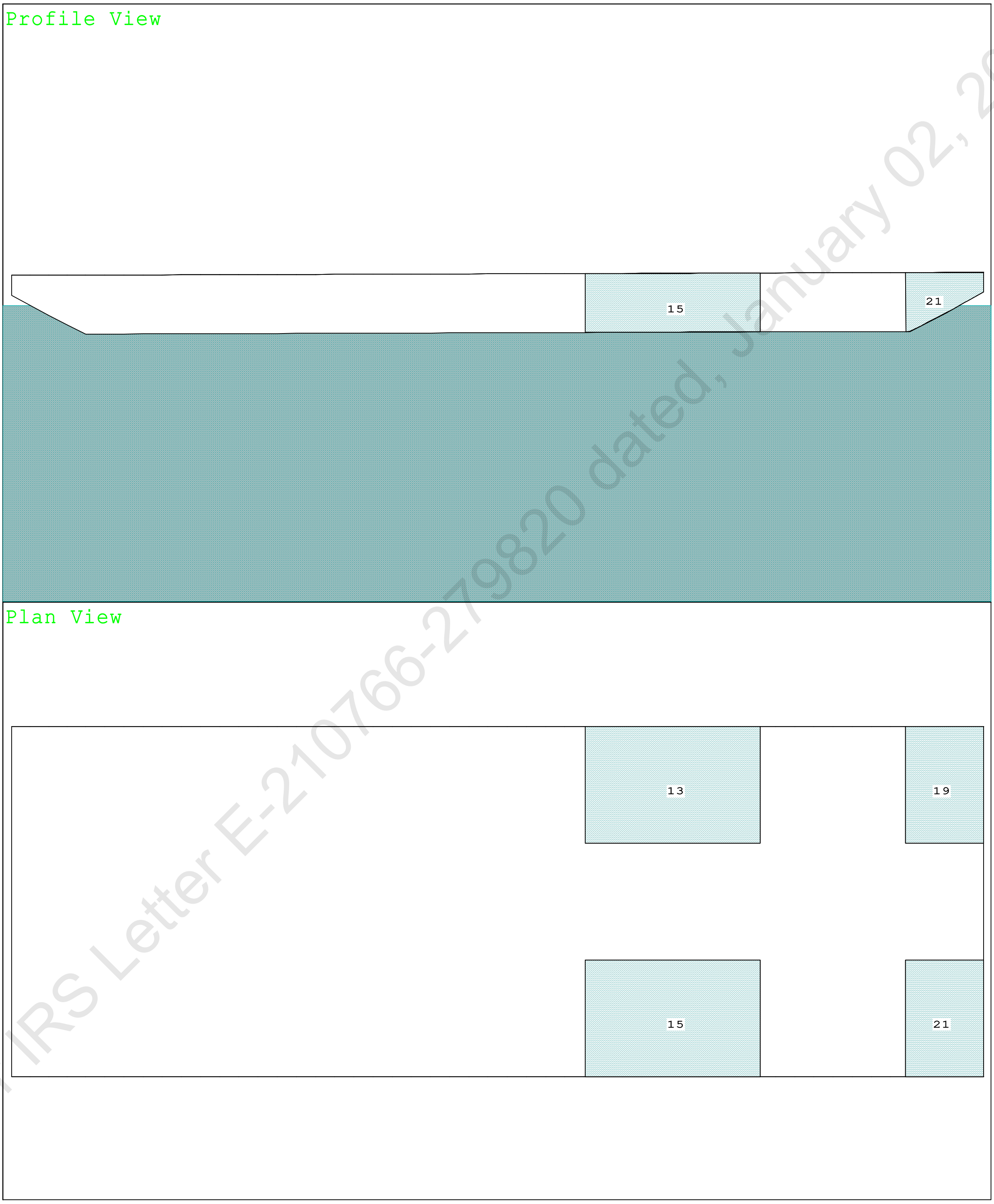
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.321 @ 50.000f, 1.480 @ 0.000 Heel: zero



Tanks
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:39:31    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
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Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.321 @ 50.00f, 1.400 @ 25.00f, 1.480 @ 0.00
                     Trim: Aft 0.159/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
Crane Load                            36.60  10.500a  0.000  76.977
  Total Fixed--------->              733.70  17.266f  0.000   7.440
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596    470.09
  Total Weight-------->            1,156.10  24.552f  0.000   5.305
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,156.07  24.538f  0.000   0.708
-----------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         861.3  24.862f  0.000    145.39  20.218*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.83             32.55    140.79  15.621
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      250.00 MT  Crane                        36.60 MT  Crane Load

DRAFT AT FP (M) =    1.321
DRAFT AT MS (M) =    1.400
DRAFT AT AP (M) =    1.480
DRAFT AT FWD MARK (M) =    1.333
DRAFT AT AFT MARK (M) =    1.467
                          HYDROSTATIC PROPERTIES
                Trim: Aft 0.159/50.000, No Heel, VCG = 5.305

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.401     1,156.07  24.538f  0.708      8.83  24.862f   32.55  140.79  15.621
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.                                Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
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Refer IRS Letter E-210766-279820 dated January 02, 2024

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12/21/23 09:39:31    Cybermarine Technologies PTE. Ltd.
GHS 18.66                        PONTOON
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Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

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                  HEELING MOMENT specification
                      Lateral Plane Method
                  Wind pressure toward starboard
      USK draft: 1.321 @ 50.00f, 1.400 @ 25.00f, 1.480 @ 0.00
                Trim: Aft 0.159/50.000,  Heel: zero

 Part---------------LPA*SF------HCP------Arm----Pressure----------Moment
 HULL                  80.4    0.817    1.503     0.05500             6.64
 CRANE                 57.0    4.065    4.750     0.05500            14.89
   Total wind heeling moment to starboard------------->              21.54
 Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

               RESIDUAL RIGHTING ARMS vs HEEL ANGLE
    Total CG:  LCG = 24.552f  TCG = 0.000   VCG =  5.305
                         Free Surface Adjustment:   0.407
  Adjusted CG:  LCG = 24.554f  TCG = 0.000   VCG =  5.712
```

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.468	0.18a	0.00	1,156.1	0.000	-0.018	0.0000	4.969
1.468	0.18a	0.06s	1,156.1	0.000	0.000	-0.0000	4.913
1.453	0.18a	4.97s	1,156.1	0.000	1.347	0.0577	0.000
1.453	0.18a	5.06s	1,156.1	0.000	1.371	0.0598	-0.087
1.408	0.20a	10.06s	1,156.1	0.000	2.637	0.2354	-5.087
1.326	0.29a	15.06s	1,156.1	0.000	3.038	0.4904	-10.087
1.238	0.39a	20.06s	1,156.1	0.000	2.861	0.7520	-15.087
1.142	0.50a	25.06s	1,156.1	0.000	2.499	0.9872	-20.087
1.040	0.61a	30.06s	1,156.1	0.000	2.051	1.1863	-25.087
0.933	0.72a	35.06s	1,156.1	0.000	1.557	1.3441	-30.087
0.820	0.82a	40.06s	1,156.1	0.000	1.034	1.4573	-35.087
0.703	0.93a	45.06s	1,156.1	0.000	0.493	1.5241	-40.087
0.594	1.01a	49.53s	1,156.1	0.000	0.000	1.5433	-44.558
0.581	1.02a	50.06s	1,156.1	0.000	-0.059	1.5431	-45.087
0.454	1.11a	55.06s	1,156.1	0.000	-0.613	1.5138	-50.087
0.322	1.19a	60.06s	1,156.1	0.000	-1.167	1.4361	-55.087

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.54 (constant)

Refer IRS Letter: E-210766-279820 dated., January 12, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA        >   0.0800 m.-Rad 0.4904 P
(2) Angle from Equilibrium to RAzero or Flood    >    20.00  deg    49.47 P
(3) Angle from Equilibrium to 50% Dk Imm Angle   >     0.00  deg     4.91 P
-----------------------------------------------------------------------------
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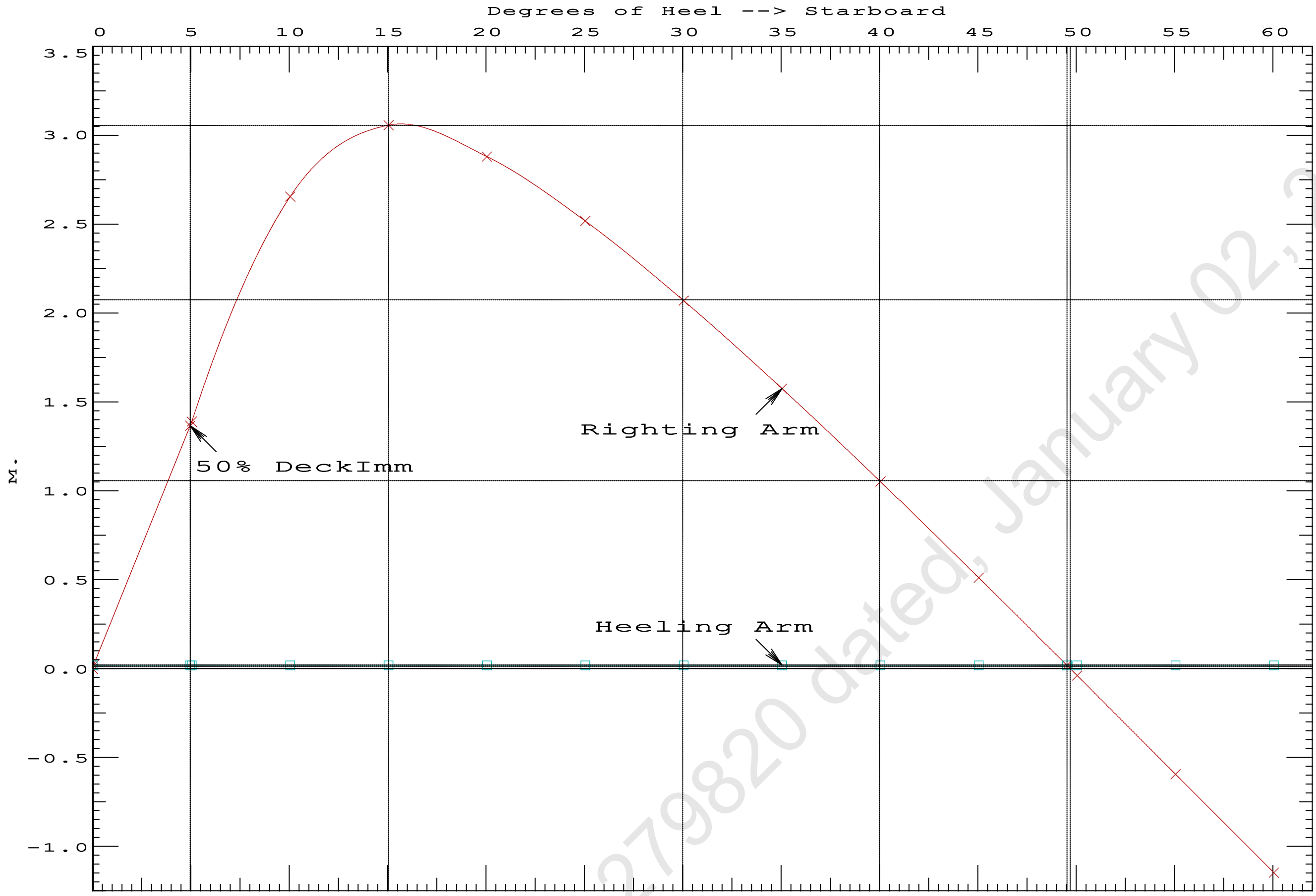
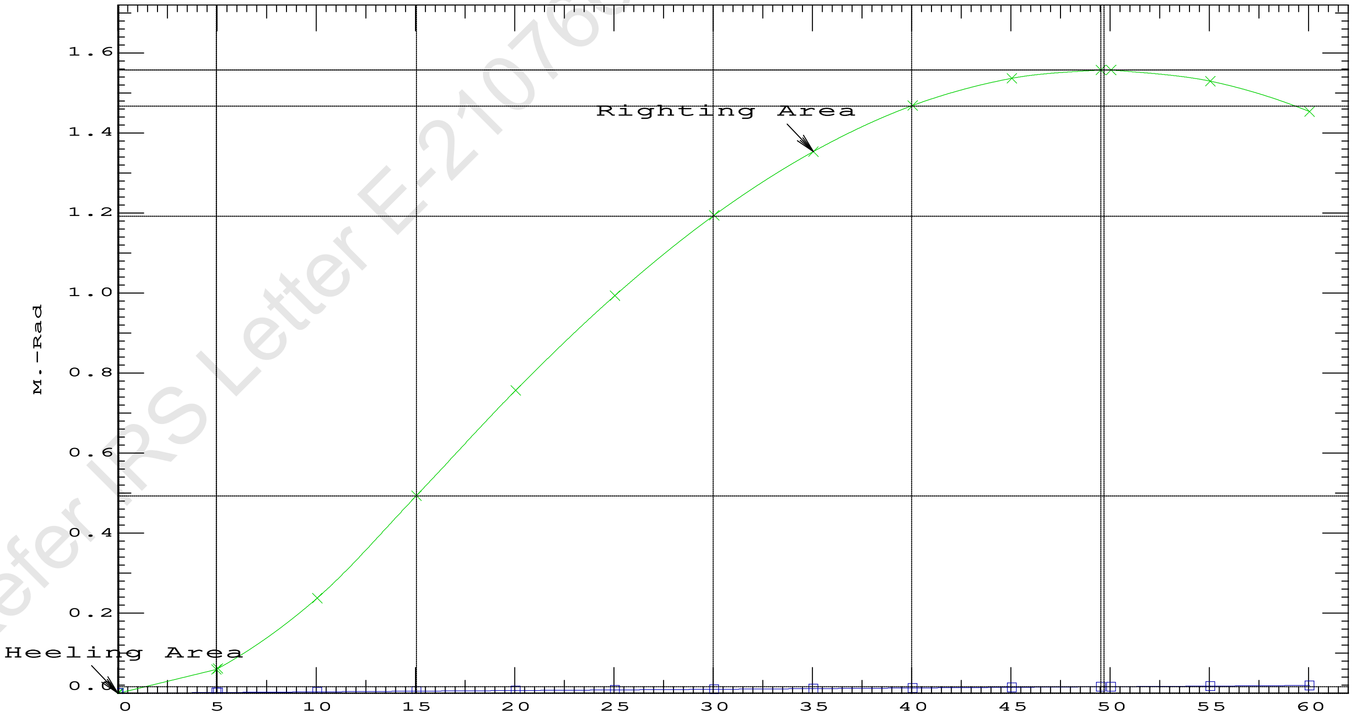
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft





Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.552f TCG = 0.000 VCG = 5.305
Free Surface Adjustment: 0.407
Adjusted CG: LCG = 24.554f TCG = 0.000 VCG = 5.712

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.468	0.18a	0.01s	1,156.1	0.000	0.000	0.0000
1.453	0.18a	5.01s	1,156.1	0.000	1.372	0.0599
1.409	0.20a	10.01s	1,156.1	0.000	2.641	0.2357
1.327	0.29a	15.01s	1,156.1	0.000	3.052	0.4916
1.239	0.39a	20.01s	1,156.1	0.000	2.878	0.7545
1.143	0.50a	25.01s	1,156.1	0.000	2.516	0.9913
1.041	0.61a	30.01s	1,156.1	0.000	2.070	1.1920
0.934	0.71a	35.01s	1,156.1	0.000	1.576	1.3514
0.821	0.82a	40.01s	1,156.1	0.000	1.053	1.4663
0.704	0.92a	45.01s	1,156.1	0.000	0.512	1.5347
0.591	1.02a	49.65s	1,156.1	0.000	0.000	1.5555
0.582	1.02a	50.01s	1,156.1	0.000	-0.039	1.5554
0.455	1.11a	55.01s	1,156.1	0.000	-0.594	1.5278
0.324	1.19a	60.01s	1,156.1	0.000	-1.147	1.4518

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

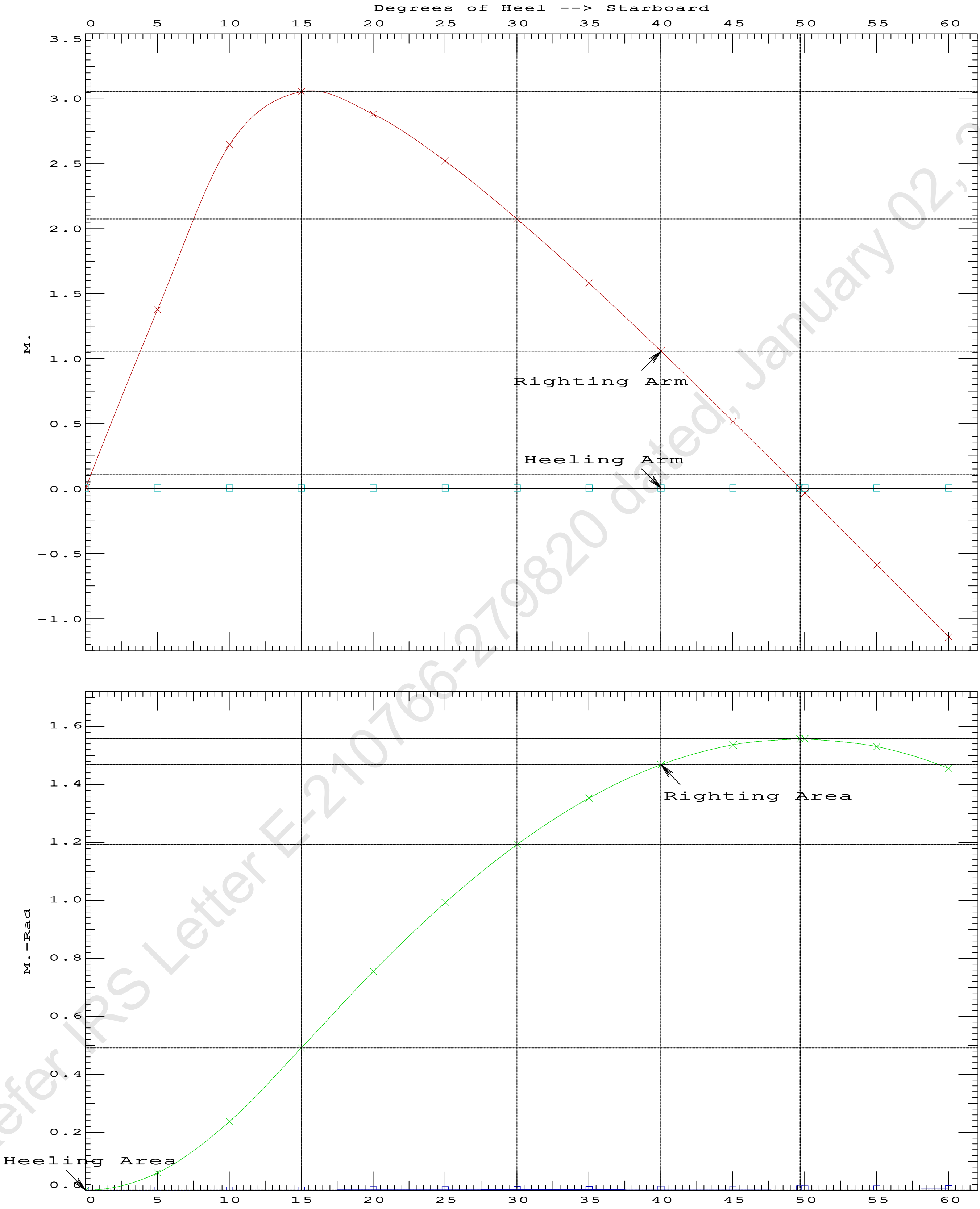
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1)	Absolute Angle at Equilibrium	<	15.00	deg	0.01	P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	70561.6	P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		416.094	P

Refer IRS Letter E210786 279800 dated January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.321 @ 50.00f, 1.400 @ 25.00f, 1.480 @ 0.00
Trim: Aft 0.159/50.000, Heel: Stbd 0.01 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Total Fixed--------->			733.70	17.266f	0.000	7.440	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,156.10	24.552f	0.000	5.305	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,156.10	24.538f	0.005s	0.708

Righting Arms:	0.000	0.004s
External Arms:	0.000	0.004s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	857.2	24.746f	0.028s	143.34	20.105*
		MT/cm				
		8.79				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:31    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

                           After hook load drop
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.478 @ 50.00f, 1.358 @ 25.00f, 1.239 @ 0.00
                  Trim: Fwd 0.238/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
  Total Fixed--------->              697.10  18.724f  0.000   3.789
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->            1,119.50  25.698f  0.000   2.962
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,119.50  25.709f  0.001s  0.687
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.001s
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          855.9  25.282f  0.019s   147.33  20.717*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.77           32.48    145.06  18.442
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 651303616.00; t3: 30.0; t: 15.00
limmarg: 244047232.00; t3: 15.0; t: 7.50
limmarg: 66295812.00; t3: 7.5; t: 3.75
limmarg: 16650869.00; t3: 3.8; t: 1.88
limmarg: 4379058.00; t3: 1.9; t: 0.94
limmarg: 1100062.13; t3: 1.0; t: 0.47
limmarg: 338464.69; t3: 0.5; t: 0.23
limmarg: 89740.80; t3: 0.3; t: 0.12
limmarg: 40720.68; t3: 0.2; t: 0.06
limmarg: 25048.34; t3: 0.1; t: 0.03
limmarg: 6661.30; t3: 0.1; t: 0.01

              Theta3 is 0.07 degrees, as demonstrated below.
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Refer IRS Letter F-210766 279820 dated January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.724f TCG = 0.000 VCG = 3.789

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.227	0.27f	0.01s	1,119.5	0.000	-0.003	0.0000
1.227	0.27f	0.00	1,119.5	0.000	0.000	-0.0000
1.227	0.27f	0.07p	1,119.6	0.000	0.027	0.0000
1.212	0.27f	5.00p	1,119.5	0.000	1.665	0.0728
1.148	0.30f	10.00p	1,119.5	0.000	3.199	0.2860
0.950	0.43f	15.00p	1,119.5	0.000	3.859	0.6003
0.793	0.53f	18.62p	1,119.7	0.000	3.932	0.8479
0.732	0.57f	20.00p	1,119.5	0.000	3.922	0.9428
0.506	0.72f	25.00p	1,119.5	0.000	3.782	1.2813
0.274	0.87f	30.00p	1,119.5	0.000	3.546	1.6018

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 697.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	67.613 P

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.724f TCG = 0.000 VCG = 3.789

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.227	0.27f	0.01s	1,119.5	0.000	-0.003	0.0000
1.227	0.27f	0.00	1,119.5	0.000	0.000	-0.0000
1.227	0.27f	0.07p	1,119.6	0.000	0.027	0.0000
1.212	0.27f	5.00p	1,119.5	0.000	1.665	0.0728
1.148	0.30f	10.00p	1,119.5	0.000	3.199	0.2860
0.950	0.43f	15.00p	1,119.5	0.000	3.859	0.6003
0.793	0.53f	18.62p	1,119.7	0.000	3.932	0.8479
0.732	0.57f	20.00p	1,119.5	0.000	3.922	0.9428
0.506	0.72f	25.00p	1,119.5	0.000	3.782	1.2813
0.274	0.87f	30.00p	1,119.5	0.000	3.546	1.6018
0.037	1.02f	35.00p	1,119.5	0.000	3.250	1.8988
-0.203	1.16f	40.00p	1,119.5	0.000	2.912	2.1680
-0.444	1.31f	45.00p	1,119.5	0.000	2.542	2.4062
-0.684	1.44f	50.00p	1,119.5	0.000	2.146	2.6109
-0.919	1.57f	55.00p	1,119.5	0.000	1.729	2.7801
-1.148	1.69f	60.00p	1,119.5	0.000	1.296	2.9122
-1.368	1.79f	65.00p	1,119.5	0.000	0.852	3.0061
-1.576	1.88f	70.00p	1,119.5	0.000	0.400	3.0608
-1.746	1.94f	74.39p	1,119.5	0.000	0.000	3.0761
-1.769	1.95f	75.00p	1,119.5	0.000	-0.056	3.0758

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

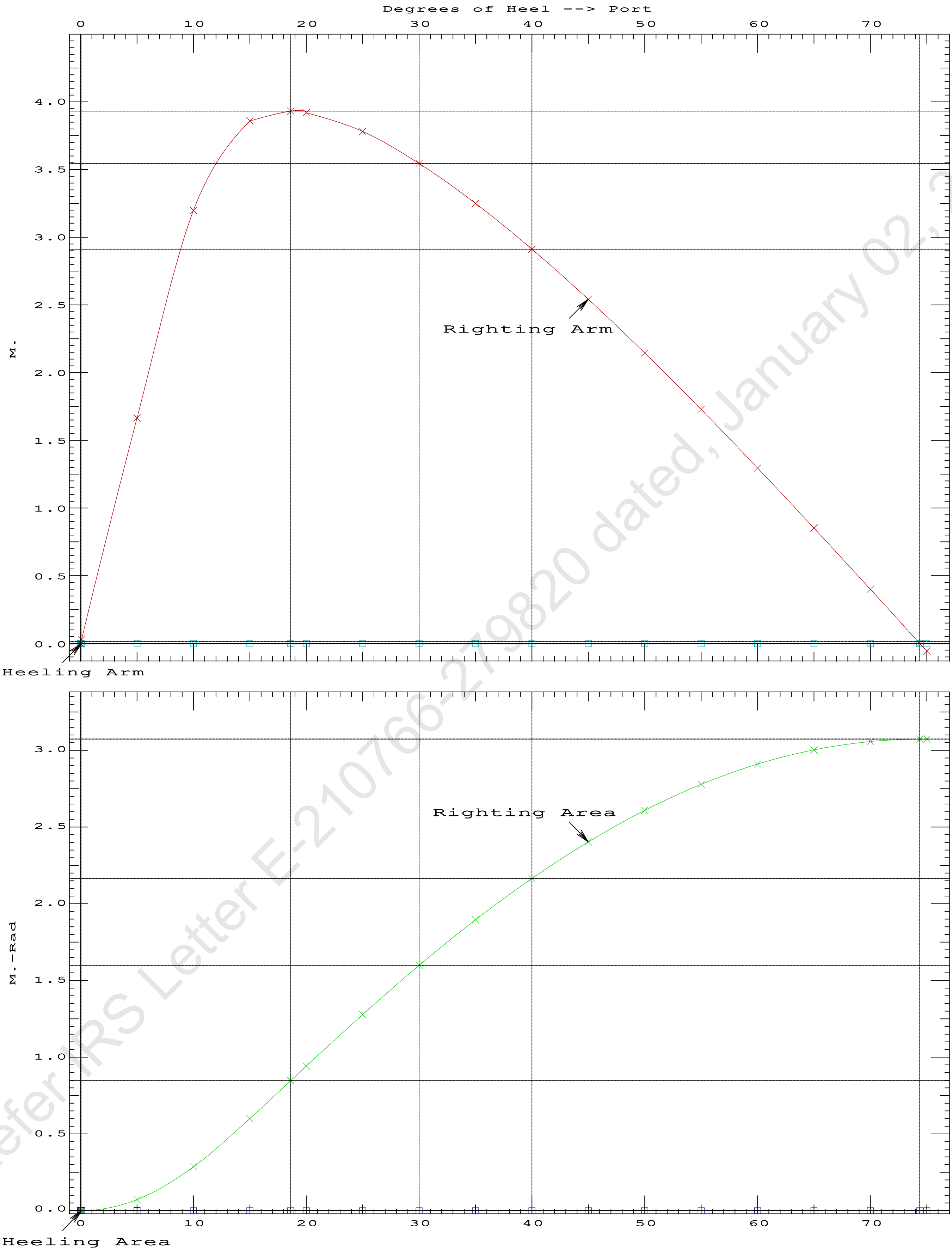
Note: The Center of Gravity shown above is for the Fixed Weight of 697.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	74.32 P

Refer IRS Letter E210766-270820 dated January 02, 2024

12/21/23 09:39:31 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

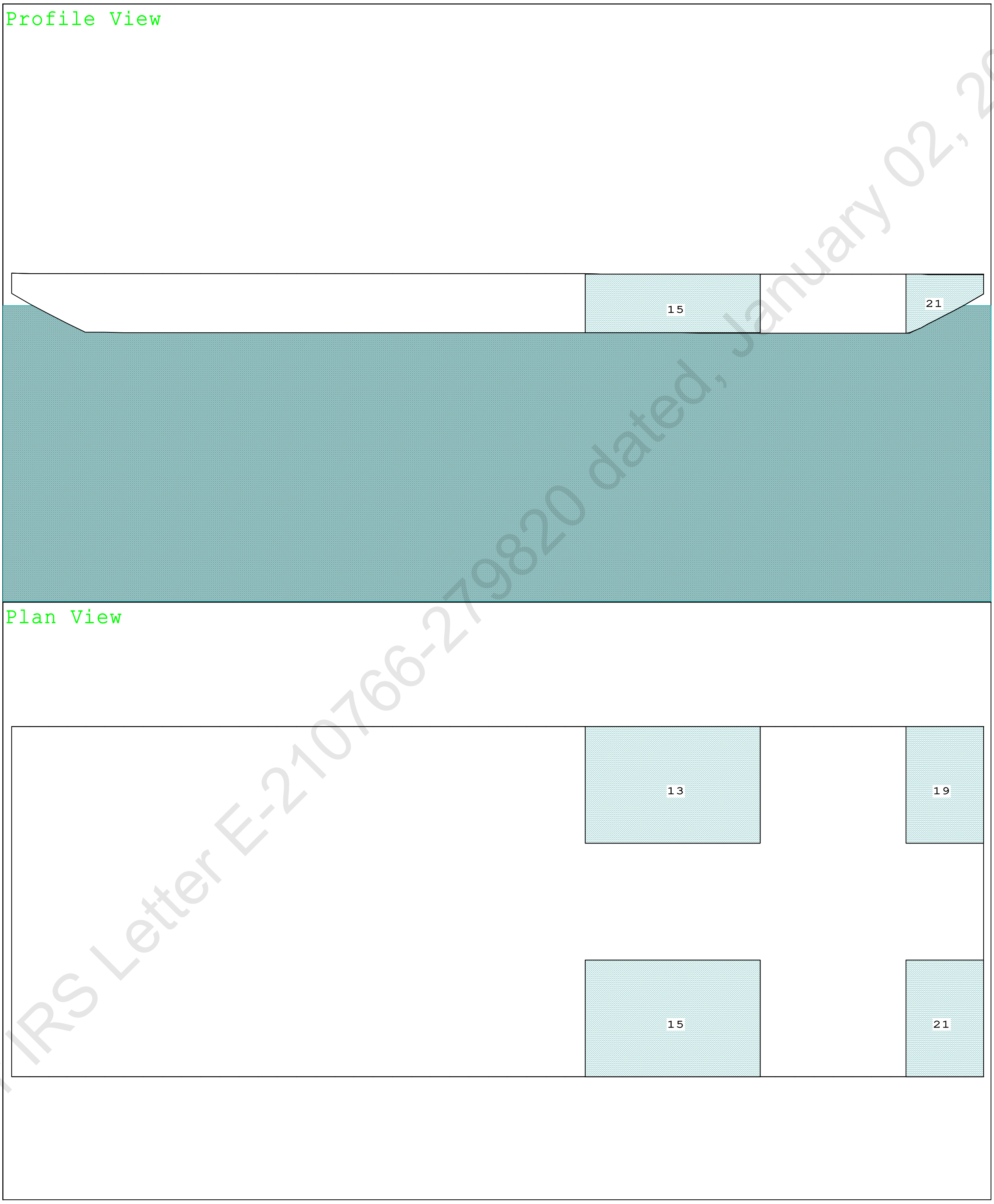
12/21/23 09:39:35 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 11 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Stbd

Condition Graphic - Draft: 1.422 @ 50.000f, 1.378 @ 0.000 Heel: zero



Refer IRS Letter E-210766-279820 dated, January 02, 2024

Tanks
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

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12/21/23 09:39:35    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
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Condition 11 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Stbd

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                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.422 @ 50.00f, 1.400 @ 25.00f, 1.378 @ 0.00
                     Trim: Fwd 0.044/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
Crane Load                            36.60   7.500f  0.000  76.977
  Total Fixed--------->              733.70  18.164f  0.000   7.440
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596    470.09
  Total Weight-------->            1,156.10  25.122f  0.000   5.305
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,156.07  25.127f  0.000   0.707
------------------------------------------------------------------------------
             Righting Arms:                   0.001f  0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         861.7  25.042f  0.000    145.61  20.249*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.83           32.60    141.01  15.651
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                             SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      250.00 MT  Crane                          36.60 MT  Crane Load

DRAFT AT FP (M) =    1.422
DRAFT AT MS (M) =    1.400
DRAFT AT AP (M) =    1.378
DRAFT AT FWD MARK (M) =    1.419
DRAFT AT AFT MARK (M) =    1.382
                            HYDROSTATIC PROPERTIES
                Trim: Fwd 0.044/50.000, No Heel, VCG = 5.305

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.400     1,156.07  25.127f  0.707     8.83  25.042f   32.60  141.01  15.651
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
 Draft is from USK.                                Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
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Refer IRS Letter E-240766 279820 dated January 02, 2024

12/21/23 09:39:35 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.422 @ 50.00f, 1.400 @ 25.00f, 1.378 @ 0.00
Trim: Fwd 0.044/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	80.4	0.817	1.502	0.05500	6.64
CRANE	57.0	4.125	4.810	0.05500	15.08
Total wind heeling moment to starboard------------->					21.72

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.122f TCG = 0.000 VCG = 5.305
Free Surface Adjustment: 0.407
Adjusted CG: LCG = 25.122f TCG = 0.000 VCG = 5.712

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.367	0.05f	0.00	1,156.0	0.000	-0.018	0.0000	5.184
1.366	0.05f	0.05s	1,156.1	0.000	0.000	-0.0000	5.135
1.352	0.05f	5.05s	1,156.1	0.000	1.368	0.0597	0.135
1.351	0.05f	5.18s	1,156.1	0.000	1.405	0.0630	0.000
1.299	0.05f	10.05s	1,156.1	0.000	2.637	0.2353	-4.865
1.165	0.08f	15.05s	1,156.1	0.000	3.042	0.4905	-9.865
1.019	0.11f	20.05s	1,156.1	0.000	2.866	0.7525	-14.865
0.865	0.14f	25.05s	1,156.1	0.000	2.503	0.9881	-19.865
0.703	0.17f	30.05s	1,156.1	0.000	2.056	1.1877	-24.865
0.536	0.20f	35.05s	1,156.1	0.000	1.562	1.3459	-29.865
0.364	0.22f	40.05s	1,156.1	0.000	1.039	1.4595	-34.865
0.189	0.25f	45.05s	1,156.1	0.000	0.497	1.5267	-39.865
0.030	0.28f	49.56s	1,156.1	0.000	0.000	1.5463	-44.376
0.012	0.28f	50.05s	1,156.1	0.000	-0.054	1.5461	-44.865
-0.164	0.30f	55.05s	1,156.1	0.000	-0.609	1.5172	-49.865
-0.339	0.33f	60.05s	1,156.1	0.000	-1.163	1.4398	-54.865

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.72 (constant)

Refer IRS Letter: E-21076279820, dated., January 12, 2024

12/21/23 09:39:35 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA        >   0.0800 m.-Rad 0.4905 P
(2) Angle from Equilibrium to RAzero or Flood    >    20.00  deg    49.51 P
(3) Angle from Equilibrium to 50% Dk Imm Angle   >     0.00  deg     5.14 P
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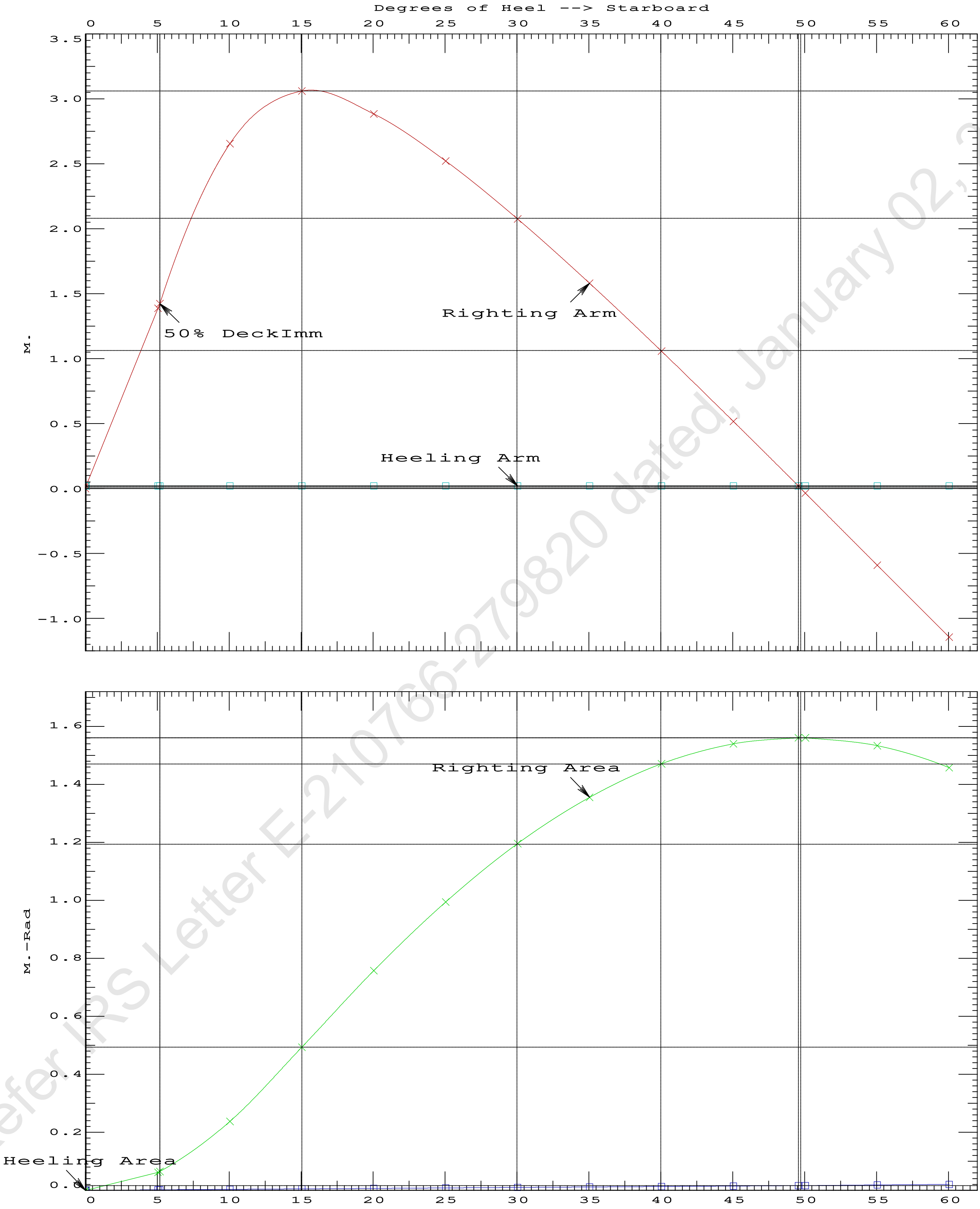
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:35 Cybermarine Technologies PTE. Ltd.
GHS 18.66 PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:35 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.122f TCG = 0.000 VCG = 5.305
Free Surface Adjustment: 0.407
Adjusted CG: LCG = 25.122f TCG = 0.014p VCG = 5.711

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.363	0.05f	2.02s	1,156.1	0.000	0.000	0.0000
1.339	0.05f	7.02s	1,156.2	0.000	1.361	0.0594
1.248	0.06f	12.02s	1,156.1	0.000	2.388	0.2254
1.158	0.08f	15.28s	1,156.3	0.000	2.503	0.3665
1.108	0.09f	17.02s	1,156.1	0.000	2.470	0.4421
0.959	0.12f	22.02s	1,156.1	0.000	2.199	0.6498
0.802	0.15f	27.02s	1,156.1	0.000	1.796	0.8251
0.638	0.18f	32.02s	1,156.1	0.000	1.327	0.9619
0.469	0.21f	37.02s	1,156.1	0.000	0.819	1.0558
0.295	0.24f	42.02s	1,156.1	0.000	0.286	1.1041
0.203	0.25f	44.65s	1,156.1	0.000	0.000	1.1107
0.120	0.26f	47.02s	1,156.1	0.000	-0.261	1.1053
-0.057	0.29f	52.02s	1,156.1	0.000	-0.815	1.0585
-0.234	0.31f	57.02s	1,156.1	0.000	-1.371	0.9631
-0.408	0.33f	62.02s	1,156.1	0.000	-1.924	0.8193

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

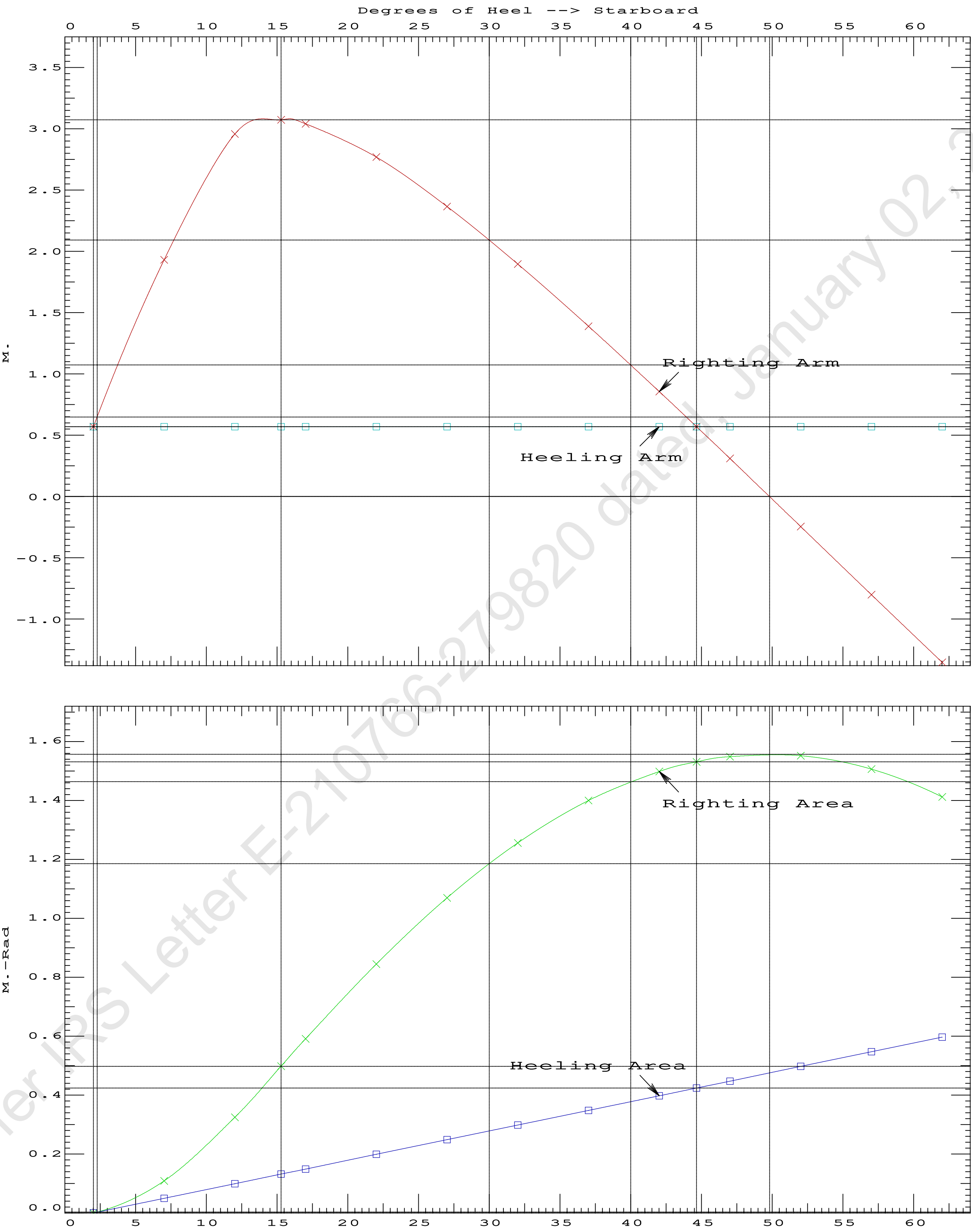
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	2.02 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	439.2 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		3.620 P

Refer IRS Letter E-21076-2-20 dated January 02, 2024

12/21/23 09:39:35 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:37 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.420 @ 50.00f, 1.398 @ 25.00f, 1.376 @ 0.00
Trim: Fwd 0.044/50.000, Heel: Stbd 2.02 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed--------->			733.70	18.164f	0.000	7.440	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,156.10	25.122f	0.000	5.305	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	1,156.10	25.126f	0.732s	0.720	

Righting Arms:	0.000	0.570s
External Arms:	0.000	0.570s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	858.9	25.038f	0.100s	144.11	20.141*
		MT/cm				
		8.80				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.02

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:37    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

                         After hook load drop
              WEIGHT and DISPLACEMENT and WATERPLANE STATUS
         USK draft: 1.478 @ 50.00f, 1.358 @ 25.00f, 1.239 @ 0.00
                Trim: Fwd 0.238/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
  Total Fixed--------->              697.10  18.724f  0.000   3.789
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->            1,119.50  25.698f  0.000   2.962
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,119.50  25.709f  0.001s  0.687
------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.001s
               External Arms:                   0.000   0.001
      Residual Righting Arms:                   0.000   0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         855.9  25.282f  0.019s   147.33  20.717*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.77           32.48    145.06  18.442
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 13394.69; t3: 30.00; t: 15.00
limmarg: 4956.62; t3: 15.00; t: 7.35
limmarg: 1328.78; t3: 7.65; t: 3.60
limmarg: 302.26; t3: 4.05; t: 1.76
limmarg: 28.90; t3: 2.29; t: 0.86
limmarg: -49.60; t3: 1.43; t: 0.42
limmarg: -15.78; t3: 1.85; t: 0.21
limmarg: 4.37; t3: 2.06; t: 0.10
limmarg: -5.50; t3: 1.96; t: 0.05
limmarg: -0.63; t3: 2.01; t: 0.02
limmarg: 1.36; t3: 2.03; t: 0.01

                   Theta3 is 2.03 degrees, as demonstrated below.
```

Refer IRS Letter E-20766 279820 dated January 02, 2024

12/21/23 09:39:37 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.724f TCG = 0.000 VCG = 3.789

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.223	0.27f	2.02s	1,119.5	0.000	-0.674	0.0000
1.227	0.27f	0.00	1,119.5	0.000	-0.000	-0.0119
1.223	0.27f	2.03p	1,119.5	0.000	0.679	0.0002
1.212	0.27f	5.00p	1,119.5	0.000	1.665	0.0609
1.148	0.30f	10.00p	1,119.5	0.000	3.199	0.2740
0.950	0.43f	15.00p	1,119.5	0.000	3.859	0.5883
0.793	0.53f	18.62p	1,119.7	0.000	3.932	0.8359
0.732	0.57f	20.00p	1,119.5	0.000	3.922	0.9308
0.506	0.72f	25.00p	1,119.5	0.000	3.782	1.2693
0.274	0.87f	30.00p	1,119.5	0.000	3.546	1.5898

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 697.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs -2 deg to abs 2	>	1.000	1.014 P

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:37 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.724f TCG = 0.000 VCG = 3.789

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.223	0.27f	2.02s	1,119.5	0.000	-0.674	0.0000
1.227	0.27f	0.00	1,119.5	0.000	-0.000	-0.0119
1.223	0.27f	2.03p	1,119.5	0.000	0.679	0.0002
1.212	0.27f	5.00p	1,119.5	0.000	1.665	0.0609
1.148	0.30f	10.00p	1,119.5	0.000	3.199	0.2740
0.950	0.43f	15.00p	1,119.5	0.000	3.859	0.5883
0.793	0.53f	18.62p	1,119.7	0.000	3.932	0.8359
0.732	0.57f	20.00p	1,119.5	0.000	3.922	0.9308
0.506	0.72f	25.00p	1,119.5	0.000	3.782	1.2693
0.274	0.87f	30.00p	1,119.5	0.000	3.546	1.5898
0.037	1.02f	35.00p	1,119.5	0.000	3.250	1.8868
-0.203	1.16f	40.00p	1,119.5	0.000	2.912	2.1560
-0.444	1.31f	45.00p	1,119.5	0.000	2.542	2.3942
-0.684	1.44f	50.00p	1,119.5	0.000	2.146	2.5989
-0.919	1.57f	55.00p	1,119.5	0.000	1.729	2.7681
-1.148	1.69f	60.00p	1,119.5	0.000	1.296	2.9002
-1.368	1.79f	65.00p	1,119.5	0.000	0.852	2.9941
-1.576	1.88f	70.00p	1,119.5	0.000	0.400	3.0488
-1.746	1.94f	74.39p	1,119.5	0.000	0.000	3.0641
-1.769	1.95f	75.00p	1,119.5	0.000	-0.056	3.0638

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

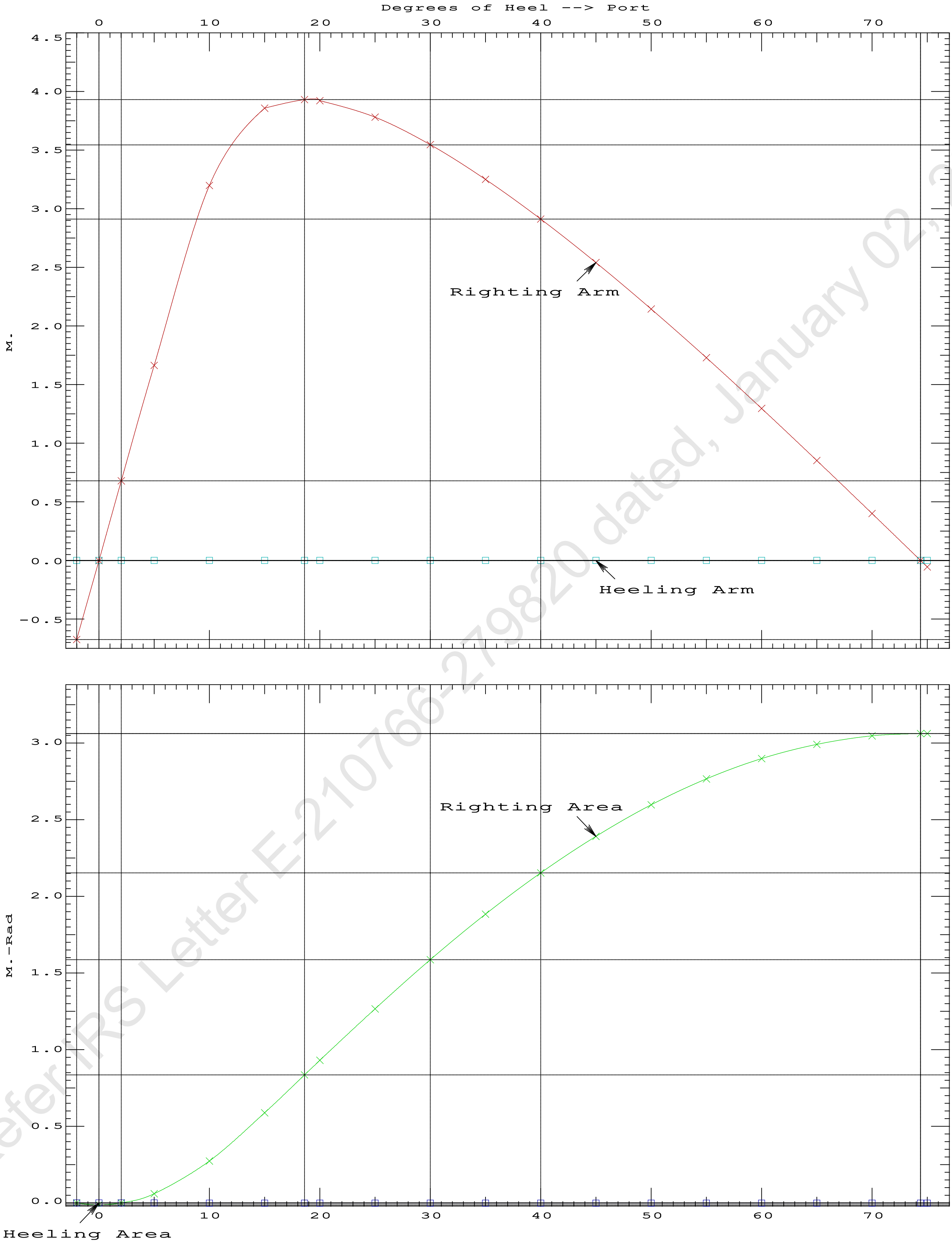
Note: The Center of Gravity shown above is for the Fixed Weight of 697.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs -2 deg to RAzero	>	1.000		259.154 P
(2) Angle from abs 2 deg to RAzero	>	20.00	deg	72.36 P

Refer IRS Letter E210766-270820 dated January 02, 2024

12/21/23 09:39:37 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

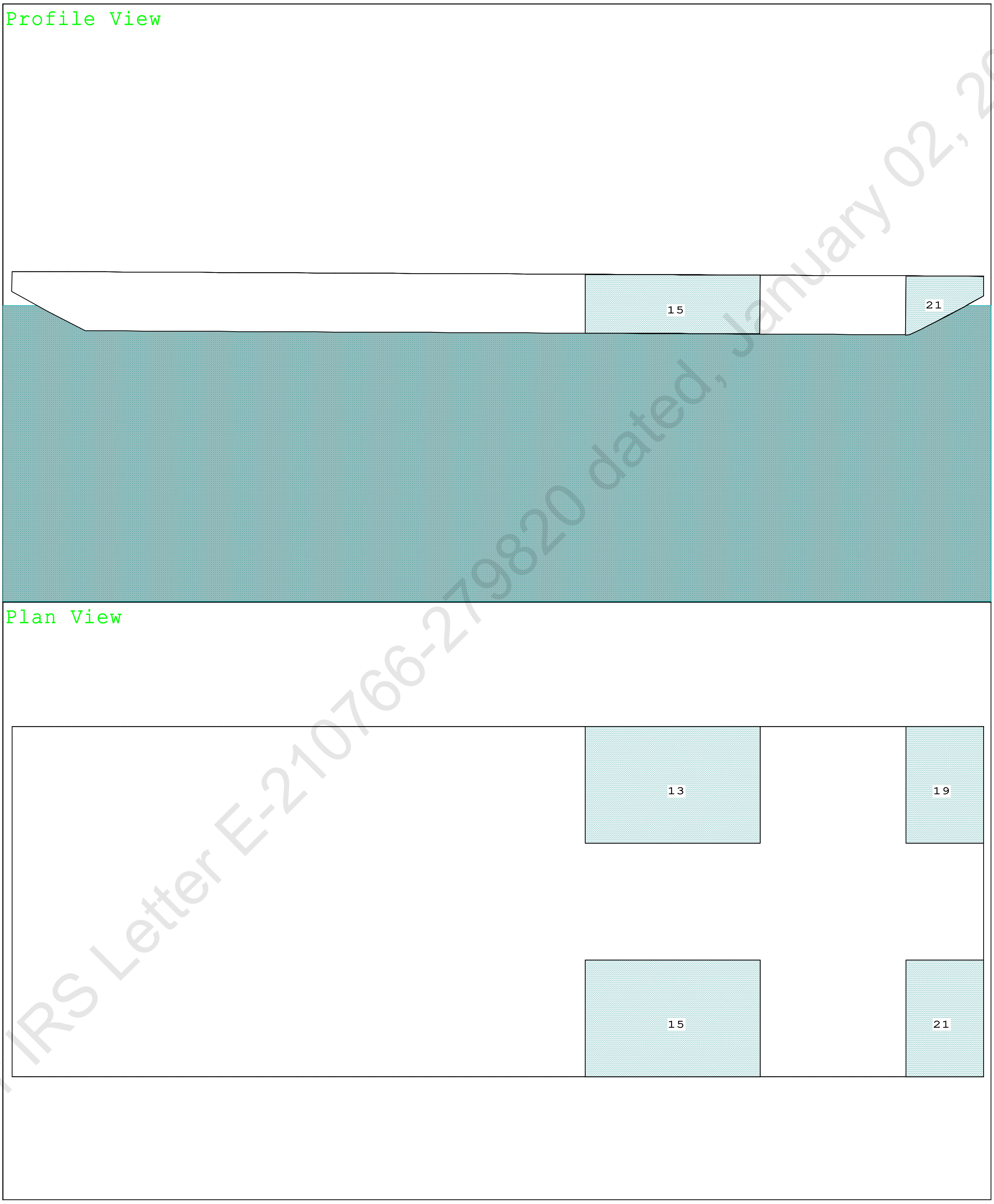
12/21/23 09:39:39 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

Condition Graphic - Draft: 1.523 @ 50.000f, 1.277 @ 0.000 Heel: zero



Tanks
15 05_WBTK.S.100% SALT WATER Intact 21 07_WBTK.S.100% SALT WATER Intact
13 05_WBTK.P.100% SALT WATER Intact 19 07_WBTK.P.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:39:39    Cybermarine Technologies PTE. Ltd.
GHS 18.66                           PONTOON
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Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

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                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.523 @ 50.00f, 1.400 @ 25.00f, 1.277 @ 0.00
                      Trim: Fwd 0.246/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                250.00   7.500f  0.000   5.200
Crane Load                            36.60  25.500f  0.000  76.977
  Total Fixed--------->              733.70  19.062f  0.000   7.440
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              422.40  37.208f  0.000   1.596    470.09
  Total Weight-------->            1,156.10  25.692f  0.000   5.305
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,156.05  25.715f  0.000   0.709
------------------------------------------------------------------------------
              Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          861.1  25.215f  0.000    145.33  20.210*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        8.83            32.54   140.73  15.614
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       412.1 Cu.M.  (16%) SALT WATER

      250.00 MT  Crane                        36.60 MT  Crane Load

DRAFT AT FP (M) =     1.523
DRAFT AT MS (M) =     1.400
DRAFT AT AP (M) =     1.277
DRAFT AT FWD MARK (M) =    1.503
DRAFT AT AFT MARK (M) =    1.297
                             HYDROSTATIC PROPERTIES
                Trim: Fwd 0.246/50.000, No Heel, VCG = 5.305

    LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.401     1,156.05  25.715f  0.709      8.83  25.215f   32.54  140.73  15.614
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                              Trim is per 50.00m.
 Draft is from USK.                                Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 470.1 m.-MT
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Refer IRS Letter E-240766 279820 dated January 02, 2024

12/21/23 09:39:39 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.523 @ 50.00f, 1.400 @ 25.00f, 1.277 @ 0.00
Trim: Fwd 0.246/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	80.4	0.819	1.505	0.05500	6.65
CRANE	57.0	4.185	4.871	0.05500	15.27
Total wind heeling moment to starboard------------->					21.92

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.692f TCG = 0.000 VCG = 5.305
Free Surface Adjustment: 0.407
Adjusted CG: LCG = 25.690f TCG = 0.000 VCG = 5.711

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.265	0.28f	0.00	1,156.0	0.000	-0.019	0.0000	4.817
1.265	0.28f	0.06s	1,156.1	0.000	0.000	-0.0000	4.754
1.251	0.28f	4.82s	1,156.1	0.000	1.305	0.0541	0.000
1.249	0.28f	5.06s	1,156.1	0.000	1.372	0.0599	-0.246
1.188	0.31f	10.06s	1,156.1	0.000	2.635	0.2355	-5.246
1.002	0.45f	15.06s	1,156.1	0.000	3.031	0.4900	-10.246
0.799	0.61f	20.06s	1,156.1	0.000	2.854	0.7510	-15.246
0.586	0.77f	25.06s	1,156.1	0.000	2.491	0.9855	-20.246
0.365	0.94f	30.06s	1,156.1	0.000	2.044	1.1840	-25.246
0.138	1.11f	35.06s	1,156.1	0.000	1.549	1.3411	-30.246
-0.093	1.27f	40.06s	1,156.1	0.000	1.026	1.4537	-35.246
-0.325	1.43f	45.06s	1,156.1	0.000	0.485	1.5198	-40.246
-0.529	1.56f	49.47s	1,156.1	0.000	0.000	1.5385	-44.654
-0.556	1.58f	50.06s	1,156.1	0.000	-0.066	1.5381	-45.246
-0.782	1.72f	55.06s	1,156.1	0.000	-0.620	1.5083	-50.246
-1.001	1.84f	60.06s	1,156.1	0.000	-1.172	1.4301	-55.246

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.92 (constant)

Refer IRS Letter: E-210766-279820, dated: January 02, 2024

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12/21/23 09:39:39    Cybermarine Technologies PTE. Ltd.
GHS 18.66                           PONTOON
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Condition 12 : Operating Condition of 250T Crane

Crane Lifting 36.6T @ 18m Radius Fwd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.4900 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg    49.41 P
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     4.75 P
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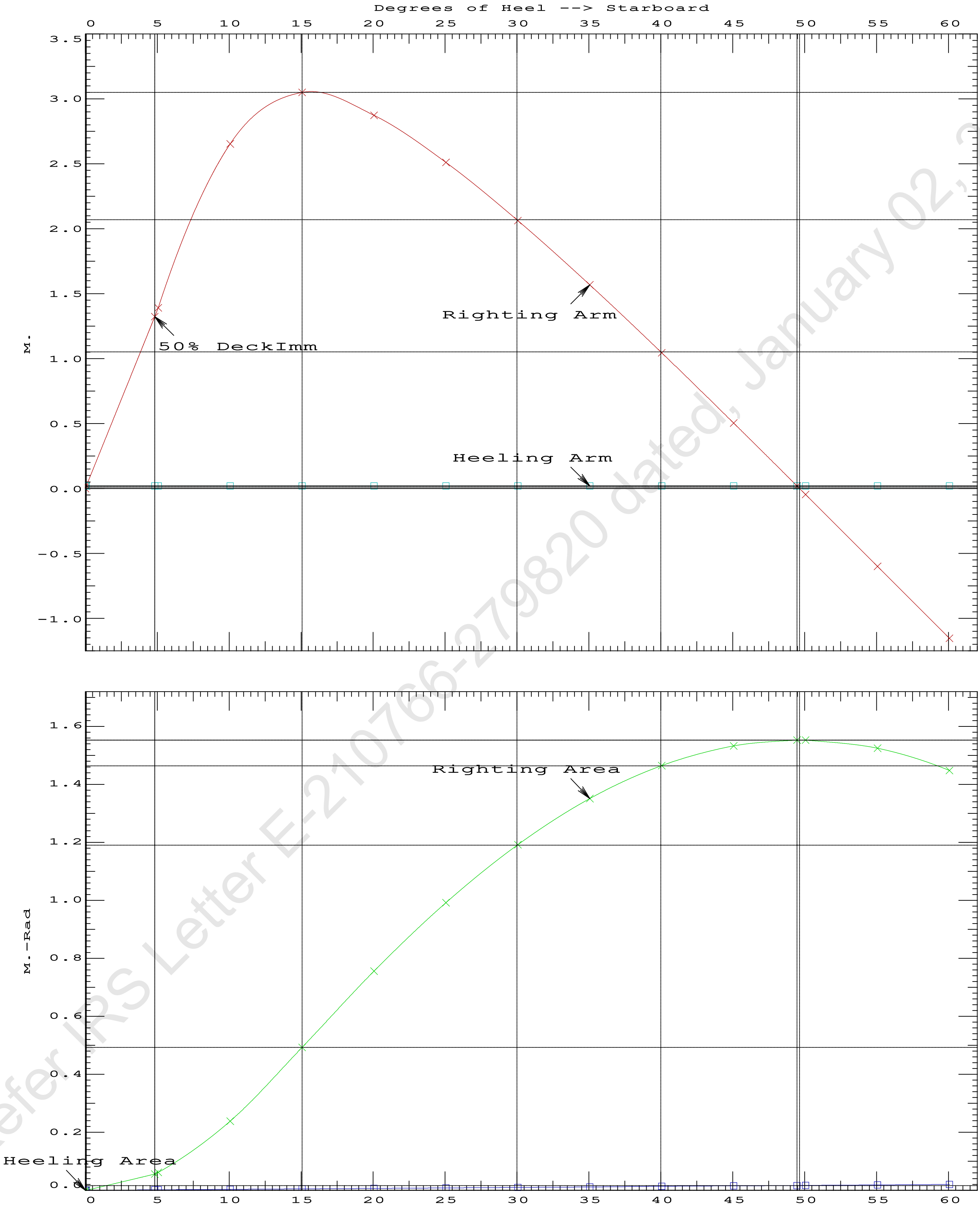
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:39 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd



Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:39 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.692f TCG = 0.000 VCG = 5.305
Free Surface Adjustment: 0.407
Adjusted CG: LCG = 25.690f TCG = 0.000 VCG = 5.711

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.265	0.28f	0.01s	1,156.1	0.000	0.000	0.0000
1.250	0.28f	5.01s	1,156.0	0.000	1.373	0.0599
1.189	0.30f	10.01s	1,156.1	0.000	2.639	0.2357
1.004	0.45f	15.01s	1,156.3	0.000	3.045	0.4911
0.801	0.61f	20.01s	1,156.1	0.000	2.871	0.7534
0.588	0.77f	25.01s	1,156.1	0.000	2.510	0.9896
0.368	0.94f	30.01s	1,156.1	0.000	2.063	1.1897
0.141	1.10f	35.01s	1,156.1	0.000	1.569	1.3485
-0.090	1.27f	40.01s	1,156.1	0.000	1.046	1.4628
-0.323	1.43f	45.01s	1,156.1	0.000	0.505	1.5306
-0.535	1.57f	49.60s	1,156.1	0.000	0.000	1.5509
-0.554	1.58f	50.01s	1,156.1	0.000	-0.045	1.5507
-0.780	1.72f	55.01s	1,156.1	0.000	-0.600	1.5226
-0.999	1.84f	60.01s	1,156.1	0.000	-1.152	1.4462

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

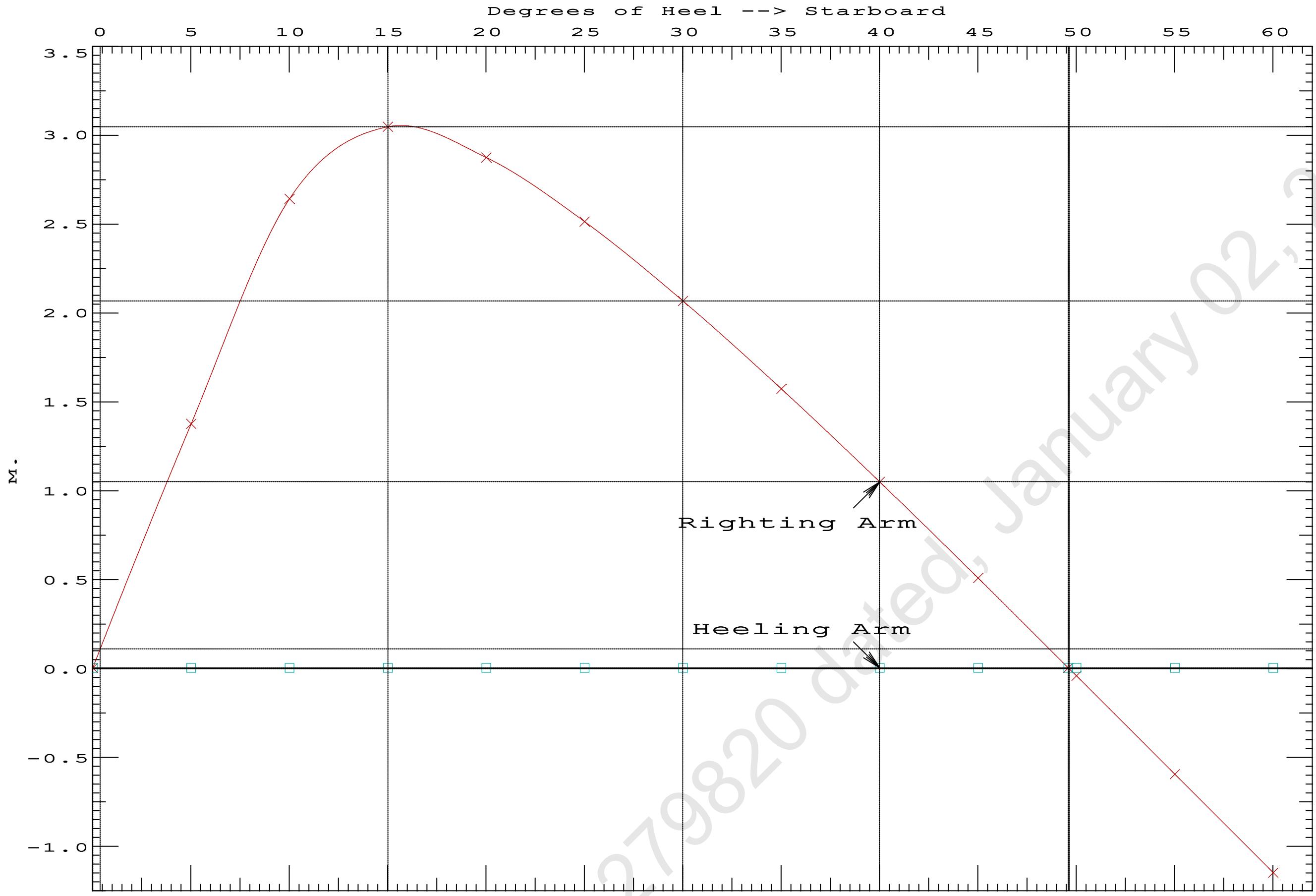
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

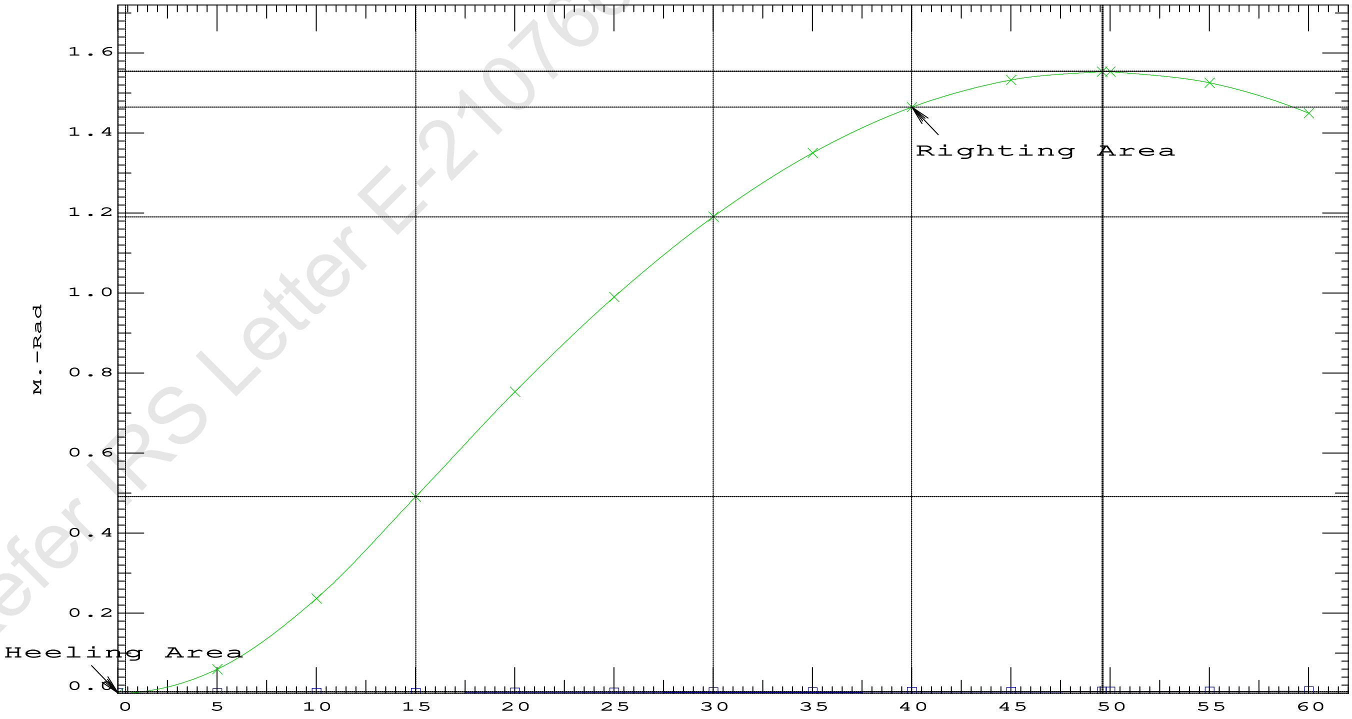
LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	70400.4	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		415.335	P

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:39 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd





Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:42 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.523 @ 50.00f, 1.400 @ 25.00f, 1.277 @ 0.00
Trim: Fwd 0.246/50.000, Heel: Stbd 0.01 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Total Fixed--------->			733.70	19.062f	0.000	7.440	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks--------->			422.40	37.208f	0.000	1.596	470.09
Total Weight-------->			1,156.10	25.692f	0.000	5.305	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,156.10	25.714f	0.005s	0.709	

Righting Arms:	0.000	0.004s
External Arms:	0.000	0.004s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	857.9	25.304f	0.024s	143.69	20.111*
		MT/cm				
		8.79				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

Refer IRS Letter E-2107662798320 dated, January 02, 2024

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12/21/23 09:39:42    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

                          After hook load drop
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.478 @ 50.00f, 1.358 @ 25.00f, 1.239 @ 0.00
                 Trim: Fwd 0.239/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          447.10  25.000f  0.000   3.000
Crane                               250.00   7.500f  0.000   5.200
  Total Fixed--------->             697.10  18.724f  0.000   3.789
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025       161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025       161.60  34.000f  5.980s  1.509    162.73*
07_WBTK.P      1.000    1.025        49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025        49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->             422.40  37.208f  0.000   1.596   470.09
  Total Weight-------->           1,119.50  25.698f  0.000   2.962
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025     1,119.50  25.710f  0.001s  0.687
-------------------------------------------------------------------------------
               Righting Arms:                   0.000    0.001s
               External Arms:                   0.000    0.001
      Residual Righting Arms:                   0.000    0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         855.9  25.283f  0.019s   147.33  20.717*
                                      MT/cm---------m.-MT/cm-------GML-----GMT
                                       8.77             32.48    145.06  18.442
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
```

```
limmarg: 353537952.00; t3: 30; t: 15.00
limmarg: 132470760.00; t3: 15; t: 7.50
limmarg: 35986560.00; t3: 8; t: 3.75
limmarg: 11599190.00; t3: 4; t: 1.88
limmarg: 2896367.75; t3: 2; t: 0.94
limmarg: 727291.56; t3: 1; t: 0.47
limmarg: 183644.13; t3: 1; t: 0.23
limmarg: 385139.91; t3: 1; t: 0.12
limmarg: 502187.84; t3: 1; t: 0.06
limmarg: 572587.50; t3: 1; t: 0.03
limmarg: 609525.13; t3: 1; t: 0.01
```

Theta3 is 0.97 degrees, as demonstrated below.

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:42 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.724f TCG = 0.000 VCG = 3.789

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.227	0.27f	0.01s	1,119.5	0.000	-0.004	0.0000
1.227	0.27f	0.00	1,119.5	0.000	-0.000	-0.0000
1.225	0.28f	0.97p	1,119.5	0.000	0.326	0.0028
1.212	0.27f	5.00p	1,119.5	0.000	1.665	0.0728
1.148	0.30f	10.00p	1,119.5	0.000	3.199	0.2860
0.950	0.43f	15.00p	1,119.5	0.000	3.859	0.6003
0.793	0.53f	18.62p	1,119.7	0.000	3.932	0.8479
0.732	0.57f	20.00p	1,119.5	0.000	3.922	0.9428
0.506	0.72f	25.00p	1,119.5	0.000	3.782	1.2813
0.274	0.87f	30.00p	1,119.5	0.000	3.546	1.6018

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 697.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------------STABILITY CRITERION		Min/Max	Attained
(1) Residual Area Ratio from abs 0 deg to abs 1	>	1.000	6096.25 P

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:42 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 18.724f TCG = 0.000 VCG = 3.789

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.227	0.27f	0.01s	1,119.5	0.000	-0.004	0.0000
1.227	0.27f	0.00	1,119.5	0.000	-0.000	-0.0000
1.225	0.28f	0.97p	1,119.5	0.000	0.326	0.0028
1.212	0.27f	5.00p	1,119.5	0.000	1.665	0.0728
1.148	0.30f	10.00p	1,119.5	0.000	3.199	0.2860
0.950	0.43f	15.00p	1,119.5	0.000	3.859	0.6003
0.793	0.53f	18.62p	1,119.7	0.000	3.932	0.8479
0.732	0.57f	20.00p	1,119.5	0.000	3.922	0.9428
0.506	0.72f	25.00p	1,119.5	0.000	3.782	1.2813
0.274	0.87f	30.00p	1,119.5	0.000	3.546	1.6018
0.037	1.02f	35.00p	1,119.5	0.000	3.250	1.8988
-0.203	1.16f	40.00p	1,119.5	0.000	2.912	2.1680
-0.444	1.31f	45.00p	1,119.5	0.000	2.542	2.4062
-0.684	1.44f	50.00p	1,119.5	0.000	2.146	2.6109
-0.919	1.57f	55.00p	1,119.5	0.000	1.729	2.7801
-1.148	1.69f	60.00p	1,119.5	0.000	1.296	2.9122
-1.368	1.79f	65.00p	1,119.5	0.000	0.852	3.0061
-1.576	1.88f	70.00p	1,119.5	0.000	0.400	3.0608
-1.746	1.94f	74.39p	1,119.5	0.000	0.000	3.0761
-1.769	1.95f	75.00p	1,119.5	0.000	-0.056	3.0758

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

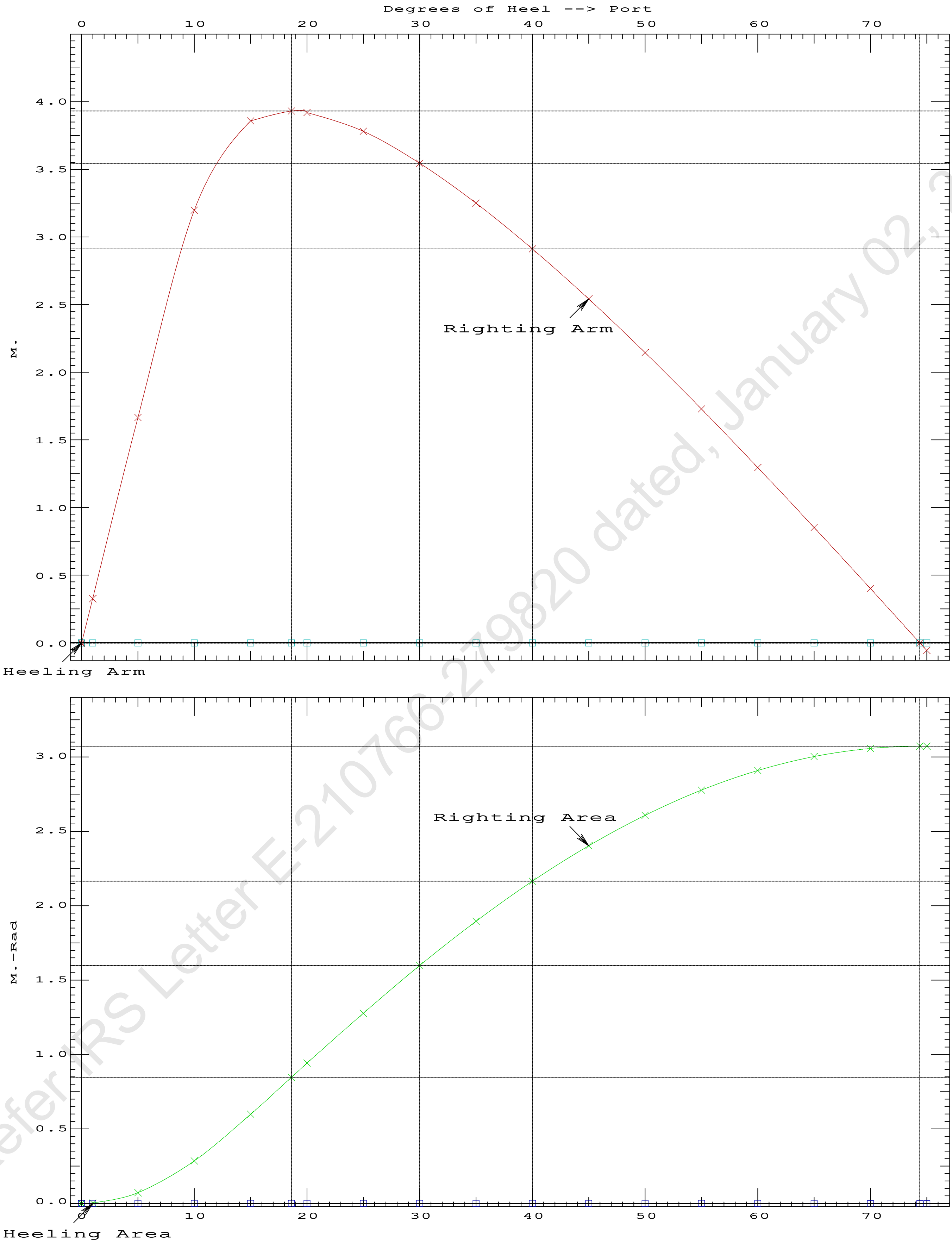
Note: The Center of Gravity shown above is for the Fixed Weight of 697.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		6789916 P
(2) Angle from abs 1 deg to RAzero	>	20.00	deg	73.42 P

Refer IRS Letter E210766-270820 dated January 02, 2024

12/21/23 09:39:42 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



Refer IRS Letter E-210766-279820 dated, January 02, 2024

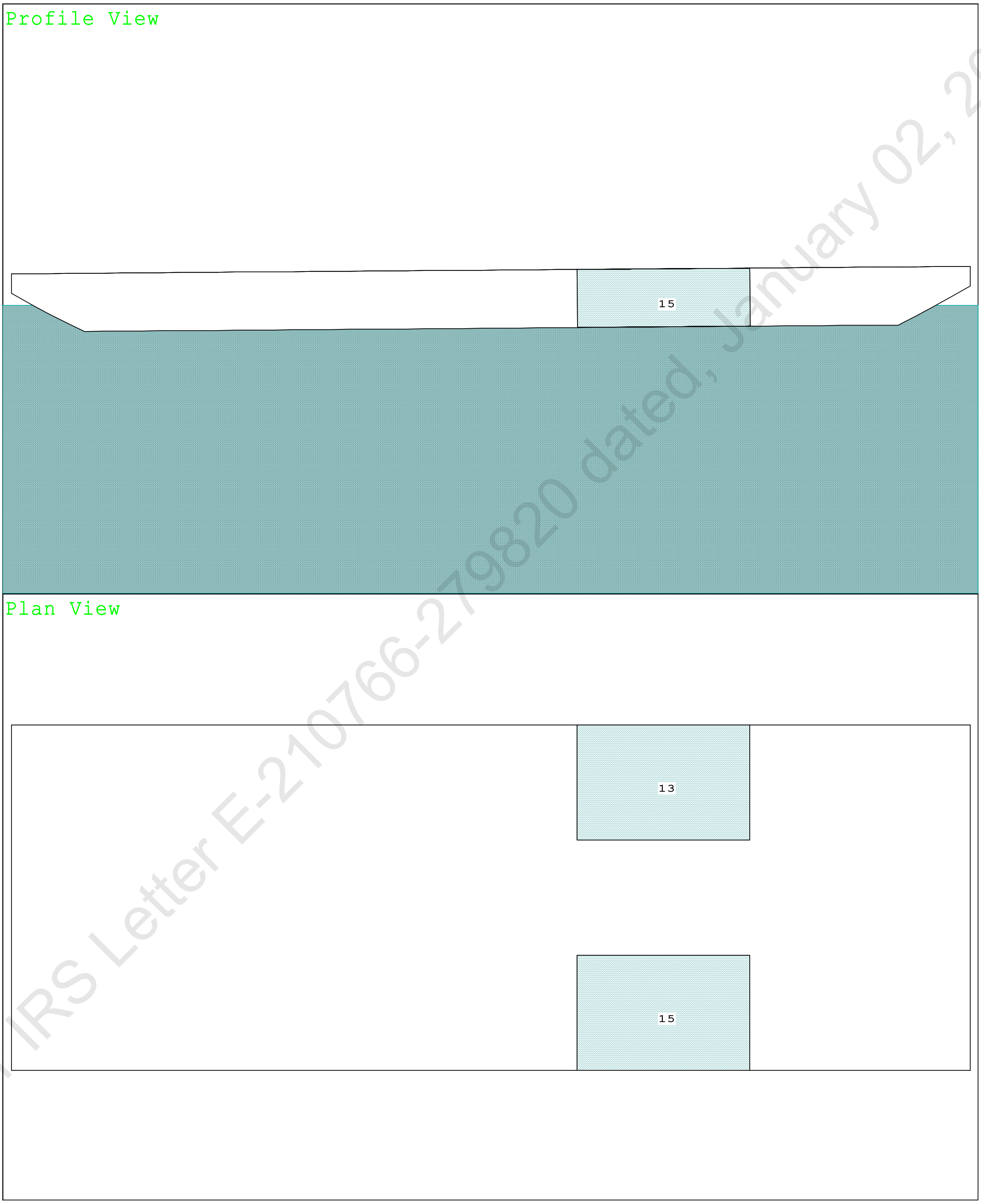
12/21/23 09:39:44 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 13 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Aft

Condition Graphic - Draft: 0.991 @ 50.000f, 1.385 @ 0.000 Heel: zero



Tanks

13 05_WBTK.P.100% SALT WATER Intact 15 05_WBTK.S.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:39:44    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
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Condition 13 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Aft

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               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 0.991 @ 50.00f, 1.188 @ 25.00f, 1.385 @ 0.00
                     Trim: Aft 0.394/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Crane Load                            51.30   4.500a  0.000  33.218
  Total Fixed--------->              648.40  18.618f  0.000   5.900
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->              323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->              971.60  23.734f  0.000   4.439
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        971.60  23.704f  0.000   0.603
-------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         847.3  24.649f  0.000    164.71  23.809*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.68            31.26    160.87  19.972
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                          SUMMARY OF LOADING
       315.3 Cu.M.  (13%) SALT WATER

      150.00 MT  Crane                        51.30 MT  Crane Load

DRAFT AT FP (M) =    0.991
DRAFT AT MS (M) =    1.188
DRAFT AT AP (M) =    1.385
DRAFT AT FWD MARK (M) =    1.023
DRAFT AT AFT MARK (M) =    1.354
                            HYDROSTATIC PROPERTIES
                Trim: Aft 0.394/50.000, No Heel, VCG = 4.439

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.191       971.60  23.704f  0.603     8.68  24.649f   31.26  160.87  19.972
 Distances in METERS.------Specific Gravity = 1.025.-----------Moment in m.-MT.
                           Trim is per 50.00m.
 Draft is from USK.                                Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 325.5 m.-MT
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Refer IRS Letter E-220766-279820 dated January 02, 2024

12/21/23 09:39:44 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 0.991 @ 50.00f, 1.188 @ 25.00f, 1.385 @ 0.00
Trim: Aft 0.394/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	90.4	0.930	1.517	0.05500	7.54
CRANE	57.0	4.206	4.793	0.05500	15.03
Total wind heeling moment to starboard------------->					22.57

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 23.734f TCG = 0.000 VCG = 4.439
Free Surface Adjustment: 0.335
Adjusted CG: LCG = 23.737f TCG = 0.000 VCG = 4.774

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel--->	Res. Area	50% DeckImm Angle
1.373	0.45a	0.00	971.59	0.000	-0.023	0.0000	5.318
1.373	0.45a	0.06s	971.60	0.000	0.000	-0.0000	5.261
1.358	0.45a	5.06s	971.59	0.000	1.746	0.0762	0.261
1.357	0.45a	5.32s	971.54	0.000	1.835	0.0843	0.000
1.309	0.51a	10.06s	971.49	0.000	3.197	0.2943	-4.739
1.171	0.68a	15.06s	971.60	0.000	3.718	0.6021	-9.739
1.128	0.76a	16.68s	971.60	0.000	3.739	0.7079	-11.364
1.036	0.92a	20.06s	971.60	0.000	3.661	0.9270	-14.739
0.898	1.16a	25.06s	971.60	0.000	3.379	1.2367	-19.739
0.758	1.40a	30.06s	971.60	0.000	2.994	1.5155	-24.739
0.616	1.64a	35.06s	971.60	0.000	2.549	1.7578	-29.739
0.474	1.87a	40.06s	971.60	0.000	2.064	1.9593	-34.739
0.331	2.10a	45.06s	971.60	0.000	1.551	2.1173	-39.739
0.188	2.32a	50.06s	971.60	0.000	1.020	2.2296	-44.739
0.044	2.53a	55.06s	971.60	0.000	0.476	2.2950	-49.739
-0.082	2.68a	59.39s	971.59	0.000	0.000	2.3130	-54.069
-0.101	2.71a	60.06s	971.60	0.000	-0.074	2.3126	-54.739

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.57 (constant)

Refer IRS Letter E-21076-279820, dated January 12, 2024

12/21/23 09:39:44   Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA         >   0.0800 m.-Rad 0.7079 P
(2) Angle from Equilibrium to RAzero or Flood     >    20.00  deg    59.33 P
(3) Angle from Equilibrium to 50% Dk Imm Angle    >     0.00  deg     5.26 P
-----------------------------------------------------------------------------
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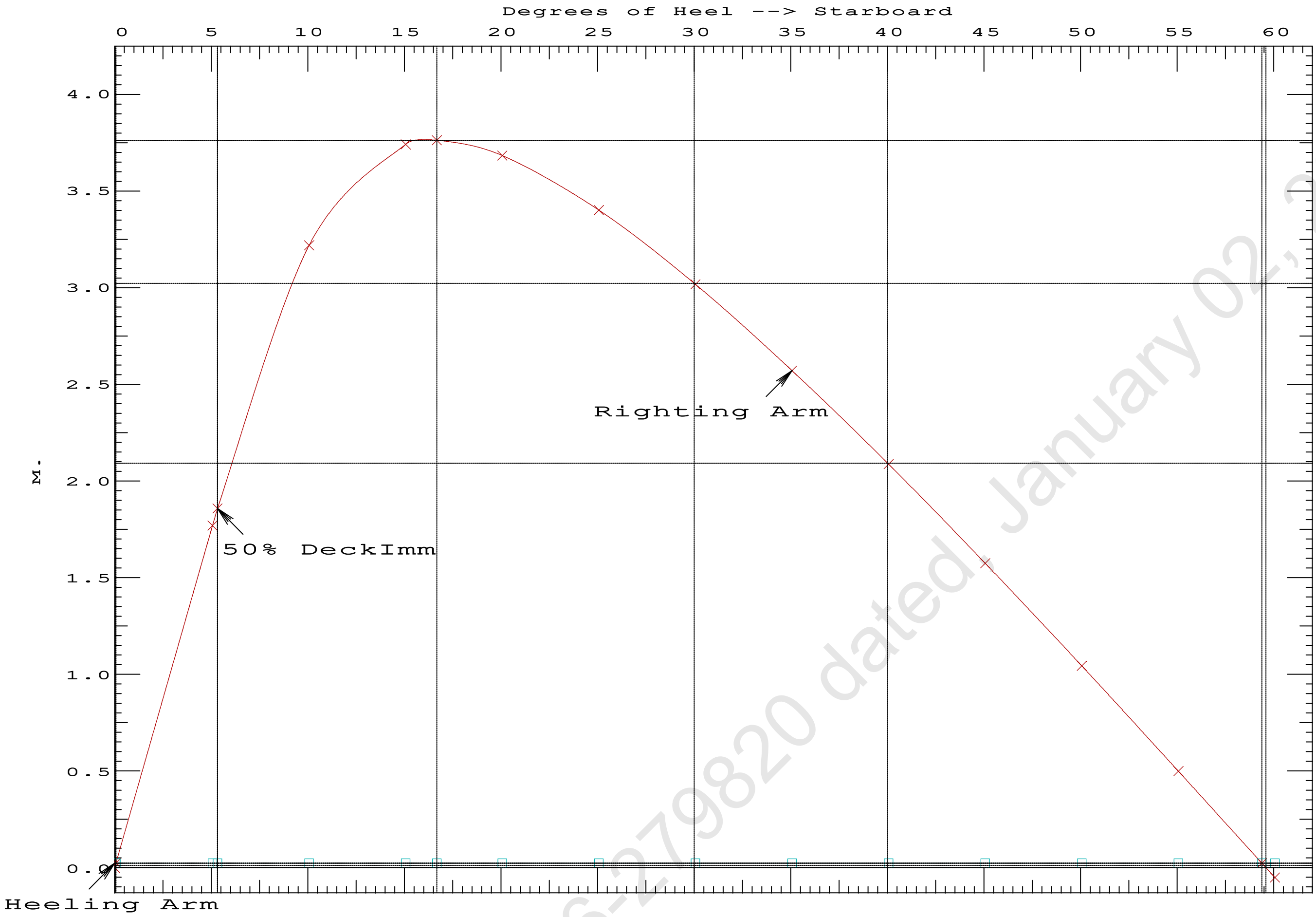
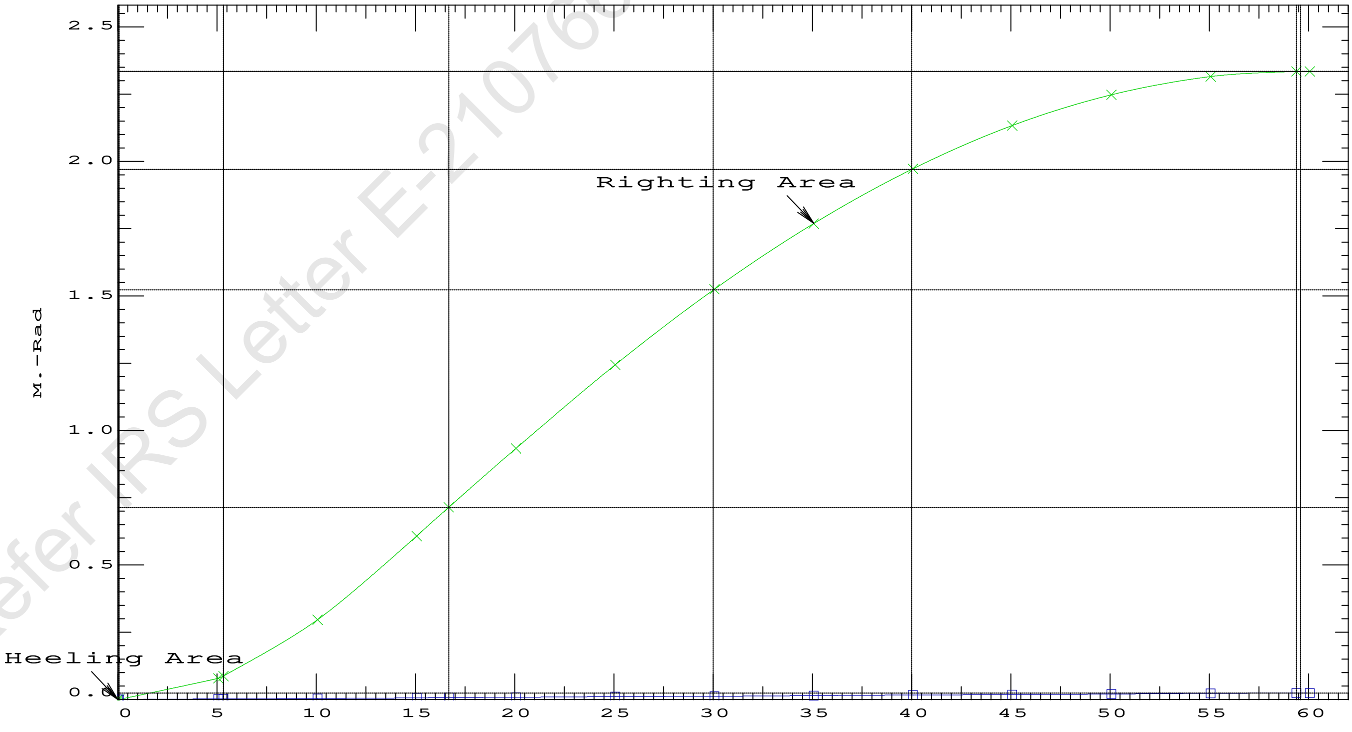
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:44 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 13 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Aft





Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:44 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.734f TCG = 0.000 VCG = 4.439
Free Surface Adjustment: 0.335
Adjusted CG: LCG = 23.737f TCG = 0.000 VCG = 4.774

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.373	0.45a	0.01s	971.60	0.000	0.000	0.0000
1.358	0.45a	5.01s	971.60	0.000	1.747	0.0762
1.310	0.51a	10.01s	971.49	0.000	3.205	0.2944
1.173	0.68a	15.01s	971.60	0.000	3.735	0.6033
1.128	0.76a	16.69s	971.60	0.000	3.757	0.7132
1.038	0.92a	20.01s	971.60	0.000	3.681	0.9299
0.899	1.15a	25.01s	971.60	0.000	3.401	1.2414
0.759	1.40a	30.01s	971.60	0.000	3.016	1.5221
0.617	1.64a	35.01s	971.60	0.000	2.571	1.7663
0.475	1.87a	40.01s	971.60	0.000	2.087	1.9699
0.332	2.10a	45.01s	971.60	0.000	1.575	2.1298
0.189	2.32a	50.01s	971.60	0.000	1.043	2.2442
0.045	2.52a	55.01s	971.60	0.000	0.500	2.3116
-0.086	2.69a	59.55s	971.60	0.000	0.000	2.3315
-0.100	2.70a	60.01s	971.60	0.000	-0.050	2.3313

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

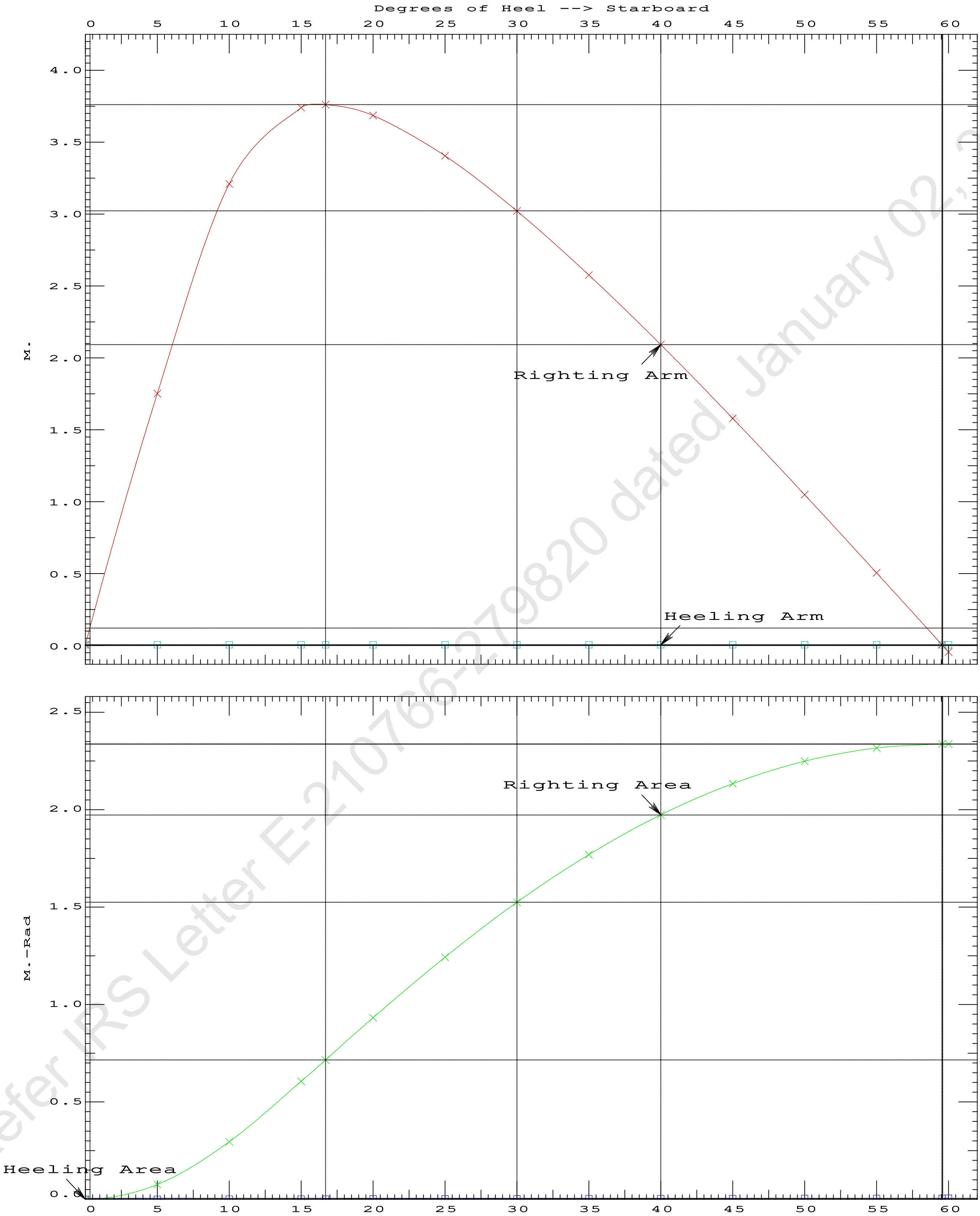
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	73002.8 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		436.944 P

Refer IRS Letter E-210766-2/0620 dated January 02, 2024

12/21/23 09:39:44 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft



Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 14 : Operating Condition of 150T Crane

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                    Crane in operation with load in place
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 0.991 @ 50.00f, 1.188 @ 25.00f, 1.385 @ 0.00
                     Trim: Aft 0.394/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Crane Load                            51.30   4.500a  0.000  33.218
  Total Fixed--------->              648.40  18.618f  0.000   5.900
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->              323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->              971.60  23.734f  0.000   4.439
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        971.59  23.704f  0.000   0.603
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
               External Arms:                  0.000   0.001s
      Residual Righting Arms:                  0.000  -0.001s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         847.3  24.649f  0.000    164.71  23.809*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.68            31.26    160.87  19.972
Distances in METERS.-----------------------------------------Moments in m.-MT.
```

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

Refer IRS Letter E-210766-279820 dated January 02, 2024

```
12/21/23 09:39:46    Cybermarine Technologies PTE. Ltd.
GHS 18.66                              PONTOON

                          After hook load drop
              WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 1.176 @ 50.00f, 1.130 @ 25.00f, 1.085 @ 0.00
                 Trim: Fwd 0.091/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
  Total Fixed--------->              597.10  20.604f  0.000   3.553
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->              323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->              920.30  25.308f  0.000   2.835
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        920.30  25.312f  0.001s  0.568
------------------------------------------------------------------------------
               Righting Arms:                 0.000   0.001s
               External Arms:                 0.000   0.001s
      Residual Righting Arms:                 0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         841.0  25.054f  0.016s   170.04  24.886*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.62            30.88   167.77  22.618
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

               Theta3 is 0.07 degrees, as demonstrated below.

                     RESIDUAL RIGHTING ARMS vs HEEL ANGLE
          Fixed CG:  LCG = 20.604f  TCG = 0.000   VCG =  3.553

        Origin    Degrees of   Displacement    Residual Arms       Res.
         Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
         1.072   0.10f   0.00       920.29      0.000    0.000    0.0000
         1.073   0.10f   0.07s      920.35      0.000    0.028    0.0000
         1.058   0.10f   5.00s      920.30      0.000    2.021    0.0882
         0.968   0.12f  10.00s      920.29      0.000    3.679    0.3395
         0.719   0.15f  15.00s      920.30      0.000    4.388    0.6985
         0.468   0.20f  19.34s      920.30      0.000    4.514    1.0387
         0.429   0.21f  20.00s      920.30      0.000    4.512    1.0905
```

Refer IRS Letter F-210766-279820 dated January 02, 2024

12/21/23 09:39:46   Cybermarine Technologies PTE. Ltd.
GHS 18.66   **PONTOON**

```
  0.136   0.26f  25.00s      920.30     0.000    4.392    1.4816
 -0.160   0.31f  30.00s      920.30     0.000    4.156    1.8554
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.
```

Note: The Center of Gravity shown above is for the Fixed Weight of 597.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1  >    1.000          LARGE
-----------------------------------------------------------------------------
```

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.604f TCG = 0.000 VCG = 3.553

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.072	0.10f	0.00	920.29	0.000	0.000	0.0000
1.073	0.10f	0.07s	920.35	0.000	0.028	0.0000
1.058	0.10f	5.00s	920.30	0.000	2.021	0.0882
0.968	0.12f	10.00s	920.29	0.000	3.679	0.3395
0.719	0.15f	15.00s	920.30	0.000	4.388	0.6985
0.468	0.20f	19.34s	920.30	0.000	4.514	1.0387
0.429	0.21f	20.00s	920.30	0.000	4.512	1.0905
0.136	0.26f	25.00s	920.30	0.000	4.392	1.4816
-0.160	0.31f	30.00s	920.30	0.000	4.156	1.8554
-0.455	0.36f	35.00s	920.30	0.000	3.849	2.2053
-0.747	0.42f	40.00s	920.30	0.000	3.491	2.5259
-1.035	0.47f	45.00s	920.30	0.000	3.094	2.8136
-1.315	0.51f	50.00s	920.30	0.000	2.666	3.0651
-1.586	0.56f	55.00s	920.30	0.000	2.212	3.2781
-1.845	0.60f	60.00s	920.30	0.000	1.738	3.4506
-2.089	0.64f	65.00s	920.30	0.000	1.248	3.5810
-2.318	0.67f	70.00s	920.30	0.000	0.747	3.6681
-2.528	0.69f	75.00s	920.30	0.000	0.239	3.7111
-2.619	0.70f	77.34s	920.30	0.000	0.000	3.7159
-2.718	0.71f	80.00s	920.30	0.000	-0.272	3.7096

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

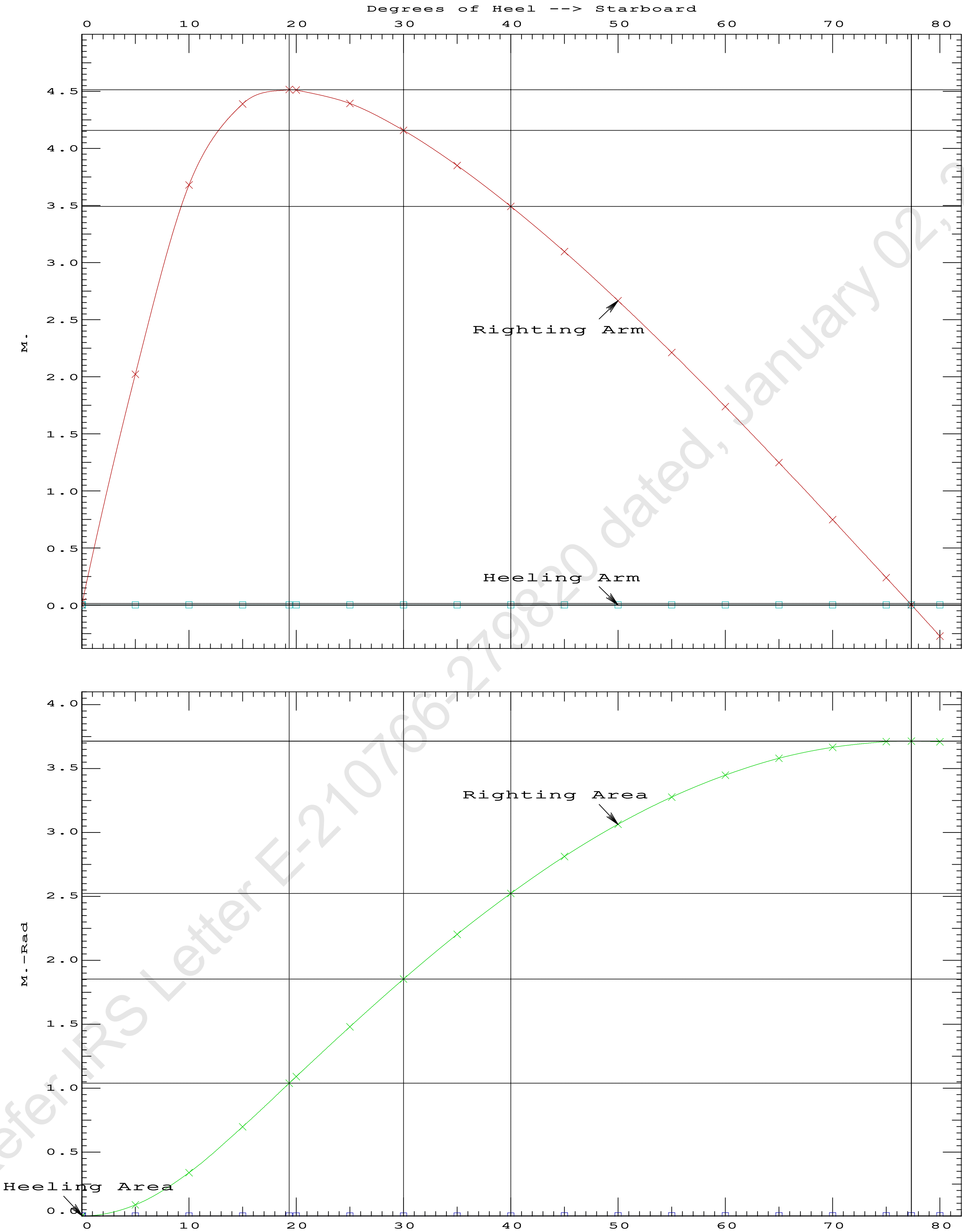
Note: The Center of Gravity shown above is for the Fixed Weight of 597.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	77.27 P

Refer IRS Letter E210766-270820 dated January 02, 2024

12/21/23 09:39:46 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



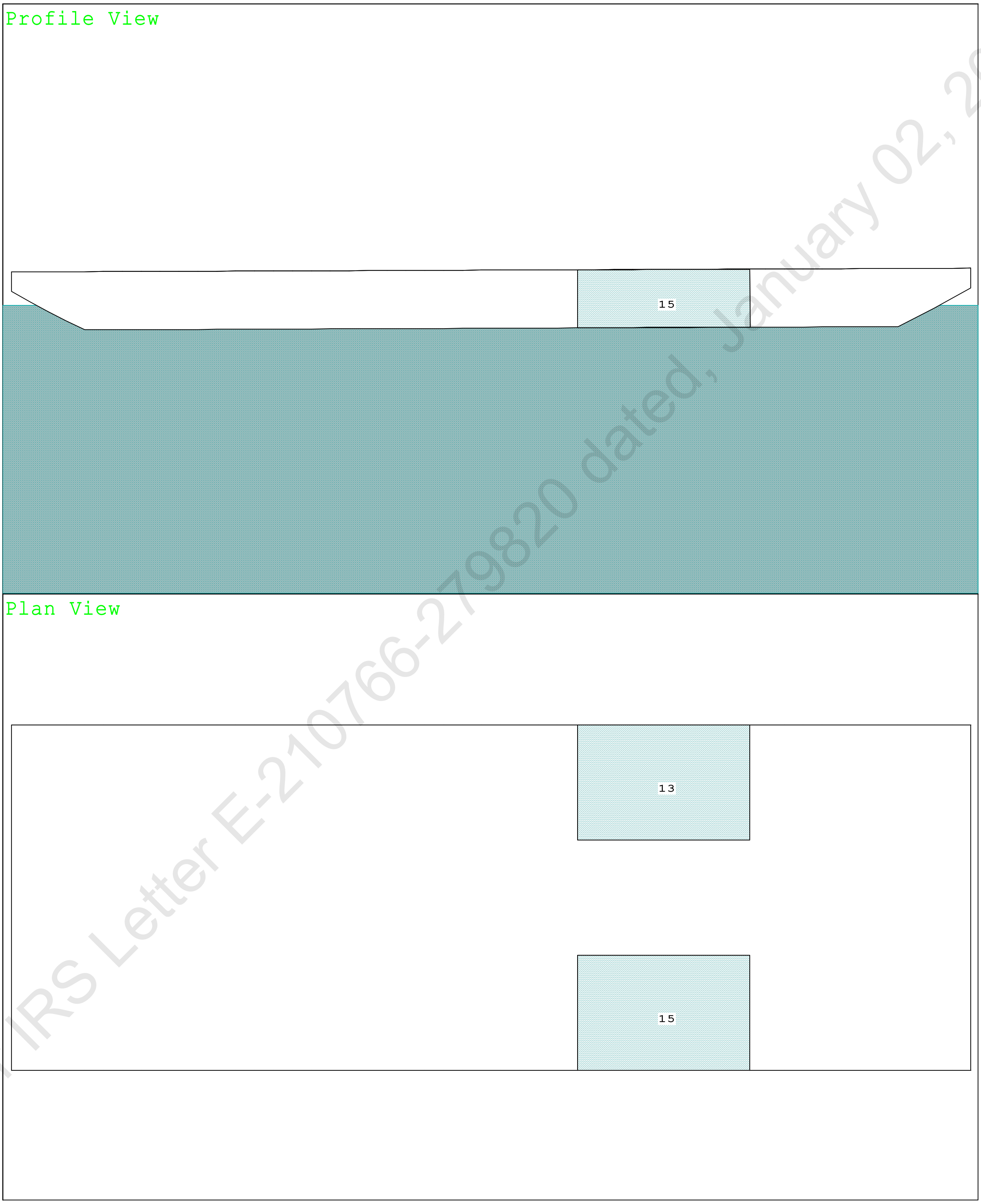
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

Condition Graphic - Draft: 1.091 @ 50.000f, 1.288 @ 0.000 Heel: zero



Tanks
13 05_WBTK.P.100% SALT WATER Intact 15 05_WBTK.S.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:39:48    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
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Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

```
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.091 @ 50.00f, 1.189 @ 25.00f, 1.288 @ 0.00
                      Trim: Aft 0.197/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                150.00   7.500f  0.000   5.200
Crane Load                            51.30   7.500f  0.000  33.218
  Total Fixed--------->              648.40  19.567f  0.000   5.900
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025        161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025        161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->              323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->              971.60  24.368f  0.000   4.439
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        971.56  24.352f  0.000   0.599
------------------------------------------------------------------------------
               Righting Arms:                  0.001   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         847.0  24.825f  0.000    164.54  23.789*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.68           31.23    160.70  19.949
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           SUMMARY OF LOADING
       315.3 Cu.M.  (13%) SALT WATER

      150.00 MT  Crane                        51.30 MT  Crane Load

DRAFT AT FP (M) =     1.091
DRAFT AT MS (M) =     1.189
DRAFT AT AP (M) =     1.288
DRAFT AT FWD MARK (M) =     1.107
DRAFT AT AFT MARK (M) =     1.272
                          HYDROSTATIC PROPERTIES
                 Trim: Aft 0.197/50.000, No Heel, VCG = 4.439

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.190       971.56  24.352f  0.599     8.68  24.825f   31.23  160.70  19.949
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.                              Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 325.5 m.-MT
```

Refer IRS Letter E-220766-279820 dated January 02, 2024

12/21/23 09:39:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.091 @ 50.00f, 1.189 @ 25.00f, 1.288 @ 0.00
Trim: Aft 0.197/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	90.3	0.927	1.511	0.05500	7.51
CRANE	57.0	4.264	4.848	0.05500	15.20
Total wind heeling moment to starboard------------->					22.71

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.368f TCG = 0.000 VCG = 4.439
Free Surface Adjustment: 0.335
Adjusted CG: LCG = 24.369f TCG = 0.000 VCG = 4.774

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.276	0.23a	0.00	971.60	0.000	-0.023	0.0000	5.749
1.275	0.23a	0.06s	971.60	0.000	0.000	-0.0000	5.687
1.260	0.22a	5.06s	971.60	0.000	1.749	0.0763	0.687
1.256	0.23a	5.75s	971.57	0.000	1.983	0.0987	0.000
1.199	0.26a	10.06s	971.57	0.000	3.208	0.2956	-4.313
1.022	0.34a	15.06s	971.60	0.000	3.741	0.6039	-9.313
0.962	0.38a	16.69s	971.60	0.000	3.761	0.7103	-10.938
0.836	0.46a	20.06s	971.60	0.000	3.683	0.9307	-14.313
0.645	0.58a	25.06s	971.60	0.000	3.402	1.2424	-19.313
0.452	0.70a	30.06s	971.60	0.000	3.016	1.5232	-24.313
0.258	0.82a	35.06s	971.60	0.000	2.571	1.7674	-29.313
0.064	0.94a	40.06s	971.60	0.000	2.085	1.9709	-34.313
-0.129	1.05a	45.06s	971.60	0.000	1.572	2.1307	-39.313
-0.320	1.16a	50.06s	971.60	0.000	1.040	2.2448	-44.313
-0.508	1.26a	55.06s	971.60	0.000	0.494	2.3118	-49.313
-0.673	1.34a	59.53s	971.60	0.000	0.000	2.3311	-53.784
-0.692	1.35a	60.06s	971.60	0.000	-0.059	2.3308	-54.313

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.71 (constant)

Refer IRS Letter E-210766-279820 dated, January 12, 2024

12/21/23 09:39:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66
PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.7103 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg    59.47 P
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     5.69 P
-----------------------------------------------------------------------------
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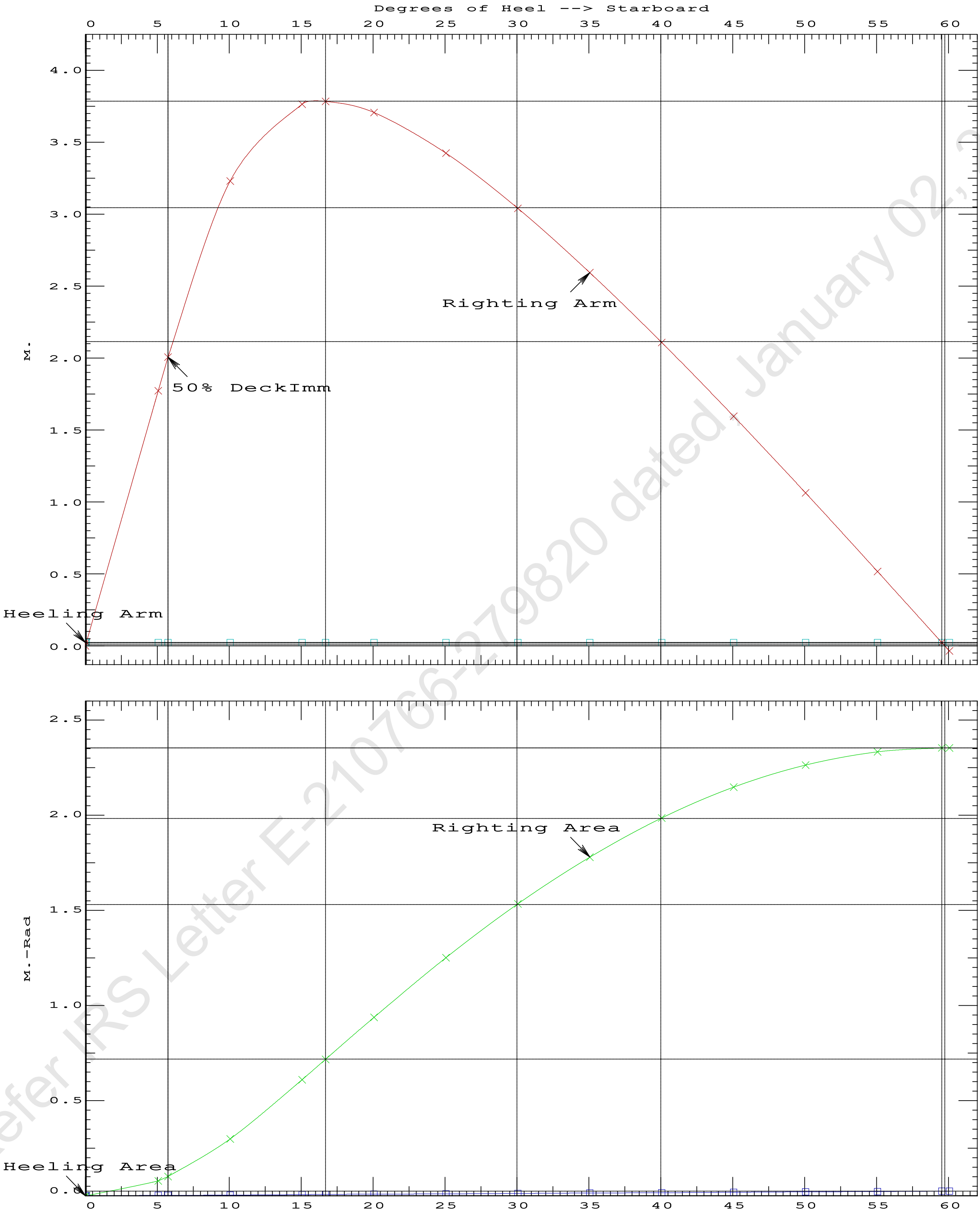
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd



Refer IRS Letter E-210766-279820 dated: January 02, 2024

12/21/23 09:39:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.368f TCG = 0.000 VCG = 4.439
Free Surface Adjustment: 0.335
Adjusted CG: LCG = 24.369f TCG = 0.010p VCG = 4.774

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.272	0.22a	1.77s	971.60	0.000	0.000	0.0000
1.249	0.23a	6.77s	971.60	0.000	1.719	0.0750
1.144	0.27a	11.77s	971.54	0.000	2.882	0.2798
0.967	0.38a	16.55s	971.60	0.000	3.161	0.5378
0.959	0.38a	16.77s	971.60	0.000	3.160	0.5496
0.771	0.50a	21.77s	971.60	0.000	3.002	0.8217
0.580	0.62a	26.77s	971.60	0.000	2.678	1.0708
0.386	0.74a	31.77s	971.60	0.000	2.268	1.2872
0.192	0.86a	36.77s	971.60	0.000	1.807	1.4654
-0.002	0.98a	41.77s	971.60	0.000	1.310	1.6017
-0.194	1.09a	46.77s	971.60	0.000	0.789	1.6935
-0.384	1.20a	51.77s	971.60	0.000	0.251	1.7390
-0.470	1.24a	54.06s	971.60	0.000	0.000	1.7440
-0.571	1.30a	56.77s	971.60	0.000	-0.299	1.7369
-0.754	1.38a	61.77s	971.60	0.000	-0.853	1.6867

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	1.77 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	498.8 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		4.016 P

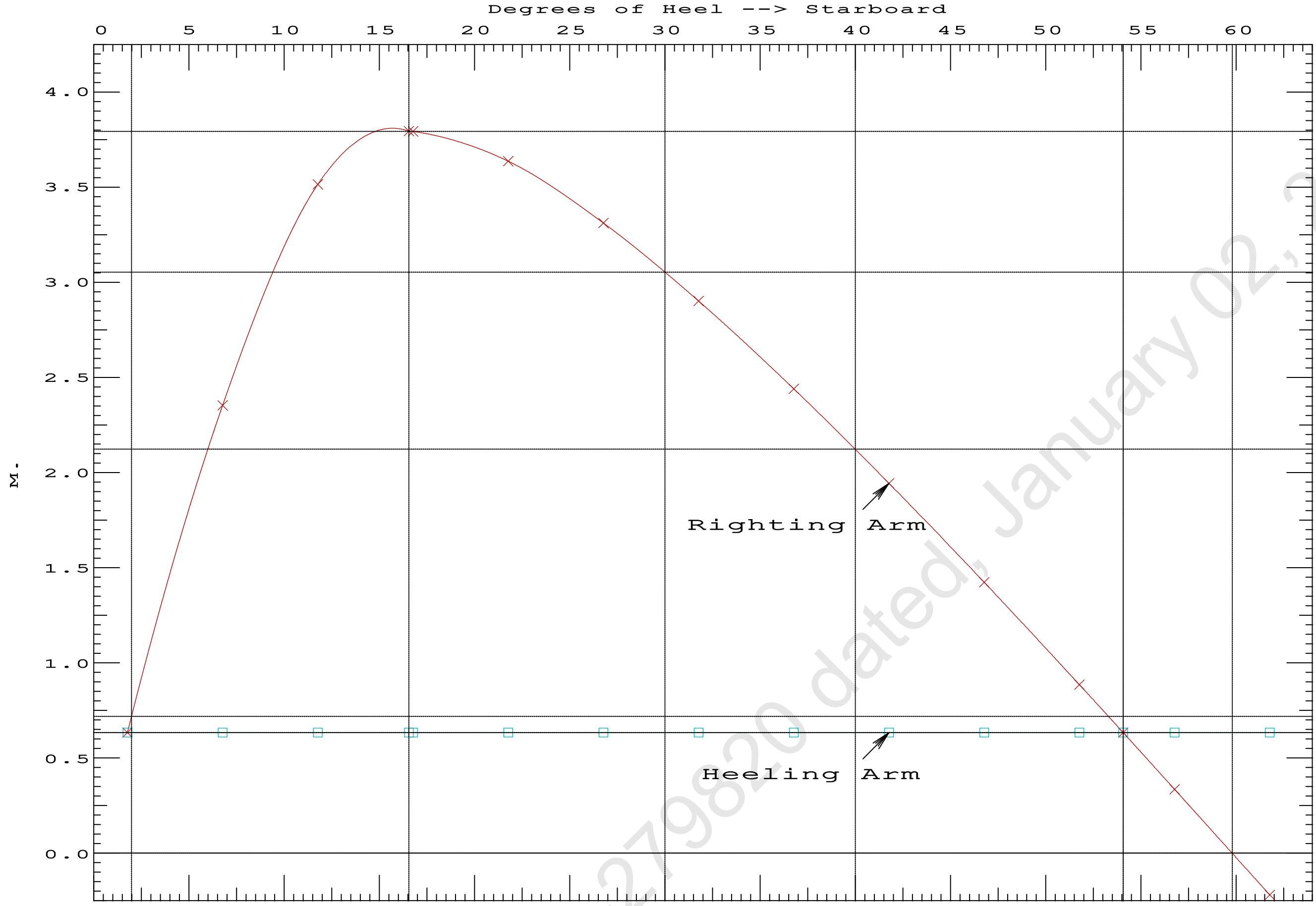
Refer IRS Letter E-210766-230620 dated January 02, 2024

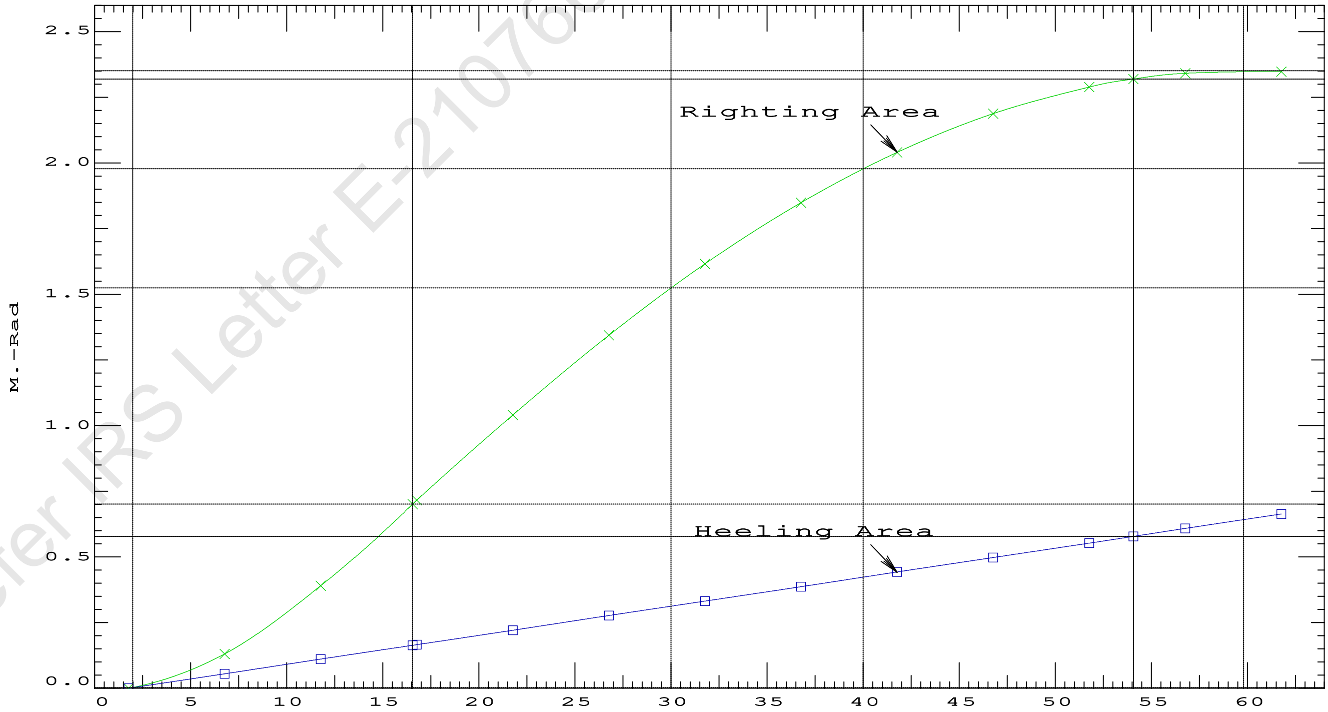
12/21/23 09:39:48 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 14 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Stbd





Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:51 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.090 @ 50.00f, 1.187 @ 25.00f, 1.284 @ 0.00
Trim: Aft 0.195/50.000, Heel: Stbd 1.77 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Total Fixed--------->			648.40	19.567f	0.000	5.900	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks--------->			323.20	34.000f	0.000	1.509	325.46
Total Weight-------->			971.60	24.368f	0.000	4.439	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	971.60	24.353f	0.752s	0.611

Righting Arms:	0.000	0.634s
External Arms:	0.000	0.634s
Residual Righting Arms:	0.000	0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	846.3	24.847f	0.102s	164.12	23.774*
		MT/cm				
		8.67				

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.77

Refer IRS Letter E-210766-270820 dated January 02, 2024

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12/21/23 09:39:51    Cybermarine Technologies PTE. Ltd.
GHS 18.66                           PONTOON

                              After hook load drop
                   WEIGHT and DISPLACEMENT and WATERPLANE STATUS
              USK draft: 1.176 @ 50.00f, 1.130 @ 25.00f, 1.085 @ 0.00
                     Trim: Fwd 0.091/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          447.10  25.000f  0.000   3.000
Crane                               150.00   7.500f  0.000   5.200
  Total Fixed--------->             597.10  20.604f  0.000   3.553
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025       161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025       161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->             323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->             920.30  25.308f  0.000   2.835
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       920.30  25.312f  0.001s  0.568
------------------------------------------------------------------------------
               Righting Arms:                0.000   0.001s
               External Arms:                0.000   0.001s
      Residual Righting Arms:                0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025        841.0  25.054f  0.016s   170.04  24.886*
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      8.62           30.88    167.77  22.618
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 16616.71; t3: 30.00; t: 15.00
limmarg: 6193.49; t3: 15.00; t: 7.50
limmarg: 1679.74; t3: 7.50; t: 3.75
limmarg: 349.34; t3: 3.75; t: 1.88
limmarg: 12.47; t3: 1.87; t: 0.94
limmarg: -72.08; t3: 0.93; t: 0.47
limmarg: -36.85; t3: 1.40; t: 0.23
limmarg: -14.46; t3: 1.63; t: 0.12
limmarg: -1.46; t3: 1.75; t: 0.06
limmarg: 5.39; t3: 1.81; t: 0.03
limmarg: 1.94; t3: 1.78; t: 0.01

               Theta3 is 1.78 degrees, as demonstrated below.

                     RESIDUAL RIGHTING ARMS vs HEEL ANGLE
        Fixed CG:  LCG = 20.604f  TCG = 0.000   VCG =  3.553

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.071   0.10f   1.77s      920.28     0.000   -0.721    0.0000
       1.073   0.10f   0.00       920.29     0.000    0.000   -0.0111
       1.071   0.10f   1.78p      920.30     0.000    0.728    0.0002
       1.058   0.10f   5.00p      920.29     0.000    2.022    0.0776
       0.969   0.12f  10.00p      920.29     0.000    3.681    0.3286
       0.719   0.15f  15.00p      920.30     0.000    4.390    0.6877
       0.468   0.20f  19.34p      920.30     0.000    4.516    1.0283
```

Refer IRS Letter E2107/66-279820 dated January 02, 2024

12/21/23 09:39:51 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

0.429	0.21f	20.00p	920.30	0.000	4.514	1.0799
0.136	0.26f	25.00p	920.30	0.000	4.395	1.4712
-0.159	0.31f	30.00p	920.30	0.000	4.159	1.8453

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 597.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -1.8 deg to abs 1.8	>	1.000	1.019 P

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:51 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.604f TCG = 0.000 VCG = 3.553

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.071	0.10f	1.77s	920.28	0.000	-0.721	0.0000
1.073	0.10f	0.00	920.29	0.000	0.000	-0.0111
1.071	0.10f	1.78p	920.30	0.000	0.728	0.0002
1.058	0.10f	5.00p	920.29	0.000	2.022	0.0776
0.969	0.12f	10.00p	920.29	0.000	3.681	0.3286
0.719	0.15f	15.00p	920.30	0.000	4.390	0.6877
0.468	0.20f	19.34p	920.30	0.000	4.516	1.0283
0.429	0.21f	20.00p	920.30	0.000	4.514	1.0799
0.136	0.26f	25.00p	920.30	0.000	4.395	1.4712
-0.159	0.31f	30.00p	920.30	0.000	4.159	1.8453
-0.454	0.36f	35.00p	920.30	0.000	3.852	2.1953
-0.747	0.42f	40.00p	920.30	0.000	3.494	2.5162
-1.034	0.47f	45.00p	920.30	0.000	3.097	2.8040
-1.315	0.51f	50.00p	920.30	0.000	2.668	3.0558
-1.586	0.56f	55.00p	920.30	0.000	2.214	3.2690
-1.845	0.60f	60.00p	920.30	0.000	1.740	3.4417
-2.089	0.64f	65.00p	920.30	0.000	1.250	3.5723
-2.318	0.67f	70.00p	920.30	0.000	0.749	3.6596
-2.528	0.69f	75.00p	920.30	0.000	0.241	3.7028
-2.620	0.70f	77.36p	920.30	0.000	0.000	3.7078
-2.718	0.71f	80.00p	920.30	0.000	-0.270	3.7016

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

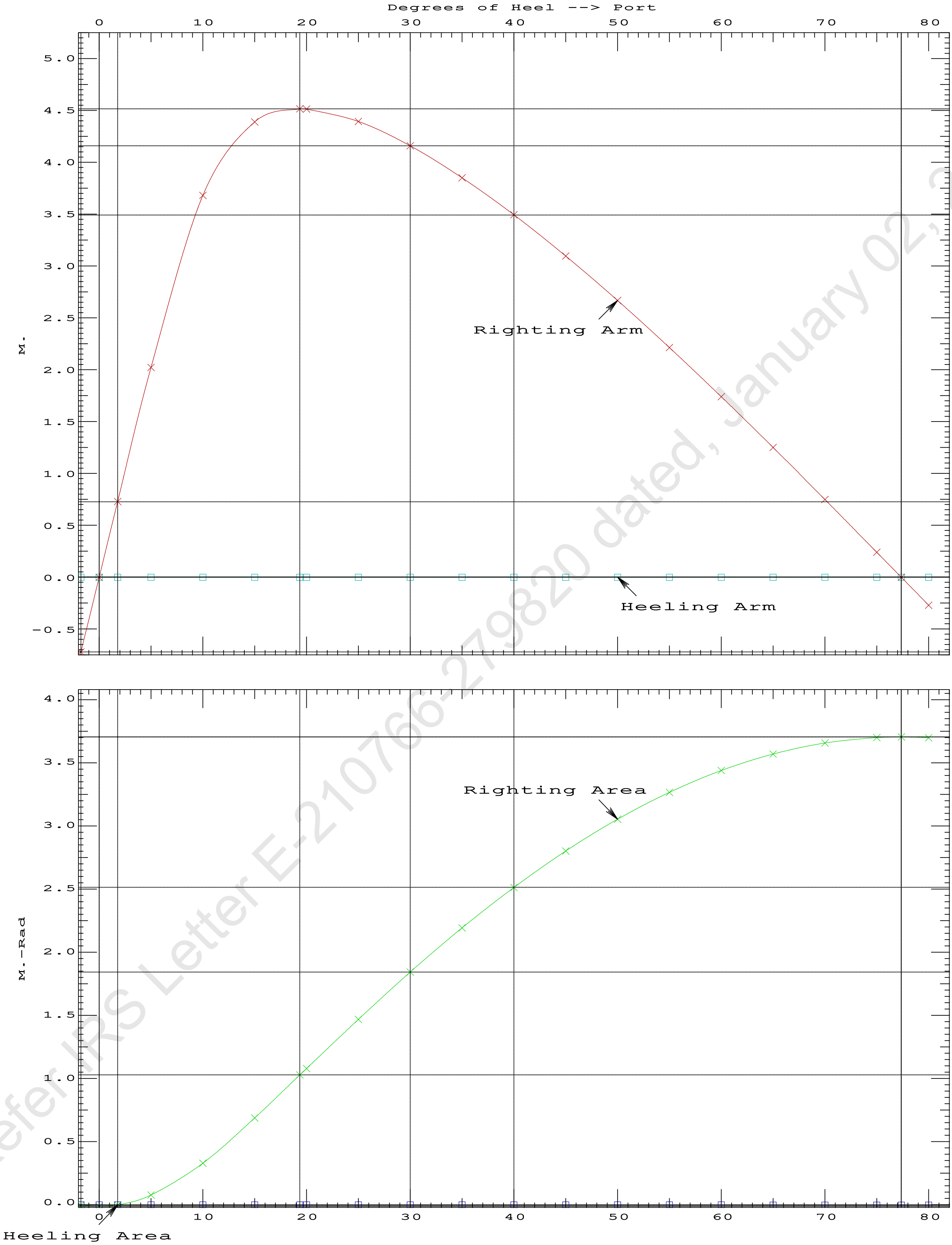
Note: The Center of Gravity shown above is for the Fixed Weight of 597.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -1.8 deg to RAzero	>	1.000		334.863 P
(2) Angle from abs 1.8 deg to RAzero	>	20.00	deg	75.58 P

Refer IRS Letter E-210766-270820 dated January 02, 2024

12/21/23 09:39:51 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**



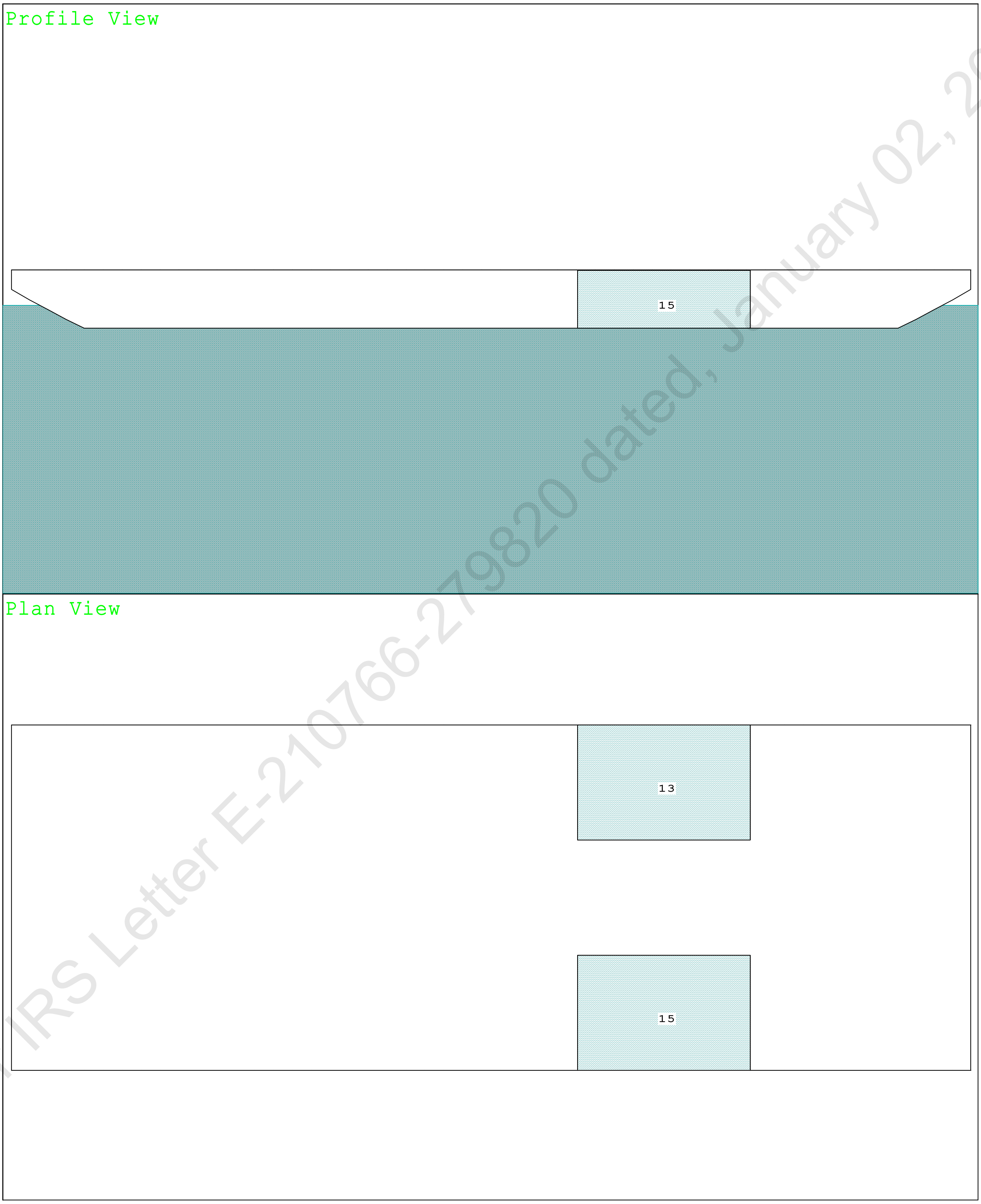
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

Condition Graphic - Draft: 1.190 @ 50.000f, 1.190 @ 0.000 Heel: zero



Tanks
13 05_WBTK.P.100% SALT WATER Intact 15 05_WBTK.S.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.190 @ 50.00f, 1.190 @ 25.00f, 1.190 @ 0.00
Trim: 0.000/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Total Fixed--------->			648.40	20.516f	0.000	5.900	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks--------->			323.20	34.000f	0.000	1.509	325.46
Total Weight-------->			971.60	25.002f	0.000	4.439	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	971.59	25.001f	0.000	0.598

Righting Arms: 0.001a 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	846.9	25.000f	0.000	164.48	23.782*

MT/cm	m.-MT/cm	GML	GMT
8.68	31.22	160.64	19.941

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane 51.30 MT Crane Load

DRAFT AT FP (M) = 1.190
DRAFT AT MS (M) = 1.190
DRAFT AT AP (M) = 1.190
DRAFT AT FWD MARK (M) = 1.190
DRAFT AT AFT MARK (M) = 1.190

HYDROSTATIC PROPERTIES
Trim: 0.000/50.000, No Heel, VCG = 4.439

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.190	971.59	25.001f	0.598	8.68	25.000f	31.22	160.64	19.941

Distances in METERS.------Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

Refer IRS Letter E-230766-279820 dated January 02, 2024

12/21/23 09:39:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 1.190 @ 50.00f, 1.190 @ 25.00f, 1.190 @ 0.00
Trim: 0.000/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	90.3	0.926	1.509	0.05500	7.50
CRANE	57.0	4.322	4.905	0.05500	15.38
Total wind heeling moment to starboard------------->					22.87

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 25.002f TCG = 0.000 VCG = 4.439
Free Surface Adjustment: 0.335
Adjusted CG: LCG = 25.002f TCG = 0.000 VCG = 4.774

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.177	0.00	0.00	971.52	0.000	-0.023	0.0000	6.233
1.177	0.00	0.06s	971.59	0.000	0.000	-0.0000	6.169
1.162	0.00	5.06s	971.60	0.000	1.751	0.0764	1.169
1.154	0.00	6.23s	971.56	0.000	2.144	0.1161	0.000
1.088	0.00	10.06s	971.67	0.000	3.211	0.2963	-3.831
0.873	0.00	15.06s	971.76	0.000	3.747	0.6051	-8.831
0.793	0.00	16.76s	971.60	0.000	3.768	0.7166	-10.528
0.635	0.00	20.06s	971.60	0.000	3.690	0.9327	-13.831
0.392	0.00	25.06s	971.60	0.000	3.409	1.2450	-18.831
0.147	0.00	30.06s	971.60	0.000	3.024	1.5264	-23.831
-0.100	0.00	35.06s	971.60	0.000	2.578	1.7713	-28.831
-0.346	0.00	40.06s	971.60	0.000	2.092	1.9753	-33.831
-0.589	0.00	45.06s	971.60	0.000	1.579	2.1357	-38.831
-0.828	0.00f	50.06s	971.59	0.000	1.046	2.2504	-43.831
-1.061	0.00f	55.06s	971.60	0.000	0.500	2.3180	-48.831
-1.264	0.00f	59.58s	971.60	0.000	0.000	2.3377	-53.348
-1.285	0.00f	60.06s	971.60	0.000	-0.054	2.3375	-53.831

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.87 (constant)

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12/21/23 09:39:53    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON
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Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

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LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.7166 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg    59.52 P
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     6.17 P
-----------------------------------------------------------------------------
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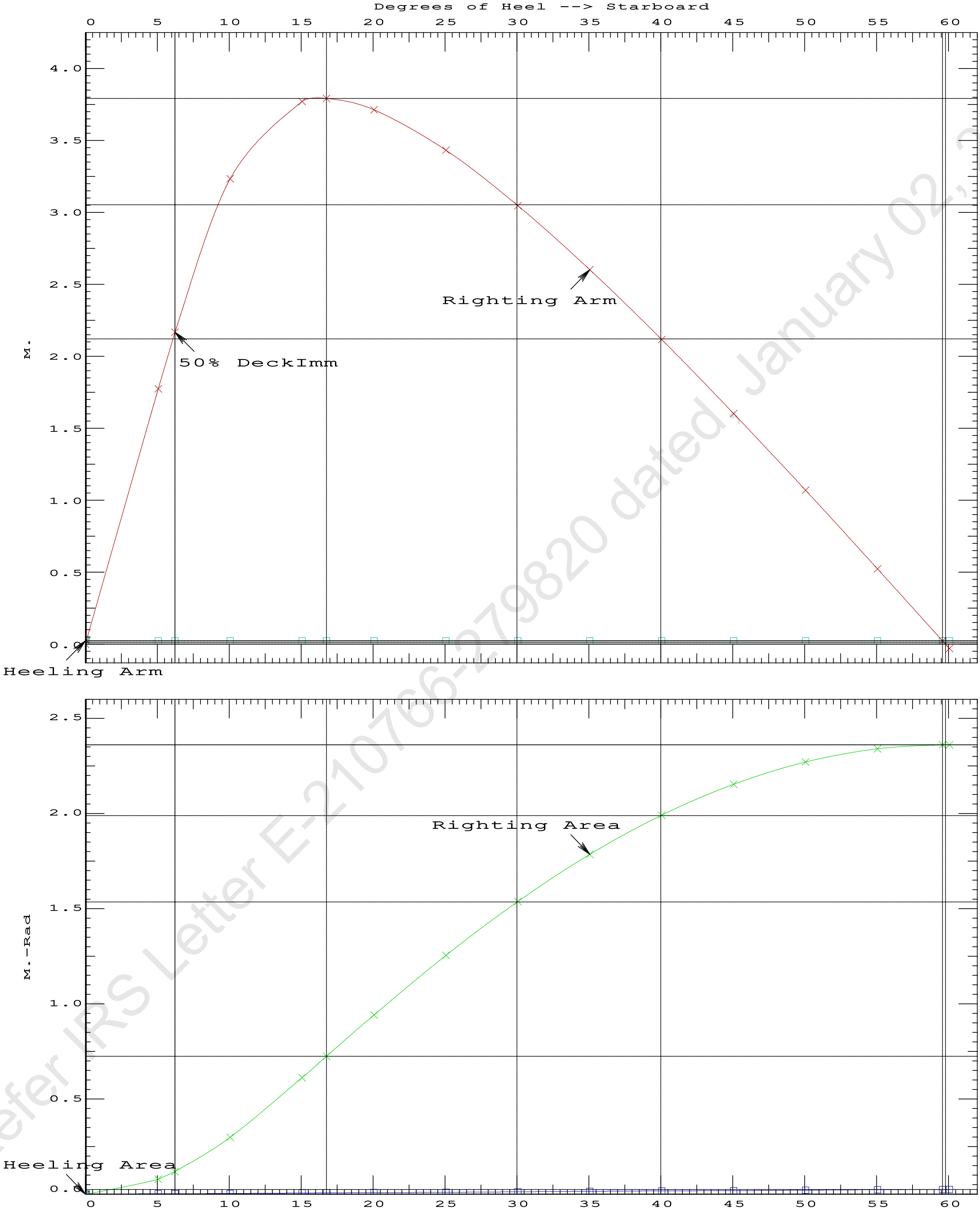
Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 15 : Operating Condition of 150T Crane

Crane Lifting 51.3T @ 12m Radius Fwd



Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.002f TCG = 0.000 VCG = 4.439
Free Surface Adjustment: 0.335
Adjusted CG: LCG = 25.002f TCG = 0.000 VCG = 4.774

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.177	0.00	0.00	971.60	0.000	0.000	0.0000
1.163	0.00	5.00s	971.54	0.000	1.751	0.0764
1.090	0.00	10.00s	971.60	0.000	3.221	0.2954
0.876	0.00	15.00s	971.76	0.000	3.768	0.6071
0.792	0.00	16.77s	971.60	0.000	3.790	0.7238
0.638	0.00	20.00s	971.60	0.000	3.715	0.9367
0.395	0.00	25.00s	971.60	0.000	3.435	1.2512
0.150	0.00	30.00s	971.60	0.000	3.051	1.5350
-0.097	0.00	35.00s	971.60	0.000	2.606	1.7823
-0.343	0.00	40.00s	971.60	0.000	2.121	1.9888
-0.586	0.00	45.00s	971.59	0.000	1.608	2.1517
-0.825	0.00f	50.00s	971.60	0.000	1.075	2.2689
-1.058	0.00f	55.00s	971.60	0.000	0.529	2.3390
-1.273	0.00f	59.78s	971.60	0.000	0.000	2.3611
-1.282	0.00f	60.00s	971.60	0.000	-0.024	2.3611

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

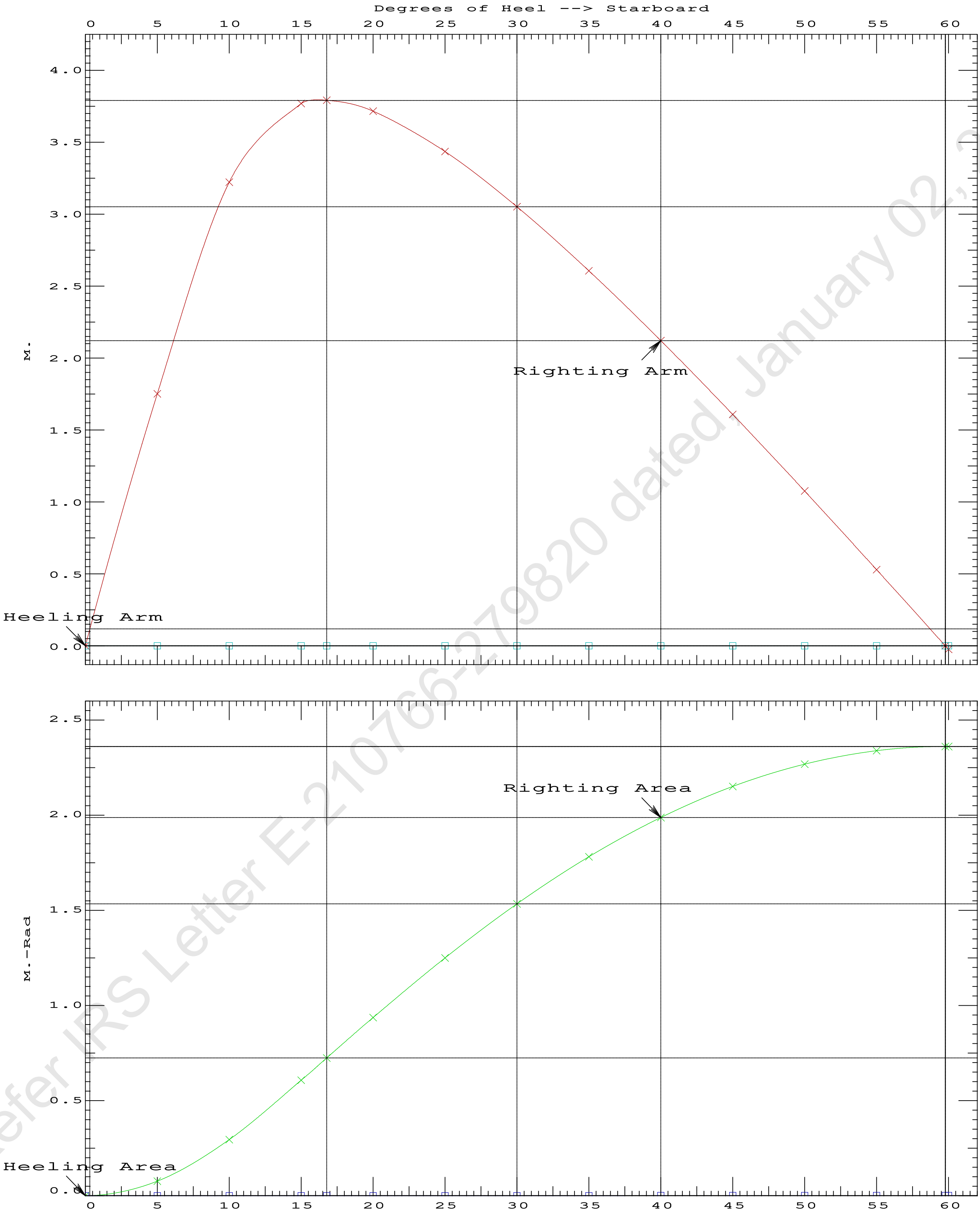
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained	
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00	P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	368273	P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		2199.71	P

Refer IRS Letter E-210766-23020 dated January 02, 2024

12/21/23 09:39:53 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd



Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:55 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.190 @ 50.00f, 1.190 @ 25.00f, 1.189 @ 0.00
Trim: Fwd 0.001/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			447.10	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Total Fixed--------->			648.40	20.516f	0.000	5.900	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks--------->			323.20	34.000f	0.000	1.509	325.46
Total Weight-------->			971.60	25.002f	0.000	4.439	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	971.60	25.002f	0.001s	0.598	

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.6	25.000f	0.026s	162.58	23.635*
		MT/cm		**m.-MT/cm**	**GML**	**GMT**
		8.65		30.85	158.74	19.794

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

Refer IRS Letter E-210766-270820 dated January 02, 2024

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12/21/23 09:39:55    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

                        After hook load drop
              WEIGHT and DISPLACEMENT and WATERPLANE STATUS
         USK draft: 1.176 @ 50.00f, 1.130 @ 25.00f, 1.085 @ 0.00
                Trim: Fwd 0.091/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          447.10  25.000f  0.000   3.000
Crane                               150.00   7.500f  0.000   5.200
  Total Fixed--------->             597.10  20.604f  0.000   3.553
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
05_WBTK.P      1.000    1.025       161.60  34.000f  5.980p  1.509    162.73*
05_WBTK.S      1.000    1.025       161.60  34.000f  5.980s  1.509    162.73*
  Total Tanks--------->             323.20  34.000f  0.000   1.509    325.46
  Total Weight-------->             920.30  25.308f  0.000   2.835
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       920.30  25.312f  0.001s  0.568
------------------------------------------------------------------------------
               Righting Arms:                0.000   0.001s
               External Arms:                0.000   0.001s
      Residual Righting Arms:                0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         841.0  25.054f  0.016s   170.04  24.886*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       8.62           30.88    167.77  22.618
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01

              Theta3 is 0.97 degrees, as demonstrated below.

                     RESIDUAL RIGHTING ARMS vs HEEL ANGLE
         Fixed CG:  LCG = 20.604f  TCG = 0.000   VCG =  3.553

      Origin    Degrees of   Displacement    Residual Arms       Res.
       Depth---Trim----Heel----Weight(MT)---in Trim--in Heel---> Area
       1.072   0.10f   0.00       920.30     0.000    0.000    0.0000
       1.072   0.10f   0.97s      920.30     0.000    0.396    0.0033
       1.058   0.10f   5.00s      920.31     0.000    2.021    0.0884
       0.968   0.12f  10.00s      920.29     0.000    3.679    0.3397
       0.719   0.15f  15.00s      920.30     0.000    4.388    0.6986
       0.468   0.20f  19.34s      920.30     0.000    4.514    1.0389
       0.429   0.21f  20.00s      920.30     0.000    4.512    1.0906
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Refer IRS Letter E-2/0766-279820 dated January 02, 2024

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12/21/23 09:39:55    Cybermarine Technologies PTE. Ltd.
GHS 18.66                          PONTOON

       0.136   0.26f  25.00s      920.30     0.000    4.392    1.4818
      -0.160   0.31f  30.00s      920.30     0.000    4.156    1.8556
      Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

  Note:  The Center of Gravity shown above is for the Fixed Weight of
         597.10 MT.  As the tank load centers shift with heel and
         trim, the total Center of Gravity varies.  The righting arms
         shown above include the effect of the C.G. variation.

  Note:  The Residual Righting Arms shown above are in excess of the
         overturning arms derived from these moments (in m.-MT):
                         Stbd heeling moment = 1.00

LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Residual Area Ratio from abs 0 deg to abs 1    >    1.000           LARGE
-----------------------------------------------------------------------------
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Refer IRS Letter E-210766-279820 dated, January 02, 2024

12/21/23 09:39:55 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.604f TCG = 0.000 VCG = 3.553

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.072	0.10f	0.00	920.30	0.000	0.000	0.0000
1.072	0.10f	0.97s	920.30	0.000	0.396	0.0033
1.058	0.10f	5.00s	920.31	0.000	2.021	0.0884
0.968	0.12f	10.00s	920.29	0.000	3.679	0.3397
0.719	0.15f	15.00s	920.30	0.000	4.388	0.6986
0.468	0.20f	19.34s	920.30	0.000	4.514	1.0389
0.429	0.21f	20.00s	920.30	0.000	4.512	1.0906
0.136	0.26f	25.00s	920.30	0.000	4.392	1.4818
-0.160	0.31f	30.00s	920.30	0.000	4.156	1.8556
-0.455	0.36f	35.00s	920.30	0.000	3.849	2.2054
-0.747	0.42f	40.00s	920.30	0.000	3.491	2.5261
-1.035	0.47f	45.00s	920.30	0.000	3.094	2.8137
-1.315	0.51f	50.00s	920.30	0.000	2.666	3.0653
-1.586	0.56f	55.00s	920.30	0.000	2.212	3.2783
-1.845	0.60f	60.00s	920.30	0.000	1.738	3.4508
-2.089	0.64f	65.00s	920.30	0.000	1.248	3.5811
-2.318	0.67f	70.00s	920.30	0.000	0.747	3.6682
-2.528	0.69f	75.00s	920.30	0.000	0.239	3.7113
-2.619	0.70f	77.34s	920.30	0.000	0.000	3.7161
-2.718	0.71f	80.00s	920.30	0.000	-0.272	3.7098

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

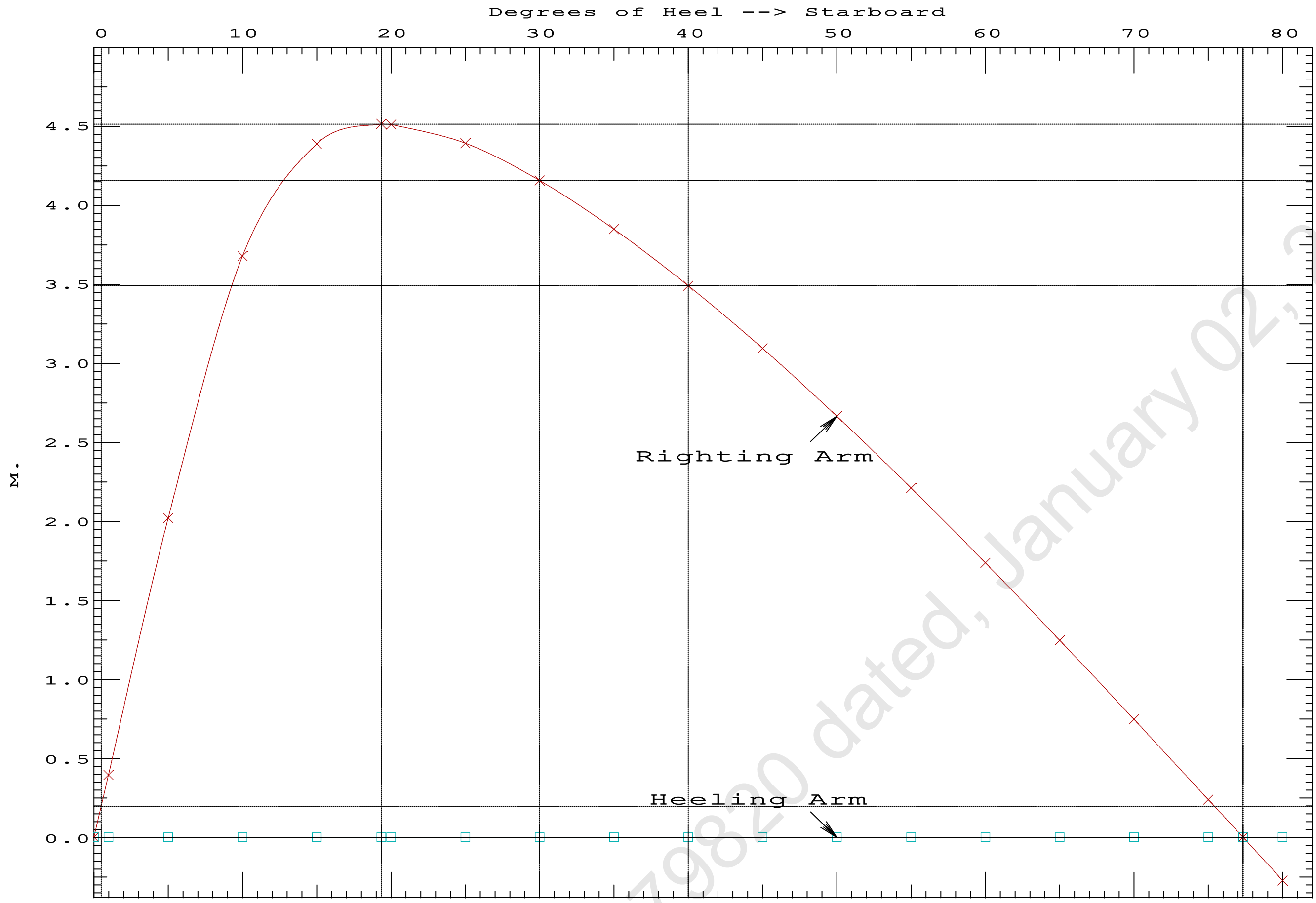
Note: The Center of Gravity shown above is for the Fixed Weight of 597.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

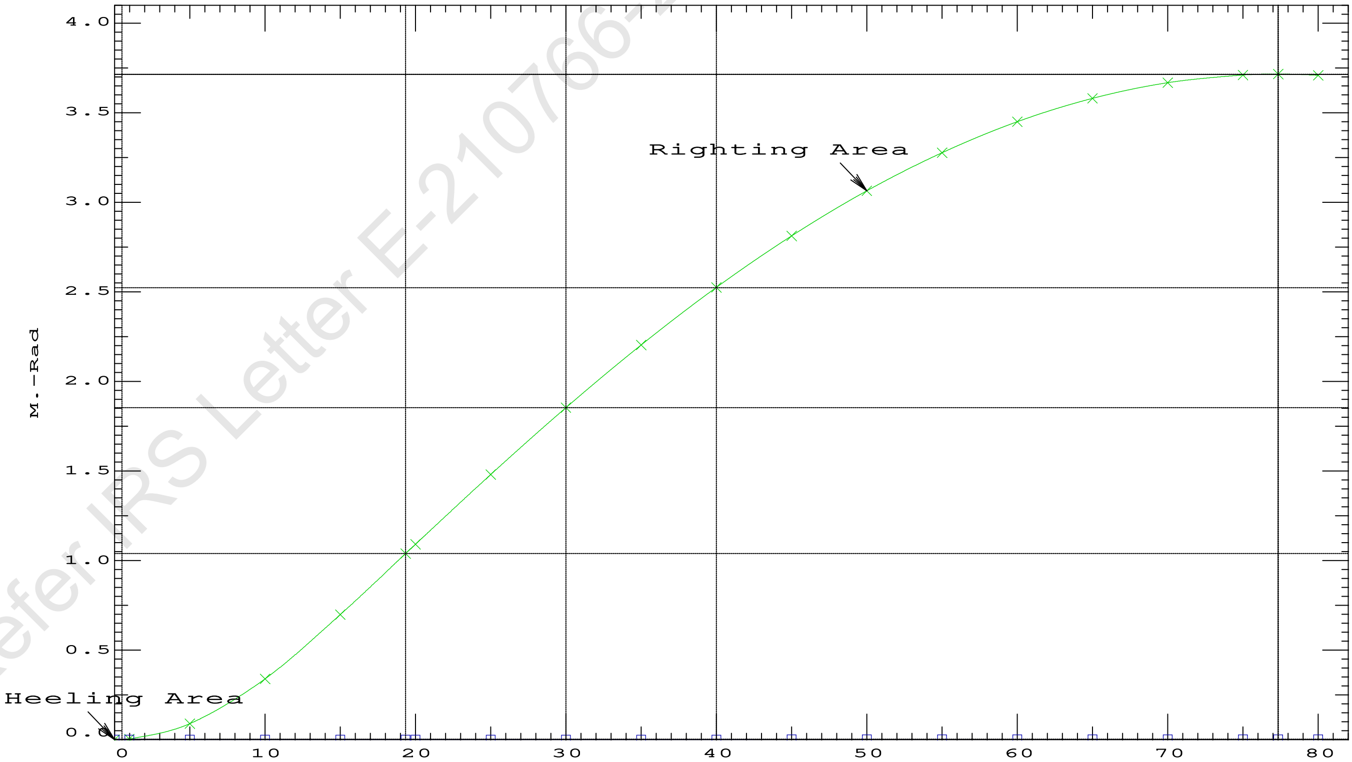
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 1 deg to RAzero	>	20.00	deg	76.37 P

Refer IRS Letter E210766-270820 dated January 02, 2024

12/21/23 09:39:55 Cybermarine Technologies PTE. Ltd.
GHS 18.66 **PONTOON**





Refer IRS Letter E-210766-279820 dated, January 02, 2024

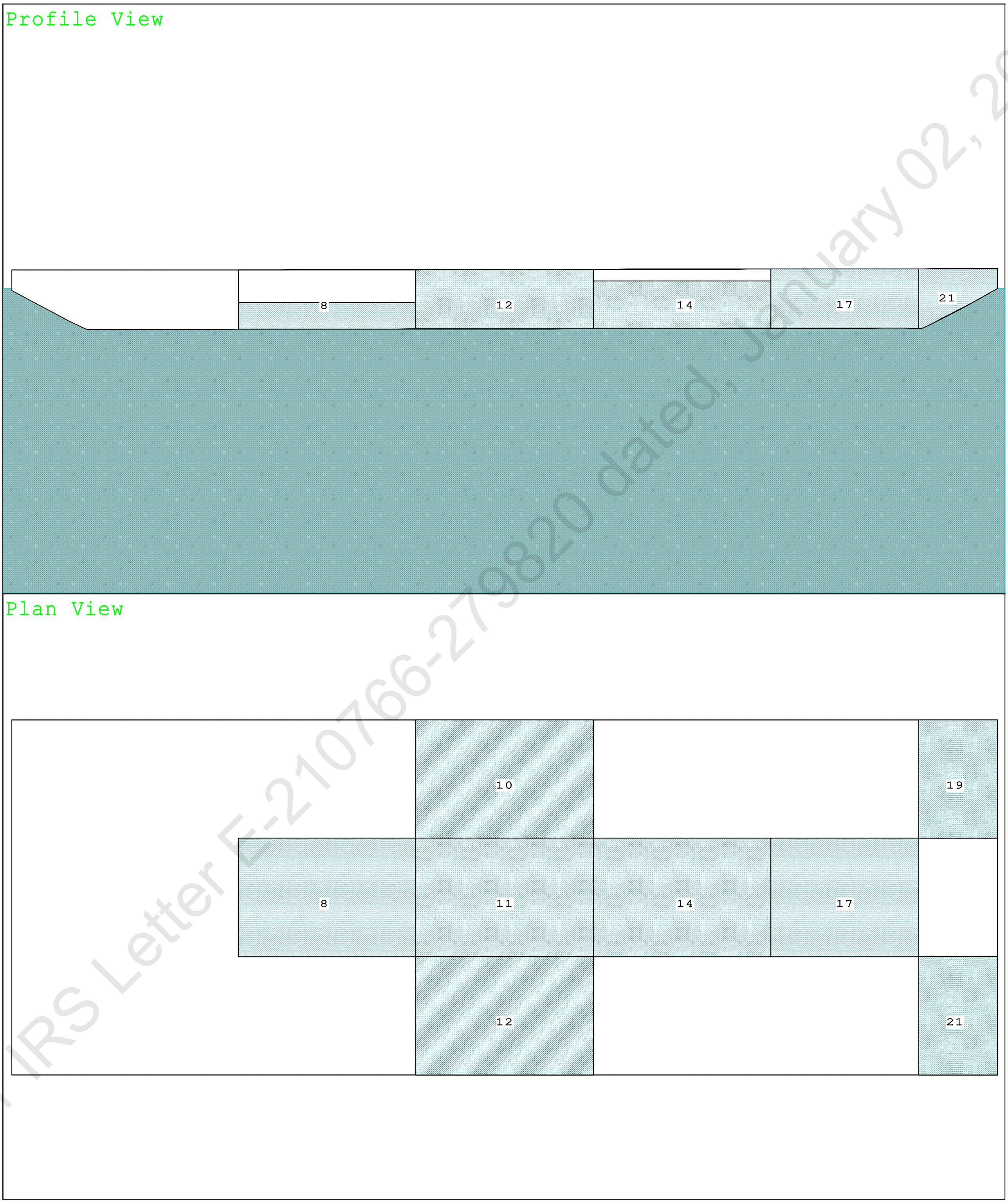
12/21/23 09:39:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

Condition Graphic - Draft: 2.040 @ 50.000f, 2.120 @ 0.000 Heel: zero



Tanks

	11 04_WBTK.C.100% SALT WATER Intact	17 06_WBTK.C.100% SALT WATER Intact
8 03_WBTK.C..45% SALT WATER Intact	12 04_WBTK.S.100% SALT WATER Intact	19 07_WBTK.P.100% SALT WATER Intact
10 04_WBTK.P.100% SALT WATER Intact	14 05_WBTK.C..80% SALT WATER Intact	21 07_WBTK.S.100% SALT WATER Intact

Refer IRS Letter E-210766-279820 dated, January 02, 2024

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12/21/23 09:39:57    Cybermarine Technologies PTE. Ltd.
GHS 18.66                        PONTOON
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Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

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                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 2.040 @ 50.00f, 2.080 @ 25.00f, 2.120 @ 0.00
                     Trim: Aft 0.080/50.000,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           447.10  25.000f  0.000   3.000
Crane                                400.00  11.500f  0.000   6.325
  Total Fixed--------->              847.10  18.625f  0.000   4.570
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_WBTK.C      0.450    1.025         73.23  15.992f  0.000   0.675    162.73
04_WBTK.P      1.000    1.025        161.60  25.000f  5.980p  1.509    162.73*
04_WBTK.C      1.000    1.025        162.73  25.000f  0.000   1.500    162.73*
04_WBTK.S      1.000    1.025        161.60  25.000f  5.980s  1.509    162.73*
05_WBTK.C      0.800    1.025        130.18  33.996f  0.000   1.200    162.73
06_WBTK.C      1.000    1.025        135.67  42.251f  0.000   1.499    135.61*
07_WBTK.P      1.000    1.025         49.60  47.660f  5.971p  1.880     72.32*
07_WBTK.S      1.000    1.025         49.60  47.660f  5.971s  1.880     72.32*
  Total Tanks--------->              924.21  30.518f  0.000   1.436   1093.89
  Total Weight-------->            1,771.31  24.831f  0.000   2.935
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,771.31  24.828f  0.000   1.063
-------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          900.2  24.998f  0.000    107.92  13.454*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        9.23             37.57    106.05  11.582
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           SUMMARY OF LOADING
       901.7 Cu.M.  (36%) SALT WATER

      400.00 MT  Crane

DRAFT AT FP (M) =     2.040
DRAFT AT MS (M) =     2.080
DRAFT AT AP (M) =     2.120
DRAFT AT FWD MARK (M) =    2.047
DRAFT AT AFT MARK (M) =    2.114
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Refer IRS Letter E-21076627980 dated January 02, 2024

12/21/23 09:39:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

HYDROSTATIC PROPERTIES
Trim: Aft 0.080/50.000, No Heel, VCG = 2.935

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.080	1,771.31	24.828f	1.063	9.23	24.998f	37.57	106.05	11.582

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 1093.9 m.-MT

HEELING MOMENT specification
Lateral Plane Method
Wind pressure toward starboard
USK draft: 2.040 @ 50.00f, 2.080 @ 25.00f, 2.120 @ 0.00
Trim: Aft 0.080/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	47.1	0.471	1.481	0.05500	3.84
CRANE	57.0	3.408	4.418	0.05500	13.85
Total wind heeling moment to starboard------------->					17.69

Distances in METERS.-----Pressure in MT/Sqm.------------Moment in m.-MT

Refer IRS Letter E-210766-279820 dated January 02, 2024

12/21/23 09:39:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.831f TCG = 0.000 VCG = 2.935
Free Surface Adjustment: 0.618
Adjusted CG: LCG = 24.832f TCG = 0.000 VCG = 3.553

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.108	0.09a	0.00	1,771.3	0.000	-0.010	0.0000	2.868
2.108	0.09a	0.05s	1,771.3	0.000	0.000	-0.0000	2.822
2.105	0.09a	2.87s	1,771.3	0.000	0.573	0.0141	-0.000
2.097	0.09a	5.05s	1,771.3	0.000	1.009	0.0442	-2.178
2.161	0.13a	10.05s	1,771.3	0.000	1.681	0.1640	-7.178
2.359	0.17a	15.05s	1,771.3	0.000	1.848	0.3217	-12.178
2.413	0.18a	16.19s	1,771.3	0.000	1.853	0.3587	-13.324
2.595	0.22a	20.05s	1,771.3	0.000	1.799	0.4823	-17.178
2.816	0.28a	25.05s	1,771.3	0.000	1.631	0.6331	-22.178
3.016	0.34a	30.05s	1,771.3	0.000	1.408	0.7661	-27.178
3.194	0.40a	35.05s	1,771.3	0.000	1.153	0.8780	-32.178
3.349	0.46a	40.05s	1,771.3	0.000	0.879	0.9668	-37.178
3.479	0.51a	45.05s	1,771.3	0.000	0.591	1.0311	-42.178
3.584	0.57a	50.05s	1,771.3	0.000	0.295	1.0698	-47.178
3.659	0.61a	54.94s	1,771.3	0.000	0.000	1.0824	-52.073
3.660	0.62a	55.05s	1,771.3	0.000	-0.006	1.0824	-52.178
3.709	0.66a	60.05s	1,771.3	0.000	-0.309	1.0687	-57.178

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 1093.9 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 17.69 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.3587	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	54.89	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.82	P

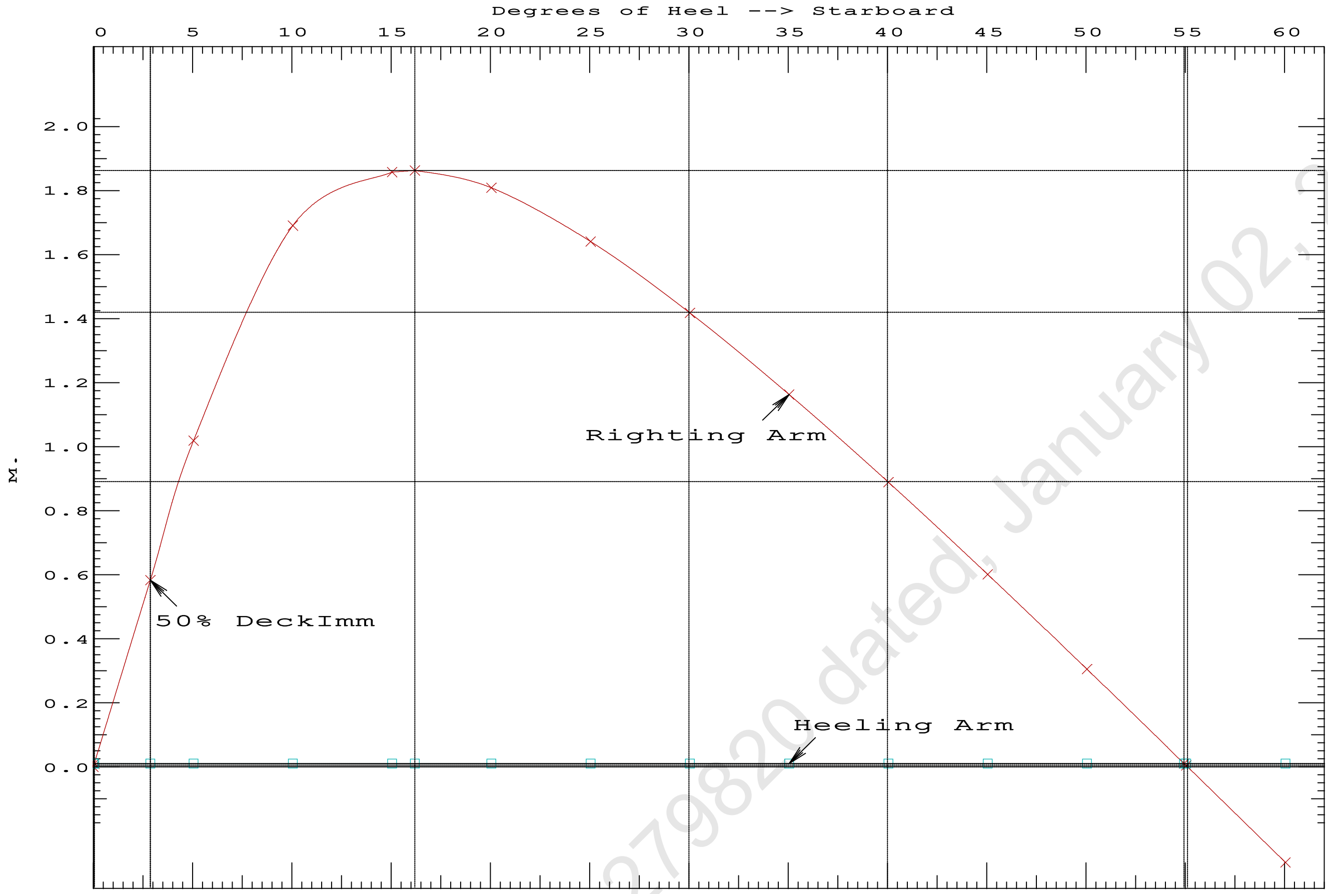
Refer IRS Letter E-2107 66 270820 dated, January 02, 2024

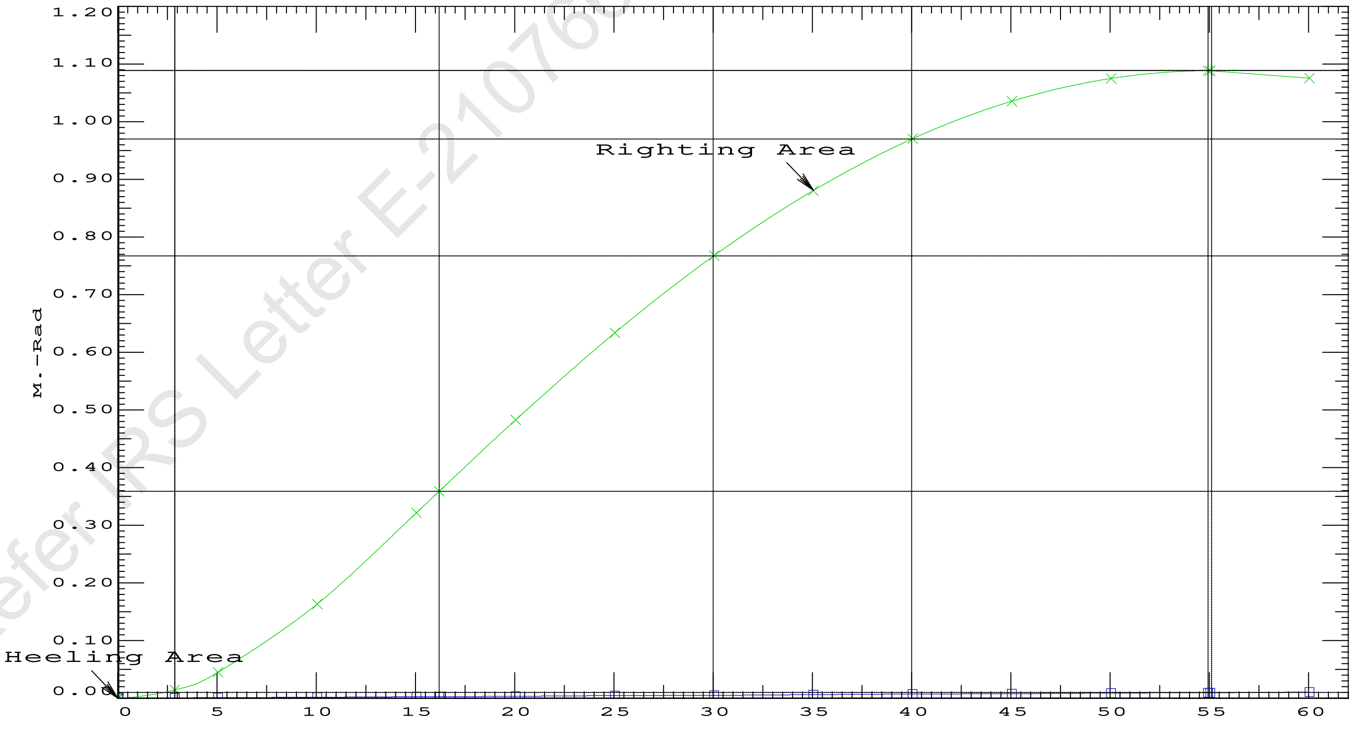
12/21/23 09:39:57 Cybermarine Technologies PTE. Ltd.
GHS 18.66

PONTOON

Condition 16 : Lightship + Crane + Ballast

TOWING CONDITION





Refer IRS Letter E-210766-279820 dated, January 02, 2024

LIGHTSHIP PARTICULARS

Lightship Weight	=	447.10	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)

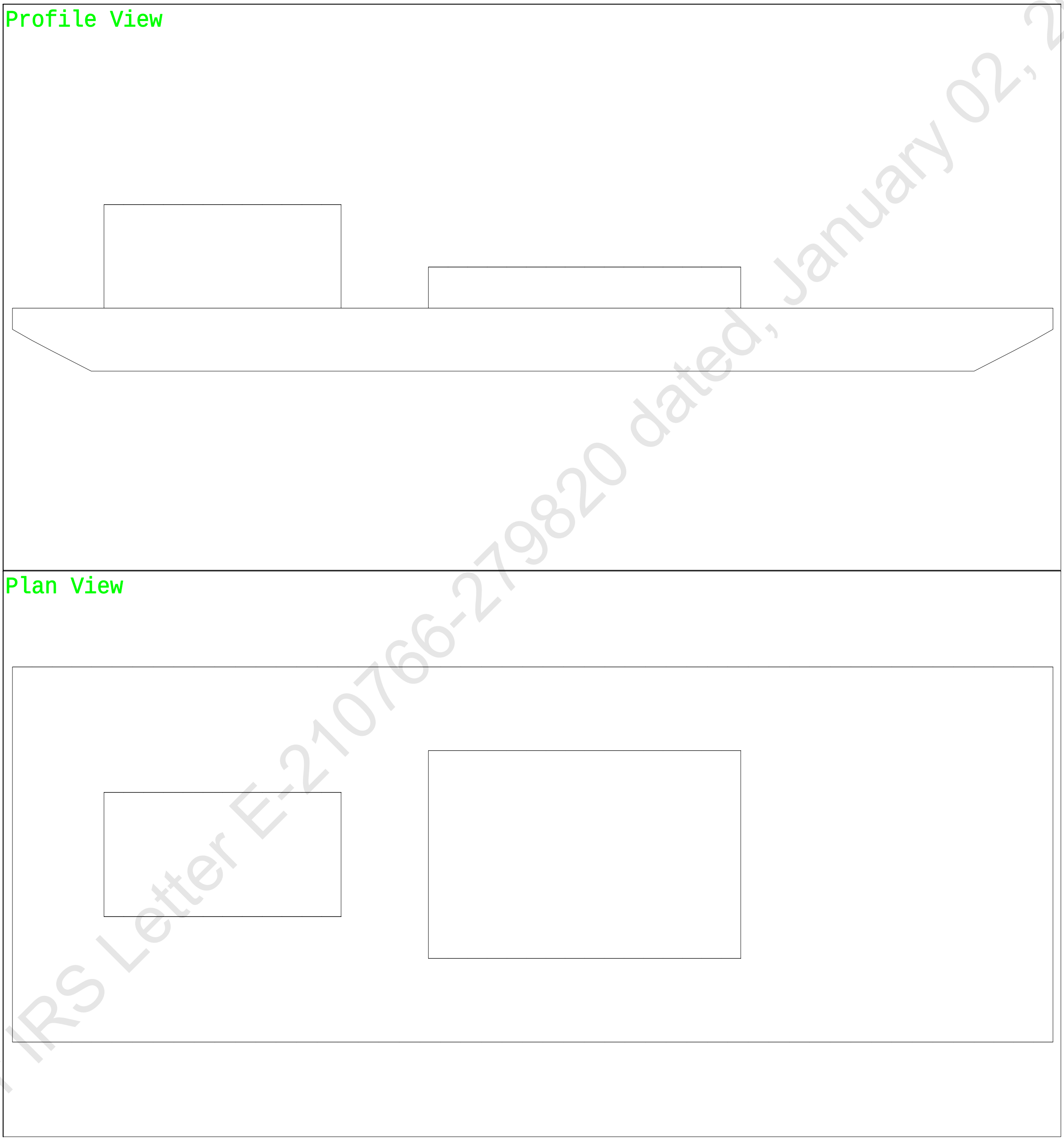
Refer IRS Letter E-210766-279820 dated, January 02, 2024

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic



Refer IRS Letter E-210766-279820 dated, January 02, 2024

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

LATERAL PLANE STATUS
USK draft: 0.512 @ 50.00f, 0.512 @ 25.00f, 0.512 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	22.3	25.000f	-0.253	121.4	25.000f	1.281
CRANE				57.0	10.100f	5.000
DECKCARGO.C				29.4	27.500f	3.500
Total Lateral Plane->	22.3	25.000f	-0.253	207.8	21.267f	2.615

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.612 @ 50.00f, 0.612 @ 25.00f, 0.612 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.8	25.000f	-0.302	116.9	25.000f	1.229
CRANE				57.0	10.100f	4.900
DECKCARGO.C				29.4	27.500f	3.400
Total Lateral Plane->	26.8	25.000f	-0.302	203.3	21.184f	2.572

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.712 @ 50.00f, 0.712 @ 25.00f, 0.712 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	31.3	25.000f	-0.351	112.4	25.000f	1.176
CRANE				57.0	10.100f	4.800
DECKCARGO.C				29.4	27.500f	3.300
Total Lateral Plane->	31.3	25.000f	-0.351	198.8	21.097f	2.529

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.812 @ 50.00f, 0.812 @ 25.00f, 0.812 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.9	25.000f	-0.400	107.8	25.000f	1.123
CRANE				57.0	10.100f	4.700
DECKCARGO.C				29.4	27.500f	3.200
Total Lateral Plane->	35.9	25.000f	-0.400	194.2	21.006f	2.487

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.912 @ 50.00f, 0.912 @ 25.00f, 0.912 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	40.4	25.000f	-0.450	103.3	25.000f	1.070
CRANE				57.0	10.100f	4.600
DECKCARGO.C				29.4	27.500f	3.100
Total Lateral Plane->	40.4	25.000f	-0.450	189.7	20.911f	2.445

Distances in METERS.

Refer IRS Letter E-21076 279820 dated, January 02, 2024

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

LATERAL PLANE STATUS
USK draft: 1.012 @ 50.00f, 1.012 @ 25.00f, 1.012 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	45.0	25.000f	-0.497	98.7	25.000f	1.018
CRANE				57.0	10.100f	4.500
DECKCARGO.C				29.4	27.500f	3.000
Total Lateral Plane->	45.0	25.000f	-0.497	185.1	20.809f	2.405

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.112 @ 50.00f, 1.112 @ 25.00f, 1.112 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	49.7	25.000f	-0.545	94.0	25.000f	0.966
CRANE				57.0	10.100f	4.400
DECKCARGO.C				29.4	27.500f	2.900
Total Lateral Plane->	49.7	25.000f	-0.545	180.4	20.699f	2.367

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.212 @ 50.00f, 1.212 @ 25.00f, 1.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	54.4	25.000f	-0.594	89.3	25.000f	0.914
CRANE				57.0	10.100f	4.300
DECKCARGO.C				29.4	27.500f	2.800
Total Lateral Plane->	54.4	25.000f	-0.594	175.7	20.584f	2.329

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.312 @ 50.00f, 1.312 @ 25.00f, 1.312 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	59.1	25.000f	-0.642	84.5	25.000f	0.863
CRANE				57.0	10.100f	4.200
DECKCARGO.C				29.4	27.500f	2.700
Total Lateral Plane->	59.1	25.000f	-0.642	170.9	20.462f	2.291

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.412 @ 50.00f, 1.412 @ 25.00f, 1.412 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	63.9	25.000f	-0.691	79.8	25.000f	0.810
CRANE				57.0	10.100f	4.100
DECKCARGO.C				29.4	27.500f	2.600
Total Lateral Plane->	63.9	25.000f	-0.691	166.2	20.333f	2.255

Distances in METERS.

Refer IRS Letter E-210766279820 dated, January 02, 2024

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

LATERAL PLANE STATUS
USK draft: 1.512 @ 50.00f, 1.512 @ 25.00f, 1.512 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	68.7	25.000f	-0.739	75.0	25.000f	0.759
CRANE				57.0	10.100f	4.000
DECKCARGO.C				29.4	27.500f	2.500
Total Lateral Plane->	68.7	25.000f	-0.739	161.4	20.194f	2.220

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.612 @ 50.00f, 1.612 @ 25.00f, 1.612 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	73.6	25.000f	-0.786	70.1	25.000f	0.708
CRANE				57.0	10.100f	3.900
DECKCARGO.C				29.4	27.500f	2.400
Total Lateral Plane->	73.6	25.000f	-0.786	156.5	20.044f	2.188

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.712 @ 50.00f, 1.712 @ 25.00f, 1.712 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	78.5	25.000f	-0.834	65.2	25.000f	0.657
CRANE				57.0	10.100f	3.800
DECKCARGO.C				29.4	27.500f	2.300
Total Lateral Plane->	78.5	25.000f	-0.834	151.6	19.883f	2.157

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.812 @ 50.00f, 1.812 @ 25.00f, 1.812 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	83.4	25.000f	-0.882	60.3	25.000f	0.607
CRANE				57.0	10.100f	3.700
DECKCARGO.C				29.4	27.500f	2.200
Total Lateral Plane->	83.4	25.000f	-0.882	146.7	19.712f	2.128

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.912 @ 50.00f, 1.912 @ 25.00f, 1.912 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	88.3	25.000f	-0.930	55.4	25.000f	0.556
CRANE				57.0	10.100f	3.600
DECKCARGO.C				29.4	27.500f	2.100
Total Lateral Plane->	88.3	25.000f	-0.930	141.8	19.529f	2.100

Distances in METERS.

Refer IRS Letter E-21076279820 dated, January 02, 2024

01-08-20 21:20:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22 **PONTOON**

LATERAL PLANE STATUS
USK draft: 2.012 @ 50.00f, 2.012 @ 25.00f, 2.012 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	93.2	25.000f	-0.978	50.5	25.000f	0.505
CRANE				57.0	10.100f	3.500
DECKCARGO.C				29.4	27.500f	2.000
Total Lateral Plane->	93.2	25.000f	-0.978	136.9	19.333f	2.073

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.112 @ 50.00f, 2.112 @ 25.00f, 2.112 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	98.2	25.000f	-1.025	45.5	25.000f	0.455
CRANE				57.0	10.100f	3.400
DECKCARGO.C				29.4	27.500f	1.900
Total Lateral Plane->	98.2	25.000f	-1.025	131.9	19.118f	2.050

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	103.2	25.000f	-1.073	40.5	25.000f	0.405
CRANE				57.0	10.100f	3.300
DECKCARGO.C				29.4	27.500f	1.800
Total Lateral Plane->	103.2	25.000f	-1.073	126.9	18.887f	2.029

Distances in METERS.

Refer IRS Letter E-210766-279820 dated, January 02, 2024

MEMORANDUM OF FREEBOARDS
1 December 2023

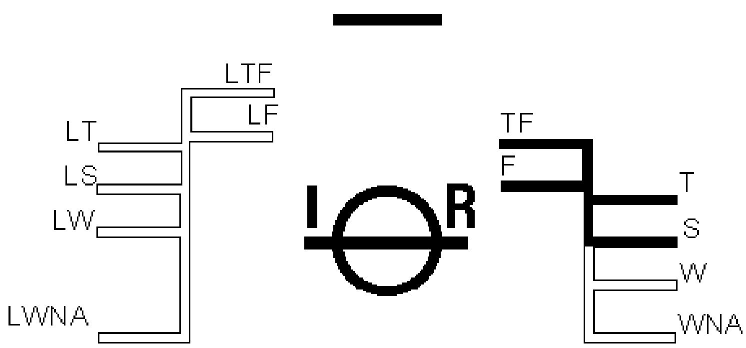
Chisti Marine Engineering Works.	Type B freeboard		
Yard No. CHISTI-036	Freeboard Length	:	48.000 m
	Summer WL Extreme Draught - load line amidships	:	2.212 m

The upper edge of the deckline from which these freeboards are measured is 0 **mm below top of * ~~Wood~~/ * steel Main Deck * at side / *~~continued to side~~. (*Delete words not applicable).**

Freeboards from Deck Line			**Seasonal, Fresh Water, or Timber allowances**	
Tropical Fresh Water	713	**mm.**	97	**mm. above summer**
Fresh Water	759	**mm.**	51	**mm. above summer**
Tropical	764	**mm.**	46	**mm. above summer**
Summer	810	**mm.**	**Upper edge of line through centre**	
Winter	Not	**mm.**	Assigned	**mm. below summer**
Winter North Atlantic	Not	**mm.**	Assigned	**mm. below summer**

Timber Tropical fresh water	—	**mm.**	—	**mm. above timber summer**
Timber fresh water	—	**mm.**	—	**mm. above timber summer**
Timber Tropical	—	**mm.**	—	**mm. above timber summer**
Timber Summer	—	**mm.**	—	**mm. above centre of ring**
Timber Winter	—	**mm.**	—	**mm. below timber summer**
Timber Winter North Atlantic	—	**mm.**	—	**mm. below timber summer**

Valid only for unmanned operation while under tow along the Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.

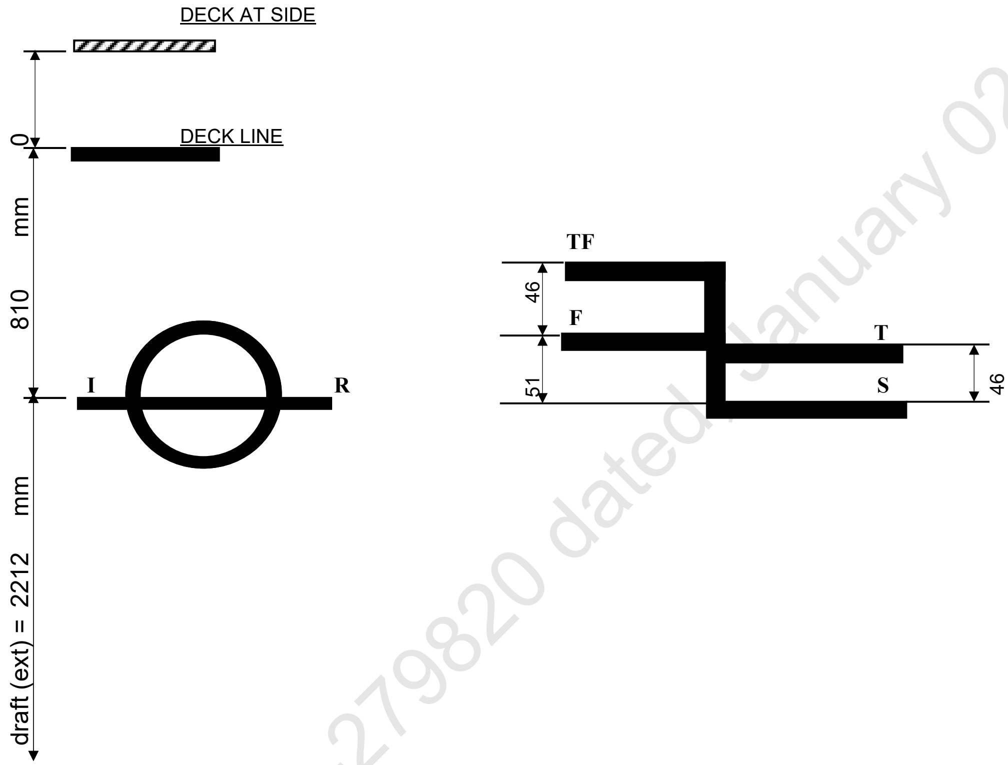


At Fr. 52

Form No. 4111 (07-83)

Refer IRS Letter E-210766-279820 dated January 02, 2024

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1893.06	1445.96
FRESH WATER	759	2.263	1892.81	1445.71
TROPICAL	764	2.258	1935.52	1488.42
TROPICAL FRESH WATER	713	2.309	1934.23	1487.13

Deck Plate Thickness = 10 mm
Keel Plate Thickness = 12 mm

Refer IRS Letter E-210766-279820 dated: January 02, 2024

52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244

Refer IRS Letter E-210766-279820 dated, January 02, 2024

Our ref: E-210469-279290

26-December-2023

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC003020
Project name Specific Yard No(s):	Chisti Marine Engineering Works - CHISTI-036 to CHISTI-038 CHISTI-036
Project Manager	Varun Yalvatkar
Project In-Charge	Amit Waje

Document Details	
Plan No.	CM23-VI-110-0207-A
Plan Title	Draft Survey Report_036
Rev.No.	0
Upload.No.	2
Submitted Date	21-December-2023
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

   Based on the draft survey conducted on 19-December-2023, the approved lightship particulars of the subject vessel are as follows:

   Lightship Weight : 447.100 T
   Lightship LCG : 25.000 m forward of AP (Frame no.0)
   Lightship VCG : 3.000 m above baseline (VCG taken at the level of main deck)
   Lightship TCG : 0.000 m off centreline

   Final stability booklet prepared based on the approved lightship particulars should be submitted for approval.

   Copy of the approved draft survey report should form part of the final stability booklet.

Thanking you,
Yours Sincerely,
For INDIAN REGISTER OF SHIPPING

Vimal R
Surveyor

This is an electronically published document and does not require signature.

APPROVED
DATE: 26 Dec 2023
INDIAN REGISTER OF SHIPPING
MUMBAI

AS AMENDED
ON PAGE NO. 2

SEE LETTER E-210469-279290

Refer IRS Letter E-210469-279290 dated December 26, 2023

DRAFT SURVEY REPORT
OF
ARCADIA SUPARSHVA
(OFFICIAL NO. JMR-0024)

LIGHTSHIP PARTICULARS:

Lightship Displacement - 447.10 T
Lightship L.C.G - 25.000 m from AP (Fr.0)
Lightship V.C.G - 3.000 m above BL (VCG taken at the level of main deck)
Lightship T.C.G - 0.000 m off-centerline

Rev.	Description	Date	Prepared	Review

Issued for : **Approval**

Client:	Project:
SMT. PARUL A. SHAH	**PONTOON (YARD NO. CHISTI-036)**
Builder:	
CHISTI MARINE ENGINEERING WORKS	

Doc #				Document Title:	Prepared	**OM**
CM23-VI-110-0207-A				**DRAFT SURVEY REPORT**	Checked	**MG**
Scale	Size	Sheet	Date		Review 1	**SR**
NTS	A4	1 of 14	20-12-2023		Review 2	

Cybermarine
www.cybermarine.net

10 Bukit Batok Crescent
#12-01 The Spire
Singapore 658 079

501 Technocity IT Premises
Plot No X-5/3 Mahape
Navi Mumbai 400 710 India

GENERAL PARTICULARS

Name of the Vessel	ARCADIA SUPARSHVA
Client	SMT. PARUL A. SHAH
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI-036
Type of Vessel	PONTOON
Officila No.	JMR-0024
Class Notation	卐SU,PONTOON INDIAN COASTAL SERVICE FOR CRANE OPERATION DECK STRENGTHENED FOR 10T/M^2

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	3.000 M
Draft (Mld.)	2.200 M

References

AP	at Fr.0
FP	at Fr.100
Midship	at Fr.50

Frame spacing

Throughout	500 mm

DRAFT SURVEY REPORT : ARCADIA SUPARSHVA

Location of Test	Sikka,Jamnagar
Date	19/Dec/23
Weather	Calm
Wind	Light
Specific Gravity	1.025
Restraining Line	Loose
Witnessed by	Mr.Rahul Parmar (KB Shipping Rep.)
	Mr.Sanjeev Sharma (IRS Surveyor)

Drafts :

(w.r.t keel line)

Location		Draft Port (m)	Draft Stbd (m)	Mean Draft (m)
Fwd	(Fr.92)	0.570	0.570	0.570
Mid	(Fr.50)	0.570	0.570	0.570
Aft	(Fr.8)	0.570	0.570	0.570

Hydrostatics

Correction for Drafts

To correct for observed drafts at marks and obtain Drafts at perps.

LBP	-	50.000	m

Observed Drafts (wrt keel line)

Fwd	-	0.570	m
Aft	-	0.570	m

Refer IRS Letter E-21004766962799820 dated, December 28, 2023

Correction to Drafts (wrt Keel line)

Location	At Fwd (Fr.92)	At Aft (Fr.8)	
Draft at Marks	0.570	0.570	m
Trim at Marks		0.000	m
Distance of Marks from PP	4.000	4.000	m
Distance between Marks		42.000	m
Draft correction factor		0.000	
Correction at PP	0.000	0.000	
Corrected Draft at PP	0.570	0.570	
Trim		0.000	m
Mean Draft		0.570	m
From Hydrostatics against above Mean Draft and Trim			
Displacement at density(T/M^3)=1.025 T/m^3		447.10	T
LCB		25.000	m
KMt		50.038	m
LCG		25.000	m
TCG		0.000	m

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

Refer IRS Letter E-21076S279920 dated December 26, 2023

```
12/20/23 10:29:31    Cybermarine Technologies PTE. Ltd.
GHS 18.66                            PONTOON

                         HYDROSTATIC PROPERTIES
                  No Trim, No Heel, Fixed VCG = 0.000

 Draft@  Displacement   Buoyancy-Ctr.  Weight/          Moment/
 25.000f---Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----KML-----KMT
  0.570       447.10  25.000f  0.278     8.24  25.000f   27.39  306.35  50.038
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.
```

Refer IRS Letter E-21047662799820 dated December 26, 2023

Weights to go off Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.00	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Weights to go on Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.000	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lightship Particulars

Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
Particulars as Draft Survey	447.10	25.000	11177.500	3.000	1341.300	0.000	0.000
Weights to go Off board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Weights to go on board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Lightship Particulars	447.10	25.000	11177.50	3.000	1341.30	0.000	0.00

Lightship Particulars

Displacement	**447.10**	**T**
LCG	**25.000**	**m From AP**
TCG	**0.000**	**m Off-Centreline**
VCG	**3.000**	**m above BL**

NOTE
1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

Refer IRS Letter ES21047669627992 dated December 26, 2023

LIGHT-SHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part I

Pre requisites for Conducting Light-Ship Survey
(To be provided /confirmed by Builder/Owner/Operator/Consultant prior commencement of Light-Ship Survey)

Name Of Builder/owner rep conducting trial	Chisti marin Eng. Works.
Hull Number/Vessel name	Chisti /036 - ARCADIA SUPARSHVA
Type of Vessel (Cargo/passenger)	Pontoon.
Location of Vessel	Sikka, Jamnagar.
Date of Proposed Light-Ship Survey	19/12) 2023
Draft of Vessel and trim	0.57 and 'O'
Depth of Water	5.0M
Anticipated tide on date of survey	5.0 M
Weather prediction for proposed date	CALM
Availability of following reports from builder/Owner/Owners rep	
Copy of Approved Inclining experiment report	☐ YES ☐ NO NA.
Details of Periodical Draft Survey report/any Draft Survey report conducted after the approval of Final TSB.	☑ YES ☐ NO
Declaration of weights to go on	☐ YES ☐ NO NA.
Declaration of weights to go off -tank contents - Loose items	☐ YES ☐ NO NA.

Refer IRS Letter E-21017662799820 dated December 28, 2023

Facilities/Arrangement for conduct of Light-Ship Survey	
Availability of boat to take draft reading	☑ YES ☐ NO
Availability of draft mark plan/approved document like Prelim TSB/Final TSB for confirming location of draft marks	☑ YES ☐ NO
Safety Requirements- lighting, gas freeing, accessibility of areas requiring inspection as part of Survey	☑ YES ☐ NO
Availability of sounding tapes, hydrometer etc	☑ YES ☐ NO
Closing/isolating cross connected tanks	☐ YES ☐ NO NA.
Removal of gangway during draft reading/experiment	☑ YES ☐ NO
Permanent ballast used ? ; if any and its details	☐ YES ☑ NO
If vessel has any list, reason for the list is established.	☐ YES ☑ NO
A trial is conducted to see whether the vessel's draft corresponding to the loading condition is matching with the approved hydrostatics.	☐ YES ☑ NO

NOTE : The responsibility of the IRS Surveyor is to verify that the survey is conducted according to accepted procedures and that all basic measurements and data are correctly taken and recorded. The responsibility for making preparation, conducting lightship survey and preparing report rest with owner/operator.

Confirm Readiness for test

Name & Signature of Shipyard /Owners rep

K. B. SHIPPING & Co. JAMNAGAR

INDIAN REGISTER OF SHIPPING

Refer IRS Letter E2210466279982 dated, December 26, 2023

<u>Sheet for recording Weights to Go-on, Go-off, relocated and Tank Soundings</u>

Weights To Go-on						
Sl No.	Weight Description	Weight (t)	L.C.G (m)	V.C.G (m)	T.C.G (m)	Surveyor Verification
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Refer IRS Letter E-21047662799280 dated, December 22, 2023

Weights To Go-off

Sl No.	Weight Description	Weight (t)	L.C.G (m)	V.C.G (m)	T.C.G (m)	Surveyor Verification
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐
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						☐
						☐
						☐
						☐
						☐
						☐
						☐
						☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Tank Soundings

Sl No.	Tank Name	Sounding (cm /m)	Density	Surveyor Endorsement
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
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				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐

Note: Tank name should be mentioned as per the approved TSB / Capacity Plan.

Refer IRS Letter E-22104/0627/280 dated, December 26, 2024

Weights To Be Relocated

Sl No.	Weight Description	Weight (t)	L.C.G (m)		V.C.G (m)		T.C.G (m)		Surveyor Verification
			Initial	Final	Initial	Final	Initial	Final	
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐
									☐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

Refer IRS Letter E-21047662799820 dated December 26, 2023

LIGHTSHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part II

REPORTING FORM FOR LIGHTSHIP SURVEY

Attending Surveyor should witness Light-Ship Survey only if the Part-I submitted by Owner/Builder/Operator/Consultant is found satisfactory

General Particulars

Name of Vessel	ARCADIA SUPARSHVA
Builder/ Owner	CHISTI MARINE ENGG. WOR
Yard Number/ IR Number	CHISTI /036
Type of Vessel — Passenger	☐ PONTOON
Non Passenger	☑
Date(dd/mm/yyyy);Local Time of inspection/test	19/12/2023, 1500
Wind / Weather condition	5 KN / CALM
Location of Test	SIKKA
Depth of water where vessel is moored, m	5.0M
Method of reading Draft(By Boat-Y/N)	BOAT

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	Fr. 08	0.570	0.570
Midship	Fr. 50	0.570	0.570
Forward	Fr. 92	0.570	0.570
Note : *Recommended to attach Photographs of draft; as record*			

Density of Water

Results of 3 samples from Forward, Aft and Mid of the vessel		
Sample 1: (ford)	1.025	**t/m3**
Sample 2: (mid)	1.025	**t/m3**
Sample 3: (aft)	1.025	**t/m3**

Refer IRS Letter [illegible] dated December 26, 2023

INDIAN REGISTER OF SHIPPING

Verification of weights to go on / go off /re-location	*SEE PART 1*
Note: *See Part 1 (the sheet for recording the above data) , verify and endorse.*	

Special Comments on Light Ship Survey, if any.
DRAFT SURVEY COMPLETED SATISFACTORILY AT SIKKA.

Attending surveyor to forward both forms Part 1 and Part 2 together with completed attachments 1 and 2 to HO.

Name & designation of Attending Surveyor: -- Sanjeev Singh
19/12/2023
At SIKKA

INDIAN REGISTER OF SHIPPING

Refer IRS Letter E-2210462779820 dated, December 26, 2023

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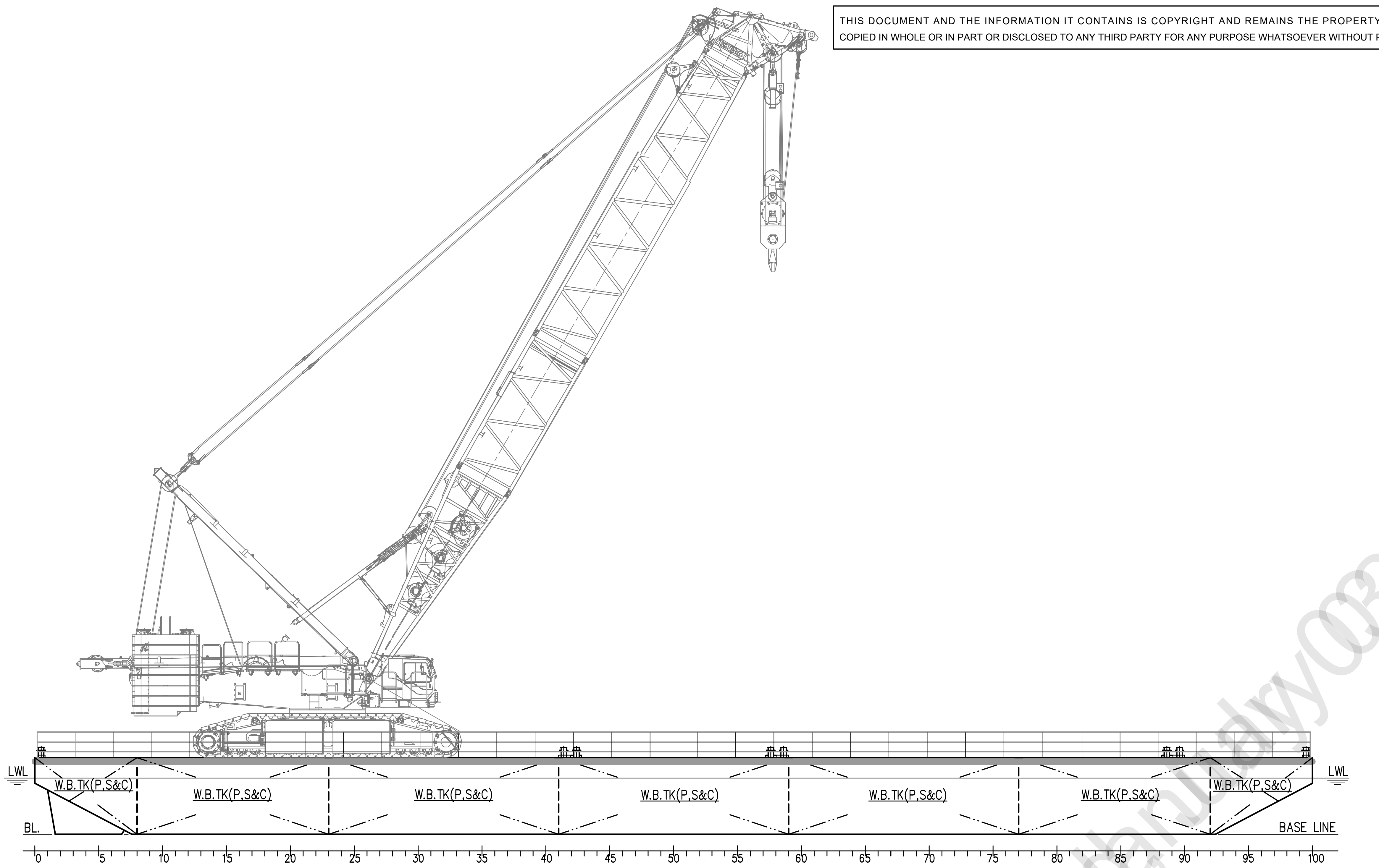
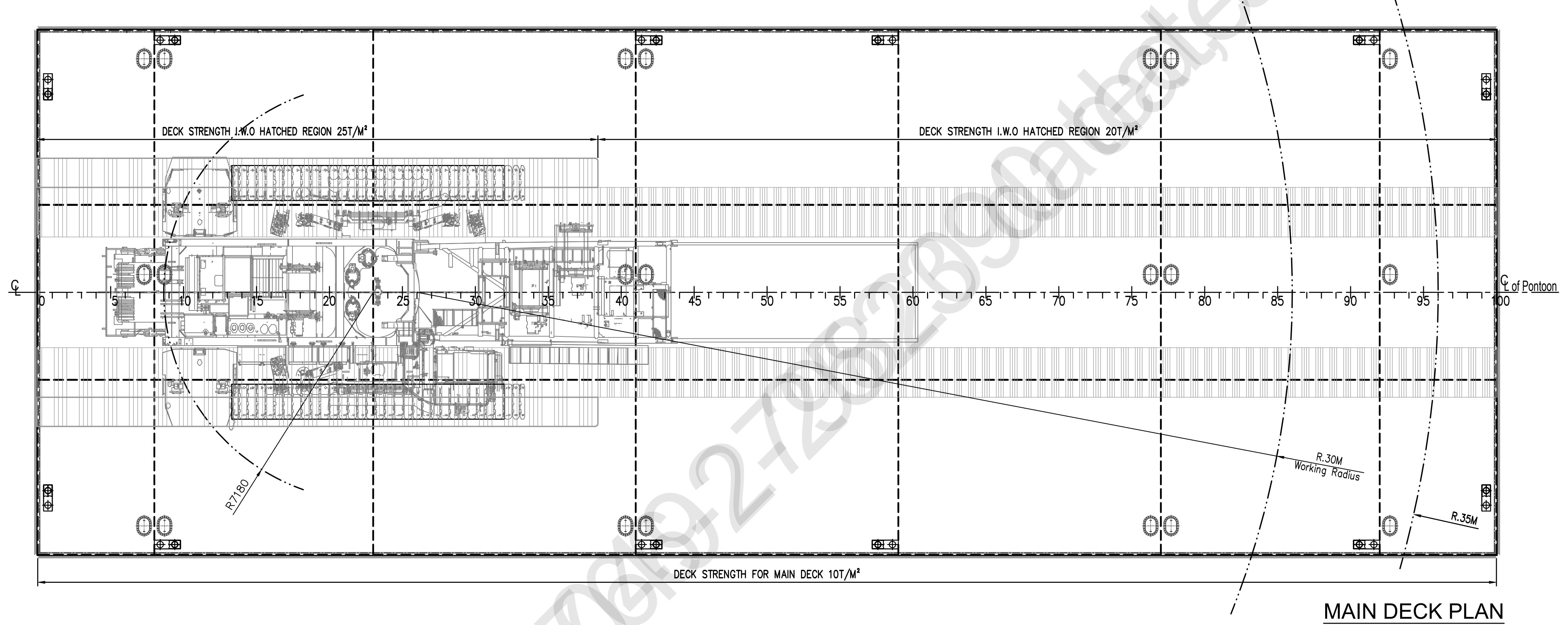
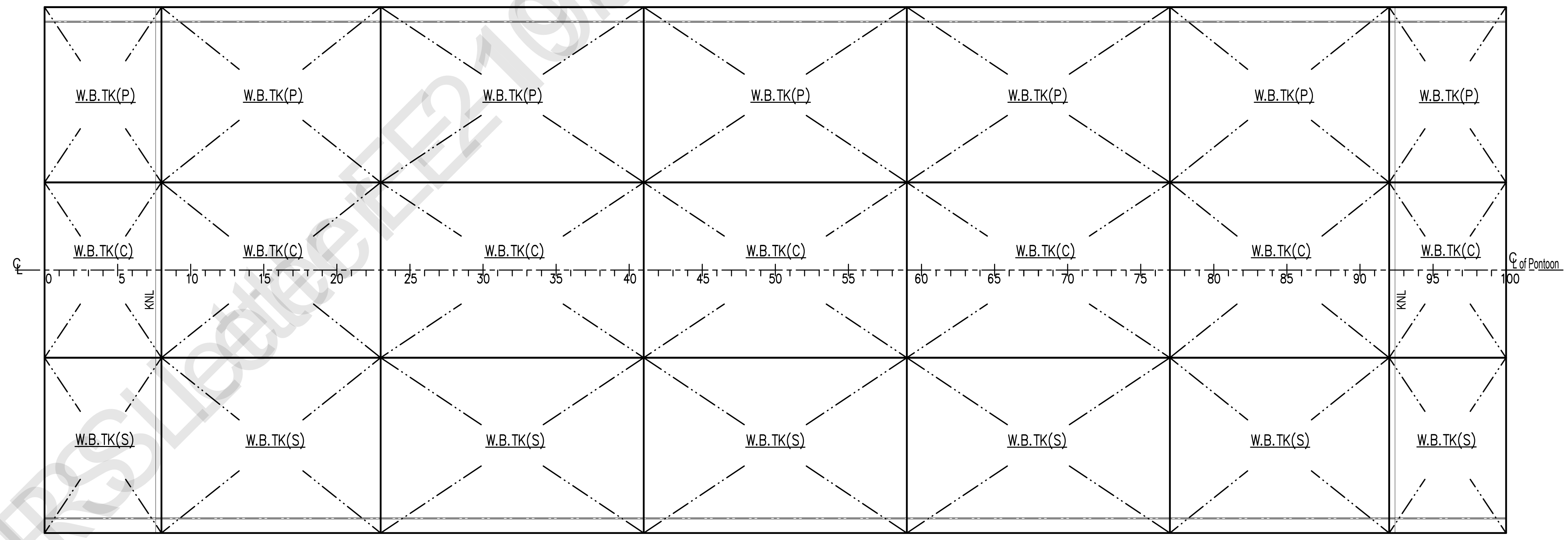


PROFILE



MAIN DECK PLAN



BELOW DECK PLAN

PRINCIPAL PARTICULARS:-

LENGTH O.A	:	50.00 M
BREADTH MLD.	:	18.00 M
DEPTH MLD.	:	3.00 M
DESIGN DRAFT	:	2.20 M
SCANTLING DRAFT	:	2.60 M
FRAME SPG.	:	500 mm

CRANE SPECIFICATION:-

CRAINE WEIGHT	=	400T
CRANE CAPACITY	=	40 T
WORKING RADIUS	=	30 M

SEE LETTER E-190492-252390

INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 03-Jul-2023
NOTED FOR INFORMATION

CLASS NOTATION:-

卐 SU, PONTOON, "INDIAN COASTAL SERVICE"
"DECK STRENGTHENED FOR 10 T/M^2 ",
"FOR CRANE OPERATION"

NOTE:

1) ALL DIMENSION ARE IN MM.
2) CRANE WILL OPERATE ONLY IN THE HATCHED REGION.

Rev.#	Description	Date	Drawn	Review

Issued for : Approval

Client: K.B. SHIPPING & CO.

Project: PONTOON 036-038 (YARD NO. CHISTI - ~~036 & 037~~)

Builder: CHISTI MARINE ENGINEERING WORKS

Drg.#	Rev.#	Drawing Title	Drawn	ISHWAR
CM23-VI-110-0001	00	GENERAL ARRANGEMENT	Checked	JASIM
Scale: 1:175 / Size: A2	Date: 29.05.2023	Sheet: 1 OF 1	Review-1	MOHIT
			Review-2	SUDHEESH

Cybermarine
www.cybermarine.net

10 Bukit Batok Crescent
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Singapore 658 079
Tel : +65 6316 6513

501 Technocity IT Premises
Plot No X-5/3 Mahape
Navi Mumbai 400 710 India
Tel : +91 22 61462400 (10 Lines)

Refer IRS Letter E-190492-252390 dated July 03., 2023